MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

Project Customer: Mansourah Massarah Gold Project
MA'ADEN GOLD AND BASE METALS (MA'ADEN)

Title:
CALCULATION
SCREEN BUILDING
DYNAMIC ANALYSIS OF BUILDING STRUCTURE

Document Number: MD-216-3120-EG-CV-CAL-1008
Revision: D01
Revision Date: 01-02-2024
Status: AS BUILT

Outotec Document Number:	900707	GAU01	ZC01	MED06	31201	OU100755136	5
L&T Document Number:	O19045	100	C	K	24776	001	5

REVISION HISTORY

Rev	Date	Revision Text	Prepared By	Reviewed By	Approved By	Employer
A01	07-06-2022	ISSUED FOR REVIEW	DPG	SBY	AG	
A02	04-07-2022	ISSUED FOR REVIEW	DPG/RPK	SBY	AG	
A03	08-08-2022	ISSUED FOR REVIEW	DPG/RPK	SBY	AG	
A04	23-08-2022	ISSUED FOR REVIEW	DPG/RPK	SBY	AG	
C01	01-02-2024	ISSUED FOR CONSTRUCTION	DPG/RPK	SBY	AG	
D01	01-02-2024	AS BUILT	DPG/RPK	SBY	AG	

PREFACE

VIBRATION PROBLEM – SCREEN BUILDING VIBRO FEEDER FLOOR

A. The Problem:

The screen building houses three screen and three feeders. When Tertiary Vibro-feeder 3130-FDR-002 was run individually at 100% speed, without material, vibration was observed at the floor slab adjoining Vibro-feeder 3130-FDR-002. The vibrations were also observed during trial runs in loaded condition. The vibrations were measured using EXTECH 407860: Heavy Duty Vibration Meter and the results were found to be above the project specific limit of 3mm/sec. The performance of all the three screens is meeting the project specified vibration limit of 3mm/s and no perceptible vibration was observed anywhere else on the structure.

B. Root Cause Analysis

Static analysis of the screen building was carried out in Staad Software during execution stage along with Modal analysis to check the frequency separation from the exciting frequency of the feeders as well as screen. The root cause can be as follows:

- Misinterpretation of the forced vibration nature of the equipment
- Misinterpretation of the feeder operation process where the feeding rate is controlled by a range of speed but not only one, and
- Incompliance with project requirements about being away from the range of excitation frequencies and vibration amplitude limit.
- Forced vibration analysis of the structure to determine the vibration velocity was not performed earlier.

C. The Probable Solutions:

The different alternatives available to control the vibrations are:

- Stiffening of the Building Structure to reduce the vibration velocity
- Reduction of dynamic load of equipment
- Vibration Isolation System for the equipment

Several options were evaluated to strengthen the structure so that the vibration velocity can be brought down to the project specified limit of 3mm/s. The strengthening of structure considerably reduces the vibration velocities at most part of the structure, however the same exceeds the limit at few corners. Also, the proposed strengthening measures have interface issues with other disciplines hence ruled out. The various strengthening options are given in ***Appendix 2*** to the report.

The possibility of the reduction of dynamic loads for the Tertiary feeders 3130—FDR-001 & 002 was investigated by the equipment supplier and the same was not possible without reduction in equipment capacity which will advertently affect the overall capacity of the plant and hence ruled out.
The secondary screen feeder is supported on double spring to suppress vibration similar vibration isolation system in tertiary feeders will either eliminate all the dynamic loads of the tertiary screen feeders or reduce them drastically. A detailed dynamic analysis of the modified

PROPRIETARY & CONFIDENTIAL
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

structure has been carried out to ascertain the same. The loads provided by the supplier of vibration isolation system are used for the calculations. Vibrations at all points of both feeder and screen floor are well within the project specified limits of 3mm/sec with some modifications to floors under tertiary feeders and screens.
However, for those column and beam joints subject to excessive vibration during the commissioning of original tertiary vibration feeder, non-destructive tests will be undertaken to make sure the concrete structure is in good condition. Strengthening of the joints will be necessary if the test results show possible defective joints.

D. Conclusion:

On the basis of the above analysis, we propose to go ahead with introduction of vibration isolation system for the Tertiary screen feeders 3130-FDR-001 & 002 as this option reduces the vibration level at all points of the building to project specified limit of 3mm/sec. The schematic arrangement of the feeder with spring supports are given in ***Appendix 5*** to the report. Also strengthening at few points (**Appendix 1**) using extra steel bracings and beams will eliminate any possible vibrations in the feeder and screen floor. All these options eliminated the strengthening measures for the existing feeder floor concrete beams as the same is not required from static or dynamic calculations.
However, for those column and beam joints subject to excessive vibration during the commissioning of original tertiary vibration feeder non-destructive tests will be undertaken to make sure the concrete structure is in good condition. Strengthening to the joints will be necessary if the test results show possible defective joints.

PROPRIETARY & CONFIDENTIAL
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

List of Contents **Pages**

1. Introduction 5
2. Design Basis 7
   - 2.1. Geometry of the Model 7
   - 2.2. Material Properties 7
   - 2.3. Damping Property 8
   - 2.4. Lump Mass Modelling of Feeder 9
   - 2.5. Lump Mass Modelling of Screen Floor 16
   - 2.6. Lump Mass Modelling Below Screen Floor 34
   - 2.7. Lump mass model of steel structure 40
3. Modelling of Dynamic Loading 40
   - 3.1. Dynamic Load, 100% Operational Speed for screens and Variable speed for feeders 46
   - 3.2. Philosophy of the Operational Dynamic Loadings 46
   - 3.3. Boundary Conditions 52
4. Analysis of the Modified Structure 52
   - 4.1. Complete Model 53
   - 4.2. Mesh 54
   - 4.3. Modal Analysis 54
   - 4.4. Harmonic Analysis 63
5. Conclusions 77
6. References 78
7. Appendix 1: Modified Feeder & Screen Floor 79
8. Appendix 2: Various Strengthening options 82
9. Appendix 3: Measuring Points 86
10. Appendix 4: Summary of RMS Velocity at measuring points 88
11. Appendix 5: Analysis with additional frequency 99
12. Appendix 6: Vibration Isolation System 104

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

1. Introduction

The screen building houses 1 no. secondary screen and 2 nos. of tertiary screen at 922.75 m level, all the screens will operate at constant speed. The screens are being fed from vibro feeders placed at 926.5 m level. To regulate the feeding rate, the speed of the secondary vibro feeder will vary from 650 to 980 RPM while the speed for the tertiary feeders will vary from 550 to 900 RPM.

During testing of Feeders of the screen building, excessive vibrations of the floor adjacent to tertiary screen feeder 3130-FDR-002 was observed at site. During subsequent investigations it was found that that very high dynamic loads of the tertiary feeder is causing this excessive vibration. Secondary screen feeder though much larger, is imparting very less dynamic loads to the building due to robust isolation system.

After observing vibrations on the feeder floor at the area adjacent to the feeder 3130-FDR-002, a quick analysis of the structure was done, and as a solution, a structural modification scheme was proposed on 13.05.2022 to reduce the beam span between grids 3120/k & L. The same is given in ***Appendix 1*** of the report. Vibration with reduced amplitude was observed even after these modifications and thereafter a detailed dynamic analysis of the modified structure was carried out for the different scenarios:

- All three Vibro-feeder operating at 100% speed with & without material

In this condition it is found that the calculated vibration of the structure is exceeding the allowable limit at area adjacent to the feeder 3130-FDR-002. ***This is the reason why high vibration is being felt on the slab when the feeders are run at 100% speed.***

Further detailed dynamic analysis of the modified structure has been carried out subsequently, for the following scenarios:

- All Vibro-feeder operating at 50% speed with/without material
- All Vibro-feeder operating at 60% speed with/without material
- All Vibro-feeder operating at 70% speed with/without material

For all these conditions it is found that the calculated vibration of the structure is within allowable limit and matches the measured values obtained during various runs at site for without material condition. The runs were continued for the plant at 50% speed with material loads with continuous measurement.

This report is a verification of the modified screen building, using finite element analysis in Ansys software. The main purpose of this analysis is to investigate Steady state dynamic behavior of the modified system with harmonic analysis to ensure that potential excitation forces will not cause vibration that might jeopardize safe operation of the equipment and building. Two sets of harmonic analyses are performed considering empty and loaded conditions of the feeder and screen floors. To arrive at the optimal solution to reduce the vibrations levels to the project specified limit, the following alternate options have also been evaluated.

- ***Stiffening of the Building Structure to reduce the vibration velocity***: - Several options were evaluated to strengthen the structure so that the vibration velocity can be brought down to the project specified limit of 3mm/s. The various strengthening options are given in ***Appendix 2*** to the report. The strengthening of structure considerably reduces the vibration velocities at most part of the structure, however

PROPRIETARY & CONFIDENTIAL
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

the same exceeds the limit at few corners. Also, the proposed strengthening measures have interface issues with other disciplines hence ruled out.

- ***Reduction of dynamic load of equipment:*** -The possibility of the reduction of dynamic loads for the Tertiary feeders 3130—FDR-001 & 002 was investigated by the equipment supplier. The reduction is not possible without compromising in equipment capacity which will advertently affect the overall capacity of the plant and hence unacceptable from operational aspect.

- ***Vibration Isolation System for the equipment***: - The secondary screen feeder is supported on double spring to suppress vibration similar vibration isolation system will either eliminate all the dynamic loads of the tertiary screen feeders or reduce them drastically. A detailed dynamic analysis of the modified structure has been carried out to ascertain the same. The loads provided by the supplier of vibration isolation system are used for the calculations. Vibrations at all points of both feeder and screen floor are well within the project specified limits of 3mm/sec as elaborated in subsequent sections.

PROPRIETARY & CONFIDENTIAL
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

2. Design Basis

2.1. Geometry of the Model

Screen building consists of concrete columns, beams, and slabs.

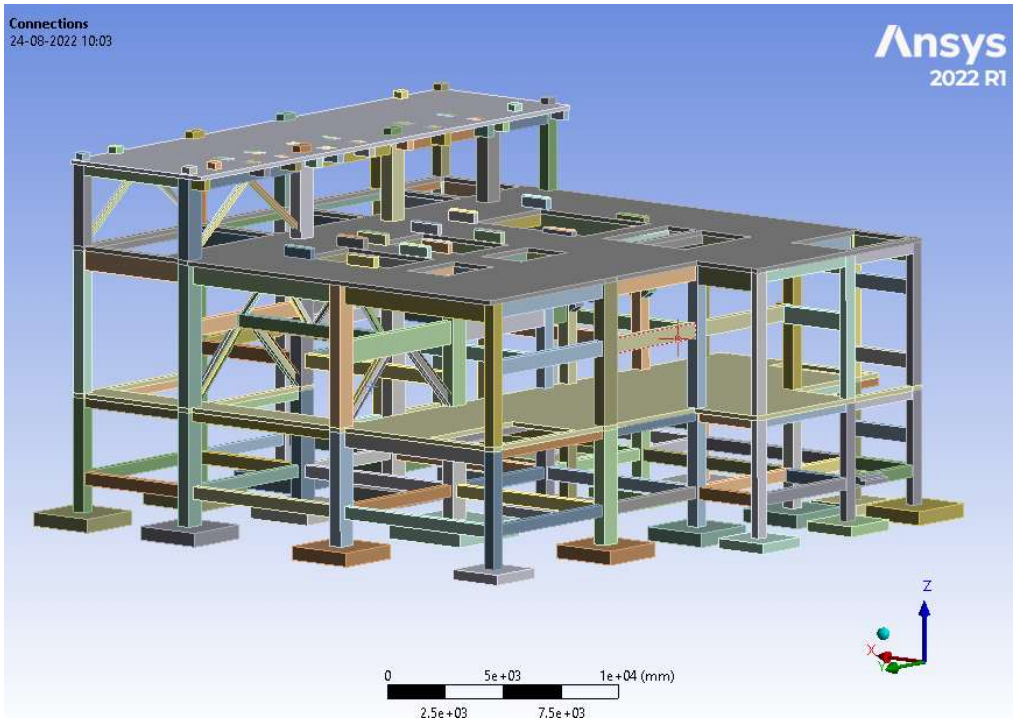


Figure 2a. Geometry of Screen building

The mass from steel superstructure is transferred to the support locations of columns and the steel members were not modelled to reduce computation time.

2.1.1. Modified Structure

The modified structure with installation of bracing and steel beam at feeder floor and below tertiary screen supporting beam has been attached in **Appendix 1**.

2.2. Material Properties

The material properties of concrete shown in Table 2a.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Properties of Outline Row 3: Concrete C35

	A	B	
1	Property	Value	
2	Material Field Variables	Table	
3	Density	2500	kg m^-3
4	Isotropic Secant Coefficient of Thermal Expansion		
6	Isotropic Elasticity		
7	Derive from	Young's Modulus and...	
8	Young's Modulus	2.7805E+10	Pa
9	Poisson's Ratio	0.18	
10	Bulk Modulus	1.4482E+10	Pa
11	Shear Modulus	1.1782E+10	Pa

Table 2a. Material properties of concrete

2.3. Damping Property

Damping of 5% is considered for dynamic analysis, shown in Table 2b.

Details of "Analysis Settings"

Options	
Rotordynamics Controls	
Output Controls	
Damping Controls	
Constant Damping Ratio	5.e-002
Stiffness Coefficient Define By	Direct Input
Stiffness Coefficient	0.
Mass Coefficient	0.
Analysis Data Management	

Table 2b. Damping properties for harmonic analysis

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

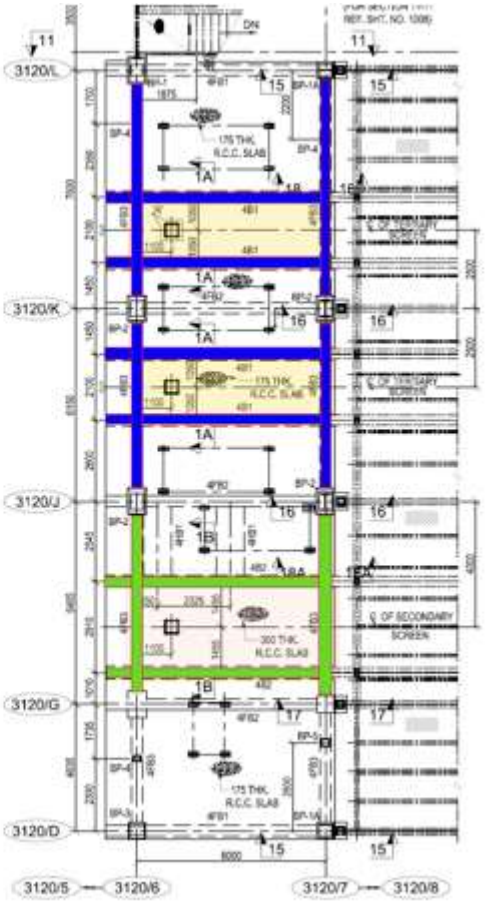
2.4. Lump Mass Modelling of Feeder

Mass model of the Feeder, listed in Table 2c, is considered as equivalent amount mass of material for simplification of the model in finite element analysis. Modelling is done taking information from load data given by equipment manufacturer shown A, B, 1, & 2 loading as follows:

3130-FDR-002
TERTIARY SCREEN-2
VIBRATING FEEDER

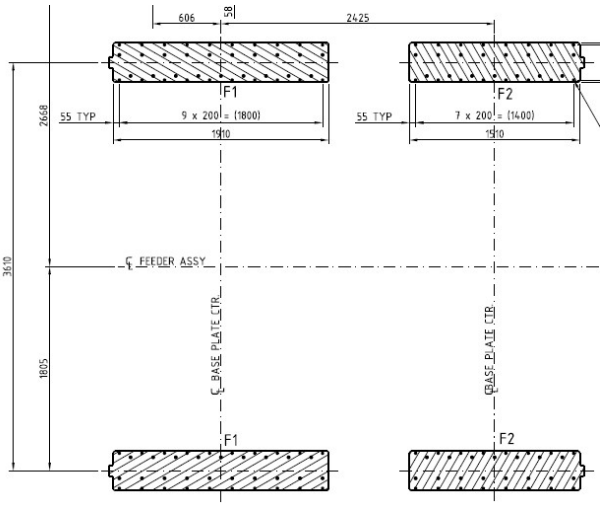
3130-FDR-001
TERTIARY SCREEN-1
VIBRATING FEEDER

3120-FDR-001
SECONDARY SCREEN
VIBRATING FEEDER



PROPRIETARY & CONFIDENTIAL
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

2.4.1. Lump Mass Modelling of Secondary Screen Vibrating Feeder (3120-FDR-001)



FEEDER LOADS AT EACH SUPPORT POINT
IN NEWTONS, UNDER ISOLATION FRAME
INCLUDES FACTOR OF SAFETY - 10% STATIC & 20% DYNAMIC

WITH MAT'L	WITHOUT MAT'L	WITHOUT MATERIAL DYNAMIC LOADS - ZERO TO PEAK											
STATIC	STATIC	DRIVE SPEED	650 RPM		732.5 RPM		815 RPM		897.5 RPM		980 RPM		MAINTENANCE
VERTICAL	VERTICAL	SPR. REACTION	FEED.F1	DISCH.F2	FEED.F1	DISCH.F2	FEED.F1	DISCH.F2	FEED.F1	DISCH.F2	FEED.F1	DISCH.F2	WHEEL LOAD
FEED END F1 : 111590 DISCH. END F2 : 87666	FEED END F1 : 57040 DISCH. END F2 : 49265 DRIVE F3 : 2354	VERTICAL	1718	1431	1050	875	729	608	544	454	426	355	VERTICAL FEED END F4 : 58399 DISCH. END F5 : 49506
		LONGITUDINAL	133	111	100	83	78	65	63	53	52	43	
		TRANSVERSE	27	22	20	17	16	13	13	11	10	9	
		MAX OVERTURNING MOMENT AT F3 : 965 Nm											

EXCITER MODEL : DF601S (1 OFF)
SETTING : X0X0X
X = WITH STEEL WEIGHT; 0 = WITHOUT WEIGHT; P = WITH LEAD WEIGHT; H = HALF
MIN. OPERATING SPEED : 650 RPM MAX. OPERATING SPEED : 980 RPM

Table 2c. Load data from [MD - 216 -3120 - EG- ME -GAD -V500-1008 Rev C01] [2]

PROPRIETARY & CONFIDENTIAL
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Empty Conditions:

- Mass of 1: 28.5 kN = around 2850 kg each
- Mass of 2: 24.5 kN = around 2450 kg each

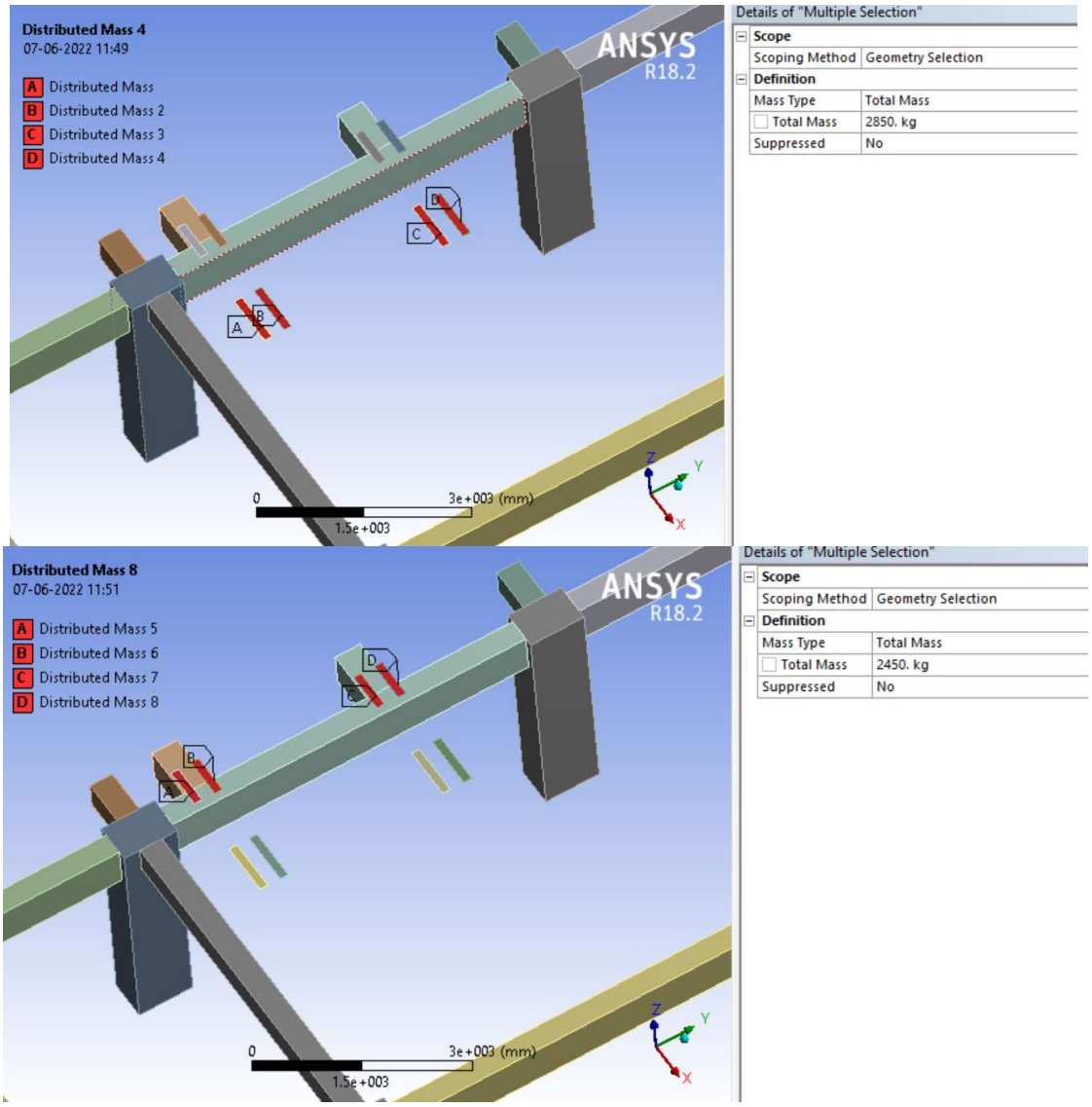


Figure 2b. Lumped mass of the Feeders at support: Unloaded

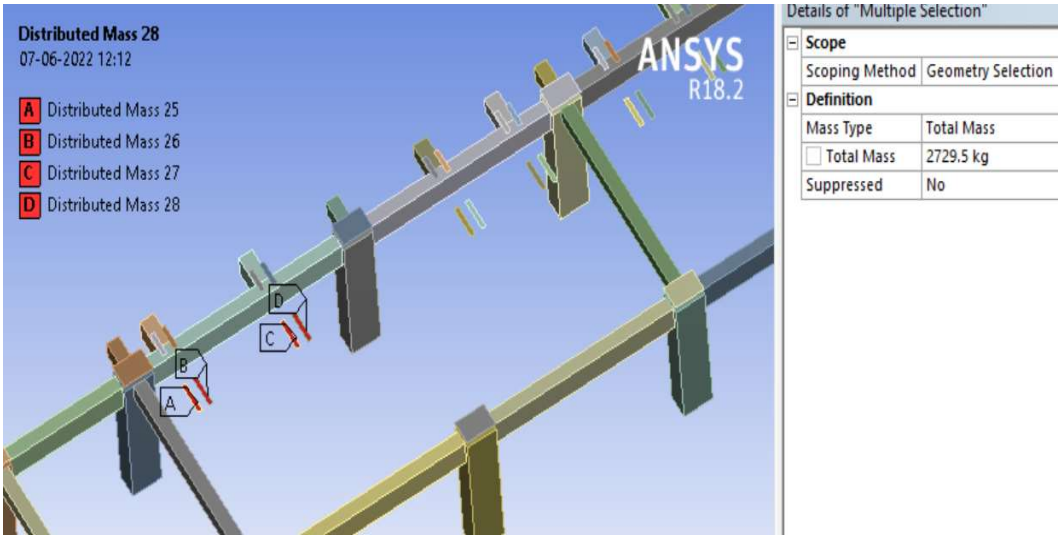
PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Material Weight:

- Mass of 1: 27.295 kN = around 2729.5 kg each
- Mass of 2: 19.20 kN = around 1920 kg each



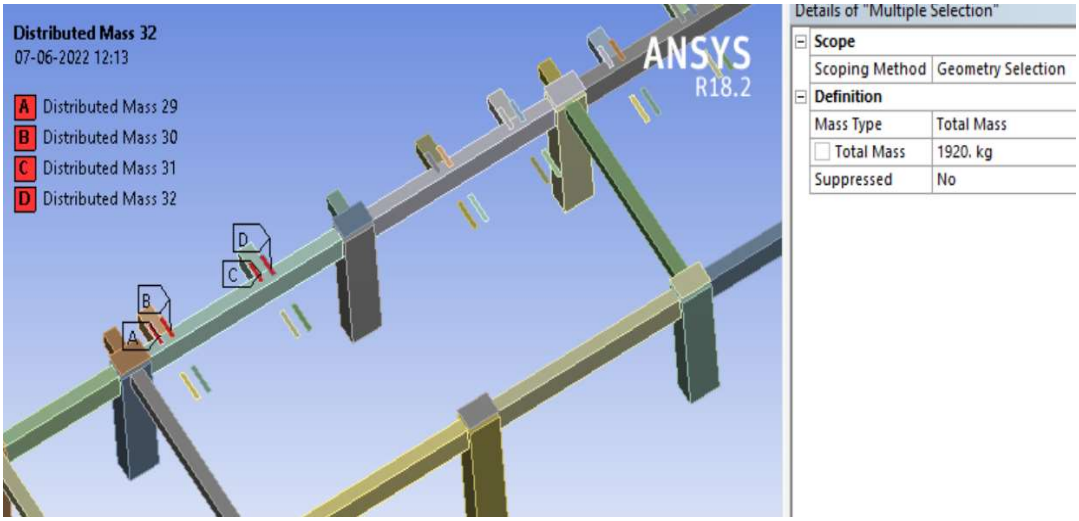


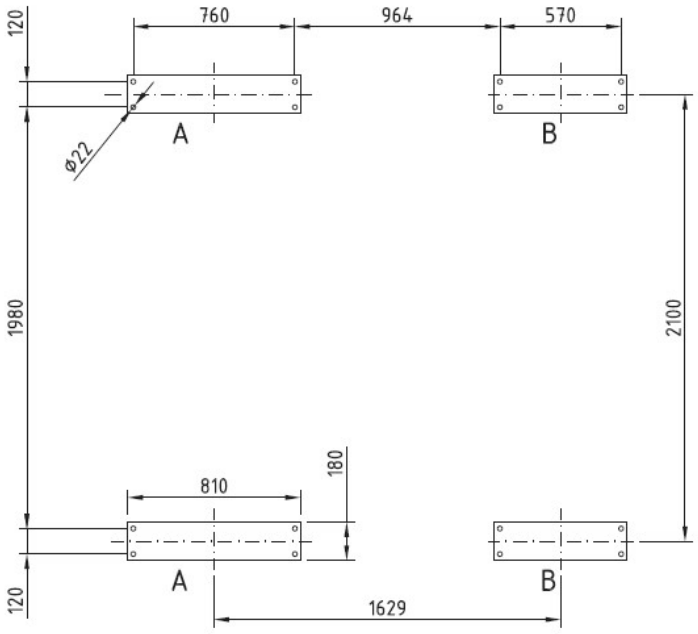
Figure 2c. Lumped mass of the Feeders at support: Material

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

2.4.2. Lump Mass Modelling of Tertiary Screen Vibrating Feeder (3130-FDR-001/002)



Tertiary Feeder Mass Table

	Empty Mass (kg)				Material Mass
Location	Machine	Stool	RCC Block	Total	Total (kg)
A	2060	575	1825	4460	6200
B	890	575	1825	3290	6180
Total	5900	2300	7300	15500	24760

Table 2d. Load data from [MD - 216 -3120 - EG- ME -GAD -V500-1001 Rev C01] [2] + Vibration isolation input

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Empty Conditions:

- Mass of A: 4460 kg = around 1115 kg at each load point
- Mass of B: 3290 kg = around 1645 kg at each load point

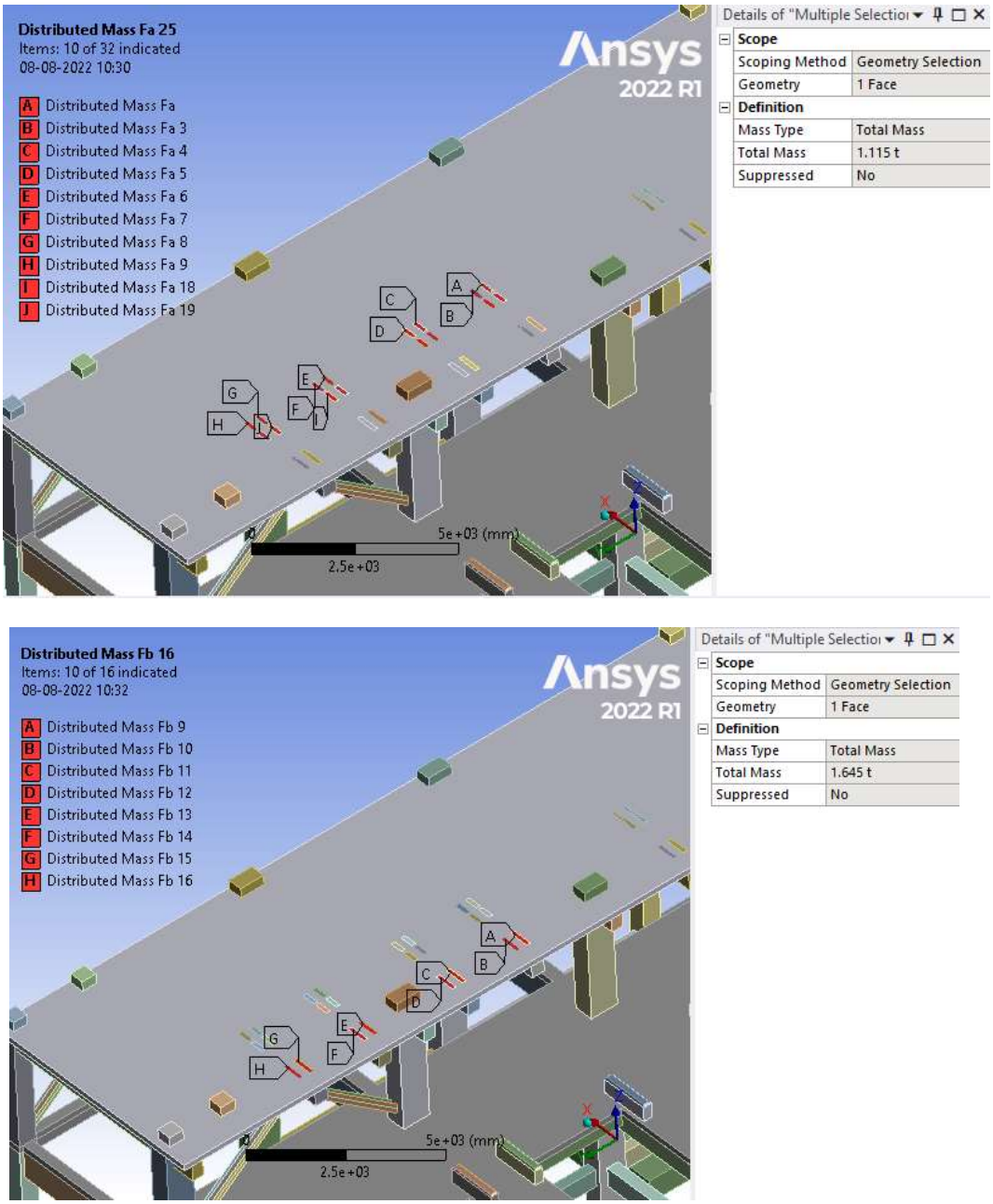


Figure 2d. Lumped mass of the Feeders at support: Unloaded

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Material Weight:

- Mass of A: 6200 kg = around 1550 kg at each load point
- Mass of B: 6180 kg = around 3090 kg at each load point

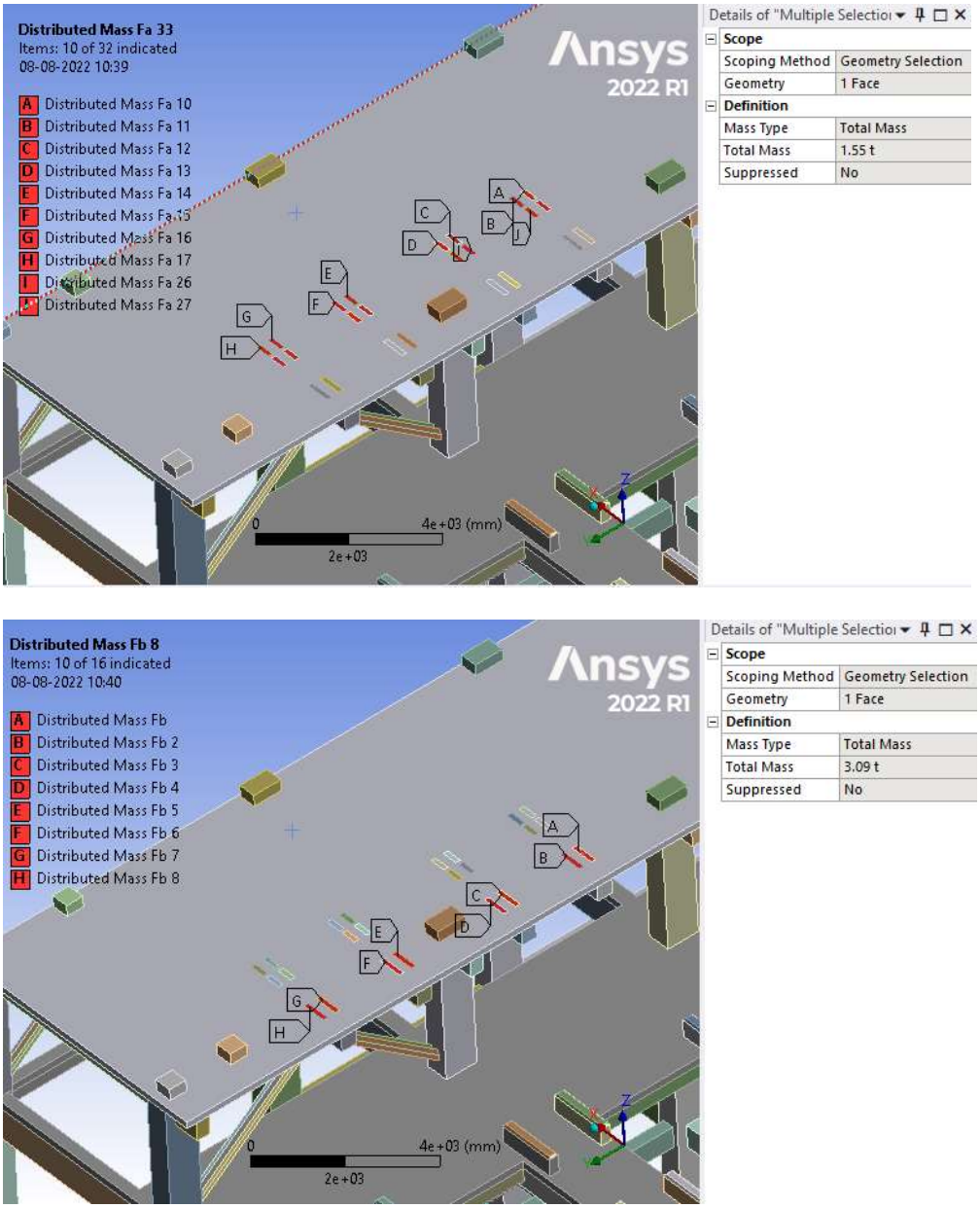
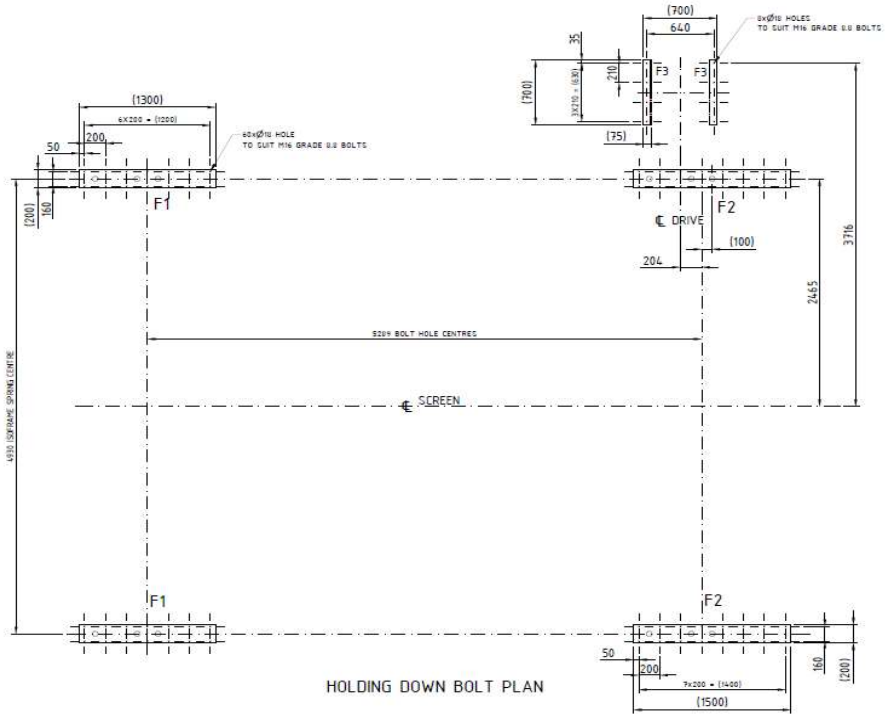


Figure 2e. Lumped mass of the Feeders at support: Material

PROPRIETARY & CONFIDENTIAL
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

2.5. Lump Mass Modelling of Screen Floor

2.5.1. Lump Mass Modelling of Secondary Screen (3120-SCR-001)



HOLDING DOWN BOLT PLAN

LIVE LOAD DUE TO MATERIAL	SCREEN LOADS AT EACH SUPPORT POINT IN NEWTONS, WITH ISOLATION FRAME & WITHOUT MATERIAL INCLUDES FACTOR OF SAFETY			
STATIC	STATIC	DYNAMIC		
VERTICAL	VERTICAL	VERTICAL	LONGITUDINAL	TRANSVERSE
FEED END F1 = 8222 DISCHARGE END F2 = 26518	FEED END F1 = 141397 DISCHARGE END F2 = 192567 DRIVE F3 = 7282	FEED END F1 = 1129 DISCHARGE END F2 = 1318 DRIVE F3 = 220	FEED END F1 = 158 DISCHARGE END F2 = 184 DRIVE F3 = 220	FEED END F1 = 79 DISCHARGE END F2 = 92 DRIVE F3 = 70
MAX OVERTURNING MOMENT AT F3: 1525 Nm				
EXCITER MODEL : DF704S VIBRATOR MOTOR MODEL : N/A SETTING : OXXXO X=WITH STEEL WT; O=WITHOUT WT; H=INNER SEGMENT ONLY; P=WITH LEAD WT. SETTING 856±10% RPM NOM.				

Table 2e. Load data from [MD-216-3120-EG-ME-GAD-V500-1005-C01 Rev C01]

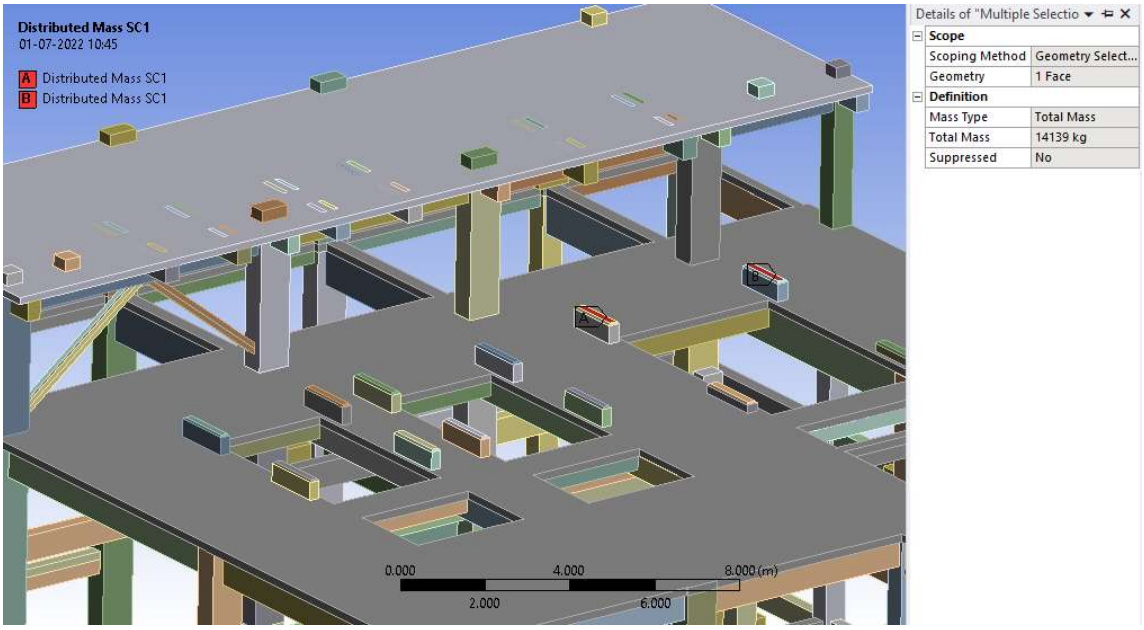
PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Empty Conditions:

- Mass of SC1(F1): 141397N = Around 14139 kg each
- Mass of SC2(F2): 192567N = Around 19256 kg each



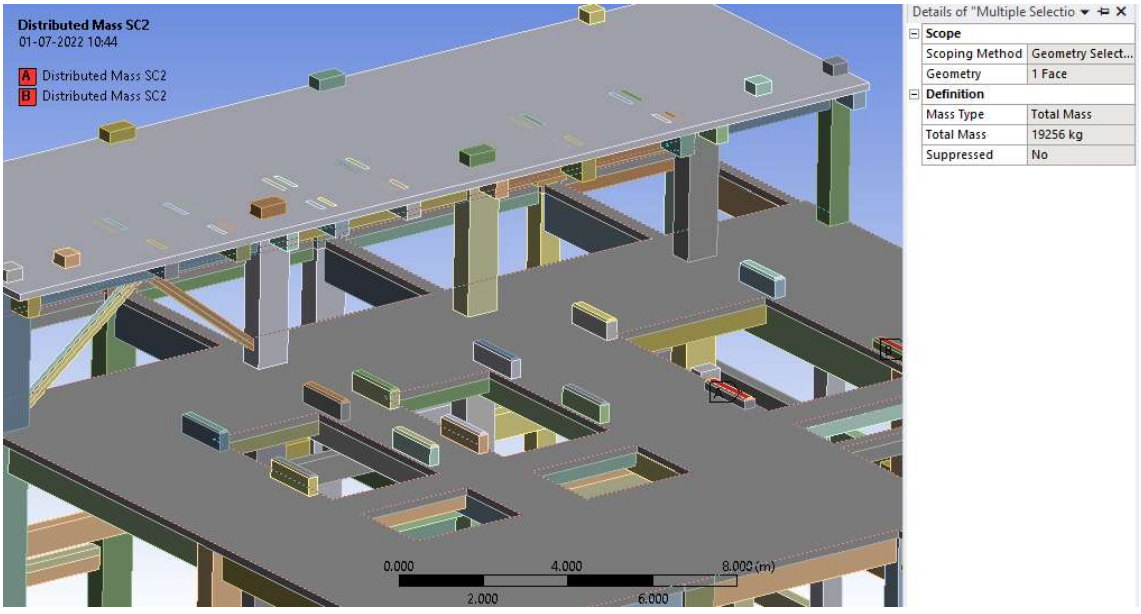


Figure 2f. Lumped mass of the Secondary Screen at support: Unloaded

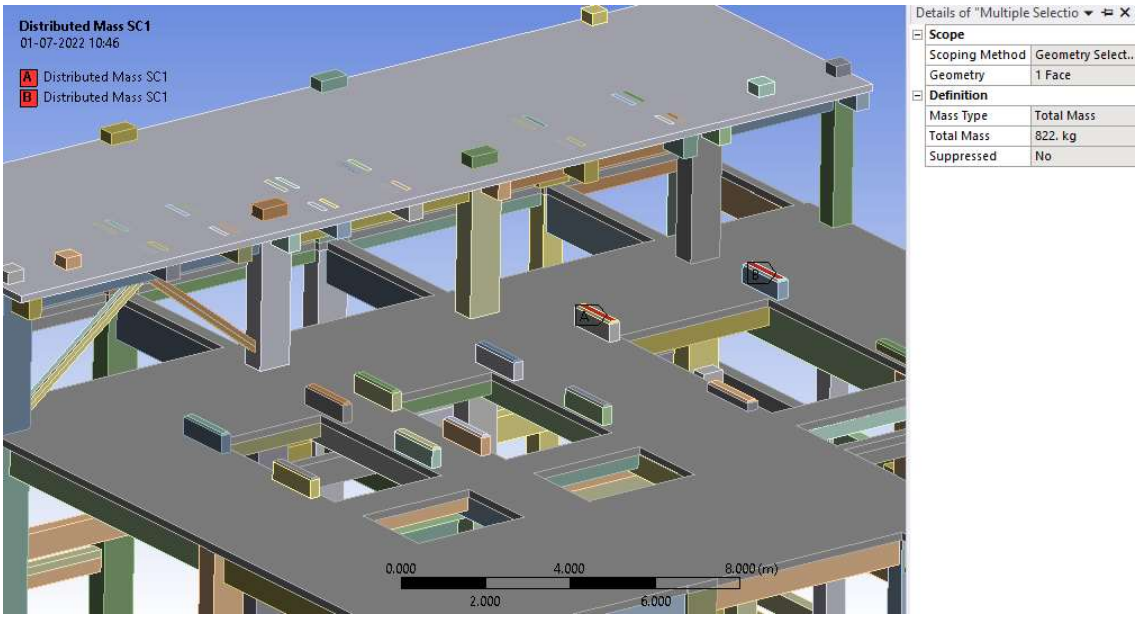
PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Material Weight:

- Mass of SC1(F1): 8222N = Around 822 kg each
- Mass of SC2(F2): 26518N = Around 2651 kg each



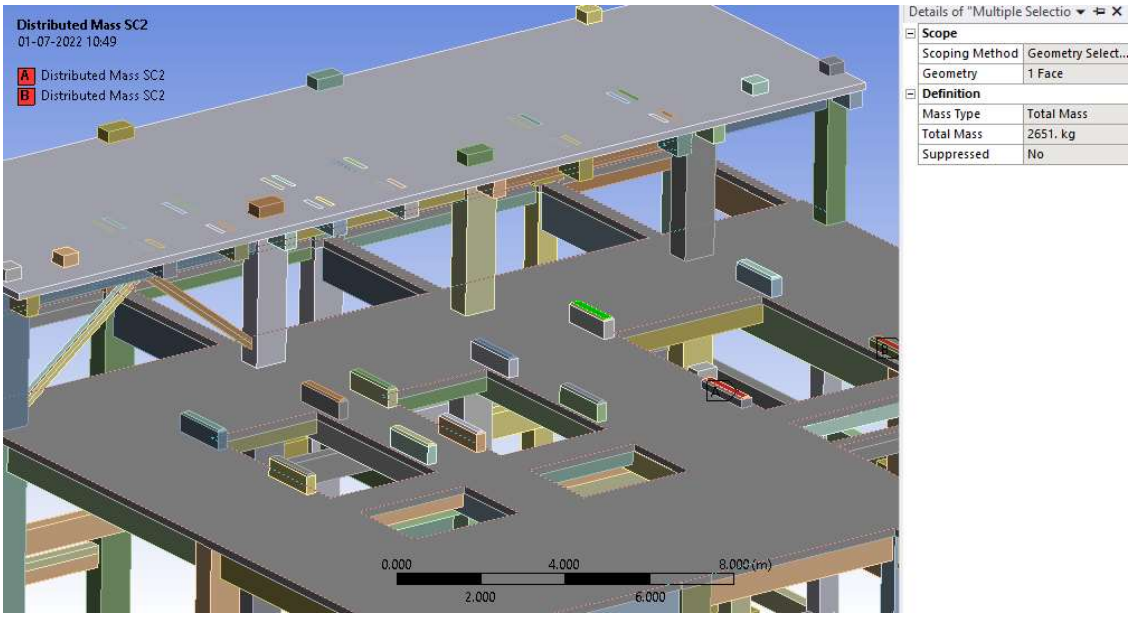


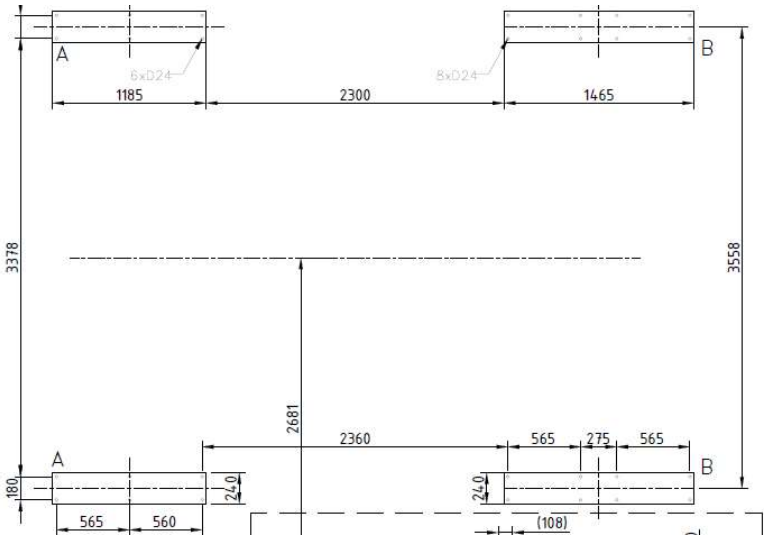
Figure 2g. Lumped mass of the Secondary Screen at support: loaded

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

2.5.2. Lump Mass Modelling of Tertiary Screen (3130-SCR-001/002)



TOTAL LOADS WITHOUT MATERIAL			
A static	41.600 N	B static	56.500 N
Dynamic load,running			
A dyn.vertical	±- 1.310 N	B dyn.vertical	±- 1.640 N
A dyn.hor/lat	±- 0.280 N	B dyn.hor/lat	±- 0.350 N
TOTAL LOADS WITH MATERIAL			
A static	44.600 N	B static	59.500 N
(Dynamic loads are same as without material)			

Table 2f. Load data from [MD-216-3130-EG-ME-GAD-V500-1005-C01 Rev C01]

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Empty Conditions:

- Mass of TS1(A): 41600N = Around 4160 kg each
- Mass of TS2 (B): 56500N = Around 5650 kg each



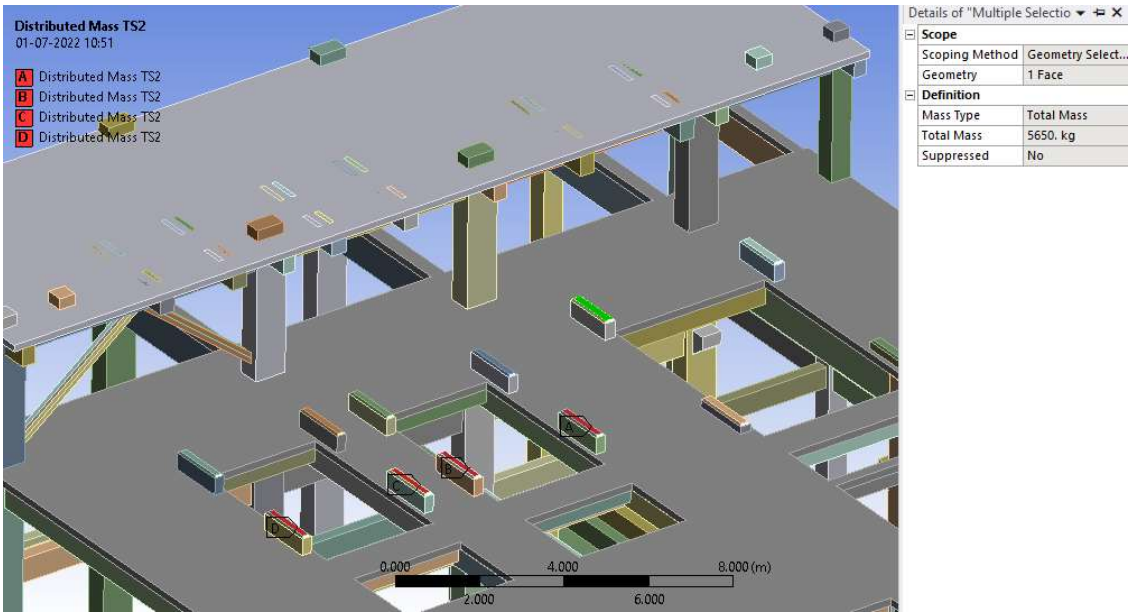


Figure 2h. Lumped mass of the Tertiary Screen at support: Unloaded

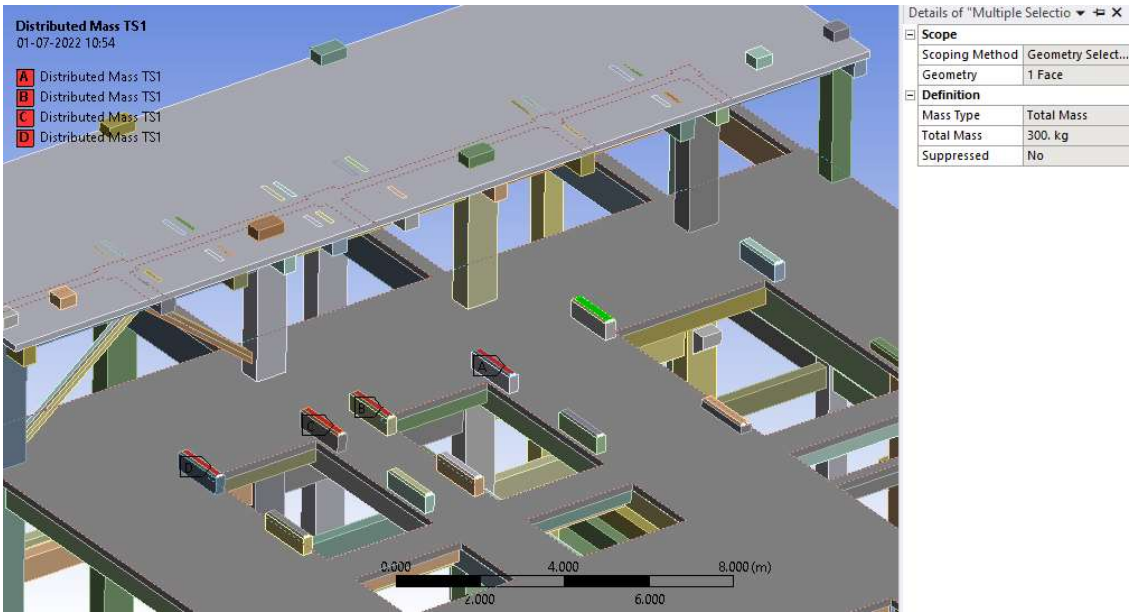
PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Material Weight:

- Mass of TS1(A): 3000N = Around 300 kg each
- Mass of TS2 (B): 3000N = Around 300 kg each



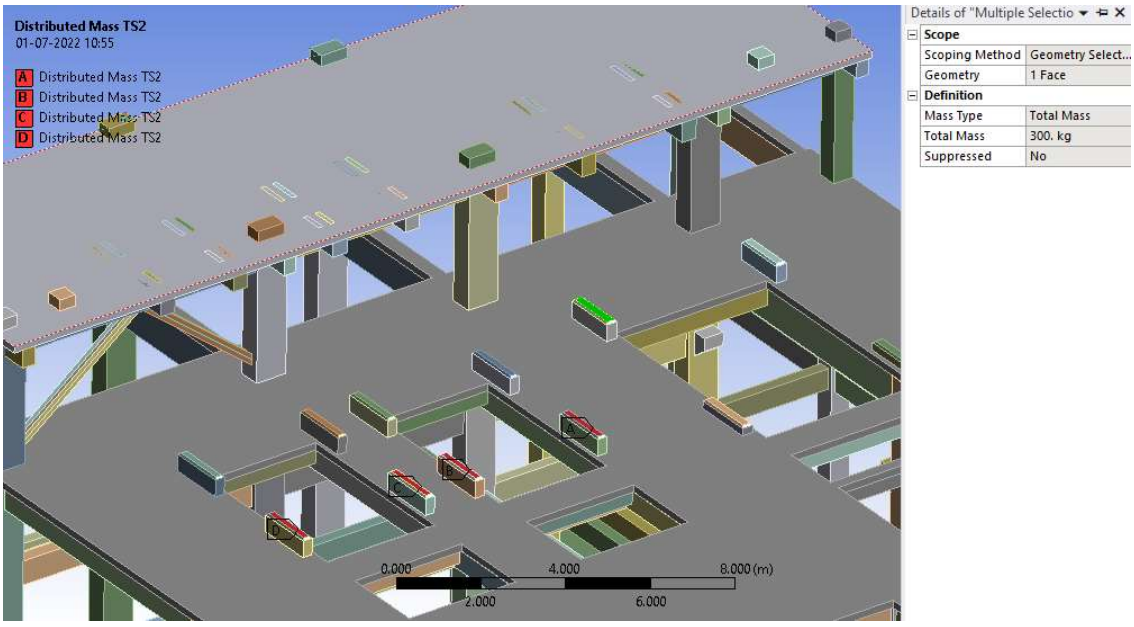


Figure 2i. Lumped mass of the Tertiary Screen at support: Loaded

PROPRIETARY & CONFIDENTIAL

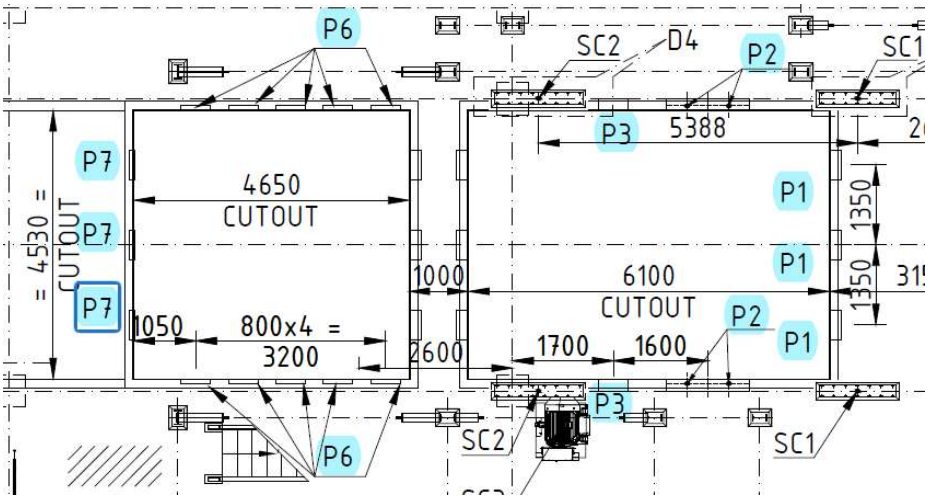
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

2.5.3. Lump Mass Modelling of Secondary Screen Chute

POINTS	ELEVATIONS IN "m"	VERTICAL LOAD (IN "t")	
		CHUTE EMPTY	CHUTE BLOCKED
P1	(+)922.750	(+)2.0	(+)4.5
P2		(+)2.0	(+)4.5
P3		(+)2.0	(+)4.5
P6		(+)1.5	(+)6.0
P7		(+)1.5	(+)6.0

Table 2g. Load data from [MD-216-3120-EG-ME-LPX-1006-Rev C01]



Empty Conditions:

- Mass of P1 : 3 x 2000 kg = Around 6000 kg each
- Mass of P2 & P3 : 2 x 2000 kg + 2000 kg = Around 6000 kg each

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

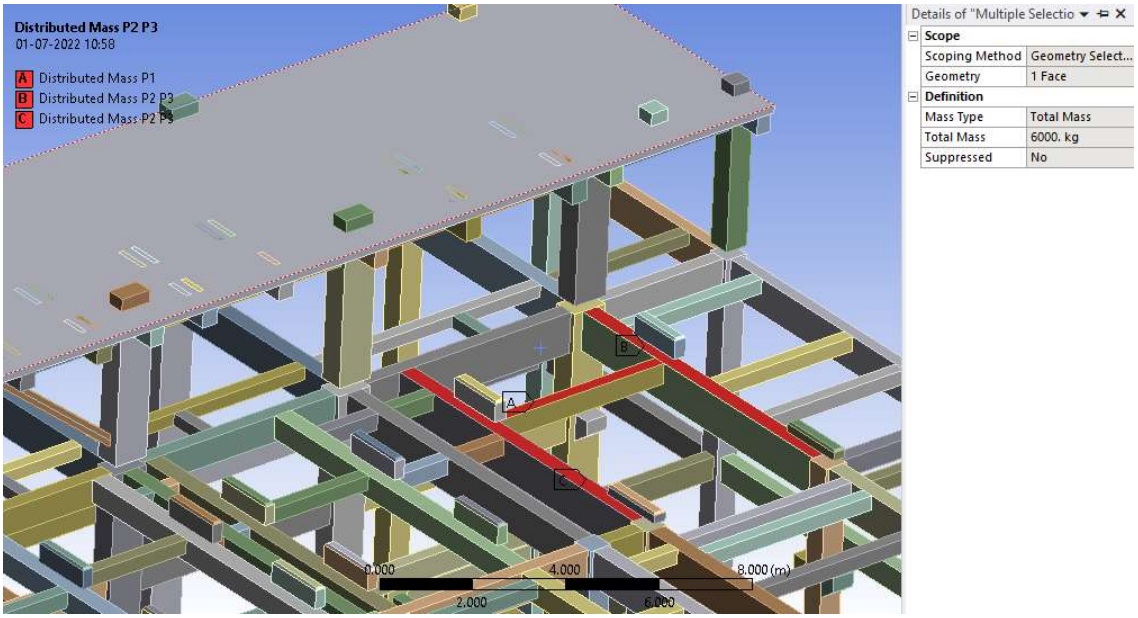
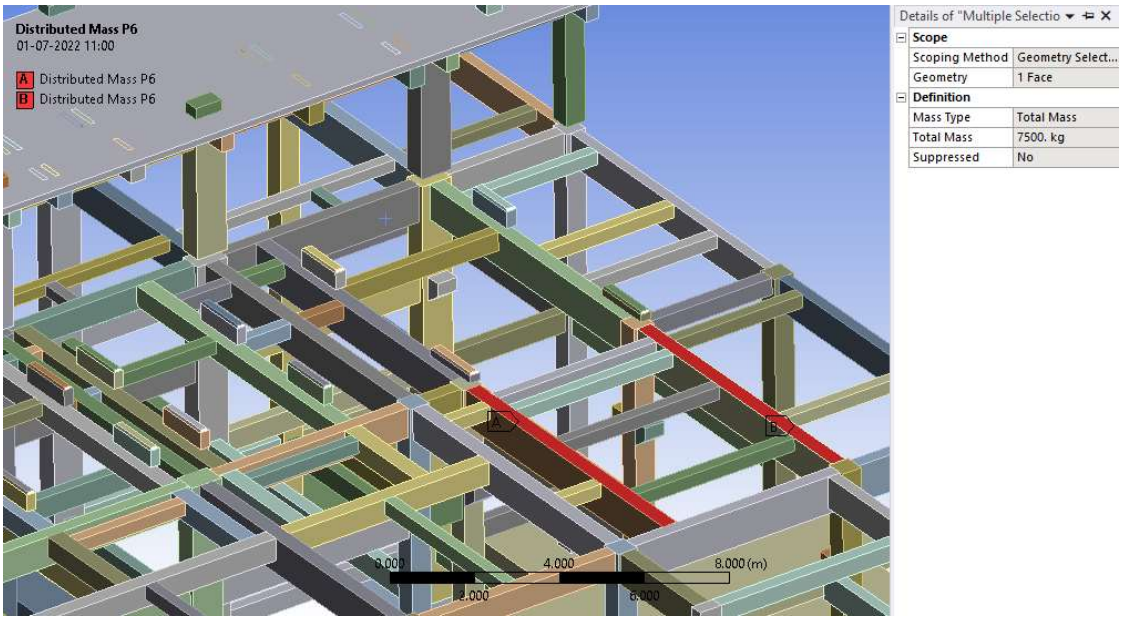


Figure 2j. Lumped mass of the Secondary Screen chute at support: Unloaded

Empty Conditions:

- Mass of P6: 5 x 1500 kg = Around 7500 kg each
- Mass of P7: 3 x 1500 kg = Around 4500 kg each



PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

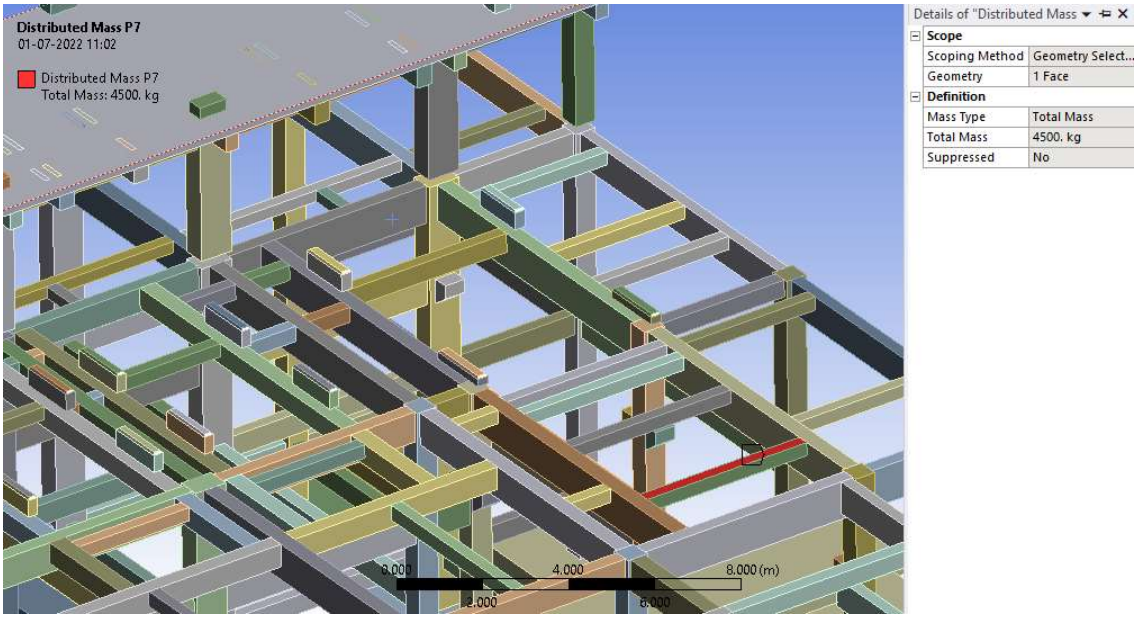
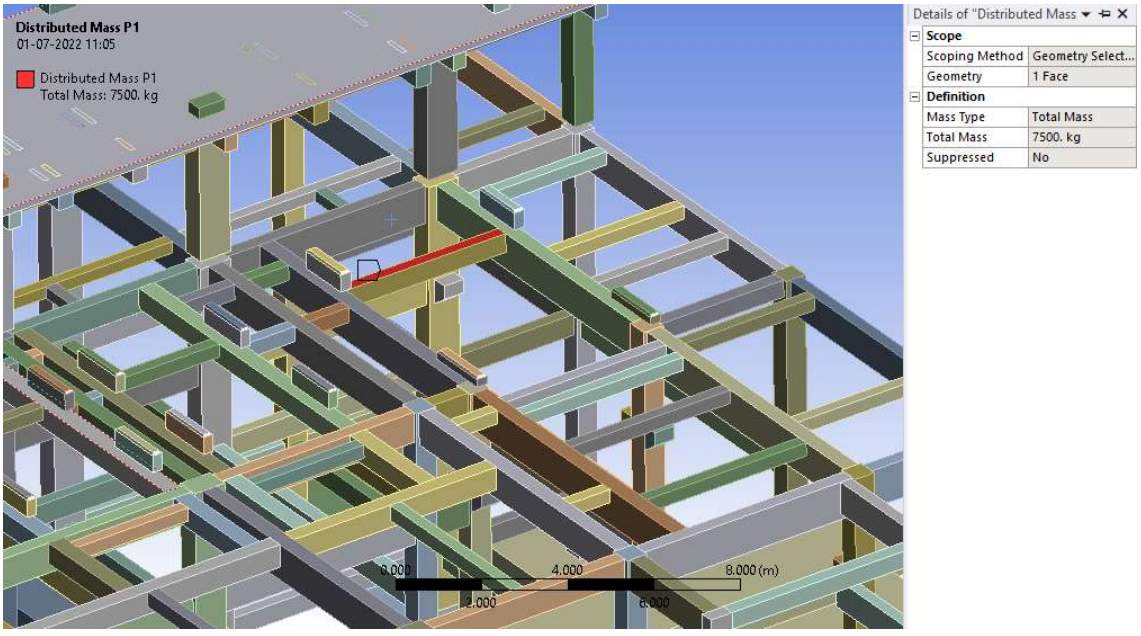


Figure 2k. Lumped mass of the Secondary Screen chute at support: Unloaded

Material Weight:

- Mass of P1 : 3 x 2500 kg = Around 7500 kg each
- Mass of P2 & P3 : 2 x 2500 kg + 2500 kg = Around 7500 kg each



PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

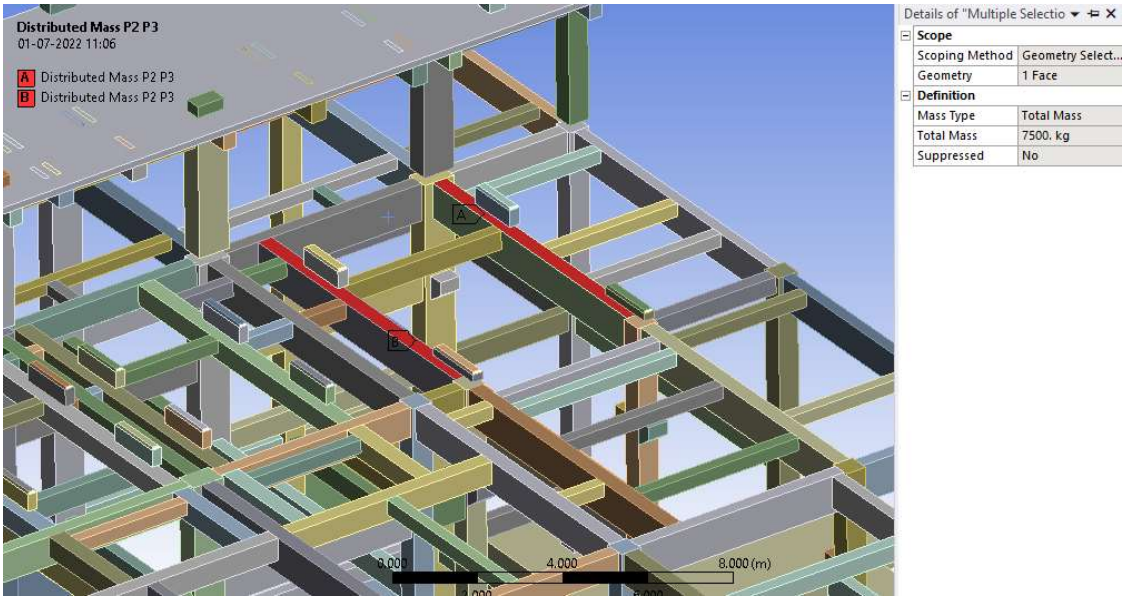
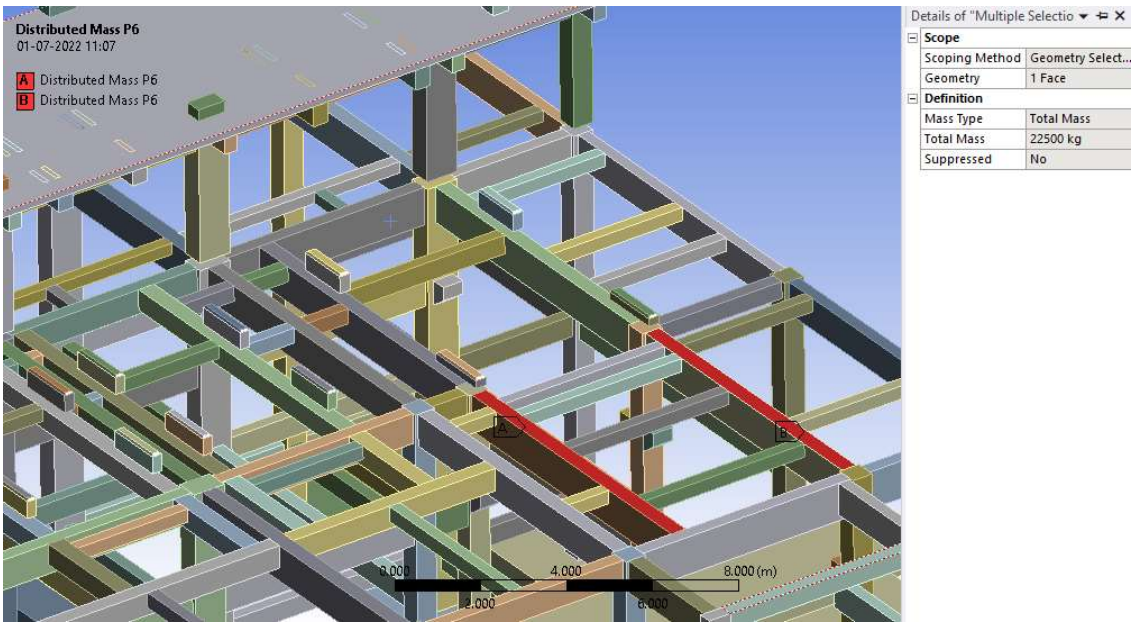


Figure 2l. Lumped mass of the Secondary Screen chute at support: Loaded

Material Weight:

- Mass of P6: 5 x 4500 kg = Around 22500 kg each
- Mass of P7: 3 x 4500 kg = Around 13500 kg each



PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

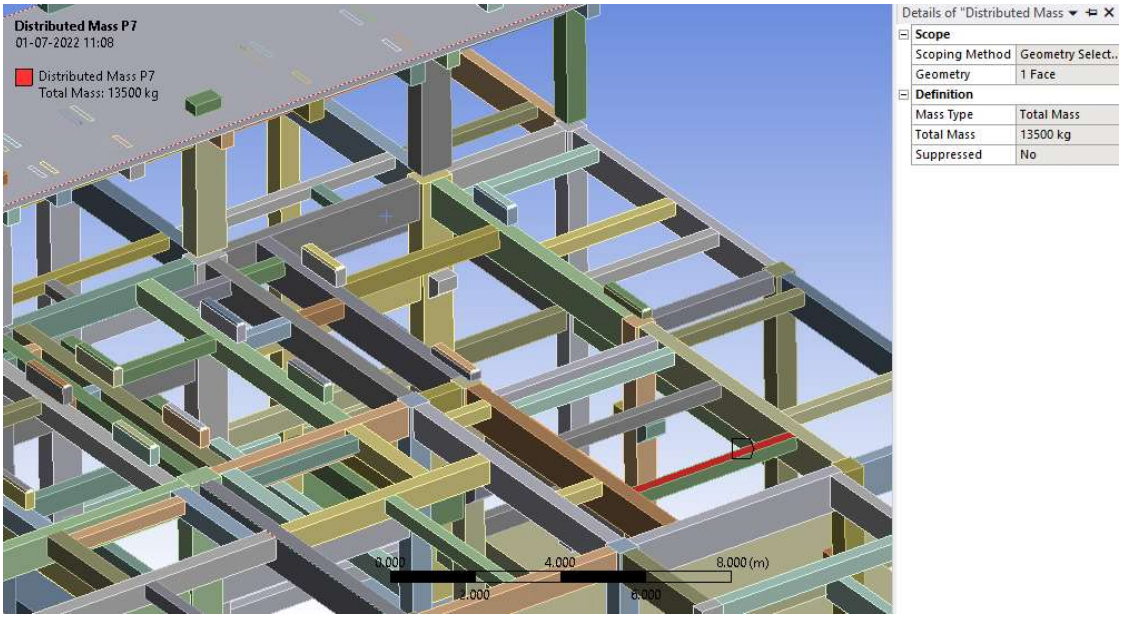


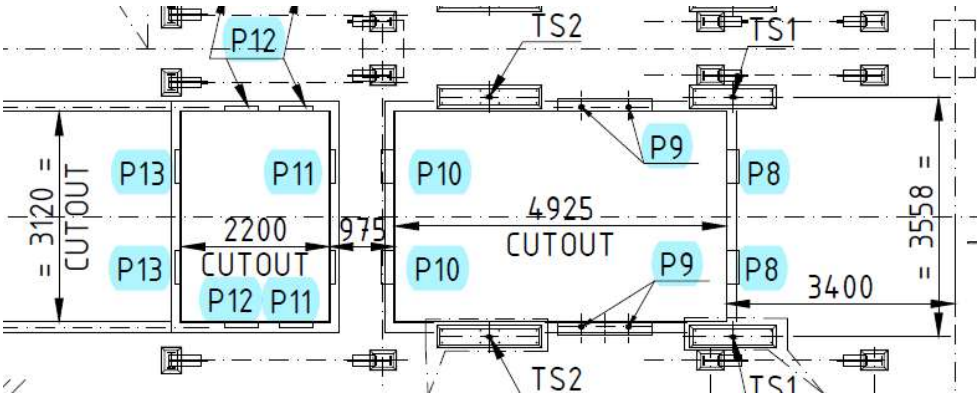
Figure 2m. Lumped mass of the Secondary Screen chute at support: Loaded

2.5.4. Lump Mass Modelling of Tertiary Screen Chute

POINTS	ELEVATIONS IN "m"	VERTICAL LOAD (IN "t")	
P8		(+)1.5	(+)5.0
P9		(+)1.5	(+)5.0
P10		(+)1.5	(+)5.0
P11		(+)1.0	(+)3.0
P12		(+)1.0	(+)3.0
P13		(+)1.0	(+)3.0

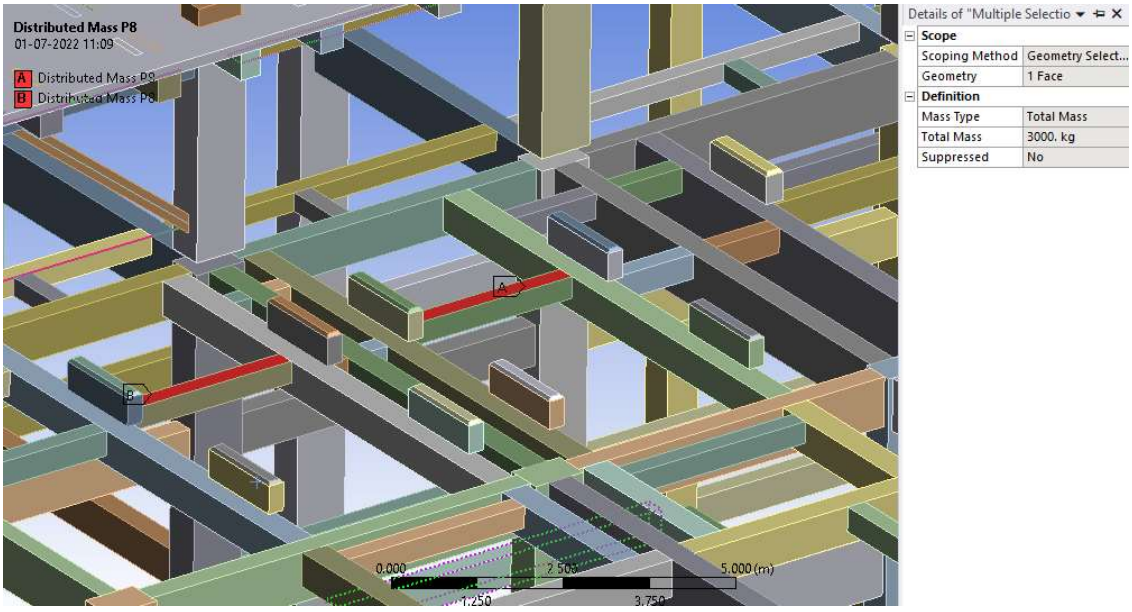
Table 2h. Load data from [MD-216-3120-EG-ME-LPX-1006-Rev C01]

PROPRIETARY & CONFIDENTIAL
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.



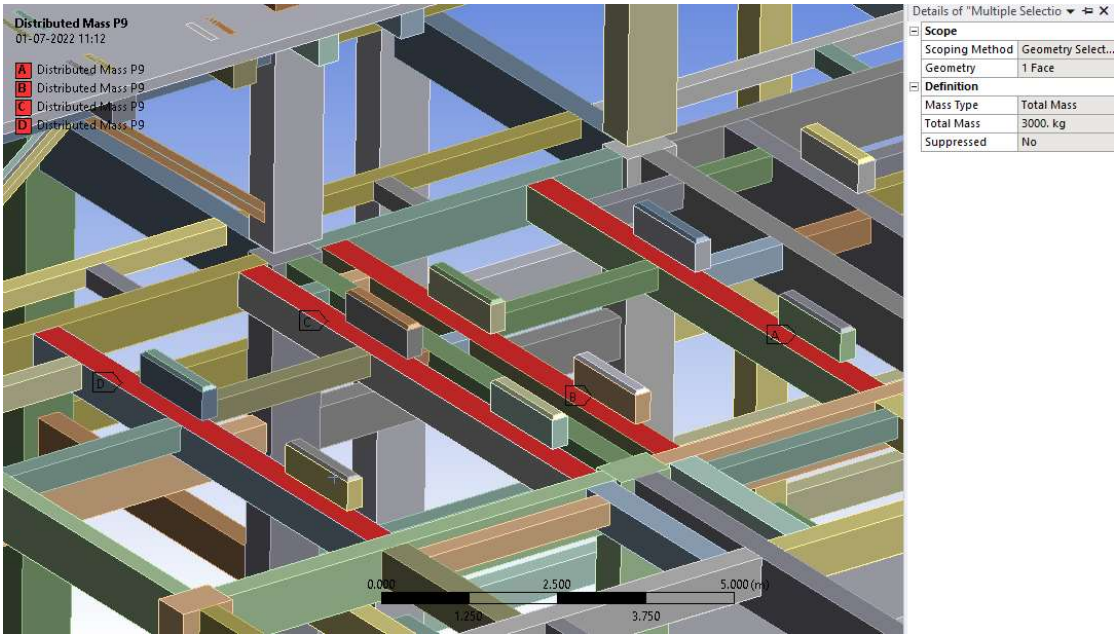
Empty Conditions:

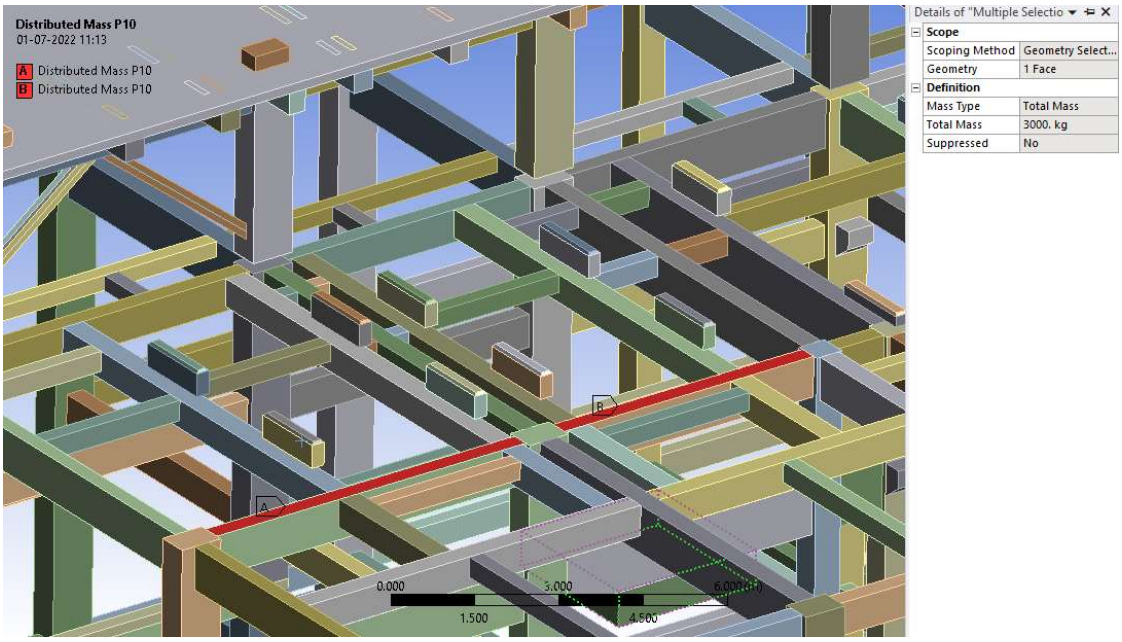
- Mass of P8: 2 x 1500 kg = Around 3000 kg each
- Mass of P9: 2 x 1500 kg = Around 3000 kg each
- Mass of P10: 2 x 1500 kg = Around 3000 kg each
- Mass of P11: 2 x 1000 kg = Around 2000 kg each
- Mass of P12: 2 x 1000 kg = Around 2000 kg each
- Mass of P13: 2 x 1000 kg = Around 2000 kg each



PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

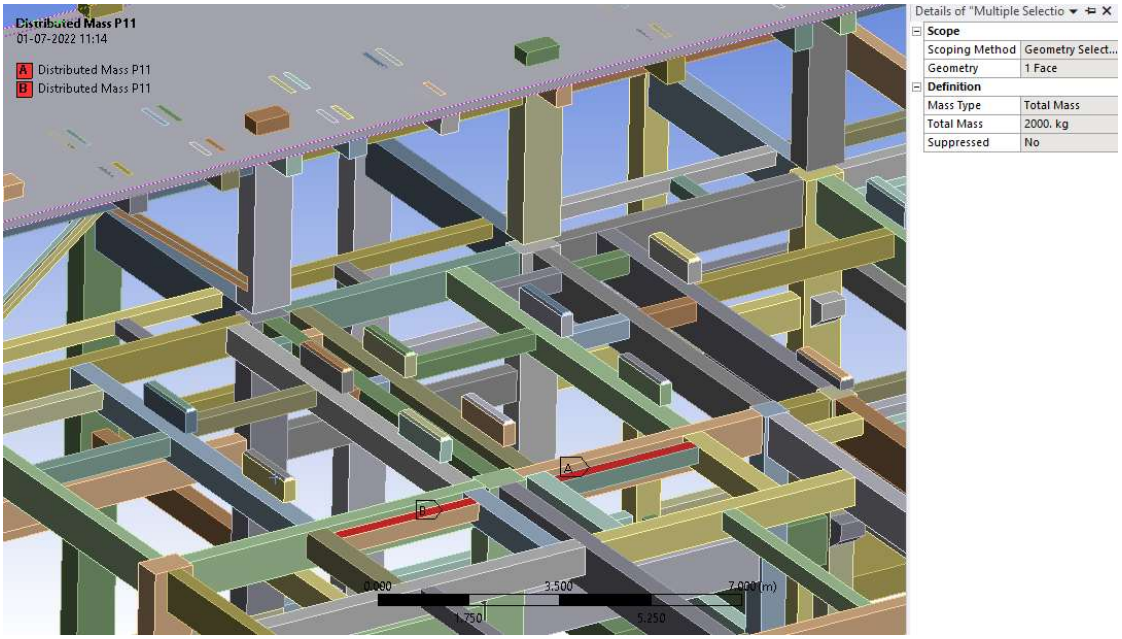


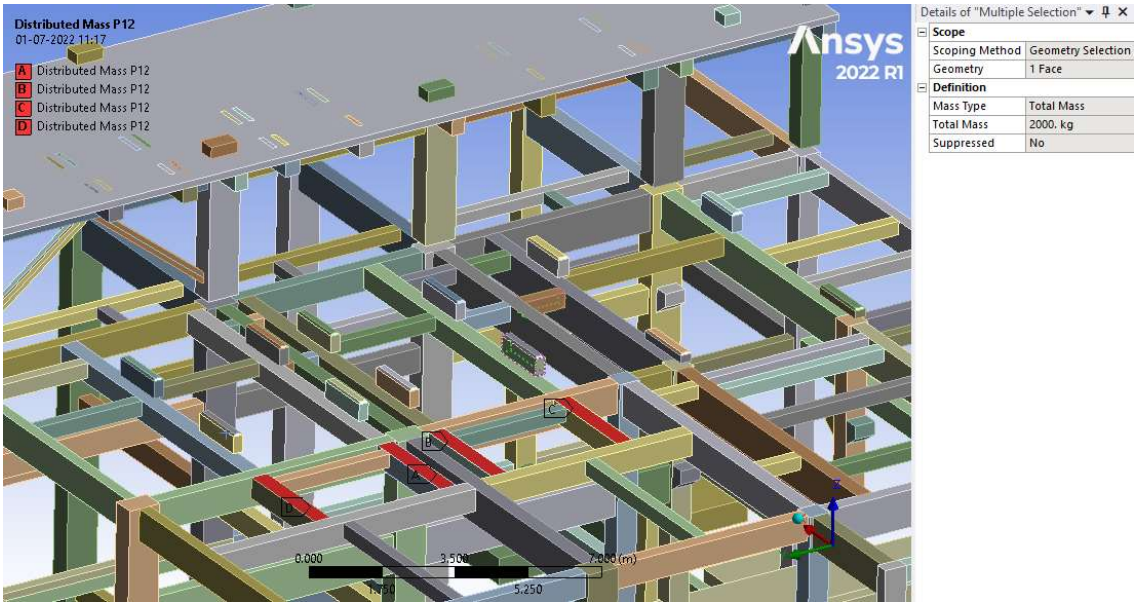


PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.





PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

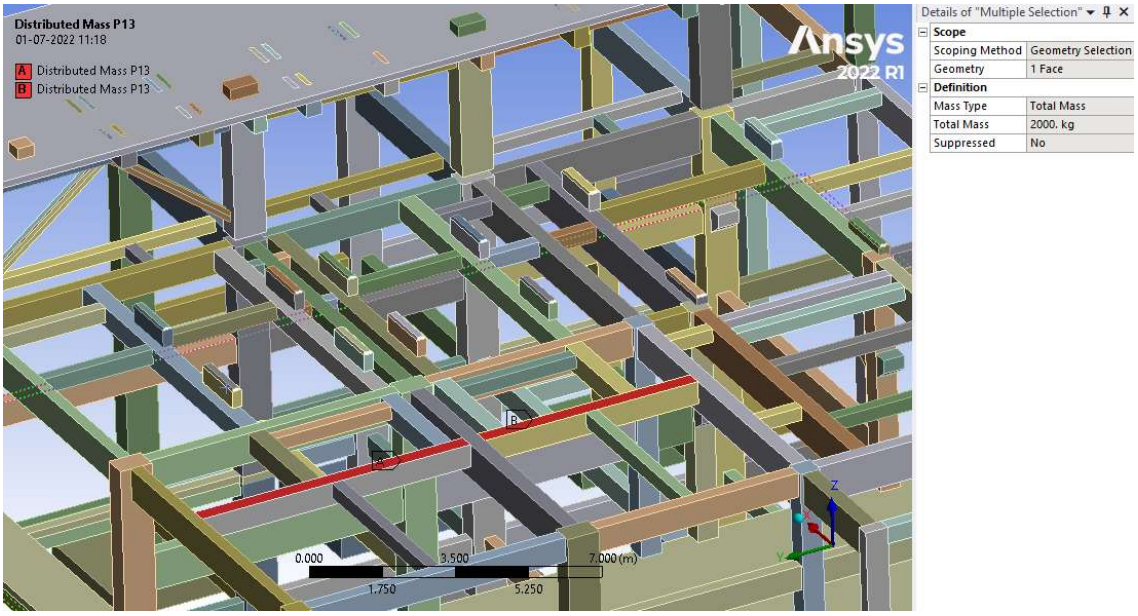


Figure 2n. Lumped mass of the Tertiary Screen chute at support: Unloaded

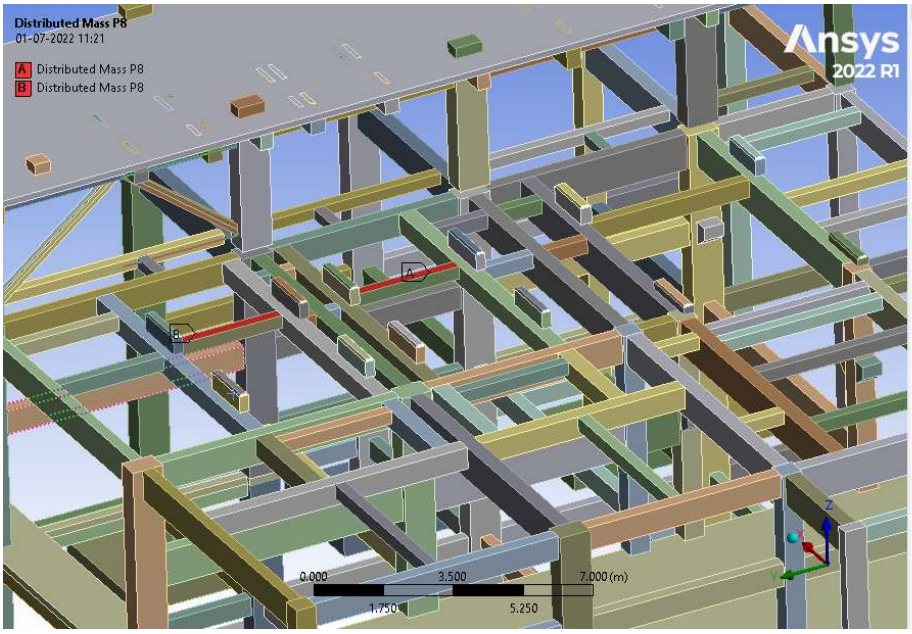
Material Weight:

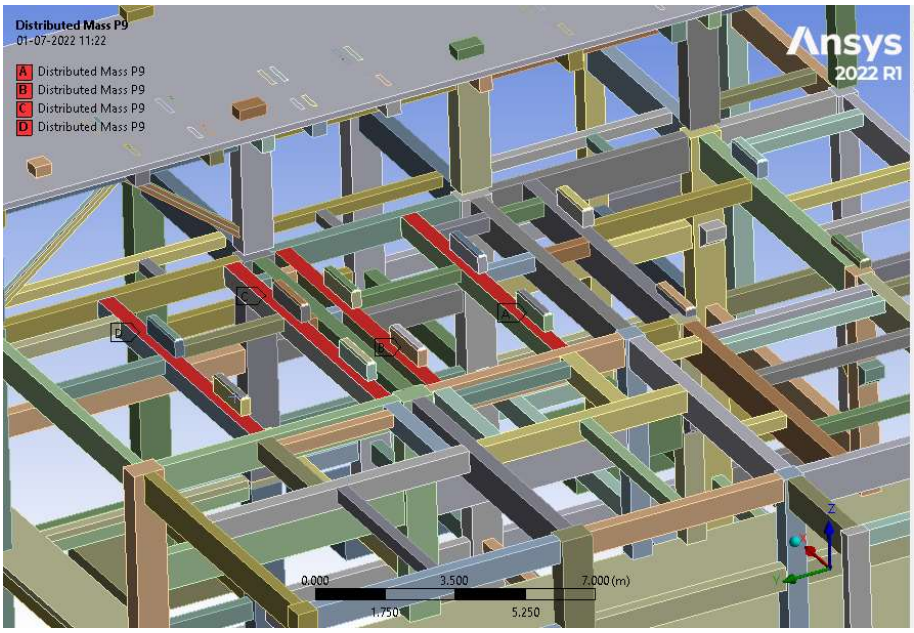
- Mass of P8: 2 x 3500 kg = Around 7000 kg each
- Mass of P9: 2 x 3500 kg = Around 7000 kg each
- Mass of P10: 2 x 3500 kg = Around 7000 kg each
- Mass of P11: 2 x 2000 kg = Around 4000 kg each
- Mass of P12: 2 x 2000 kg = Around 4000 kg each
- Mass of P13: 2 x 2000 kg = Around 4000 kg each

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

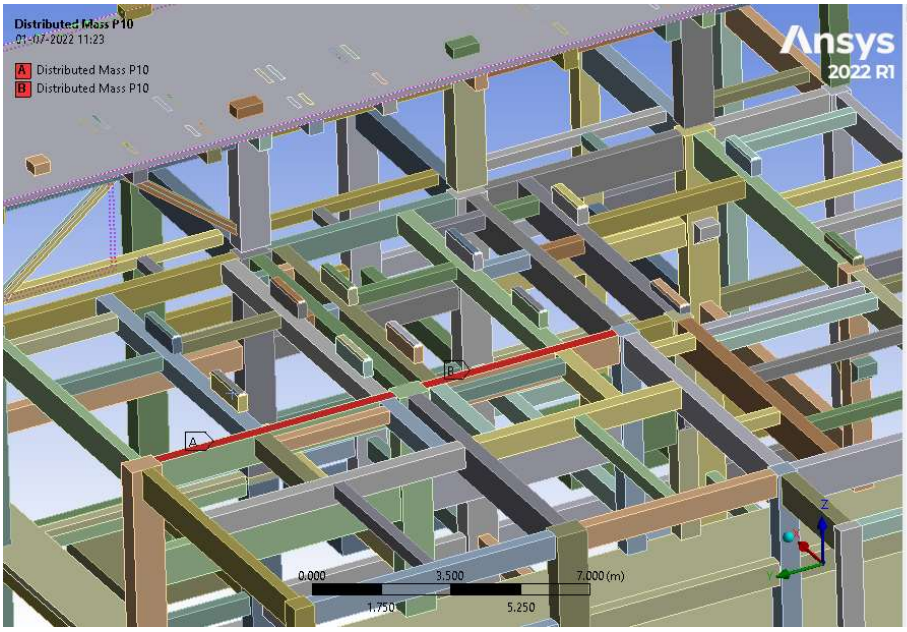


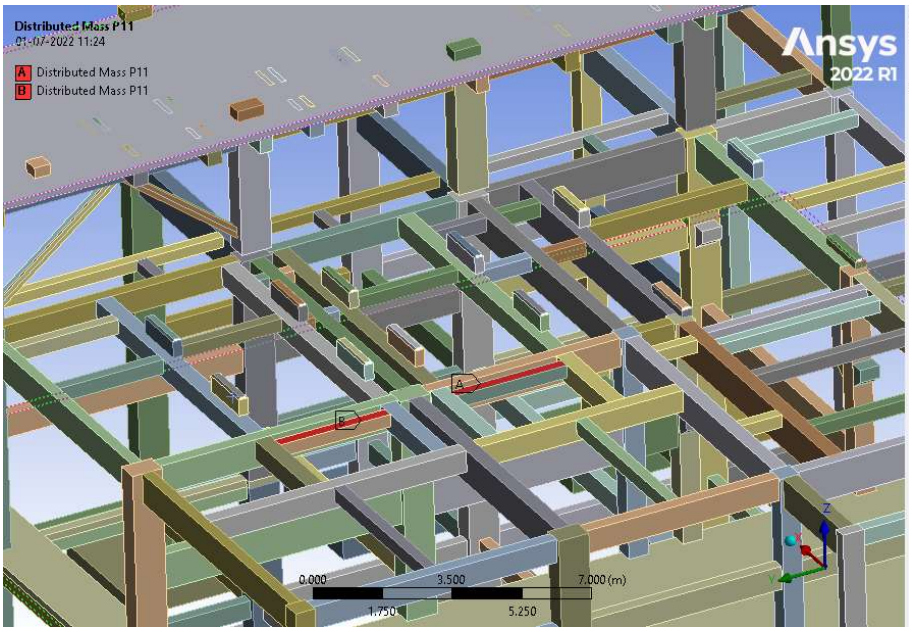


PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.





PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

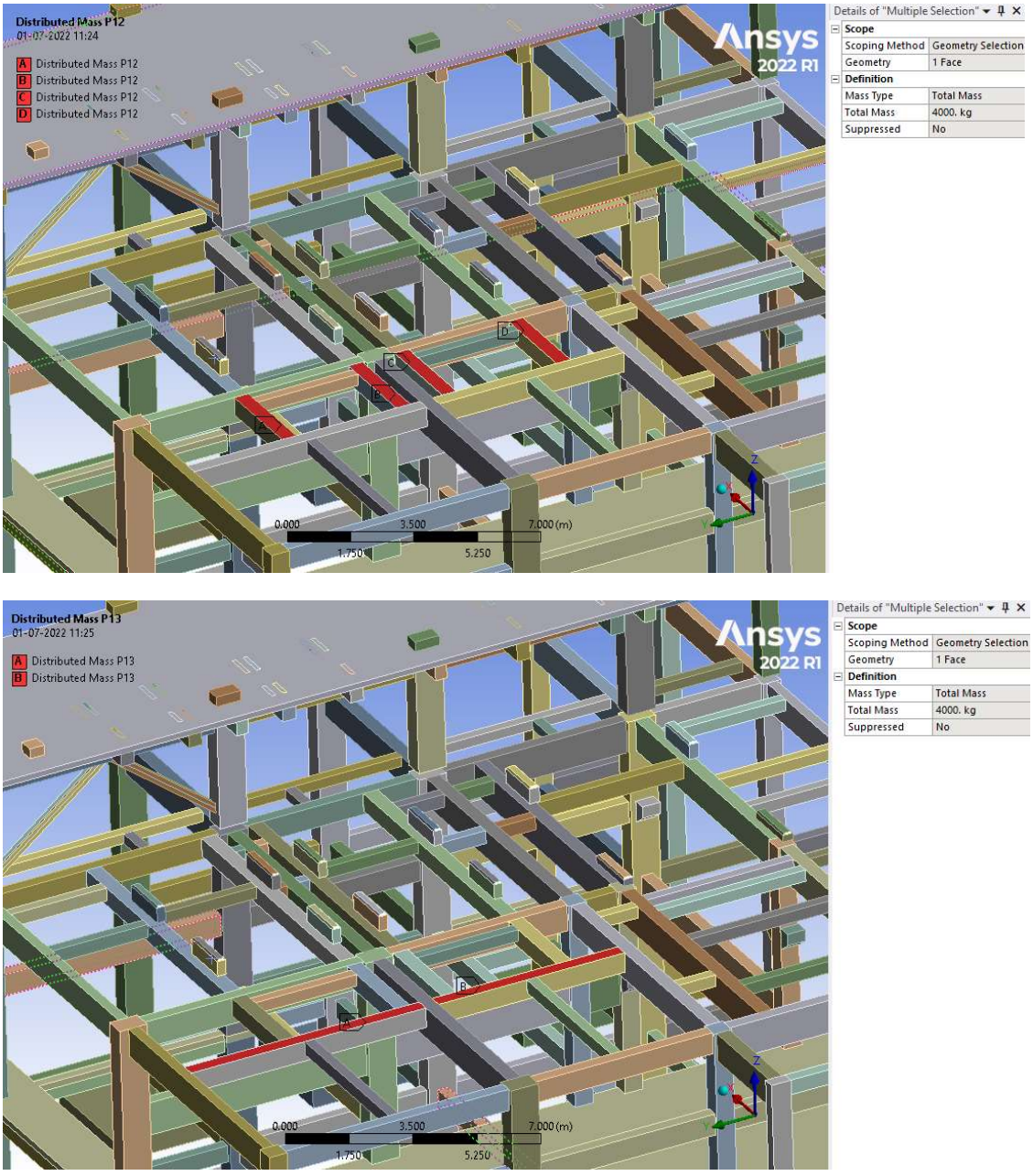


Figure 2o. Lumped mass of the Tertiary Screen chute at support: Loaded

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

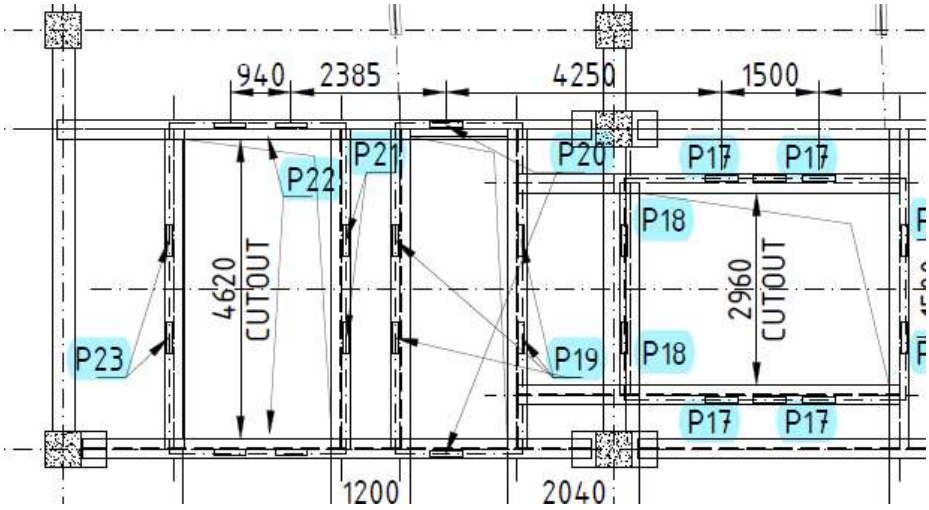
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

2.6. Lump Mass Modelling Below Screen Floor

2.6.1. Lump Mass Modelling of Secondary Screen Chute

POINTS	ELEVATIONS IN "m"	VERTICAL LOAD (IN "t")	
		CHUTE EMPTY	CHUTE BLOCKED
P16	(+)920.550	(+)0.75	(+)5.5
P17		(+)0.75	(+)5.5
P18		(+)0.75	(+)5.5
P19		(+)1.5	(+)6.0
P20		(+)1.5	(+)6.0
P21		(+)1.5	(+)6.0
P22		(+)1.5	(+)6.0
P23		(+)1.5	(+)6.0

Table 2i. Load data from [MD-216-3120-EG-ME-ELV-1009-Rev C01]



Empty Conditions:

- Mass of P16: 2 x 750 kg = Around 1500 kg each
- Mass of P17: 4 x 750 kg = Around 3000 kg each
- Mass of P18: 2 x 750 kg = Around 1500 kg each
  Total Mass of P16, P17, P18 = 6000 kg
  Mass at each point = 6000/4 = 1500 kg

- Mass of P19: 4 x 1500 kg = Around 6000 kg each
- Mass of P20: 2 x 1500 kg = Around 3000 kg each

PROPRIETARY & CONFIDENTIAL
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

- Mass of P21: 2 x 1500 kg = Around 3000 kg each
- Mass of P22: 2 x 1500 kg = Around 3000 kg each
- Mass of P23: 2 x 1500 kg = Around 3000 kg each
  Total Mass of P19, P20, P21, P22, P23 = 18000 kg
  Mass at each point = 18000/4 = 4500 kg



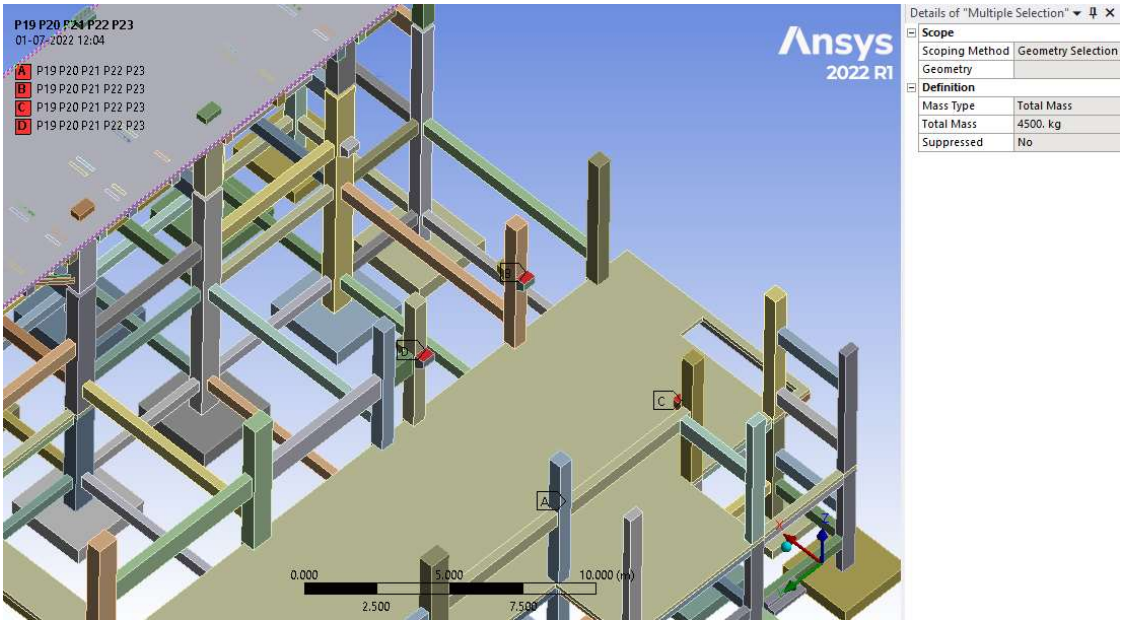


Figure 2p. Lumped mass of the Secondary Screen chute at support: Unloaded

PROPRIETARY & CONFIDENTIAL
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Material Weight:

- Mass of P16: 2 x 4750 kg = Around 9500 kg each
- Mass of P17: 4 x 4750 kg = Around 19000 kg each
- Mass of P18: 2 x 4750 kg = Around 9500 kg each

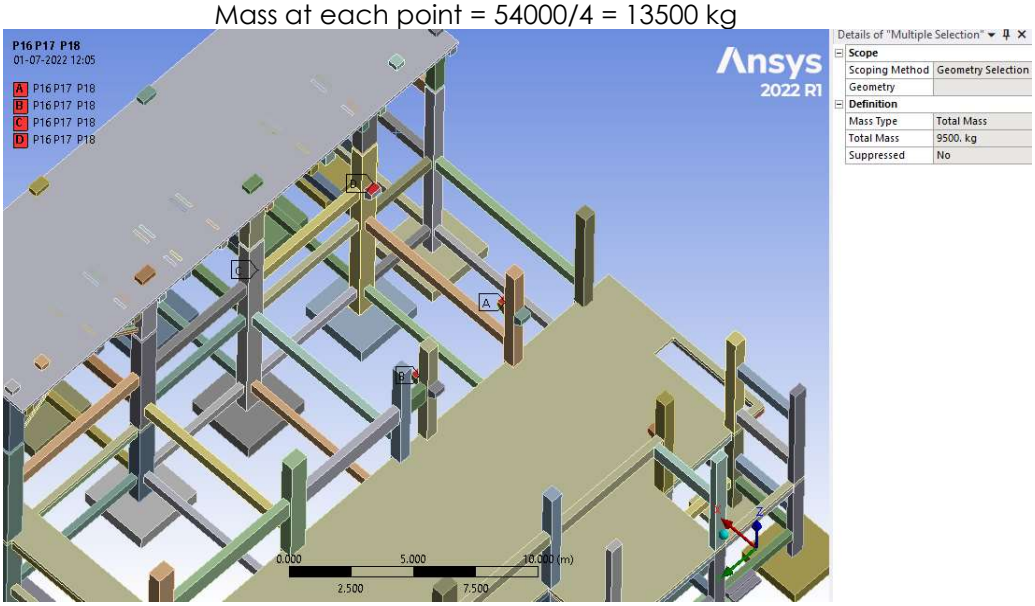
  Total Mass of P16, P17, P18 = 38000 kg

  Mass at each point = 38000/4 = 9500 kg

- Mass of P19: 4 x 4500 kg = Around 18000 kg each
- Mass of P20: 2 x 4500 kg = Around 9000 kg each
- Mass of P21: 2 x 4500 kg = Around 9000 kg each
- Mass of P22: 2 x 4500 kg = Around 9000 kg each
- Mass of P23: 2 x 4500 kg = Around 9000 kg each

  Total Mass of P19, P20, P21, P22, P23 = 54000 kg

  Mass at each point = 54000/4 = 13500 kg



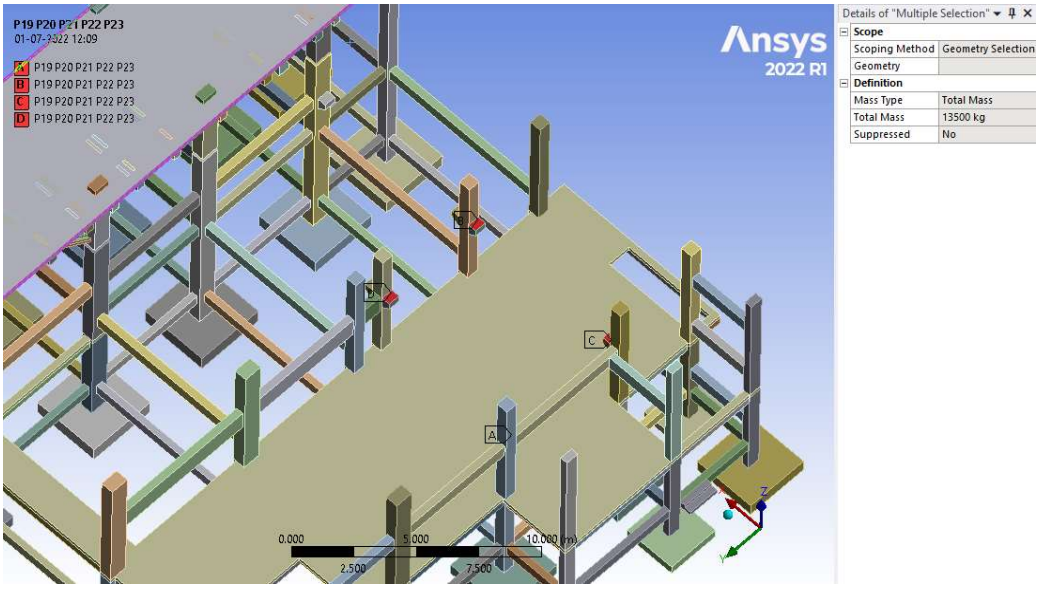


Figure 2q. Lumped mass of the Secondary Screen chute at support: Loaded

PROPRIETARY & CONFIDENTIAL

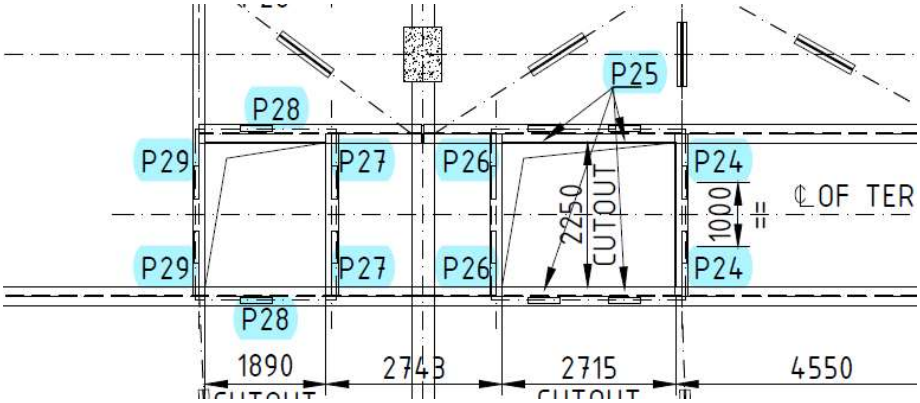
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

2.6.2. Lump Mass Modelling of Tertiary Screen Chute

POINTS	ELEVATIONS IN "m"	VERTICAL LOAD (IN "t")	
P24		(+)1.0	(+)5.0
P25		(+)1.0	(+)5.0
P26		(+)1.0	(+)5.0
P27		(+)1.5	(+)6.0
P28		(+)1.5	(+)6.0
P29		(+)1.5	(+)6.0

Table 2j. Load data from [MD-216-3120-EG-ME-ELV-1009-Rev C01]



Empty Conditions:

- Mass of P24: 2 x 1000 kg = Around 2000 kg each
- Mass of P25: 4 x 1000 kg = Around 4000 kg each
- Mass of P26: 2 x 1000 kg = Around 2000 kg each
  Total Mass of P24, P25, P26 = 8000 kg
  Mass at each point = 8000/4 = 2000 kg

- Mass of P27: 2 x 1500 kg = Around 3000 kg each
- Mass of P28: 2 x 1500 kg = Around 3000 kg each
- Mass of P29: 2 x 1500 kg = Around 3000 kg each
  Total Mass of P27, P28, P29 = 9000 kg
  Mass at each point = 9000/3 = 3000 kg

PROPRIETARY & CONFIDENTIAL
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.



Figure 2r. Lumped mass of the Tertiary Screen chute at support: Unloaded



PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.



Figure 2s. Lumped mass of the Tertiary Screen chute at support: unloaded

Material Weight:

- Mass of P24: 2 x 4000 kg = Around 8000 kg each
- Mass of P25: 4 x 4000 kg = Around 16000 kg each
- Mass of P26: 2 x 4000 kg = Around 8000 kg each
  Total Mass of P24, P25, P26 = 32000 kg
  Mass at each point = 32000/4 = 8000 kg

- Mass of P27: 2 x 4500 kg = Around 9000 kg each
- Mass of P28: 2 x 4500 kg = Around 9000 kg each
- Mass of P29: 2 x 4500 kg = Around 9000 kg each
  Total Mass of P27, P28, P29 = 27000 kg
  Mass at each point = 27000/3 = 9000 kg

PROPRIETARY & CONFIDENTIAL
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

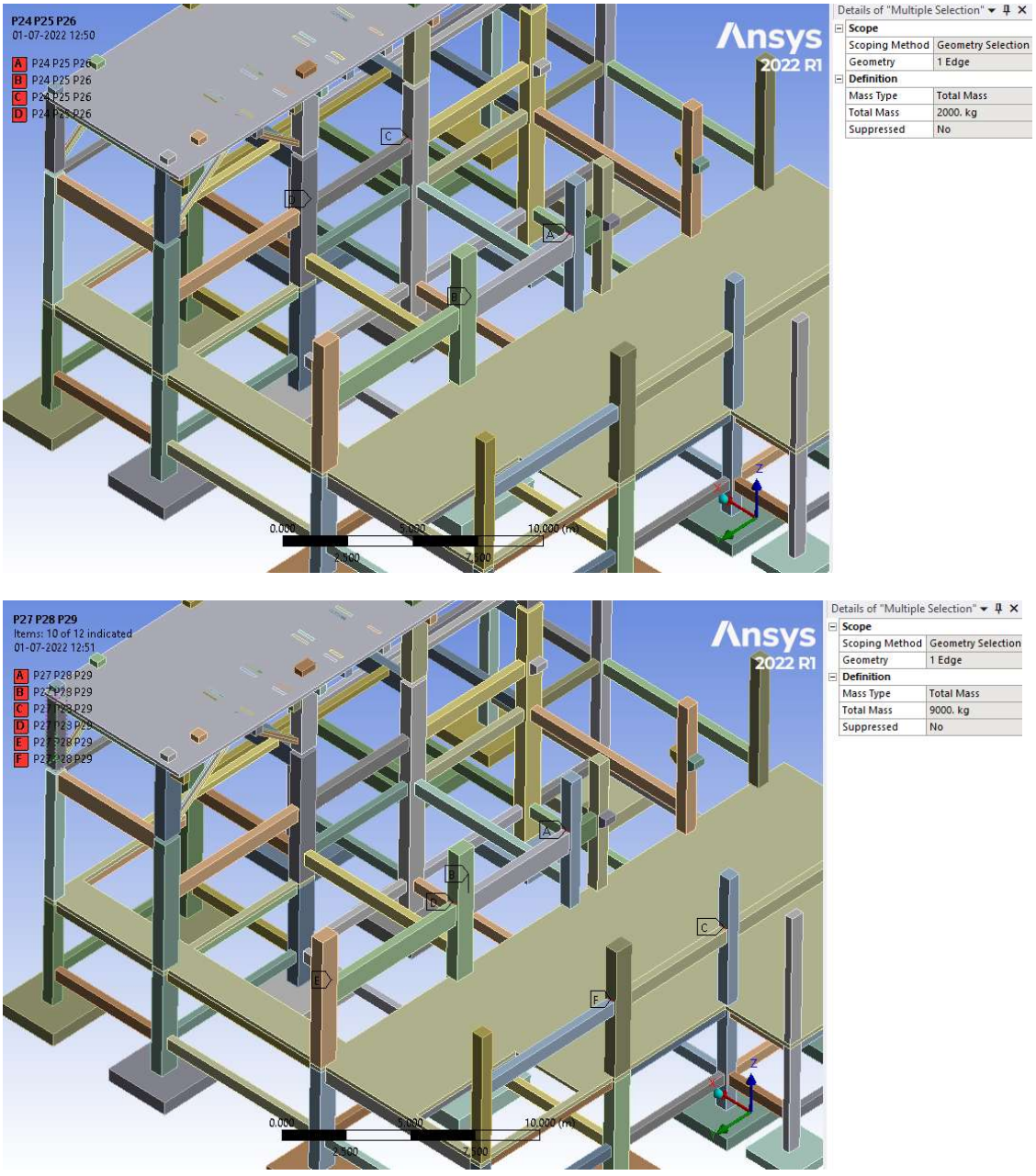


Figure 2t. Lumped mass of the Tertiary Screen chute at support: Loaded

2.7. Lump mass model of steel structure

Mass of steel structure supported at 926.500m is obtained from static analysis in Staad and it is applied on the Ansys model. Refer static analysis document numbers MD-216-3120-EG-CV-CAL-1005-C01& MD-216-3120-EG-ST-CAL-1003-C0.

3. Modelling of Dynamic Loading

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Support	L/C	Force-X kN	Force-Y kN	Force-Z kN
1	Dead Load	-40	404	27
	Live Load	-30	212	10
2	Dead Load	39	243	9
	Live Load	24	223	0
3	Dead Load	-52	731	12
	Live Load	-37	338	7
4	Dead Load	56	484	10
	Live Load	41	340	4
5	Dead Load	-47	785	-21
	Live Load	-42	356	-9
6	Dead Load	54	506	-20
	Live Load	39	357	-11
7	Dead Load	-39	285	-13
	Live Load	-29	119	3
8	Dead Load	37	204	-10
	Live Load	24	129	-1
9	Dead Load	3	177	-54
	Live Load	2	106	-28
10	Dead Load	-1	100	-19
	Live Load	-1	59	-8
11	Dead Load	-1	350	54
	Live Load	-1	137	20
12	Dead Load	2	203	29
	Live Load	0	127	16

Table 2k. Load data from steel superstructure

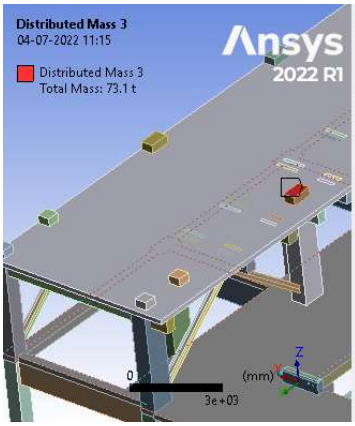




PROPRIETARY & CONFIDENTIAL

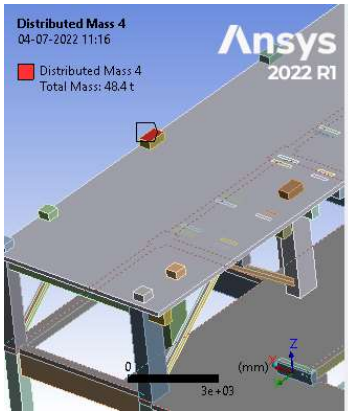
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.



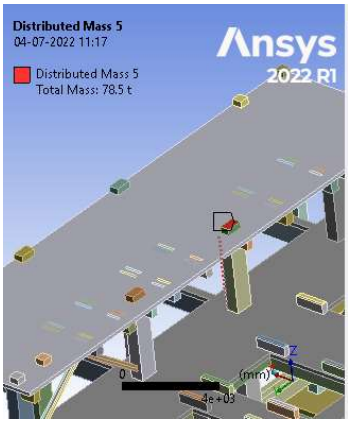
Details of "Distributed Mass"

Scope	
Scoping Method	Geometry Selec...
Geometry	1 Face
Definition	
Mass Type	Total Mass
Total Mass	73.1 t
Suppressed	No



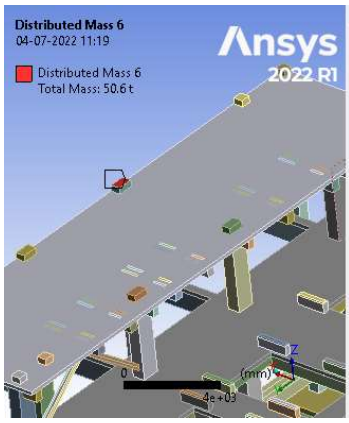
Details of "Distributed Mass"

Scope	
Scoping Method	Geometry Selec...
Geometry	1 Face
Definition	
Mass Type	Total Mass
Total Mass	48.4 t
Suppressed	No



Details of "Distributed Mass"

Scope	
Scoping Method	Geometry Selec...
Geometry	1 Face
Definition	
Mass Type	Total Mass
Total Mass	78.5 t
Suppressed	No



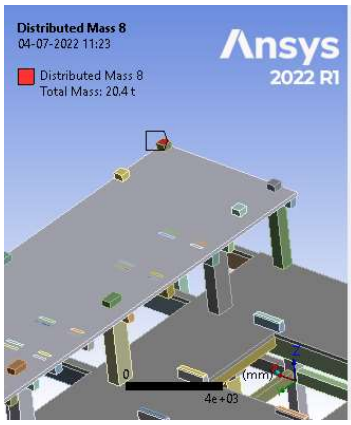
Details of "Distributed Mass"

Scope	
Scoping Method	Geometry Selec...
Geometry	1 Face
Definition	
Mass Type	Total Mass
Total Mass	50.6 t
Suppressed	No



Details of "Distributed Mass"

Scope	
Scoping Method	Geometry Selec...
Geometry	1 Face
Definition	
Mass Type	Total Mass
Total Mass	28.5 t
Suppressed	No



Details of "Distributed Mass"

Scope	
Scoping Method	Geometry Selec...
Geometry	1 Face
Definition	
Mass Type	Total Mass
Total Mass	20.4 t
Suppressed	No

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

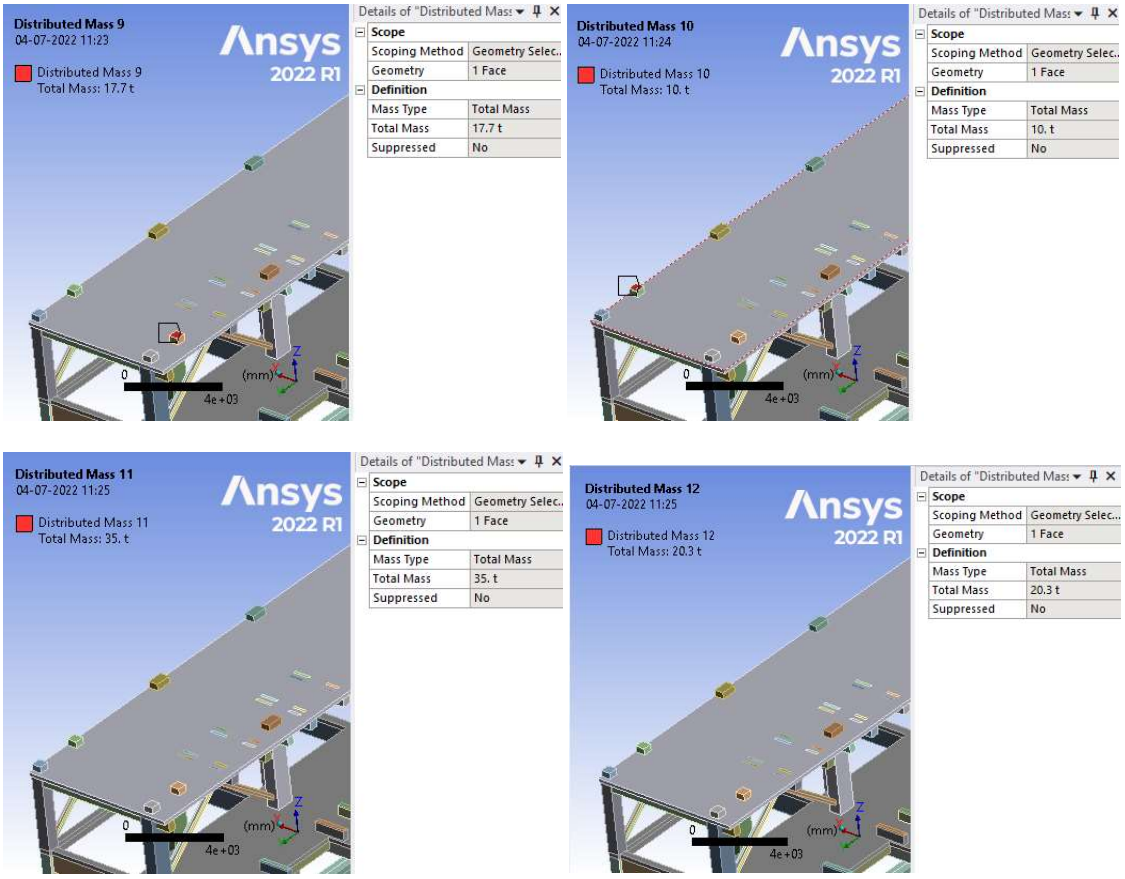


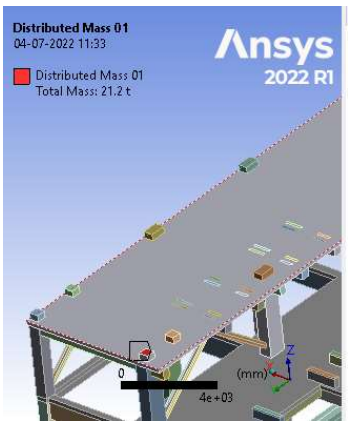
Figure 2u. Lumped mass of the Bunker at support: Unloaded

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Material Weight:



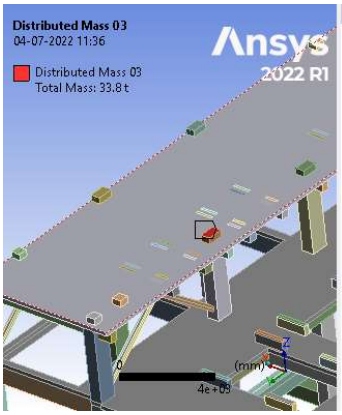
Details of "Distributed Mass"

Scope	
Scoping Method	Geometry Selec...
Geometry	1 Face
Definition	
Mass Type	Total Mass
Total Mass	21.2 t
Suppressed	No



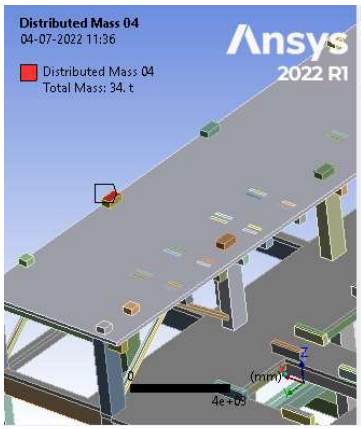
Details of "Distributed Mass"

Scope	
Scoping Method	Geometry Selec...
Geometry	1 Face
Definition	
Mass Type	Total Mass
Total Mass	22.3 t
Suppressed	No



Details of "Distributed Mass"

Scope	
Scoping Method	Geometry Selec...
Geometry	1 Face
Definition	
Mass Type	Total Mass
Total Mass	33.8 t
Suppressed	No



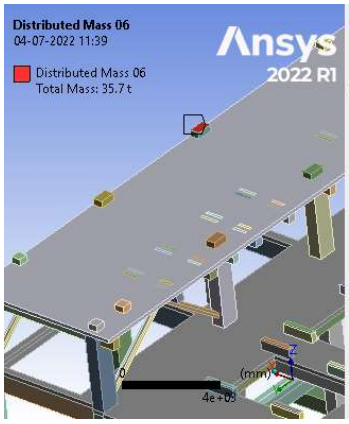
Details of "Distributed Mass"

Scope	
Scoping Method	Geometry Selec...
Geometry	1 Face
Definition	
Mass Type	Total Mass
Total Mass	34. t
Suppressed	No



Details of "Distributed Mass"

Scope	
Scoping Method	Geometry Selec...
Geometry	1 Face
Definition	
Mass Type	Total Mass
Total Mass	35.6 t
Suppressed	No



Details of "Distributed Mass"

Scope	
Scoping Method	Geometry Selec...
Geometry	1 Face
Definition	
Mass Type	Total Mass
Total Mass	35.7 t
Suppressed	No

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

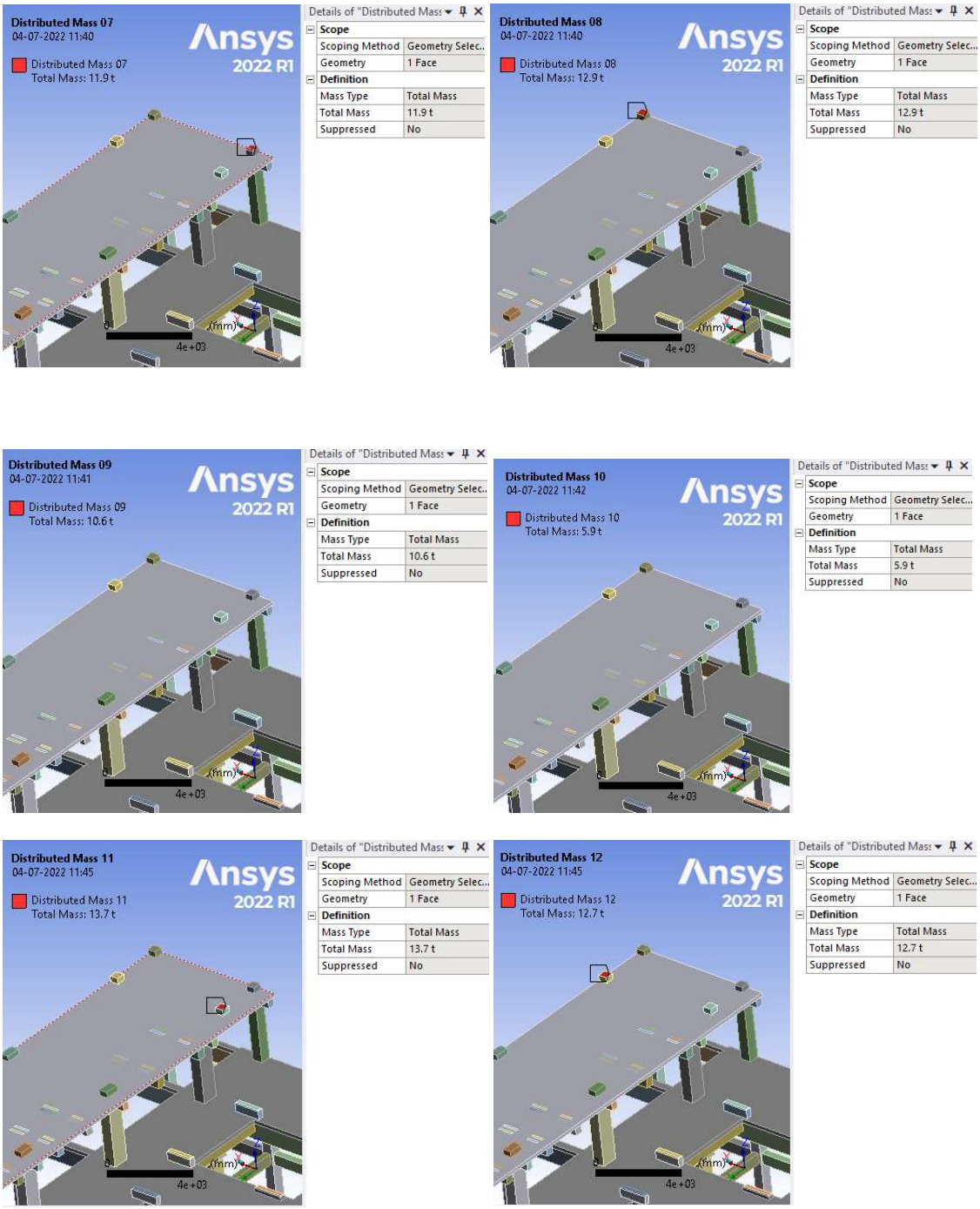


Figure 2v. Lumped mass of the Bunker at support: loaded

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

The screen building houses 1 no. secondary screen and 2 nos. of tertiary screen at 922.75 m level, all the screens will operate at constant speed. The screens are being fed from Vibrofeeders placed at 926.5 m level. The speed of the secondary Vibrofeeder will vary from 650 to 980 RPM while the speed for the tertiary feeders will vary from 550 to 900 RPM.

3.1. Dynamic Load, 100% Operational Speed for screens and Variable speed for feeders

Components of dynamic forces are taken from drawing provided by equipment manufacturer. Dynamic component is applied in the respective mass of the system. The following scenario of dynamic analysis have been followed to cater to variable speeds of feeder:

Case A - All Equipment operating at 100% Speed

Case B - Screens-100%, Ter. Feeder- 9.16 Hz, Sec. Feeder -10.83 Hz

Case C - Screens-100%, Ter. Feeder- 9.16 Hz, Sec. Feeder -12.2 Hz

Case D - Screens-100%, Ter. Feeder- 9.16 Hz, Sec. Feeder -15 Hz

Case E - Screens-100%, Ter. Feeder- 12 Hz, Sec. Feeder -10.83 Hz

Case F - Screens-100%, Ter. Feeder- 12 Hz, Sec. Feeder -12.2 Hz

Case G - Screens-100%, Ter. Feeder- 12 Hz, Sec. Feeder -15 Hz

Case H - Screens-100%, Ter. Feeder- 15 Hz, Sec. Feeder -10.83 Hz

Case I - Screens-100%, Ter. Feeder- 15 Hz, Sec. Feeder -12.2 Hz

Case J - Screens-100%, Ter. Feeder & Sec. Feeder -10.914 Hz

Case K - Screens-100%, Ter. Feeder & Sec. Feeder -11.172 Hz

Further to simulate the start/stop operation all feeders and screens are analyzed at 1.543Hz & 1.922 Hz which are the first two natural frequencies obtained from modal analysis. Refer **Appendix 5**

Similarly other natural frequencies with high mode participation factors (12.92 Hz, 13.776 Hz, 13.858 Hz) are also analyzed to assess the vibration velocities. Refer **Appendix 5**

3.2. Philosophy of the Operational Dynamic Loadings

All forces are acting in XZ plane.

All three screens and secondary feeder are in the same Phase

PROPRIETARY & CONFIDENTIAL
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

3.2.1. Secondary Screen Vibrating Feeder (3120-FDR-001)

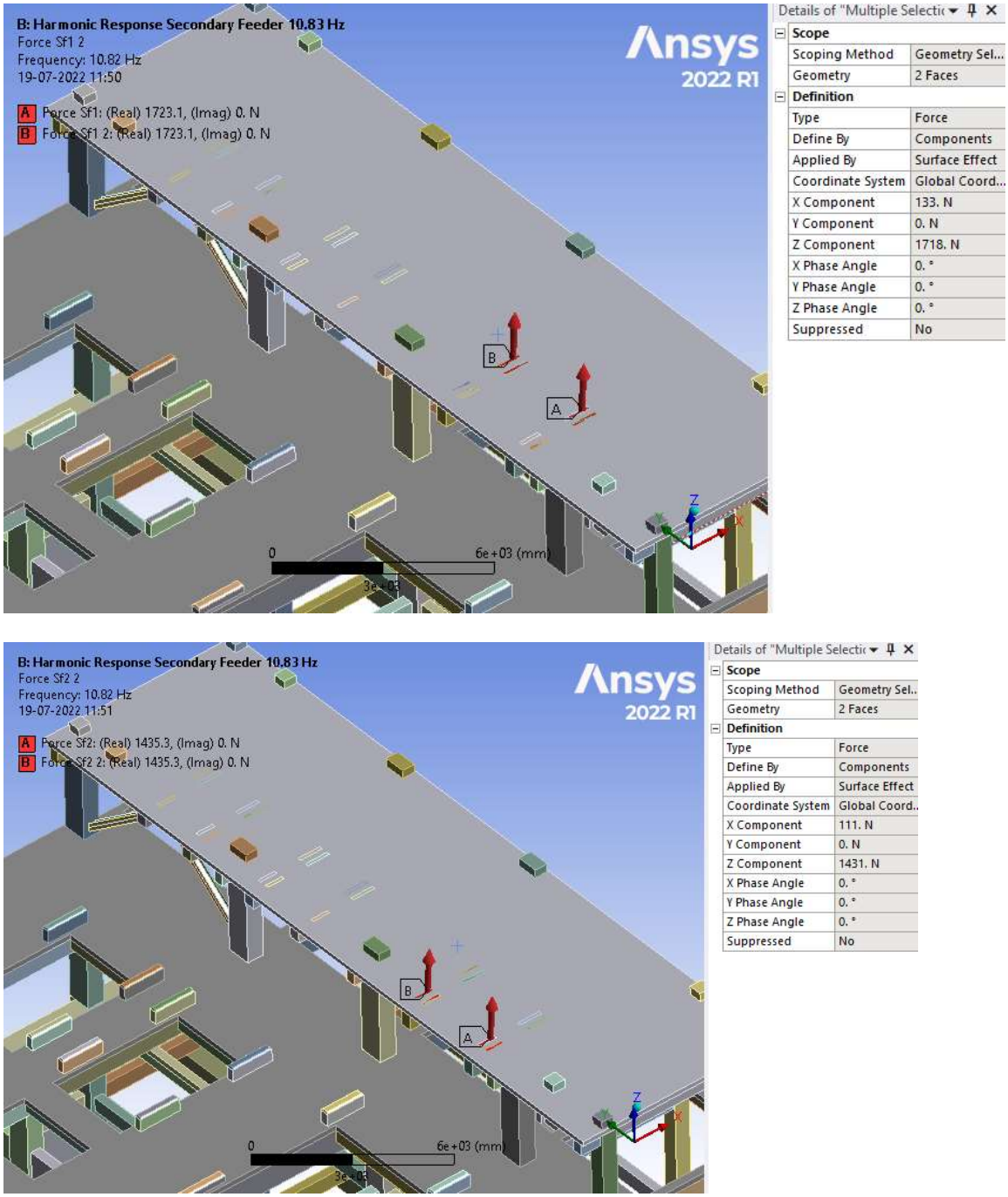


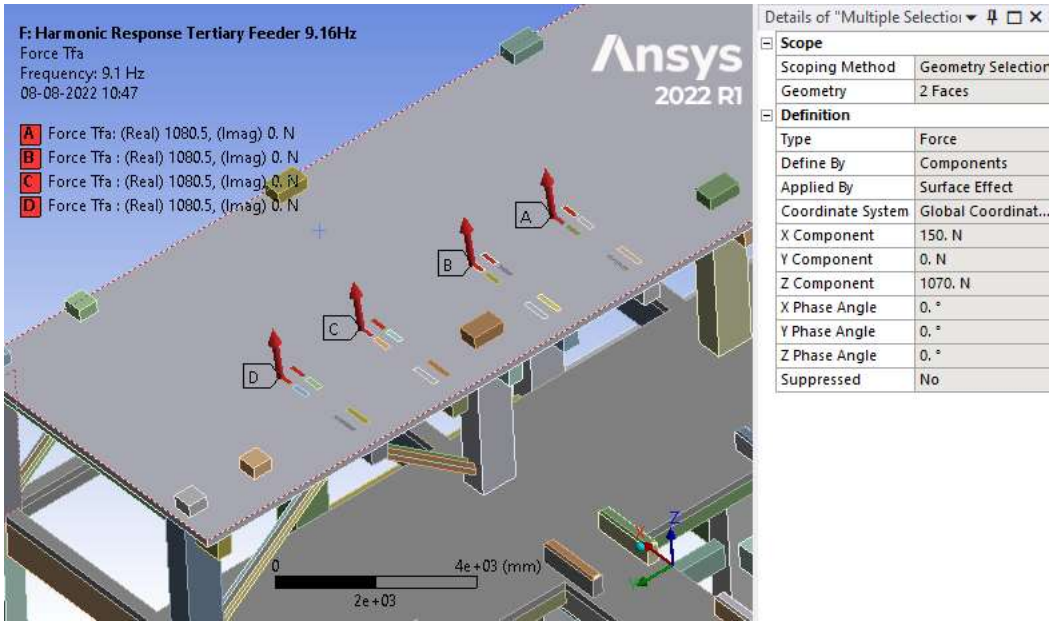
Figure 3a. Dynamic load Secondary feeder for 10.82 Hz

3.2.2. Tertiary Screen Vibrating Feeder (3130-FDR-001/002)

PROPRIETARY & CONFIDENTIAL
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

The tertiary screen vibrating feeder will be resting on vibration isolation system (refer ***Appendix 4***). The dynamic loads after installation of vibration isolation system are as follows:

Direction	Frequency [Hz]	Location	Node	Load [kN]	Location	Node	Load [kN]
X	9	Feeder	31	4.375	Below Spring	1	0.12
X	9	Feeder	34	5.825	Below Spring	4	0.15
X	9	Feeder	34	5.825	Below Spring	6	0.15
Z	9	Feeder	31	8.75	Below Spring	1	1.13
Z	9	Feeder	34	11.65	Below Spring	4	0.95
Z	9	Feeder	34	11.65	Below Spring	6	1.07
X	13	Feeder	31	4.375	Below Spring	1	0.05
X	13	Feeder	34	5.825	Below Spring	4	0.06
X	13	Feeder	34	5.825	Below Spring	6	0.06
Z	13	Feeder	31	8.75	Below Spring	1	0.51
Z	13	Feeder	34	11.65	Below Spring	4	0.42
Z	13	Feeder	34	11.65	Below Spring	6	0.47
X	15	Feeder	31	4.375	Below Spring	1	0.04
X	15	Feeder	34	5.825	Below Spring	4	0.05
Z	15	Feeder	34	5.825	Below Spring	6	0.25
Z	15	Feeder	31	8.75	Below Spring	1	0.38
Z	15	Feeder	34	11.65	Below Spring	4	0.31
Z	15	Feeder	34	11.65	Below Spring	6	0.34



PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

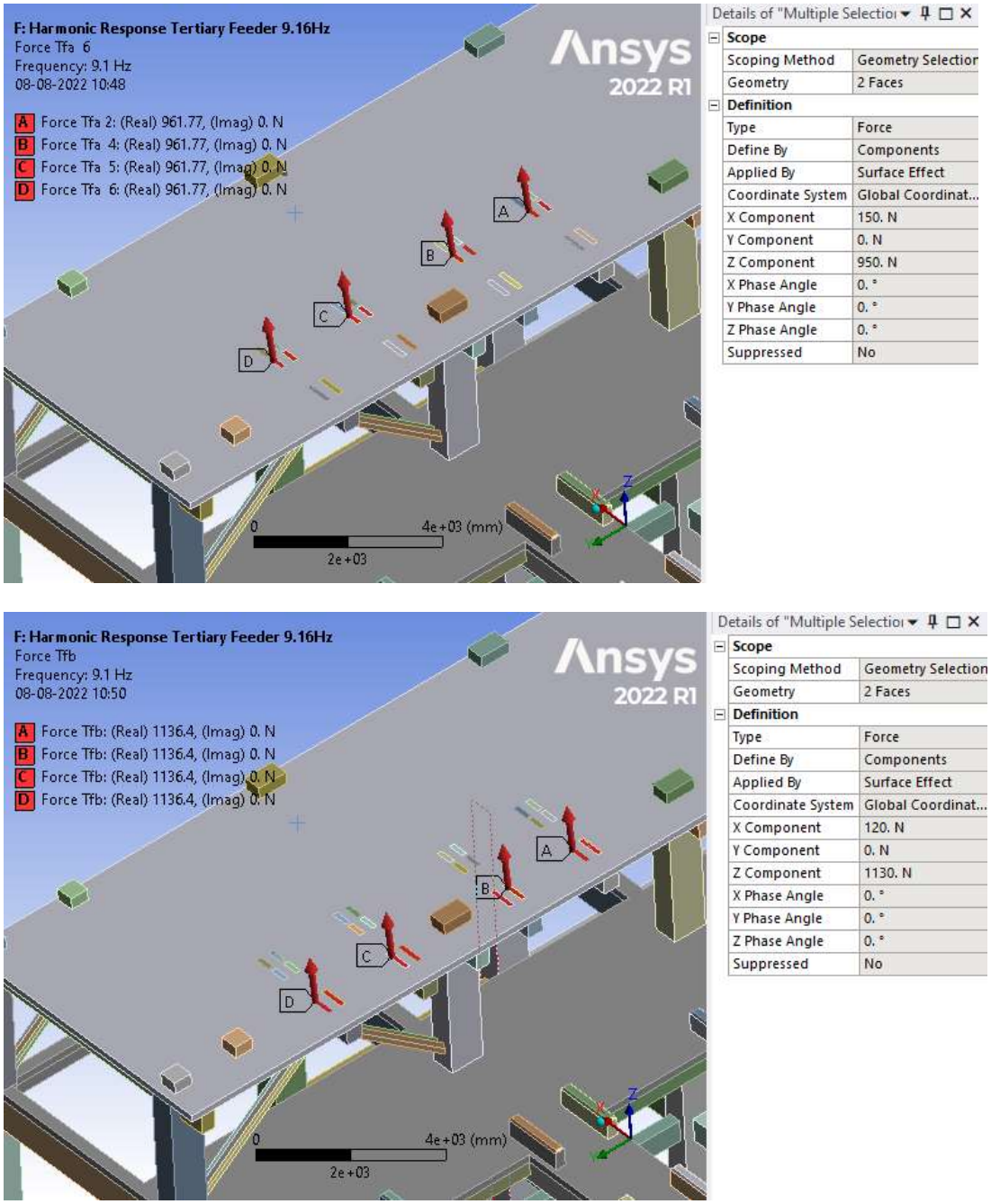


Figure 3b. Dynamic load Tertiary feeder for 9.1 Hz

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

3.2.3. Secondary Screen (3120-SCR-001)

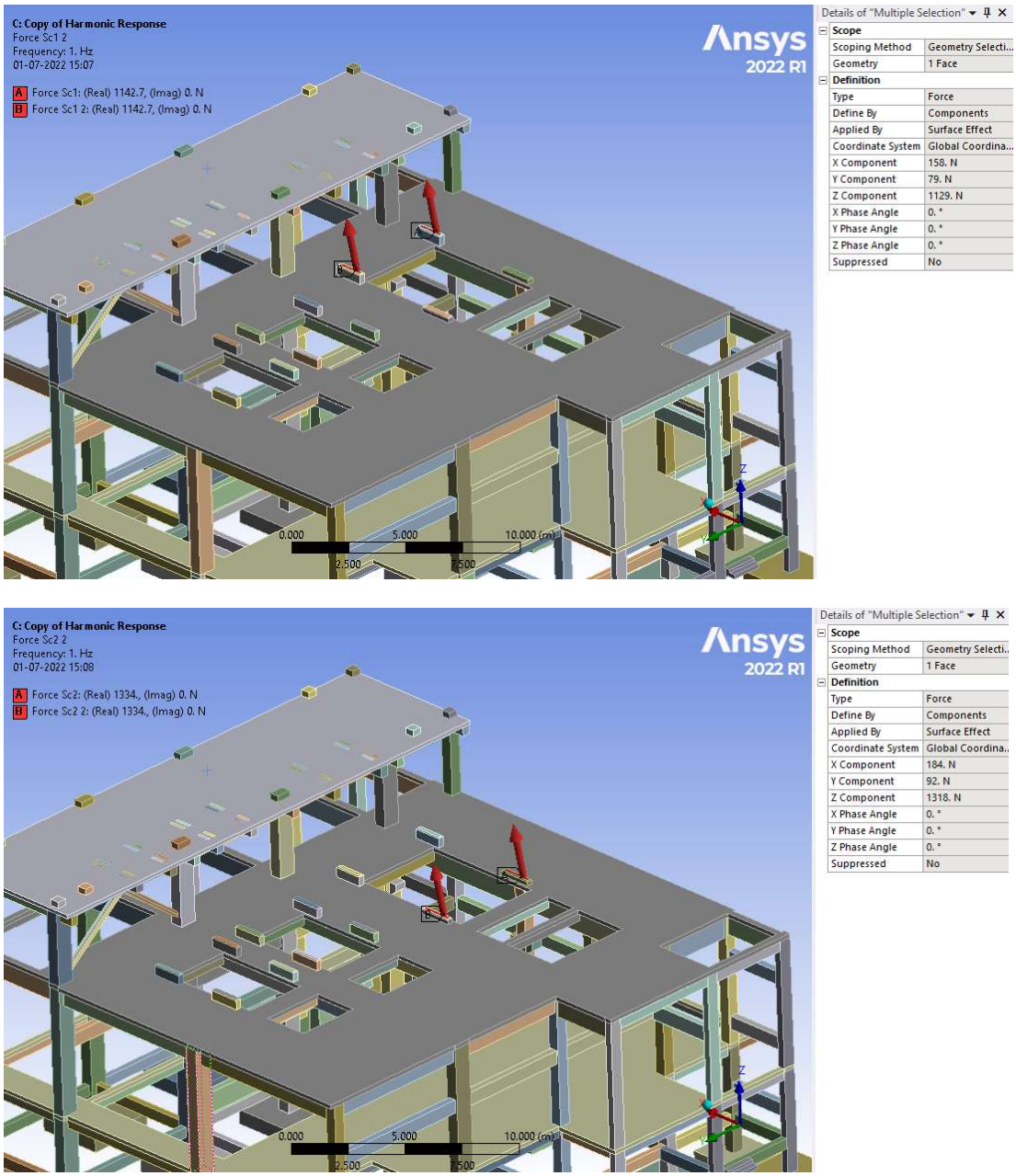


Figure 3c. Dynamic load Secondary screen

3.2.4. Tertiary Screen (3130-SCR-001/002)

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

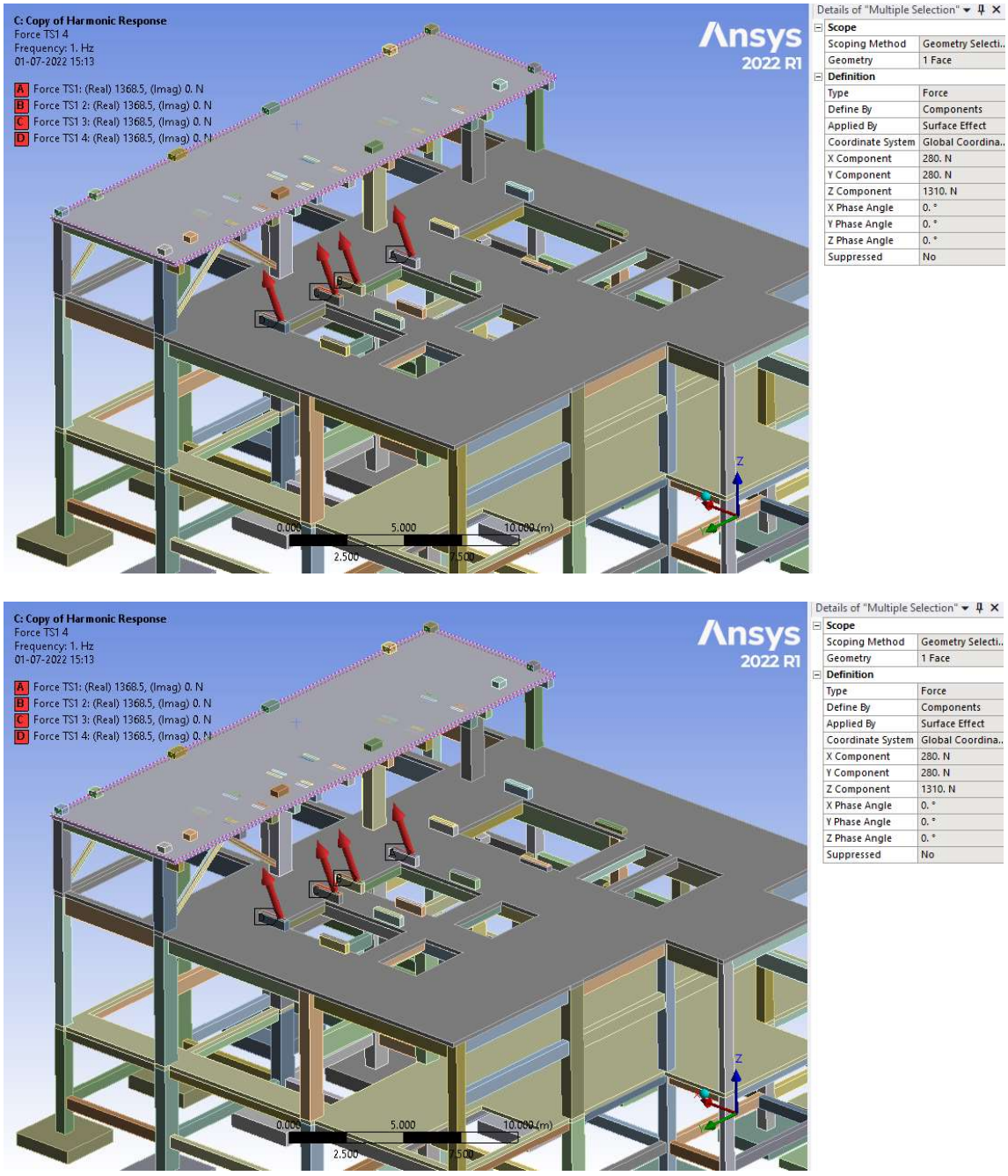


Figure 3d. Dynamic load Tertiary screen

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

3.3. Boundary Conditions

The model is attached with respective boundary condition of fixed support, shown in the Figure 3d.

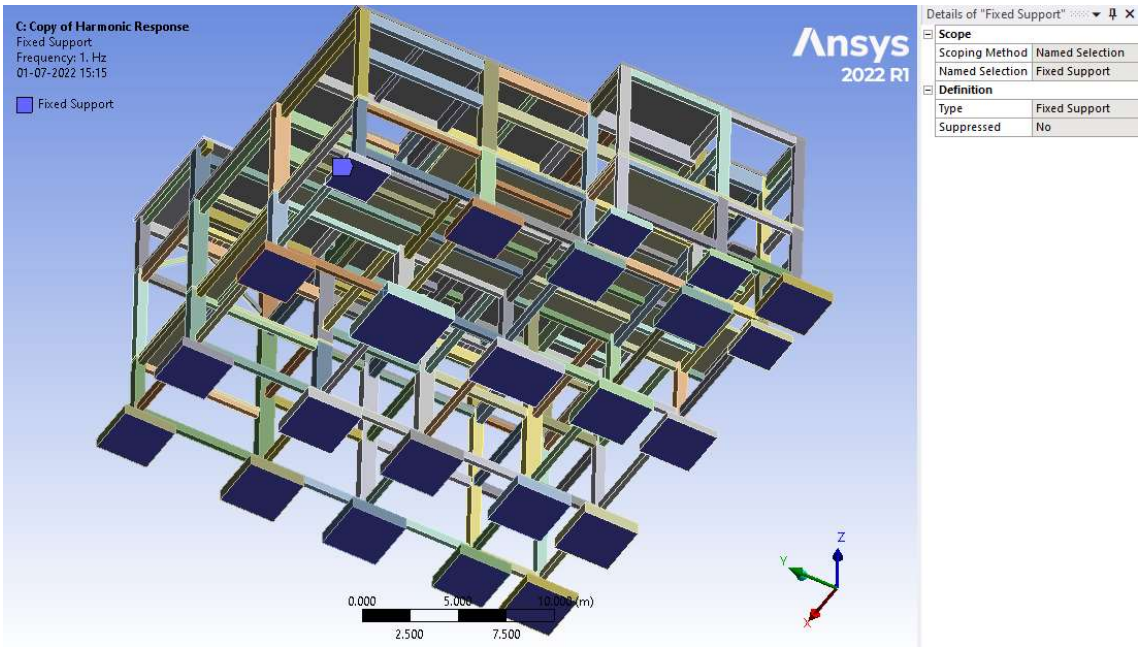


Figure 3e. Boundary condition – fixed support

4. Analysis of the Modified Structure

Main purpose of the study is to perform dynamic analysis of the modified structure to find the structural vibration during operational condition.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

4.1. Complete Model

Complete model of the system is shown in Figure 4a.

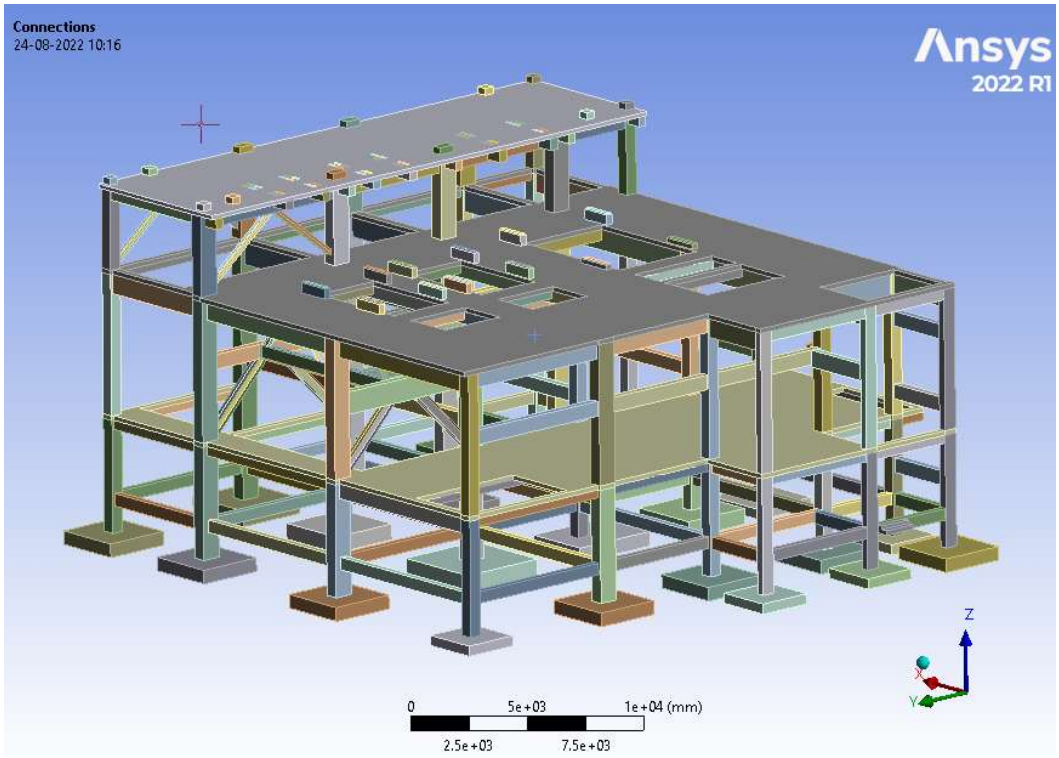


Figure 4a. Complete Model of the concrete structure of Screen building

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

4.2. Mesh

Figures 4b shows the finite element meshes of the Concrete structure. The number of elements and nodes are listed in Table 3.

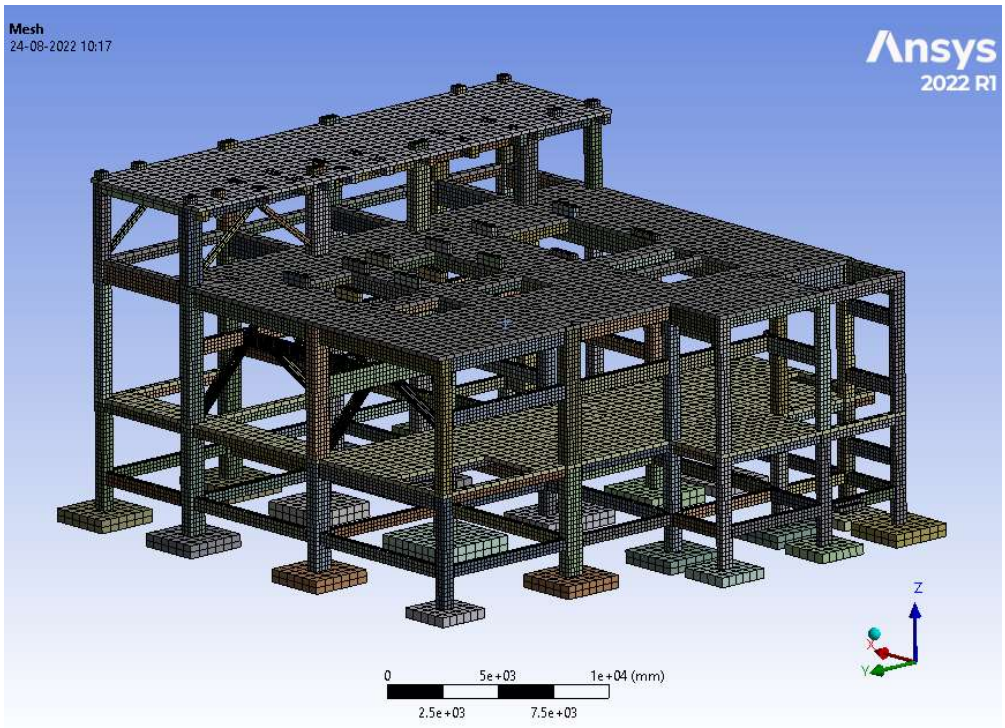


Figure 4b. Finite Element Mesh of Concrete structure

Statistics	
Nodes	379078
Elements	62968

Table 4. Finite Element Mesh Statistics

4.3. Modal Analysis

As the building has numerous degrees of freedom therefore, we have analyzed 100 such modes. The frequency separation check highlights that mode 1 to 5 lies outside frequency range of 7.5 to 30Hz. All other modes fall within the project specific frequency range of 7.5 to 30Hz. It is difficult for such a structure with so many degrees of freedom to eliminate all these modes. Hence forced analysis has been performed to evaluate the maximum vibration velocity that is getting generated on the structure. These vibration velocities are being kept within project specific limit of 3 mm/sec.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Table 4a – Modal analysis frequencies

***** PARTICIPATION FACTOR CALCULATION ***** X DIRECTION

MODE	FREQUENCY	PERIOD	PARTIC.FACTOR	RATIO	EFFECTIVE MASS	CUMULATIVE MASS FRACTION	RATIO EFF.MASS TO TOTAL MASS
1	1.54317	0.64802	47.642	1.000000	2269.77	0.898667	0.767753
2	1.92279	0.52008	4.5513	0.095532	20.7147	0.906868	0.700678E-02
3	2.90162	0.34463	-4.1474	0.087053	17.2009	0.913679	0.581821E-02
4	4.80843	0.20797	-7.4894	0.157201	56.0907	0.935886	0.189727E-01
5	5.61710	0.17803	-5.6056	0.117661	31.4228	0.948328	0.106288E-01
6	8.08609	0.12367	0.63102	0.013245	0.398185	0.948485	0.134687E-03
7	10.3846	0.96297E-01	10.818	0.227064	117.026	0.994819	0.395840E-01
8	10.9144	0.91622E-01	-0.32366	0.006794	0.104756	0.994860	0.354338E-04
9	11.1724	0.89506E-01	-0.84476	0.017731	0.713611	0.995143	0.241380E-03
10	11.3128	0.88395E-01	-0.53729	0.011278	0.288677	0.995257	0.976452E-04
11	11.8670	0.84267E-01	0.97826	0.020534	0.957000	0.995636	0.323706E-03
12	12.2759	0.81461E-01	0.18920	0.003971	0.357982E-01	0.995650	0.121088E-04
13	12.4827	0.80111E-01	-0.22289	0.004679	0.496820E-01	0.995670	0.168050E-04
14	12.9200	0.77399E-01	-0.51723	0.010857	0.267525	0.995776	0.904904E-04
15	13.5171	0.73980E-01	0.24181	0.005076	0.584725E-01	0.995799	0.197784E-04
16	13.7501	0.72727E-01	0.11617E-01	0.000244	0.134955E-03	0.995799	0.456485E-07
17	13.7757	0.72592E-01	0.24947	0.005236	0.622373E-01	0.995824	0.210518E-04
18	13.8580	0.72161E-01	-0.42870	0.008998	0.183780	0.995896	0.621638E-04
19	14.1472	0.70685E-01	-0.40201	0.008438	0.161610	0.995960	0.546648E-04
20	14.2467	0.70192E-01	-0.12098	0.002539	0.146366E-01	0.995966	0.495084E-05
21	14.3197	0.69834E-01	0.61939E-01	0.001300	0.383650E-02	0.995968	0.129770E-05
22	14.3589	0.69643E-01	0.58992E-01	0.001238	0.348001E-02	0.995969	0.117712E-05
23	14.3685	0.69597E-01	0.78322E-01	0.001644	0.613431E-02	0.995972	0.207494E-05
24	14.4157	0.69369E-01	0.82287E-01	0.001727	0.677107E-02	0.995974	0.229032E-05
25	14.7015	0.68020E-01	0.17280	0.003627	0.298608E-01	0.995986	0.101004E-04
26	14.7590	0.67755E-01	-0.80766E-01	0.001695	0.652311E-02	0.995989	0.220645E-05
27	14.9384	0.66942E-01	0.10576	0.002220	0.111845E-01	0.995993	0.378316E-05
28	14.9732	0.66786E-01	-0.11093	0.002328	0.123063E-01	0.995998	0.416262E-05
29	15.0238	0.66561E-01	0.15812E-01	0.000332	0.250009E-03	0.995998	0.845659E-07
30	15.1444	0.66031E-01	-0.38020E-01	0.000798	0.144554E-02	0.995999	0.488957E-06
31	15.2235	0.65688E-01	0.86325E-01	0.001812	0.745209E-02	0.996002	0.252068E-05
32	15.2847	0.65425E-01	0.55404E-01	0.001163	0.306961E-02	0.996003	0.103830E-05
33	15.3716	0.65055E-01	-0.10096	0.002119	0.101934E-01	0.996007	0.344794E-05
34	15.5130	0.64462E-01	0.10078E-01	0.000212	0.101564E-03	0.996007	0.343542E-07
35	15.5719	0.64218E-01	0.91186E-01	0.001914	0.831497E-02	0.996010	0.281255E-05
36	15.6573	0.63868E-01	0.46368	0.009733	0.214999	0.996095	0.727237E-04
37	15.7759	0.63388E-01	-0.49609E-01	0.001041	0.246106E-02	0.996096	0.832455E-06
38	16.0828	0.62178E-01	-0.21983	0.004614	0.483271E-01	0.996115	0.163467E-04
39	16.4499	0.60791E-01	-0.20095	0.004218	0.403818E-01	0.996131	0.136592E-04
40	16.5898	0.60278E-01	-0.14728E-01	0.000309	0.216923E-03	0.996131	0.733745E-07
41	16.6991	0.59884E-01	-0.33384	0.007007	0.111448	0.996176	0.376974E-04
42	16.8495	0.59349E-01	0.56347E-01	0.001183	0.317502E-02	0.996177	0.107395E-05
43	16.9563	0.58975E-01	-0.20157	0.004231	0.406288E-01	0.996193	0.137427E-04
44	17.1769	0.58218E-01	0.52886E-01	0.001110	0.279696E-02	0.996194	0.946075E-06
45	17.3194	0.57739E-01	-0.14527	0.003049	0.211047E-01	0.996202	0.713868E-05
46	17.4587	0.57278E-01	-0.14701	0.003086	0.216126E-01	0.996211	0.731047E-05
47	17.9980	0.55562E-01	-0.32880	0.006901	0.108110	0.996254	0.365683E-04
48	18.1101	0.55218E-01	-0.87409	0.018347	0.764038	0.996556	0.258437E-03
49	18.2642	0.54752E-01	0.21723E-01	0.000456	0.471871E-03	0.996556	0.159611E-06
50	18.5067	0.54035E-01	-0.19862	0.004169	0.394500E-01	0.996572	0.133440E-04

.

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

51	18.5651	0.53865E-01	0.26377	0.005537	0.695750E-01	0.996600	0.235338E-04
52	18.7493	0.53335E-01	1.0306	0.021631	1.06205	0.997020	0.359239E-03
53	18.8877	0.52944E-01	0.48017	0.010079	0.230561	0.997111	0.779873E-04
54	19.0548	0.52480E-01	-0.13634	0.002862	0.185881E-01	0.997119	0.628745E-05
55	19.1354	0.52259E-01	1.2027	0.025245	1.44658	0.997692	0.489306E-03
56	19.2249	0.52016E-01	0.14328	0.003007	0.205295E-01	0.997700	0.694412E-05
57	19.5101	0.51255E-01	0.76199E-01	0.001599	0.580624E-02	0.997702	0.196397E-05
58	19.7921	0.50525E-01	0.48225E-01	0.001012	0.232566E-02	0.997703	0.786656E-06
59	20.0519	0.49870E-01	-0.47944E-01	0.001006	0.229866E-02	0.997704	0.777523E-06
60	20.3170	0.49220E-01	1.4466	0.030365	2.09276	0.998532	0.707880E-03
61	20.4782	0.48833E-01	-0.93060E-01	0.001953	0.866025E-02	0.998536	0.292934E-05
62	20.5578	0.48643E-01	0.94581E-01	0.001985	0.894558E-02	0.998539	0.302585E-05
63	20.6126	0.48514E-01	0.40311	0.008461	0.162498	0.998604	0.549651E-04
64	20.7282	0.48244E-01	-0.15525	0.003259	0.241024E-01	0.998613	0.815266E-05
65	20.7576	0.48175E-01	0.10516	0.002207	0.110594E-01	0.998618	0.374087E-05
66	21.1355	0.47314E-01	0.11262	0.002364	0.126834E-01	0.998623	0.429018E-05
67	21.2448	0.47070E-01	-0.27297E-01	0.000573	0.745144E-03	0.998623	0.252046E-06
68	21.3273	0.46888E-01	0.26266E-01	0.000551	0.689920E-03	0.998623	0.233366E-06
69	21.5129	0.46484E-01	-0.48246E-01	0.001013	0.232763E-02	0.998624	0.787323E-06
70	21.7279	0.46024E-01	0.28331E-01	0.000595	0.802652E-03	0.998624	0.271498E-06
71	21.9556	0.45546E-01	0.47604E-01	0.000999	0.226612E-02	0.998625	0.766517E-06
72	22.0874	0.45275E-01	0.31295	0.006569	0.979351E-01	0.998664	0.331266E-04
73	22.1427	0.45162E-01	0.23530	0.004939	0.553658E-01	0.998686	0.187275E-04
74	22.3139	0.44815E-01	-0.26666E-01	0.000560	0.711064E-03	0.998686	0.240518E-06
75	22.4678	0.44508E-01	0.84333E-02	0.000177	0.711205E-04	0.998686	0.240566E-07
76	22.5859	0.44275E-01	-0.24908	0.005228	0.620386E-01	0.998711	0.209846E-04
77	22.7766	0.43905E-01	0.10388	0.002180	0.107909E-01	0.998715	0.365003E-05
78	22.8595	0.43746E-01	-0.42546E-01	0.000893	0.181017E-02	0.998716	0.612293E-06
79	22.9293	0.43612E-01	0.35945E-01	0.000754	0.129202E-02	0.998716	0.437028E-06
80	23.0643	0.43357E-01	-0.16840	0.003535	0.283585E-01	0.998728	0.959229E-05
81	23.4540	0.42637E-01	0.30356E-01	0.000637	0.921465E-03	0.998728	0.311686E-06
82	23.6011	0.42371E-01	-0.31720	0.006658	0.100619	0.998768	0.340344E-04
83	23.6374	0.42306E-01	0.10474	0.002199	0.109714E-01	0.998772	0.371108E-05
84	23.8100	0.41999E-01	-0.79926E-01	0.001678	0.638812E-02	0.998775	0.216079E-05
85	23.9500	0.41754E-01	-0.95363E-01	0.002002	0.909408E-02	0.998778	0.307608E-05
86	24.1685	0.41376E-01	0.32074E-01	0.000673	0.102875E-02	0.998779	0.347975E-06
87	24.4523	0.40896E-01	-0.48330E-02	0.000101	0.233577E-04	0.998779	0.790075E-08
88	24.5232	0.40778E-01	-0.13508	0.002835	0.182453E-01	0.998786	0.617149E-05
89	25.0587	0.39906E-01	1.1538	0.024218	1.33128	0.999313	0.450306E-03
90	25.1533	0.39756E-01	-0.82812E-01	0.001738	0.685777E-02	0.999316	0.231965E-05
91	25.2638	0.39582E-01	0.40590	0.008520	0.164755	0.999381	0.557284E-04
92	25.4996	0.39216E-01	0.48235E-01	0.001012	0.232661E-02	0.999382	0.786979E-06
93	25.5472	0.39143E-01	0.14790	0.003104	0.218755E-01	0.999391	0.739940E-05
94	25.5593	0.39125E-01	0.12803	0.002687	0.163923E-01	0.999397	0.554470E-05
95	25.7027	0.38906E-01	0.50094	0.010515	0.250938	0.999496	0.848800E-04
96	25.7313	0.38863E-01	0.96909	0.020341	0.939128	0.999868	0.317661E-03
97	25.7804	0.38789E-01	0.10869	0.002281	0.118138E-01	0.999873	0.399603E-05
98	25.9235	0.38575E-01	0.41935	0.008802	0.175855	0.999943	0.594831E-04
99	26.0251	0.38424E-01	-0.30379	0.006377	0.922898E-01	0.999979	0.312171E-04
100	26.0708	0.38357E-01	0.22994	0.004826	0.528736E-01	1.00000	0.178846E-04
sum					2525.71		0.854325

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

***** PARTICIPATION FACTOR CALCULATION ***** Y DIRECTION

MODE	FREQUENCY	PERIOD	PARTIC.FACTOR	RATIO	EFFECTIVE MASS	CUMULATIVE MASS FRACTION	RATIO EFF.MASS TO TOTAL MASS
1	1.54317	0.64802	-2.8820	0.064351	8.30585	0.328109E-02	0.280946E-02
2	1.92279	0.52008	44.786	1.000000	2005.75	0.795622	0.678448
3	2.90162	0.34463	19.373	0.432575	375.319	0.943886	0.126952
4	4.80843	0.20797	-4.1195	0.091982	16.9699	0.950590	0.574010E-02
5	5.61710	0.17803	3.5629	0.079555	12.6945	0.955604	0.429394E-02
6	8.08609	0.12367	4.9159	0.109765	24.1660	0.965151	0.817416E-02
7	10.3846	0.96297E-01	0.37419	0.008355	0.140018	0.965206	0.473611E-04
8	10.9144	0.91622E-01	-0.25245	0.005637	0.637287E-01	0.965231	0.215563E-04
9	11.1724	0.89506E-01	-0.69525	0.015524	0.483377	0.965422	0.163503E-03
10	11.3128	0.88395E-01	5.6675	0.126546	32.1201	0.978111	0.108647E-01
11	11.8670	0.84267E-01	-2.8120	0.062789	7.90752	0.981235	0.267473E-02
12	12.2759	0.81461E-01	-0.67593	0.015093	0.456882	0.981415	0.154541E-03
13	12.4827	0.80111E-01	-0.20878	0.004662	0.435885E-01	0.981432	0.147439E-04
14	12.9200	0.77399E-01	0.22918	0.005117	0.525245E-01	0.981453	0.177665E-04
15	13.5171	0.73980E-01	4.2171	0.094162	17.7838	0.988478	0.601539E-02
16	13.7501	0.72727E-01	0.39516	0.008823	0.156149	0.988540	0.528176E-04
17	13.7757	0.72592E-01	1.9968	0.044587	3.98736	0.990115	0.134873E-02
18	13.8580	0.72161E-01	-0.23041	0.005145	0.530881E-01	0.990136	0.179571E-04
19	14.1472	0.70685E-01	0.47279	0.010557	0.223527	0.990224	0.756083E-04
20	14.2467	0.70192E-01	-0.22146	0.004945	0.490467E-01	0.990244	0.165901E-04
21	14.3197	0.69834E-01	0.17850	0.003986	0.318636E-01	0.990256	0.107779E-04
22	14.3589	0.69643E-01	1.0436	0.023303	1.08914	0.990687	0.368402E-03
23	14.3685	0.69597E-01	1.0453	0.023340	1.09261	0.991118	0.369577E-03
24	14.4157	0.69369E-01	-1.0175	0.022719	1.03526	0.991527	0.350178E-03
25	14.7015	0.68020E-01	0.66754	0.014905	0.445610	0.991703	0.150728E-03
26	14.7590	0.67755E-01	-0.96559E-01	0.002156	0.932356E-02	0.991707	0.315371E-05
27	14.9384	0.66942E-01	-0.11488	0.002565	0.131979E-01	0.991712	0.446421E-05
28	14.9732	0.66786E-01	0.87267E-01	0.001949	0.761546E-02	0.991715	0.257594E-05
29	15.0238	0.66561E-01	1.9013	0.042453	3.61492	0.993143	0.122275E-02
30	15.1444	0.66031E-01	0.64842	0.014478	0.420446	0.993309	0.142216E-03
31	15.2235	0.65688E-01	1.7749	0.039632	3.15039	0.994554	0.106562E-02
32	15.2847	0.65425E-01	0.87664	0.019574	0.768498	0.994857	0.259945E-03
33	15.3716	0.65055E-01	1.1865	0.026493	1.40778	0.995413	0.476181E-03
34	15.5130	0.64462E-01	1.8006	0.040204	3.24201	0.996694	0.109661E-02
35	15.5719	0.64218E-01	-0.61511	0.013734	0.378357	0.996844	0.127980E-03
36	15.6573	0.63868E-01	-0.39185	0.008750	0.153548	0.996904	0.519378E-04
37	15.7759	0.63388E-01	0.79108E-01	0.001766	0.625801E-02	0.996907	0.211678E-05
38	16.0828	0.62178E-01	-0.21345	0.004766	0.455597E-01	0.996925	0.154106E-04
39	16.4499	0.60791E-01	0.48013	0.010721	0.230521	0.997016	0.779741E-04
40	16.5898	0.60278E-01	-0.28210	0.006299	0.795807E-01	0.997047	0.269182E-04
41	16.6991	0.59884E-01	0.12744E-01	0.000285	0.162422E-03	0.997047	0.549393E-07
42	16.8495	0.59349E-01	0.99368E-01	0.002219	0.987395E-02	0.997051	0.333987E-05
43	16.9563	0.58975E-01	-0.18320	0.004091	0.335614E-01	0.997064	0.113522E-04
44	17.1769	0.58218E-01	-0.83782	0.018707	0.701935	0.997342	0.237430E-03
45	17.3194	0.57739E-01	-0.28629	0.006393	0.819640E-01	0.997374	0.277244E-04
46	17.4587	0.57278E-01	0.18937	0.004228	0.358591E-01	0.997388	0.121294E-04
47	17.9980	0.55562E-01	0.28621	0.006391	0.819174E-01	0.997421	0.277087E-04
48	18.1101	0.55218E-01	-0.76292E-01	0.001704	0.582054E-02	0.997423	0.196880E-05
49	18.2642	0.54752E-01	-0.24048	0.005370	0.578296E-01	0.997446	0.195609E-04
50	18.5067	0.54035E-01	-0.38490	0.008594	0.148151	0.997504	0.501122E-04

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

51	18.5651	0.53865E-01	-0.39424	0.008803	0.155423	0.997566	0.525720E-04
52	18.7493	0.53335E-01	-0.57381	0.012812	0.329263	0.997696	0.111373E-03
53	18.8877	0.52944E-01	0.27167	0.006066	0.738055E-01	0.997725	0.249648E-04
54	19.0548	0.52480E-01	0.19542	0.004363	0.381875E-01	0.997740	0.129170E-04
55	19.1354	0.52259E-01	-0.23078	0.005153	0.532599E-01	0.997761	0.180152E-04
56	19.2249	0.52016E-01	0.44169	0.009862	0.195091	0.997838	0.659897E-04
57	19.5101	0.51255E-01	0.34315E-01	0.000766	0.117750E-02	0.997839	0.398290E-06
58	19.7921	0.50525E-01	-0.23185	0.005177	0.537562E-01	0.997860	0.181831E-04
59	20.0519	0.49870E-01	0.14832E-02	0.000033	0.219979E-05	0.997860	0.744082E-09
60	20.3170	0.49220E-01	-0.79076E-01	0.001766	0.625307E-02	0.997862	0.211511E-05
61	20.4782	0.48833E-01	0.81298E-01	0.001815	0.660941E-02	0.997865	0.223564E-05
62	20.5578	0.48643E-01	0.65698	0.014670	0.431627	0.998035	0.145998E-03
63	20.6126	0.48514E-01	0.97843E-01	0.002185	0.957325E-02	0.998039	0.323816E-05
64	20.7282	0.48244E-01	-0.52354	0.011690	0.274096	0.998147	0.927131E-04
65	20.7576	0.48175E-01	0.33505	0.007481	0.112257	0.998192	0.379709E-04
66	21.1355	0.47314E-01	-0.24257	0.005416	0.588396E-01	0.998215	0.199026E-04
67	21.2448	0.47070E-01	-0.15051E-01	0.000336	0.226522E-03	0.998215	0.766212E-07
68	21.3273	0.46888E-01	-0.34361	0.007672	0.118066	0.998262	0.399361E-04
69	21.5129	0.46484E-01	-0.36291	0.008103	0.131700	0.998314	0.445478E-04
70	21.7279	0.46024E-01	0.68343	0.015260	0.467078	0.998498	0.157990E-03
71	21.9556	0.45546E-01	1.2535	0.027989	1.57132	0.999119	0.531501E-03
72	22.0874	0.45275E-01	0.22778	0.005086	0.518838E-01	0.999140	0.175497E-04
73	22.1427	0.45162E-01	-0.17889	0.003994	0.320025E-01	0.999152	0.108249E-04
74	22.3139	0.44815E-01	-0.26918	0.006010	0.724575E-01	0.999181	0.245088E-04
75	22.4678	0.44508E-01	-0.15051E-01	0.000336	0.226542E-03	0.999181	0.766280E-07
76	22.5859	0.44275E-01	-0.66559	0.014862	0.443010	0.999356	0.149849E-03
77	22.7766	0.43905E-01	-0.18778	0.004193	0.352620E-01	0.999370	0.119274E-04
78	22.8595	0.43746E-01	-0.35909	0.008018	0.128944	0.999421	0.436155E-04
79	22.9293	0.43612E-01	0.33347	0.007446	0.111201	0.999465	0.376139E-04
80	23.0643	0.43357E-01	-0.13846	0.003092	0.191704E-01	0.999472	0.648441E-05
81	23.4540	0.42637E-01	0.33190E-01	0.000741	0.110155E-02	0.999473	0.372601E-06
82	23.6011	0.42371E-01	0.43968	0.009817	0.193318	0.999549	0.653901E-04
83	23.6374	0.42306E-01	-0.27666	0.006177	0.765387E-01	0.999579	0.258893E-04
84	23.8100	0.41999E-01	-0.92209E-02	0.000206	0.850248E-04	0.999579	0.287597E-07
85	23.9500	0.41754E-01	0.15185	0.003391	0.230583E-01	0.999588	0.779948E-05
86	24.1685	0.41376E-01	-0.10294	0.002298	0.105963E-01	0.999593	0.358420E-05
87	24.4523	0.40896E-01	-0.68718	0.015344	0.472218	0.999779	0.159728E-03
88	24.5232	0.40778E-01	0.22911	0.005116	0.524898E-01	0.999800	0.177547E-04
89	25.0587	0.39906E-01	0.17038	0.003804	0.290290E-01	0.999811	0.981908E-05
90	25.1533	0.39756E-01	-0.46527	0.010389	0.216474	0.999897	0.732226E-04
91	25.2638	0.39582E-01	0.16631	0.003714	0.276602E-01	0.999908	0.935609E-05
92	25.4996	0.39216E-01	0.24099	0.005381	0.580743E-01	0.999931	0.196437E-04
93	25.5472	0.39143E-01	0.27592	0.006161	0.761339E-01	0.999961	0.257524E-04
94	25.5593	0.39125E-01	-0.87278E-02	0.000195	0.761743E-04	0.999961	0.257660E-07
95	25.7027	0.38906E-01	0.14105	0.003149	0.198950E-01	0.999969	0.672951E-05
96	25.7313	0.38863E-01	0.14547	0.003248	0.211621E-01	0.999977	0.715810E-05
97	25.7804	0.38789E-01	0.20506	0.004579	0.420484E-01	0.999994	0.142229E-04
98	25.9235	0.38575E-01	-0.28427E-01	0.000635	0.808119E-03	0.999994	0.273347E-06
99	26.0251	0.38424E-01	0.12317	0.002750	0.151698E-01	1.00000	0.513119E-05
100	26.0708	0.38357E-01	-0.17904E-02	0.000040	0.320561E-05	1.00000	0.108430E-08
sum					2531.43		0.856258

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

***** PARTICIPATION FACTOR CALCULATION ***** Z DIRECTION

MODE	FREQUENCY	PERIOD	PARTIC.FACTOR	RATIO	EFFECTIVE MASS	CUMULATIVE MASS FRACTION	RATIO EFF.MASS TO TOTAL MASS
1	1.54317	0.64802	0.40227E-01	0.001639	0.161820E-02	0.714455E-06	0.547358E-06
2	1.92279	0.52008	0.20094	0.008187	0.403779E-01	0.185418E-04	0.136579E-04
3	2.90162	0.34463	0.90504E-01	0.003687	0.819103E-02	0.221582E-04	0.277062E-05
4	4.80843	0.20797	0.51175	0.020849	0.261887	0.137785E-03	0.885836E-04
5	5.61710	0.17803	0.46795	0.019065	0.218977	0.234466E-03	0.740691E-04
6	8.08609	0.12367	-1.2494	0.050901	1.56093	0.923636E-03	0.527986E-03
7	10.3846	0.96297E-01	3.0860	0.125727	9.52347	0.512837E-02	0.322132E-02
8	10.9144	0.91622E-01	20.090	0.818493	403.613	0.183329	0.136523
9	11.1724	0.89506E-01	24.545	1.000000	602.470	0.449327	0.203786
10	11.3128	0.88395E-01	1.4385	0.058606	2.06930	0.450240	0.699944E-03
11	11.8670	0.84267E-01	0.13925	0.005673	0.193918E-01	0.450249	0.655930E-05
12	12.2759	0.81461E-01	-0.60021	0.024453	0.360257	0.450408	0.121857E-03
13	12.4827	0.80111E-01	1.4363	0.058515	2.06288	0.451319	0.697771E-03
14	12.9200	0.77399E-01	13.571	0.552896	184.171	0.532633	0.622962E-01
15	13.5171	0.73980E-01	5.9310	0.241635	35.1767	0.548164	0.118985E-01
16	13.7501	0.72727E-01	-2.2162	0.090290	4.91145	0.550332	0.166130E-02
17	13.7757	0.72592E-01	-12.951	0.527644	167.732	0.624388	0.567357E-01
18	13.8580	0.72161E-01	-19.367	0.789023	375.072	0.789987	0.126869
19	14.1472	0.70685E-01	-0.28469	0.011599	0.810490E-01	0.790023	0.274149E-04
20	14.2467	0.70192E-01	-3.4759	0.141611	12.0818	0.795357	0.408667E-02
21	14.3197	0.69834E-01	-2.3615	0.096209	5.57653	0.797819	0.188627E-02
22	14.3589	0.69643E-01	3.5798	0.145845	12.8150	0.803477	0.433469E-02
23	14.3685	0.69597E-01	0.39923	0.016265	0.159387	0.803548	0.539130E-04
24	14.4157	0.69369E-01	5.7469	0.234134	33.0265	0.818129	0.111713E-01
25	14.7015	0.68020E-01	-0.51111	0.020823	0.261233	0.818245	0.883625E-04
26	14.7590	0.67755E-01	-9.0733	0.369657	82.3252	0.854592	0.278466E-01
27	14.9384	0.66942E-01	0.38395	0.015643	0.147418	0.854657	0.498642E-04
28	14.9732	0.66786E-01	-0.43211	0.017605	0.186721	0.854740	0.631584E-04
29	15.0238	0.66561E-01	1.4474	0.058969	2.09496	0.855665	0.708624E-03
30	15.1444	0.66031E-01	-1.8219	0.074225	3.31921	0.857130	0.112273E-02
31	15.2235	0.65688E-01	-2.1727	0.088519	4.72074	0.859215	0.159679E-02
32	15.2847	0.65425E-01	-0.59187	0.024114	0.350313	0.859369	0.118494E-03
33	15.3716	0.65055E-01	1.3014	0.053020	1.69359	0.860117	0.572857E-03
34	15.5130	0.64462E-01	-1.6731	0.068162	2.79913	0.861353	0.946809E-03
35	15.5719	0.64218E-01	-7.3187	0.298171	53.5632	0.885002	0.181178E-01
36	15.6573	0.63868E-01	2.4566	0.100083	6.03468	0.887666	0.204124E-02
37	15.7759	0.63388E-01	8.3078	0.338469	69.0196	0.918139	0.233459E-01
38	16.0828	0.62178E-01	2.2037	0.089779	4.85609	0.920283	0.164258E-02
39	16.4499	0.60791E-01	2.3843	0.097137	5.68470	0.922793	0.192285E-02
40	16.5898	0.60278E-01	-4.0072	0.163259	16.0580	0.929883	0.543163E-02
41	16.6991	0.59884E-01	2.7182	0.110740	7.38836	0.933145	0.249912E-02
42	16.8495	0.59349E-01	4.5394	0.184942	20.6065	0.942243	0.697017E-02
43	16.9563	0.58975E-01	-0.77915	0.031743	0.607071	0.942511	0.205342E-03
44	17.1769	0.58218E-01	-1.4006	0.057062	1.96168	0.943377	0.663539E-03
45	17.3194	0.57739E-01	-2.9450	0.119982	8.67291	0.947206	0.293362E-02
46	17.4587	0.57278E-01	3.8024	0.154914	14.4584	0.953590	0.489055E-02
47	17.9980	0.55562E-01	-0.94632E-01	0.003855	0.895520E-02	0.953594	0.302910E-05
48	18.1101	0.55218E-01	0.43849	0.017864	0.192272	0.953679	0.650363E-04
49	18.2642	0.54752E-01	-2.0229	0.082415	4.09215	0.955485	0.138417E-02
50	18.5067	0.54035E-01	1.4834	0.060434	2.20037	0.956457	0.744277E-03

PROPRIETARY & CONFIDENTIAL

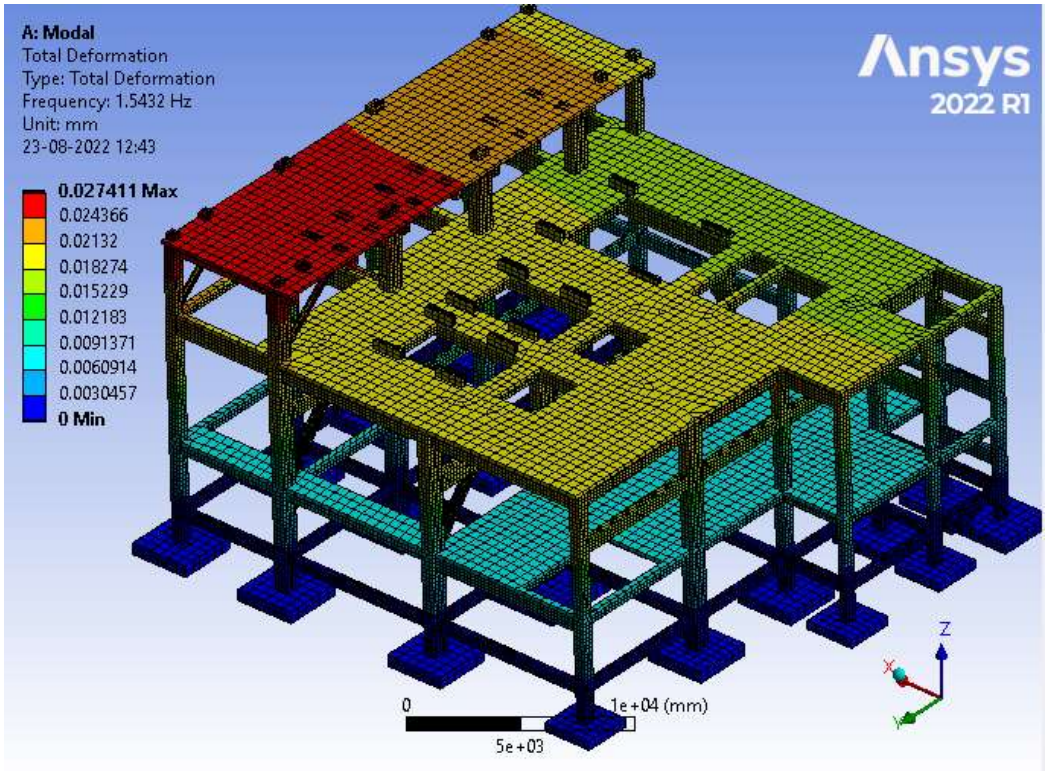
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

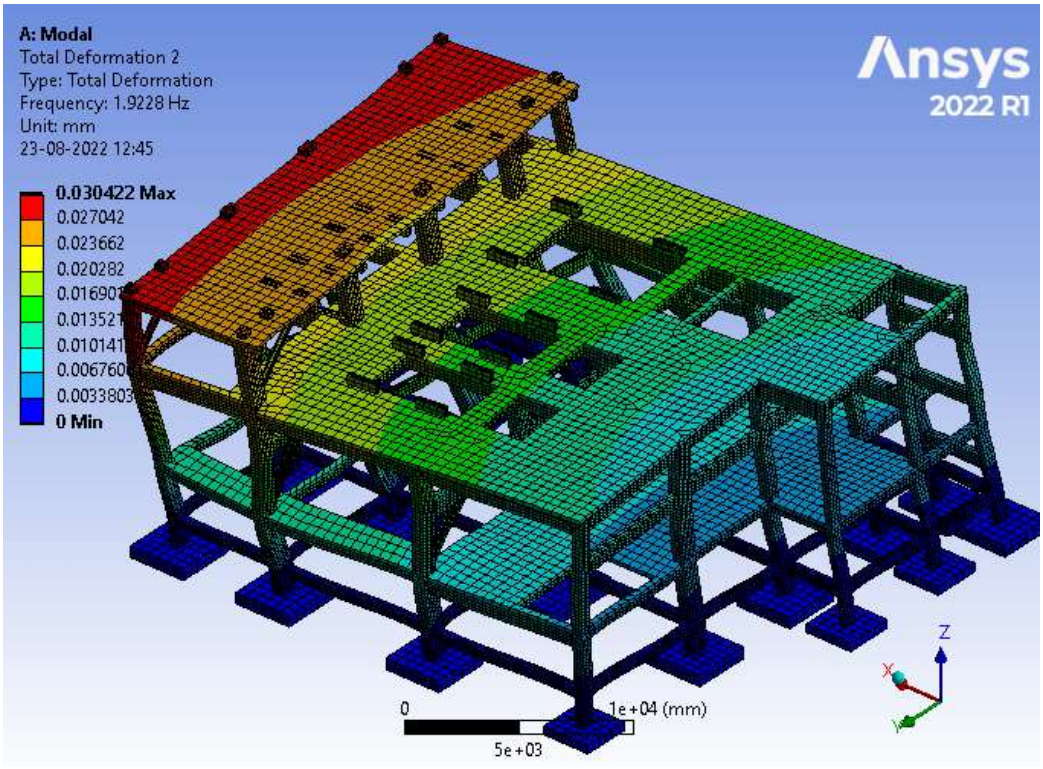
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

51	18.5651	0.53865E-01	-1.7622	0.071793	3.10529	0.957828	0.105037E-02
52	18.7493	0.53335E-01	0.35986	0.014661	0.129502	0.957885	0.438041E-04
53	18.8877	0.52944E-01	0.50686	0.020650	0.256903	0.957998	0.868977E-04
54	19.0548	0.52480E-01	-5.8931	0.240093	34.7290	0.973332	0.117471E-01
55	19.1354	0.52259E-01	-0.35968	0.014654	0.129368	0.973389	0.437589E-04
56	19.2249	0.52016E-01	0.69522E-01	0.002832	0.483334E-02	0.973391	0.163488E-05
57	19.5101	0.51255E-01	1.8850	0.076797	3.55327	0.974960	0.120190E-02
58	19.7921	0.50525E-01	-0.28851	0.011754	0.832360E-01	0.974997	0.281546E-04
59	20.0519	0.49870E-01	-1.8009	0.073369	3.24310	0.976428	0.109698E-02
60	20.3170	0.49220E-01	-0.13293	0.005416	0.176715E-01	0.976436	0.597739E-05
61	20.4782	0.48833E-01	-0.27435	0.011177	0.752656E-01	0.976469	0.254587E-04
62	20.5578	0.48643E-01	0.62286	0.025376	0.387951	0.976641	0.131225E-03
63	20.6126	0.48514E-01	0.33371	0.013596	0.111362	0.976690	0.376682E-04
64	20.7282	0.48244E-01	-1.0685	0.043533	1.14175	0.977194	0.386199E-03
65	20.7576	0.48175E-01	-0.97289	0.039637	0.946517	0.977612	0.320160E-03
66	21.1355	0.47314E-01	-0.21705	0.008843	0.471109E-01	0.977633	0.159353E-04
67	21.2448	0.47070E-01	0.75973	0.030952	0.577188	0.977888	0.195235E-03
68	21.3273	0.46888E-01	-1.4535	0.059217	2.11268	0.978820	0.714617E-03
69	21.5129	0.46484E-01	-0.69584	0.028349	0.484191	0.979034	0.163778E-03
70	21.7279	0.46024E-01	-0.23931	0.009750	0.572682E-01	0.979059	0.193710E-04
71	21.9556	0.45546E-01	0.39943	0.016273	0.159542	0.979130	0.539651E-04
72	22.0874	0.45275E-01	0.16446	0.006700	0.270477E-01	0.979142	0.914891E-05
73	22.1427	0.45162E-01	1.4720	0.059972	2.16688	0.980098	0.732948E-03
74	22.3139	0.44815E-01	0.51196E-02	0.000209	0.262107E-04	0.980098	0.886579E-08
75	22.4678	0.44508E-01	-0.13588	0.005536	0.184628E-01	0.980107	0.624507E-05
76	22.5859	0.44275E-01	-1.1494	0.046829	1.32118	0.980690	0.446891E-03
77	22.7766	0.43905E-01	0.38192	0.015560	0.145867	0.980754	0.493395E-04
78	22.8595	0.43746E-01	-0.38637	0.015741	0.149284	0.980820	0.504955E-04
79	22.9293	0.43612E-01	0.67612	0.027546	0.457137	0.981022	0.154627E-03
80	23.0643	0.43357E-01	1.1715	0.047729	1.37247	0.981628	0.464240E-03
81	23.4540	0.42637E-01	-0.24303	0.009901	0.590657E-01	0.981654	0.199790E-04
82	23.6011	0.42371E-01	-2.1332	0.086909	4.55060	0.983663	0.153925E-02
83	23.6374	0.42306E-01	2.4907	0.101473	6.20344	0.986402	0.209832E-02
84	23.8100	0.41999E-01	1.9239	0.078383	3.70147	0.988036	0.125203E-02
85	23.9500	0.41754E-01	-2.3344	0.095105	5.44935	0.990442	0.184325E-02
86	24.1685	0.41376E-01	2.7040	0.110165	7.31177	0.993671	0.247321E-02
87	24.4523	0.40896E-01	-0.88934	0.036233	0.790927	0.994020	0.267532E-03
88	24.5232	0.40778E-01	1.7263	0.070333	2.98025	0.995336	0.100807E-02
89	25.0587	0.39906E-01	-2.1221	0.086459	4.50352	0.997324	0.152332E-02
90	25.1533	0.39756E-01	0.58151	0.023691	0.338152	0.997473	0.114380E-03
91	25.2638	0.39582E-01	-1.6325	0.066509	2.66500	0.998650	0.901440E-03
92	25.4996	0.39216E-01	0.35441	0.014439	0.125605	0.998705	0.424859E-04
93	25.5472	0.39143E-01	-0.86996E-01	0.003544	0.756823E-02	0.998709	0.255996E-05
94	25.5593	0.39125E-01	0.56622	0.023069	0.320609	0.998850	0.108446E-03
95	25.7027	0.38906E-01	1.0009	0.040779	1.00188	0.999293	0.338888E-03
96	25.7313	0.38863E-01	-0.61818	0.025185	0.382146	0.999461	0.129261E-03
97	25.7804	0.38789E-01	0.75172	0.030626	0.565078	0.999711	0.191138E-03
98	25.9235	0.38575E-01	0.39011	0.015893	0.152184	0.999778	0.514765E-04
99	26.0251	0.38424E-01	-0.58714	0.023921	0.344732	0.999930	0.116606E-03
100	26.0708	0.38357E-01	-0.39747	0.016193	0.157982	1.00000	0.534376E-04
sum					2264.94		0.766118

PROPRIETARY & CONFIDENTIAL

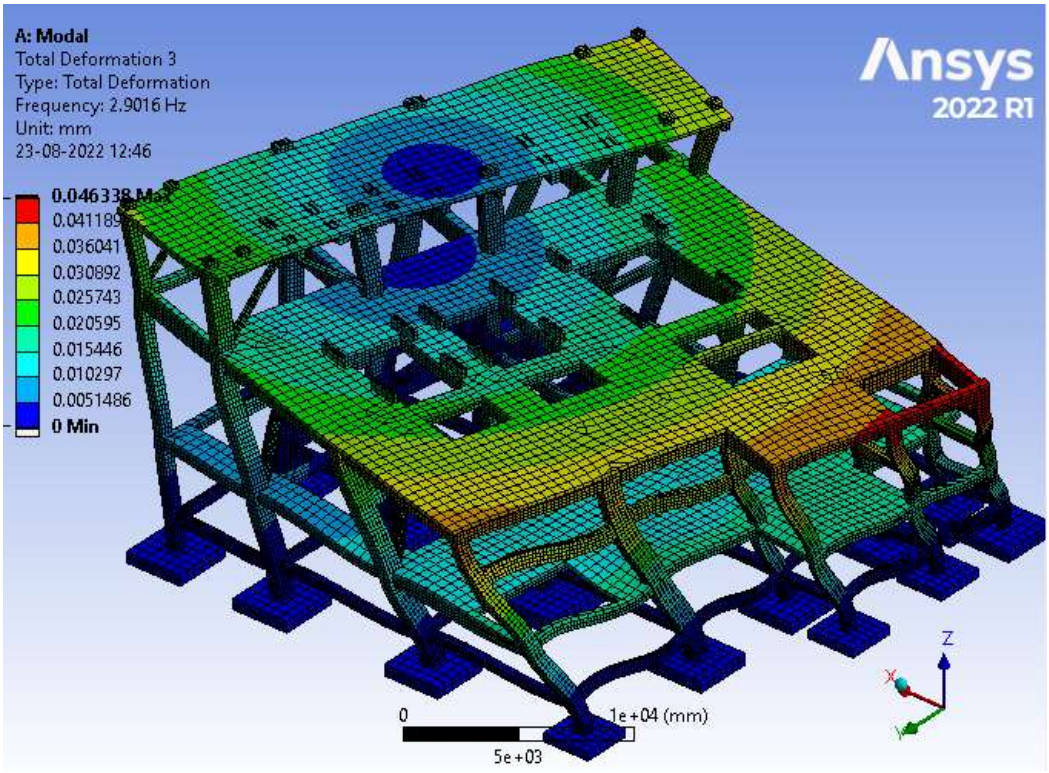
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

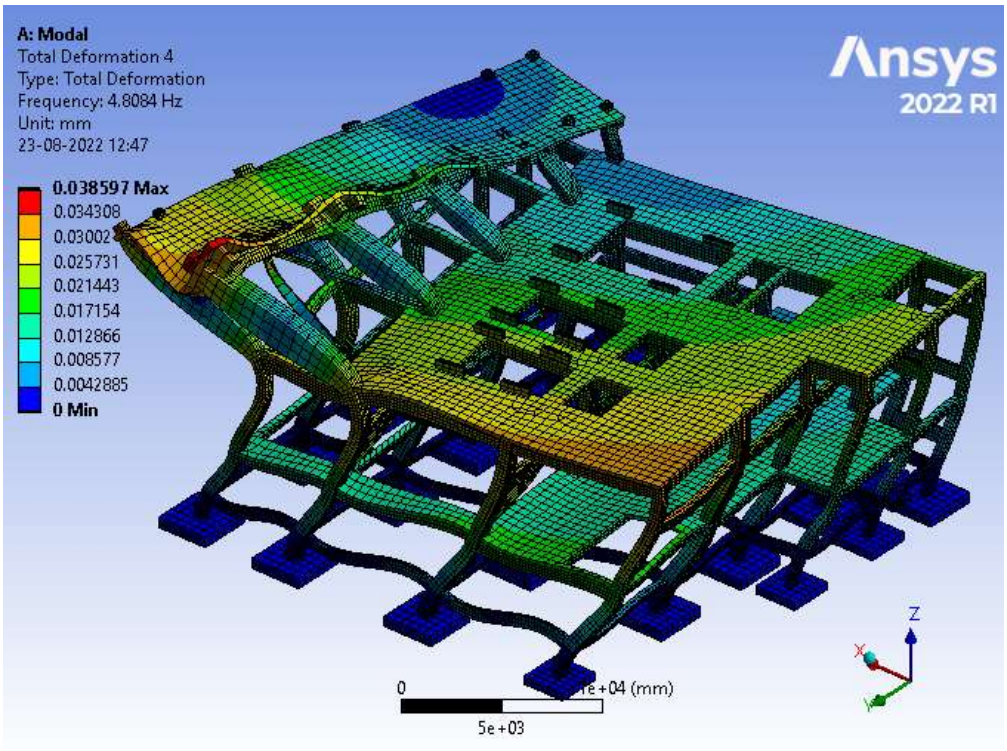




PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.





PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

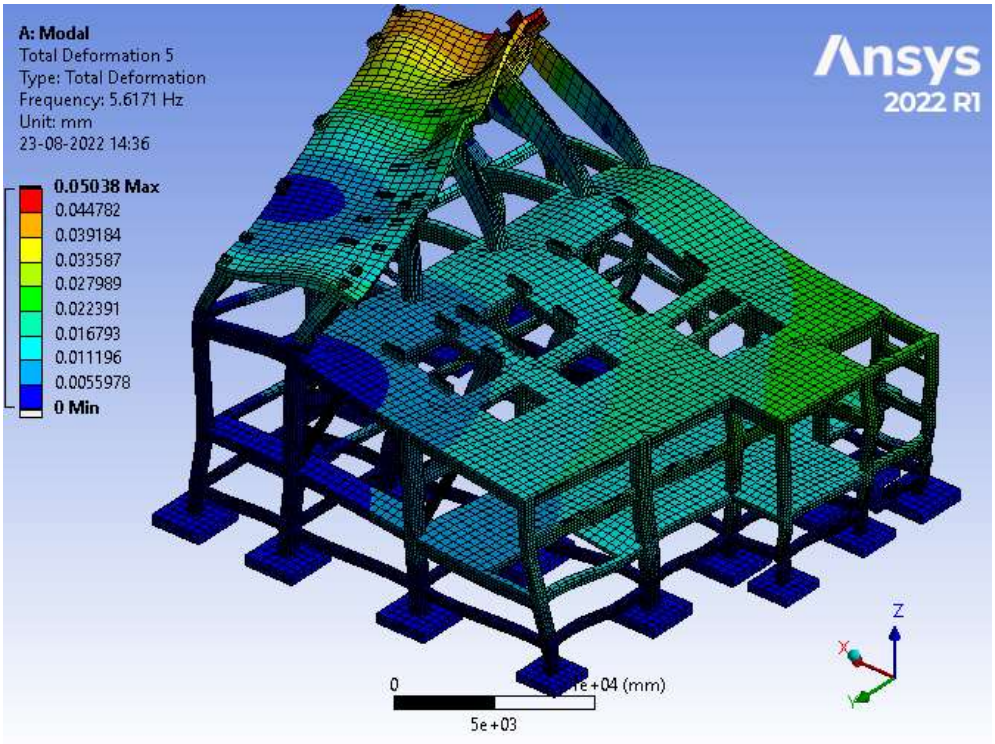


Figure 4c. Typical Mode shapes

4.4. Harmonic Analysis

Individual analysis for each running condition of Screens and feeders are analyzed and the result are added to get the final resultant response of structure at different running combination. The resultant RMS velocity at different running combination is provided in ***Appendix 4***.

4.4.1. Relationships between different terms

Relationship between peak velocity vs peak displacement vs peak accelerations is:

Peak velocity = 2 * PI * frequency * peak displacement ..... Equation (1)
Peak acceleration = 2 * PI * frequency * peak velocity ..... Equation (2)

Where for 7.5 Hz frequency, 2 * PI * 7.5 = 47.123 rad/sec
Where for 9Hz frequency, 2 * PI * 9 = 56.548 rad/sec
Where for 10.5 Hz frequency, 2 * PI * 10.5 = 65.973 rad/sec
Where for 15 Hz frequency, 2 * PI * 15 = 94.247 rad/sec

Relationship between peak velocities vs R. M. S. velocities is:

R. M. S. velocity = peak velocity / Sqrt (2) ..... Equation (3)

PROPRIETARY & CONFIDENTIAL
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

The RMS velocity has been measured at specific locations which has been attached in ***Appendix 3.***

Table 4b. RMS Velocity at measuring points on Feeder Floor

Feeder Floor			
Case	RMS Velocities(mm/s) in X, Y & Z Dir.		
Case A - All Equipments operating at 100% Speed	0.164	0.194	2.272
Case B - Screens-100%, Ter. Feeder- 9.16 Hz, Sec. Feeder -10.83 Hz	0.187	0.225	2.931
Case C - Screens-100%, Ter. Feeder- 9.16 Hz, Sec. Feeder -12.2 Hz	1.210	0.160	2.242
Case D - Screens-100%, Ter. Feeder- 9.16 Hz, Sec. Feeder -15 Hz	0.146	0.211	2.727
Case E - Screens-100%, Ter. Feeder- 12 Hz, Sec. Feeder -10.83 Hz	0.179	0.218	2.819
Case F - Screens-100%, Ter. Feeder- 12 Hz, Sec. Feeder -12.2 Hz	0.166	0.216	2.658
Case G - Screens-100%, Ter. Feeder- 12 Hz, Sec. Feeder -15 Hz	0.149	0.163	2.300
Case H - Screens-100%, Ter. Feeder- 15 Hz, Sec. Feeder -10.83 Hz	0.193	0.208	2.624
Case I - Screens-100%, Ter. Feeder- 15 Hz, Sec. Feeder -12.2 Hz	0.169	0.193	2.464
Case J -Screens-100%, Ter & Sec Feed- 10.914 Hz	0.334	0.270	2.862
Case K -Screens-100%, Ter & Sec Feed- 11.172 Hz	0.339	0.221	2.951

Following figures are results from harmonic response for ***Case A***

Analysis setting for case A:

Details of "Analysis Settings"

Step Controls	
Multiple RPMs	No
Options	
Frequency Spacing	Linear
Range Minimum	13.749 Hz
Range Maximum	13.751 Hz
Solution Intervals	3
User Defined Frequencies	Off
Solution Method	Mode Superposition
Include Residual Vector	No
Cluster Results	No
Modal Frequency Range	Program Controlled
On Demand Expansion	No
Store Results At All Frequencies	Yes
Advanced	
Output Controls	
Damping Controls	
Eqv. Damping Ratio From Modal	No
Damping Define By	Damping Ratio
Damping Ratio	5.e-002
Analysis Data Management	

For tertiary screen @ 13.75Hz

Details of "Analysis Settings"

Step Controls	
Multiple RPMs	No
Options	
Frequency Spacing	Linear
Range Minimum	14.249 Hz
Range Maximum	14.251 Hz
Solution Intervals	3
User Defined Frequencies	Off
Solution Method	Mode Superposition
Include Residual Vector	No
Cluster Results	No
Modal Frequency Range	Program Controlled
On Demand Expansion	No
Store Results At All Frequencies	Yes
Advanced	
Output Controls	
Damping Controls	
Eqv. Damping Ratio From Modal	No
Damping Define By	Damping Ratio
Damping Ratio	5.e-002
Analysis Data Management	

For Secondary screen @ 14.25Hz

Details of "Analysis Settings"

Step Controls	
Multiple RPMs	No
Options	
Frequency Spacing	Linear
Range Minimum	14.9 Hz
Range Maximum	15.1 Hz
Solution Intervals	5
User Defined Frequencies	Off
Solution Method	Mode Superposition
Include Residual Vector	No
Cluster Results	No
Modal Frequency Range	Program Controlled
On Demand Expansion	No
Store Results At All Frequencies	Yes
Advanced	
Output Controls	
Damping Controls	
Eqv. Damping Ratio From Modal	No
Damping Define By	Damping Ratio
Damping Ratio	5.e-002
Analysis Data Management	

For tertiary Feeder @ 15Hz

Details of "Analysis Settings"

Step Controls	
Multiple RPMs	No
Options	
Frequency Spacing	Linear
Range Minimum	14.9 Hz
Range Maximum	15.1 Hz
Solution Intervals	3
User Defined Frequencies	Off
Solution Method	Mode Superposition
Include Residual Vector	No
Cluster Results	No
Modal Frequency Range	Program Controlled
On Demand Expansion	No
Store Results At All Frequencies	Yes
Advanced	
Output Controls	
Damping Controls	
Eqv. Damping Ratio From Modal	No
Damping Define By	Damping Ratio
Damping Ratio	5.e-002
Analysis Data Management	

For Secondary Feeder @ 15Hz

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

4.4.2. Peak Velocity of Concrete Structure, loaded Condition: 100% Speed Screen (Secondary Screen 14.25Hz & Tertiary Screen 13.75Hz)

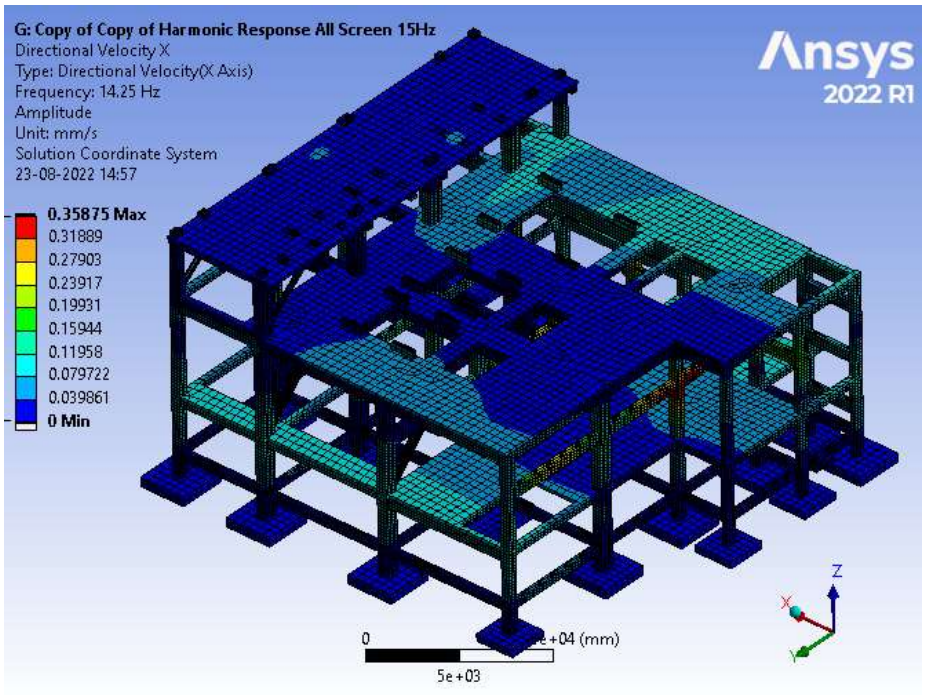


Figure 4d. i) Peak Velocity in X direction (Secondary Screen running at 14.25Hz)

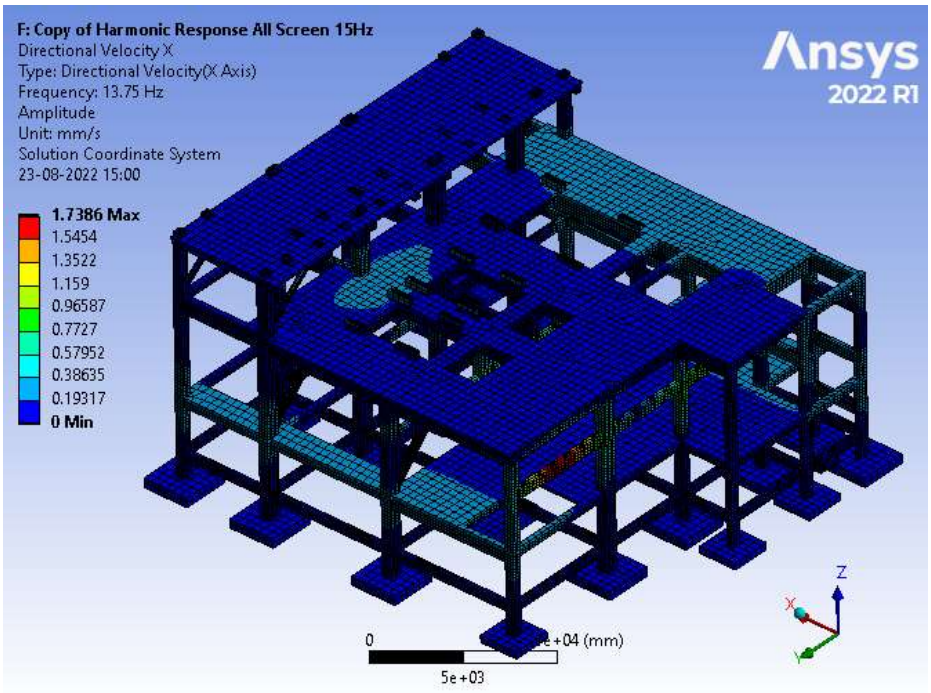


Figure 4d. ii) Peak Velocity in X direction (Tertiary Screen running at 13.75Hz)

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

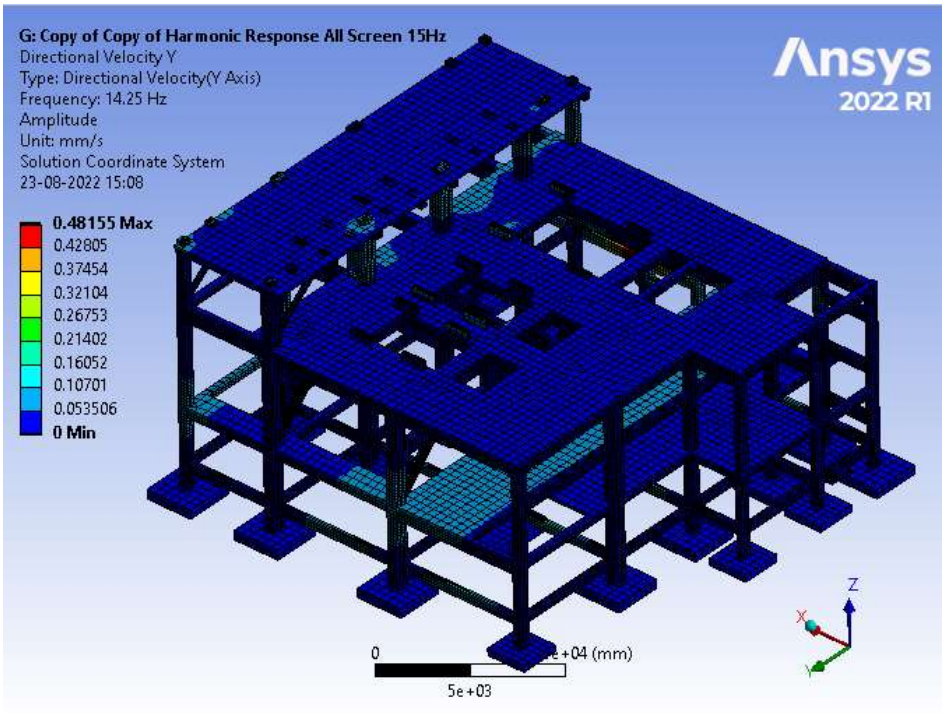


Figure 4e. i) Peak Velocity in Y direction (Secondary Screen running at 14.25Hz)

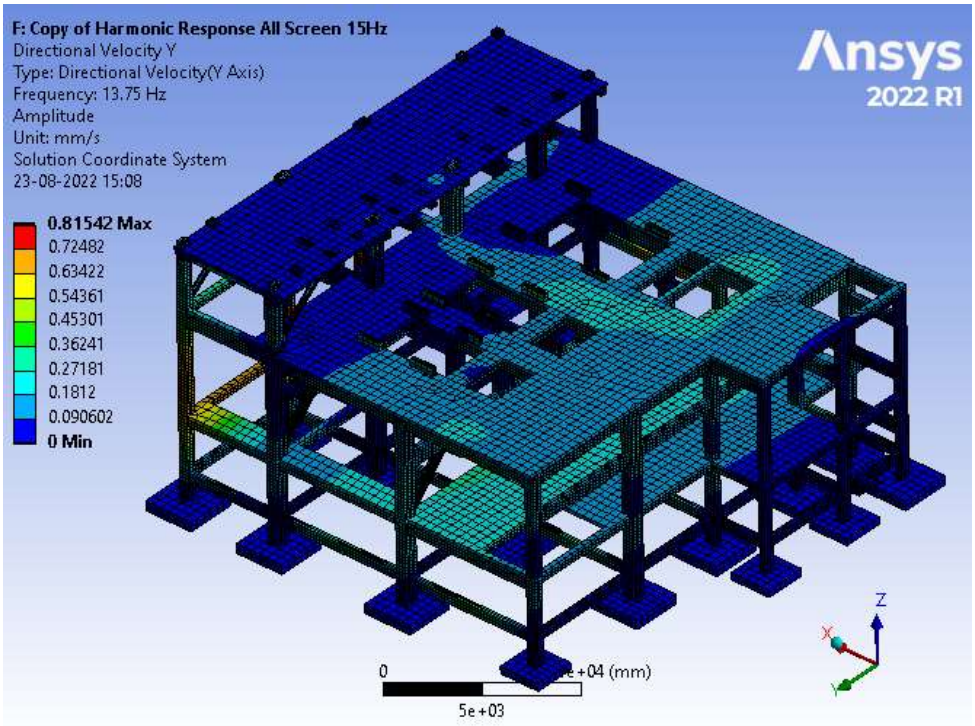


Figure 4e. ii) Peak Velocity in Y direction (Tertiary Screen running at 13.75Hz)

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

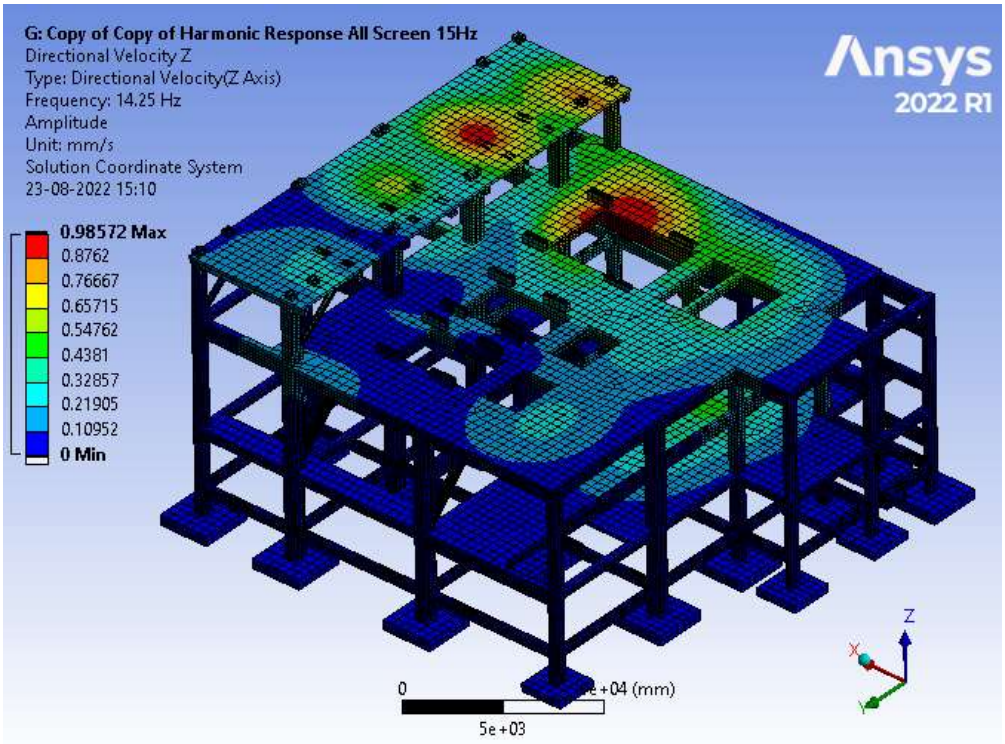


Figure 4f. i) Peak Velocity in Z direction (Secondary Screen running at 14.25Hz)

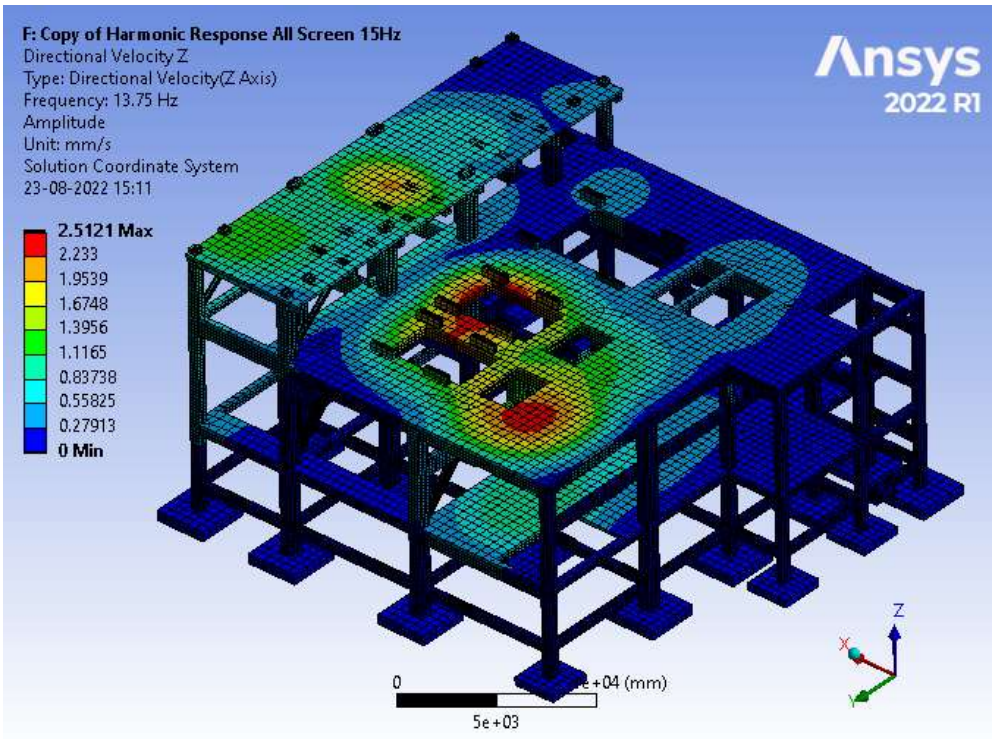


Figure 4f. ii) Peak Velocity in Z direction (Tertiary Screen running at 13.75Hz)

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

4.4.3. Peak Velocity of Concrete Structure, loaded Condition: 100% Speed Tertiary Feeder (15 Hz) – Feeder Floor

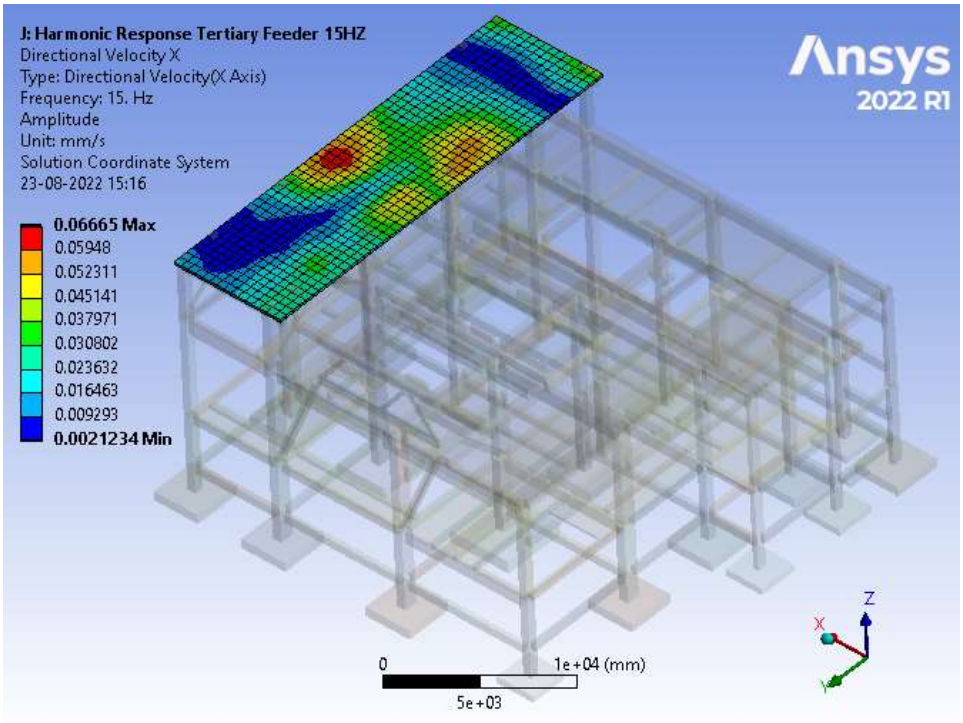


Figure 4g. Peak Velocity in X direction on feeder floor (Tertiary feeder running at 15Hz)

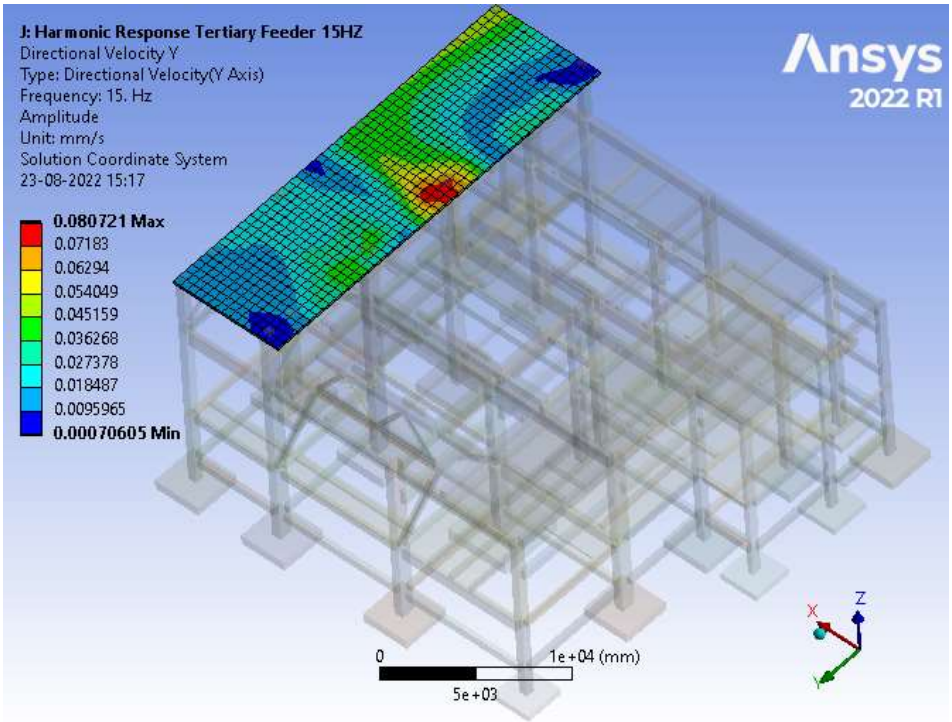


Figure 4h. Peak Velocity in Y direction on feeder floor (Tertiary feeder running at 15Hz)

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

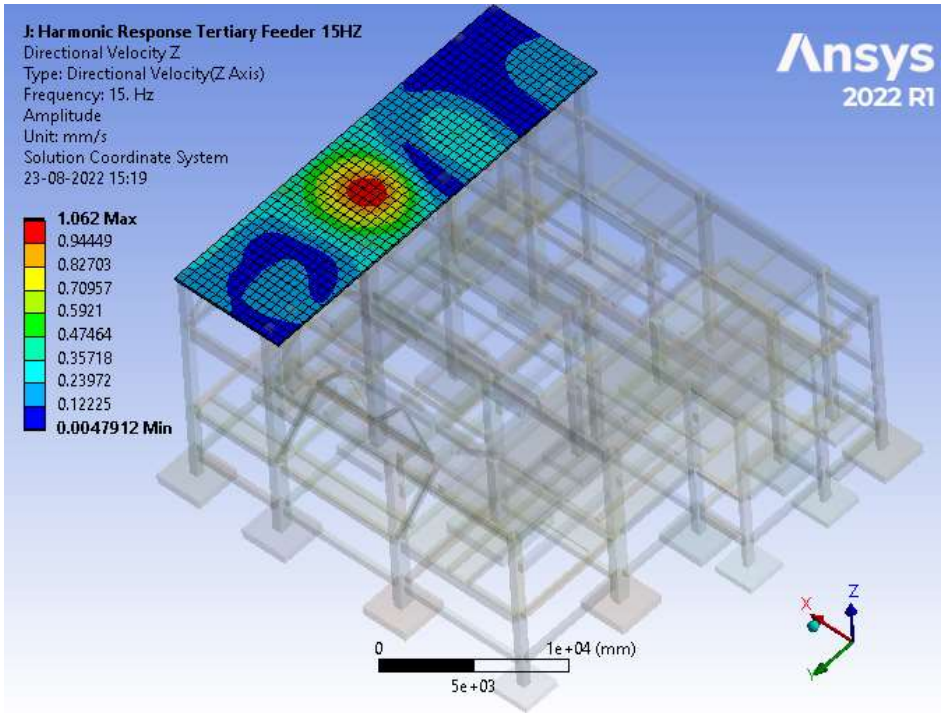


Figure 4j. Peak Velocity in Z direction on feeder floor (Tertiary feeder running at 15Hz)

4.4.4. Peak Velocity of Concrete Structure, loaded Condition: 100% Speed Secondary Feeder (15 Hz) – Feeder Floor

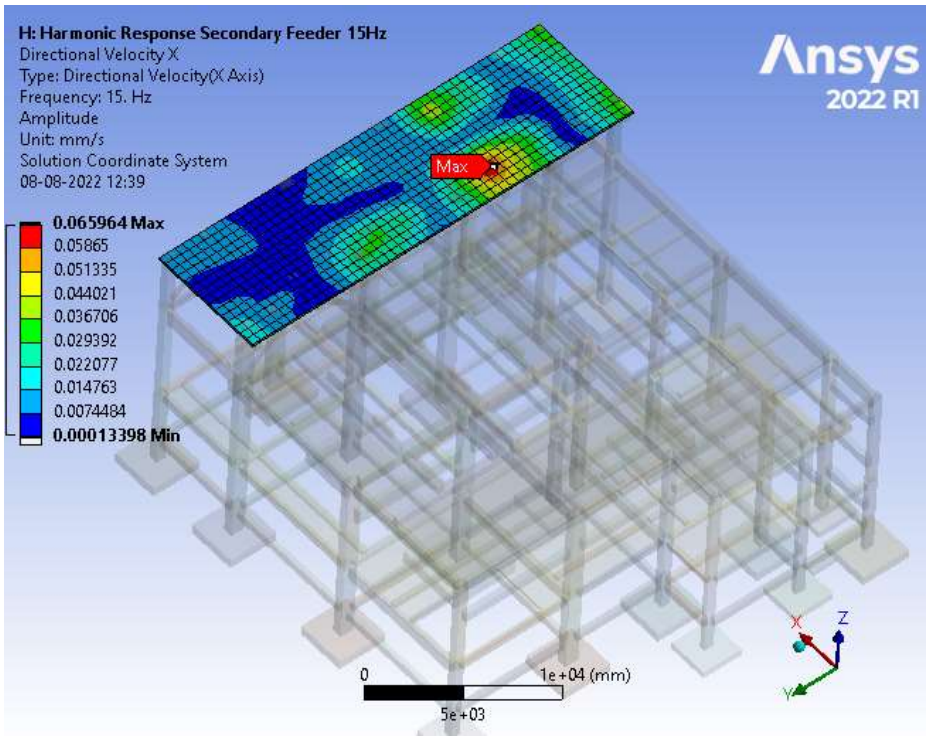


Figure 4k. Peak Velocity in X direction on feeder floor (Secondary feeder running at 15Hz)

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

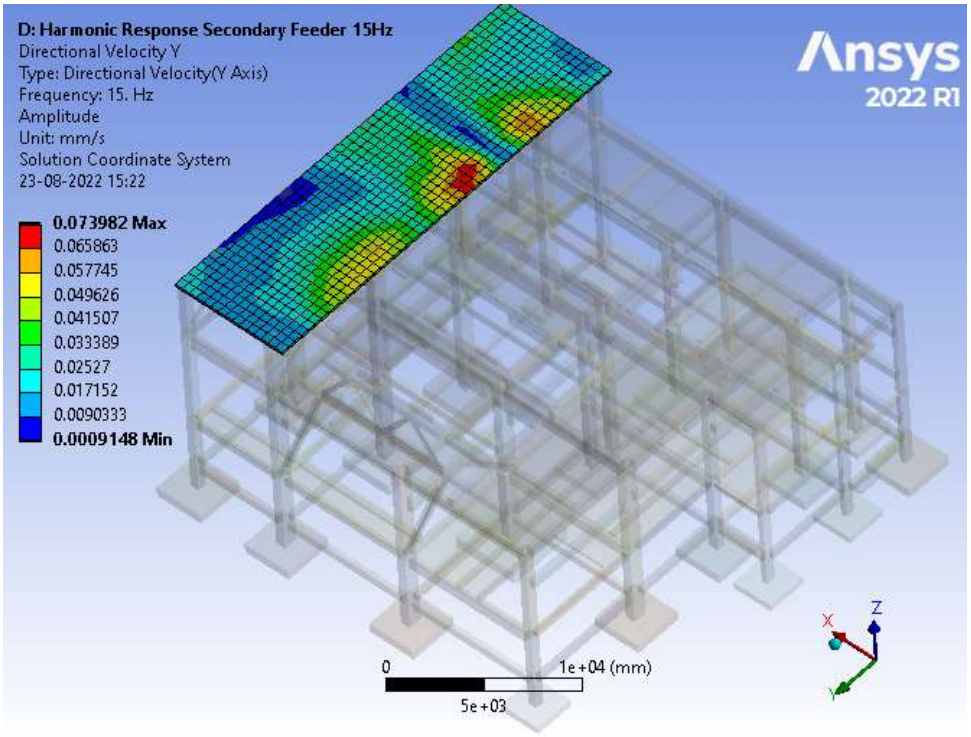


Figure 4l. Peak Velocity in Y direction on feeder floor (Secondary feeder running at 15Hz)

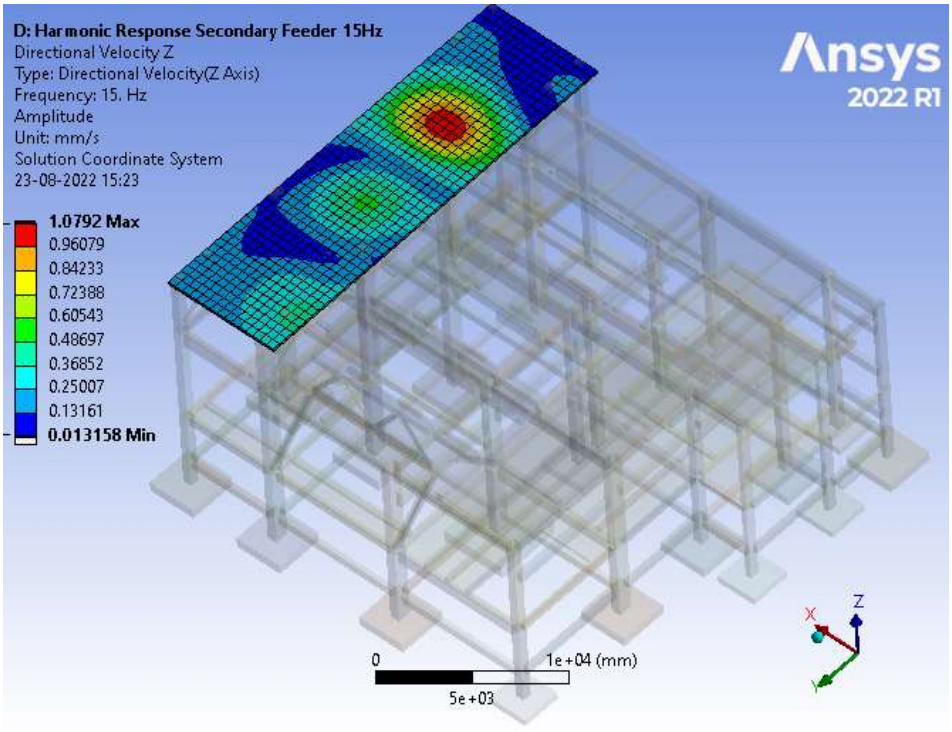


Figure 4m. Peak Velocity in Z direction on feeder floor (Secondary feeder running at 15Hz)

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Table 4c. Summary of RMS Velocity at measuring points on Screen Floor at different running condition

Screen Floor			
Case	RMS Velocities(mm/s) in X, Y & Z Dir.		
Case A - All Equipments operating at 100% Speed	0.311	0.185	1.407
Case B - Screens-100%, Ter. Feeder- 9.16 Hz, Sec. Feeder -10.83 Hz	0.294	0.177	1.868
Case C - Screens-100%, Ter. Feeder- 9.16 Hz, Sec. Feeder -12.2 Hz	0.292	0.195	1.643
Case D - Screens-100%, Ter. Feeder- 9.16 Hz, Sec. Feeder -15 Hz	0.305	0.174	1.482
Case E - Screens-100%, Ter. Feeder- 12 Hz, Sec. Feeder -10.83 Hz	0.294	0.199	1.974
Case F - Screens-100%, Ter. Feeder- 12 Hz, Sec. Feeder -12.2 Hz	0.292	0.217	1.748
Case G - Screens-100%, Ter. Feeder- 12 Hz, Sec. Feeder -15 Hz	0.305	0.196	1.587
Case H - Screens-100%, Ter. Feeder- 15 Hz, Sec. Feeder -10.83 Hz	0.300	0.188	1.793
Case I - Screens-100%, Ter. Feeder- 15 Hz, Sec. Feeder -12.2 Hz	0.297	0.206	1.567
Case J -Screens-100%, Ter & Sec Feed- 10.914 Hz	0.313	0.172	2.212
Case K -Screens-100%, Ter & Sec Feed- 11.172 Hz	0.311	0.174	2.376

*Following figures are results from harmonic response for **Case A***

4.4.5. Peak Velocity of Concrete Structure, loaded Condition: 100% Speed Screen (Secondary Screen 14.25Hz & Tertiary Screen 13.75Hz)

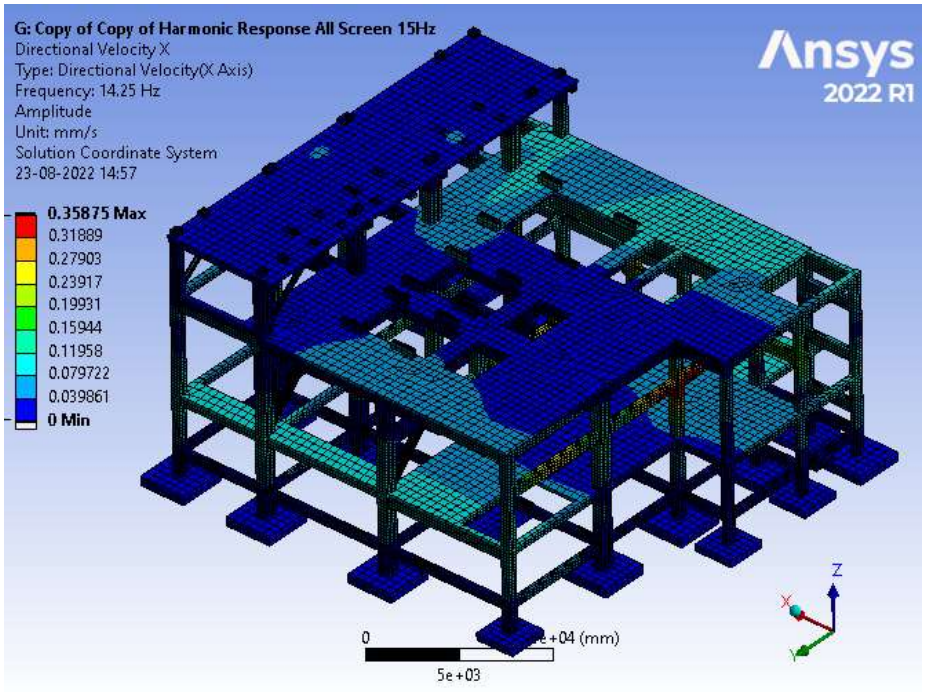


Figure 4n. i) Peak Velocity in X direction (Secondary Screen running at 14.25Hz)

PROPRIETARY & CONFIDENTIAL
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

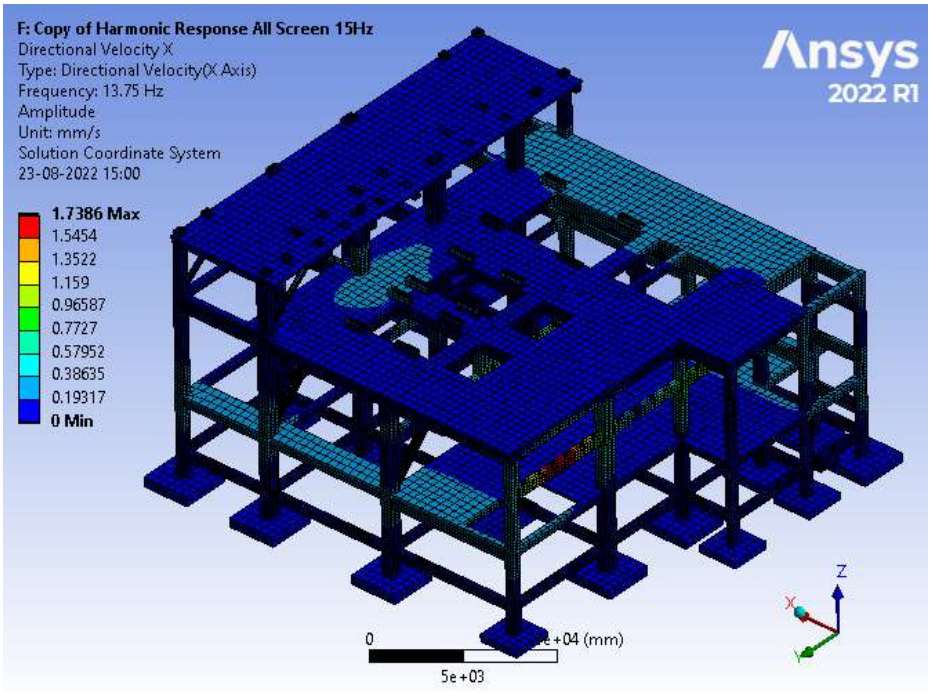


Figure 4n. ii) Peak Velocity in X direction (Tertiary Screen running at 13.75Hz)

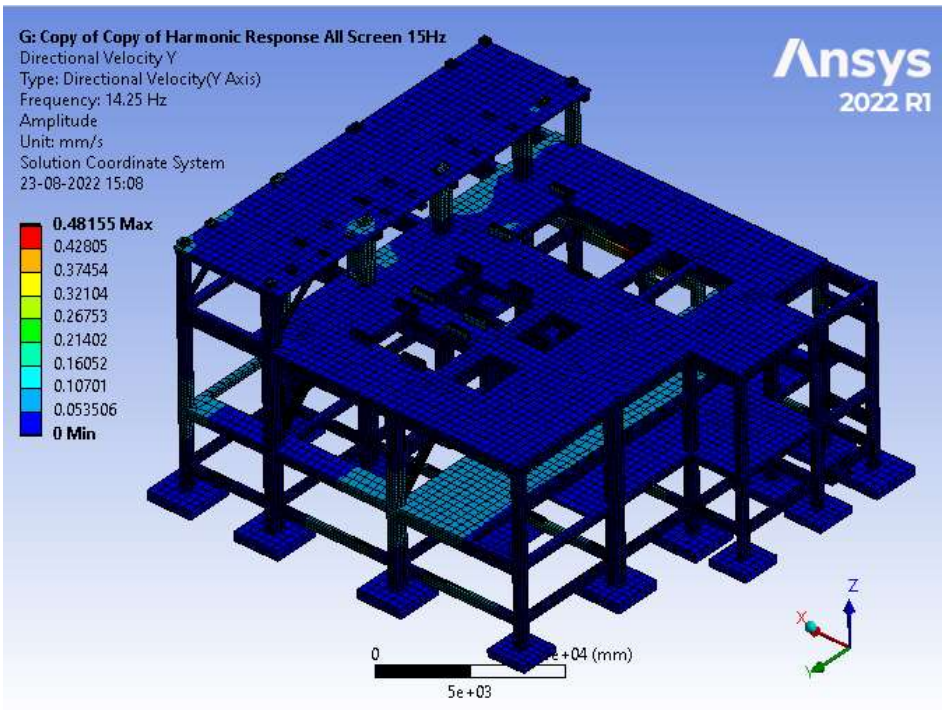


Figure 4o. i) Peak Velocity in Y direction (Secondary Screen running at 14.25Hz)

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

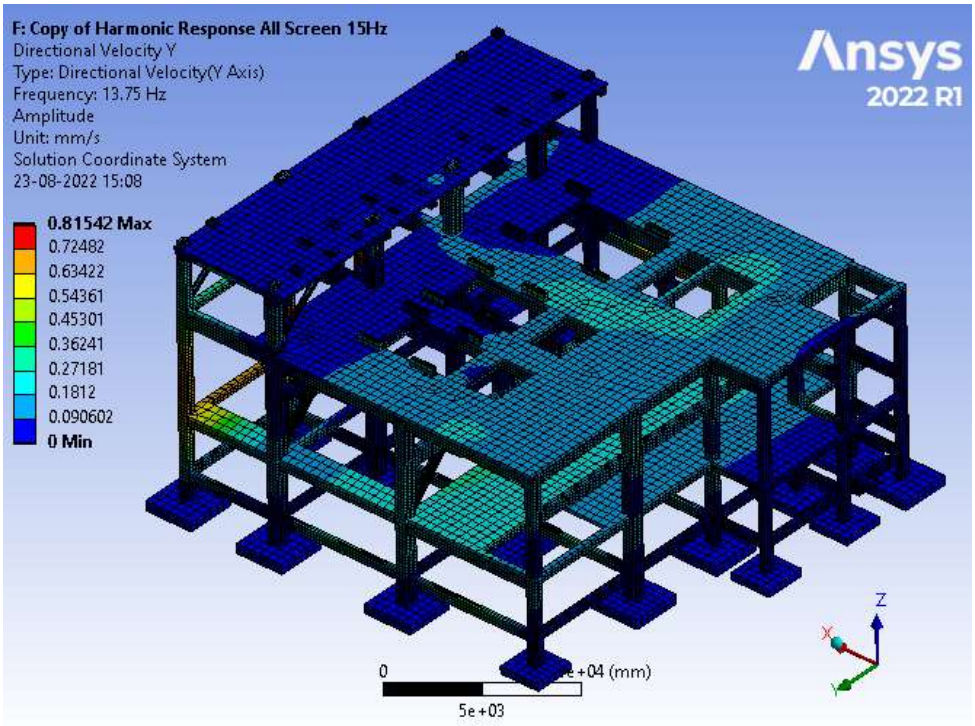


Figure 4o. ii) Peak Velocity in Y direction (Tertiary Screen running at 13.75Hz)

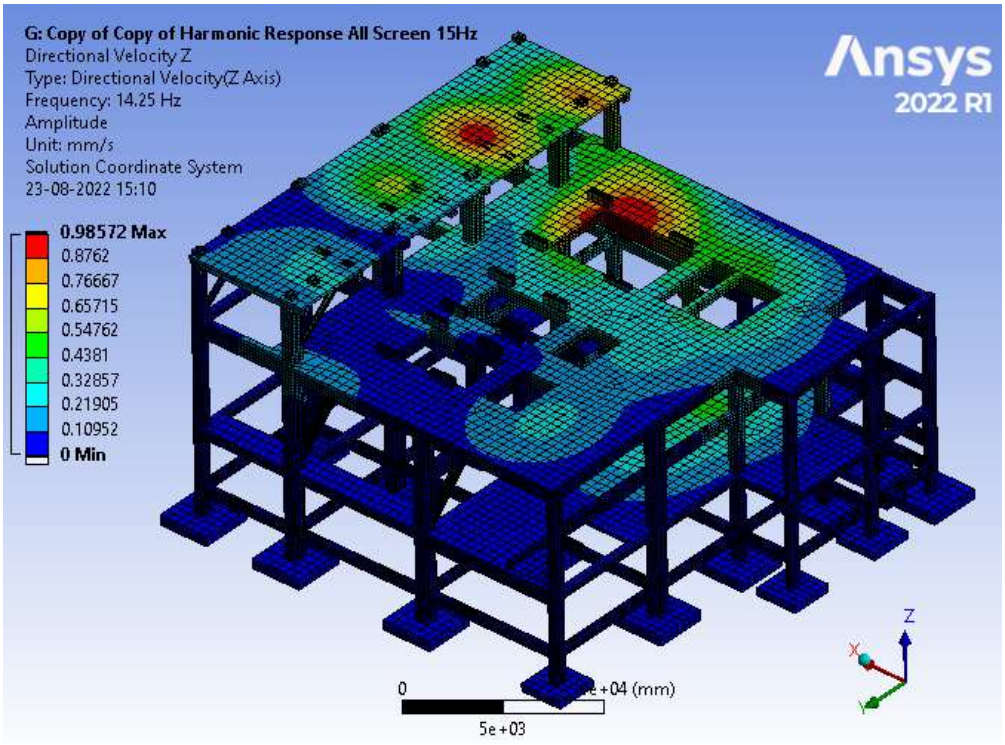


Figure 4p. i) Peak Velocity in Z direction (Secondary Screen running at 14.25Hz)

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

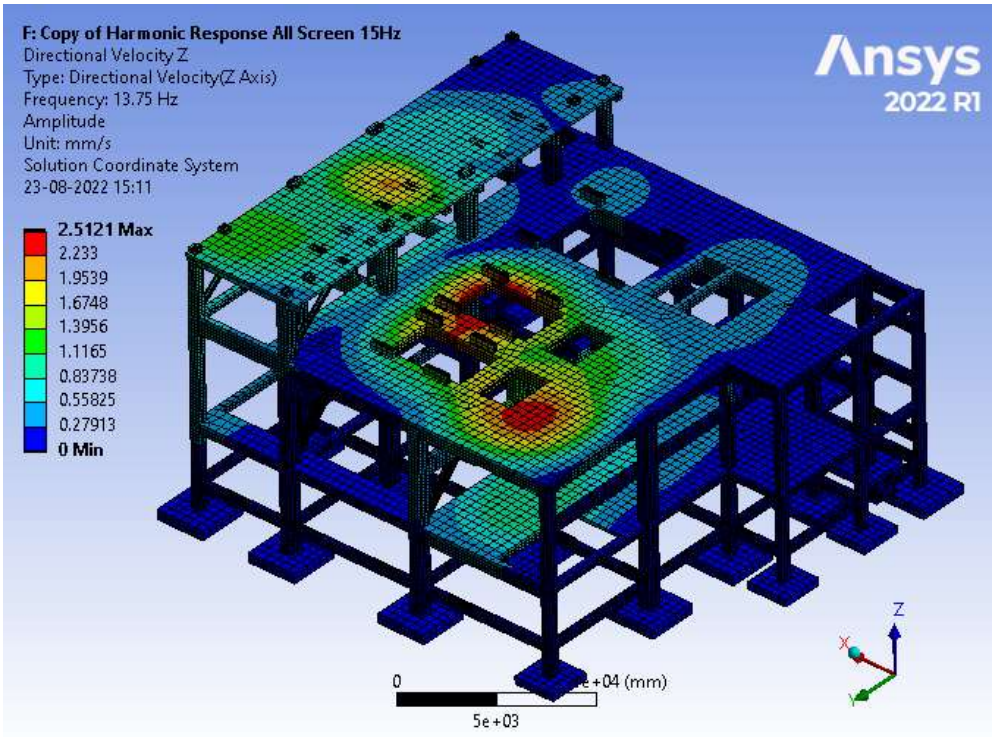
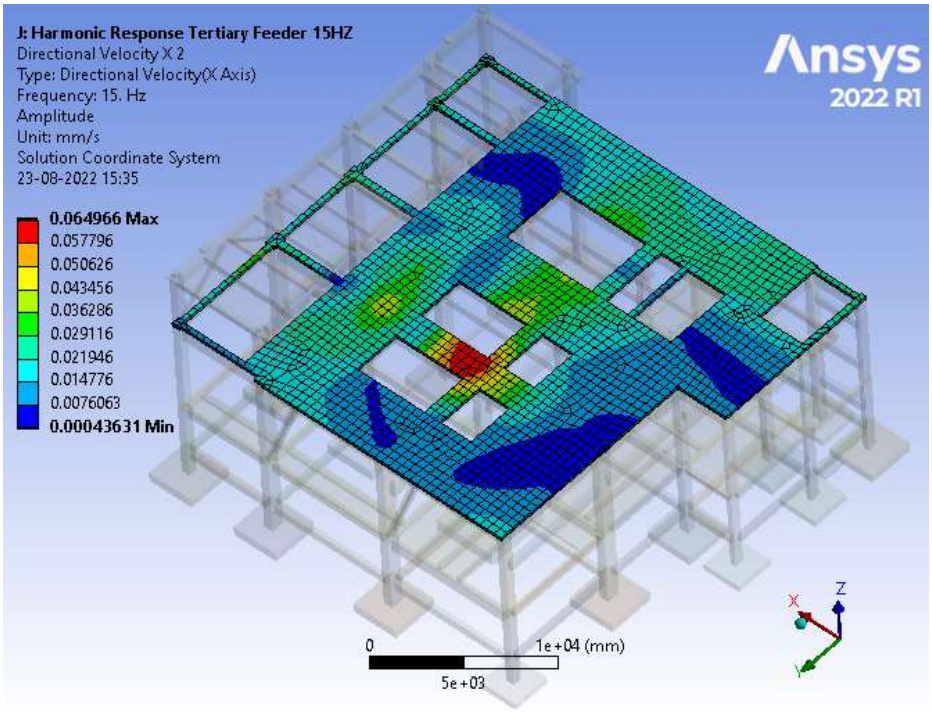


Figure 4p. ii) Peak Velocity in Z direction (Tertiary Screen running at 13.75Hz)

4.4.6. Peak Velocity of Concrete Structure, loaded Condition: 100% Speed Tertiary Vibrofeeder (15 Hz) – Screen Floor



PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Figure 4q. Peak Velocity in X direction on Screen floor (Tertiary feeder running at 15Hz)

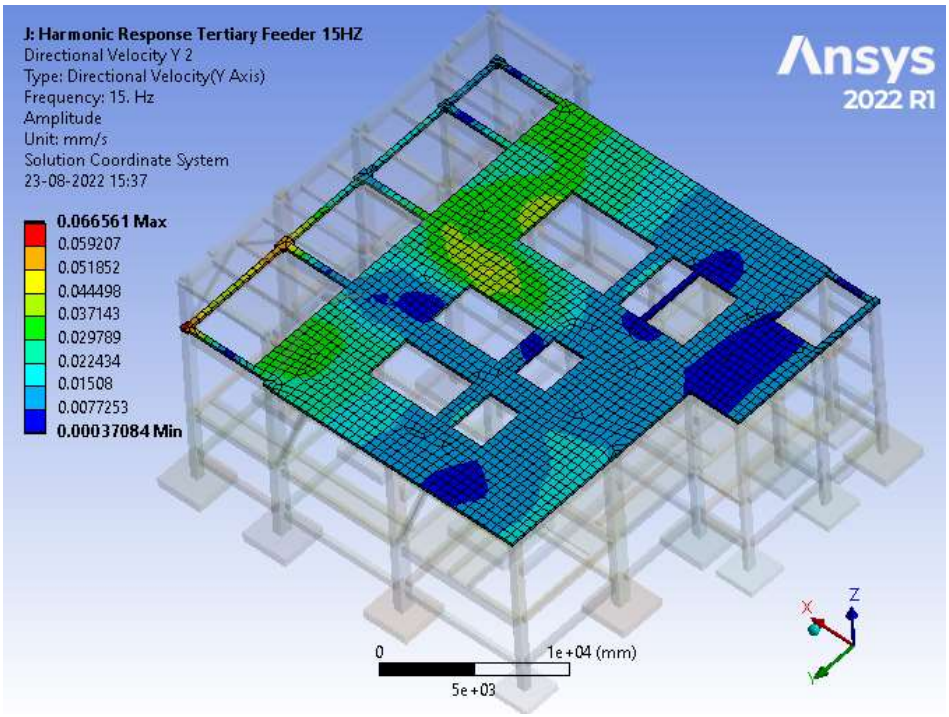


Figure 4r. Peak Velocity in Y direction on Screen floor (Tertiary feeder running at 15Hz)

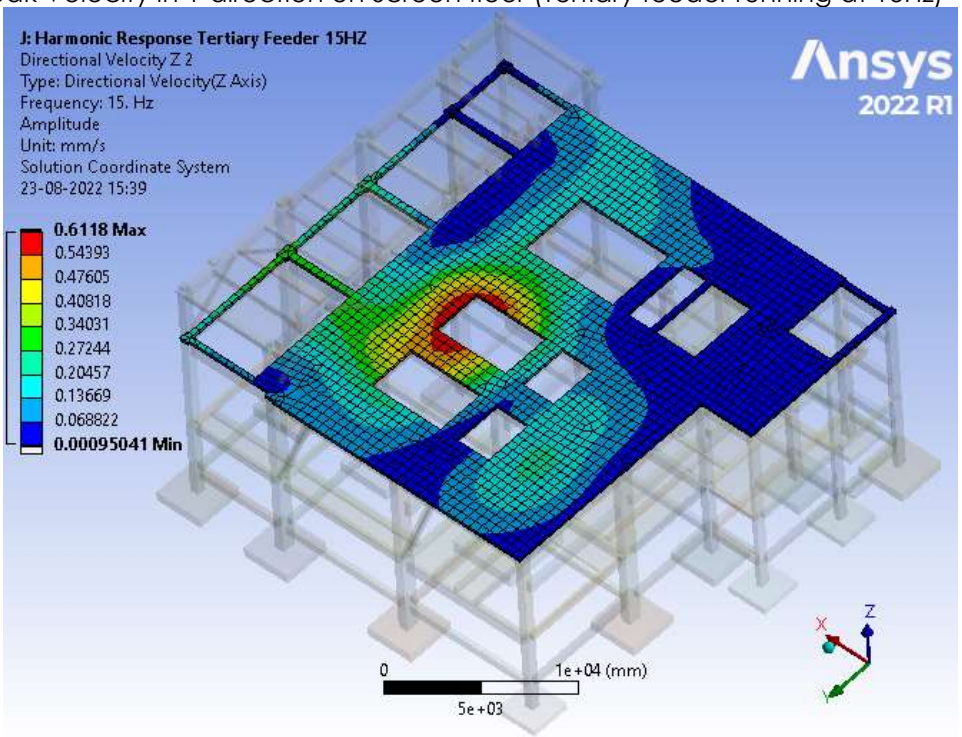


Figure 4s. Peak Velocity in Z direction on Screen floor (Tertiary feeder running at 15Hz)

4.4.7. Peak Velocity of Concrete Structure, loaded Condition: 100% Speed Secondary Vibrofeeder (15 Hz) – Screen Floor

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

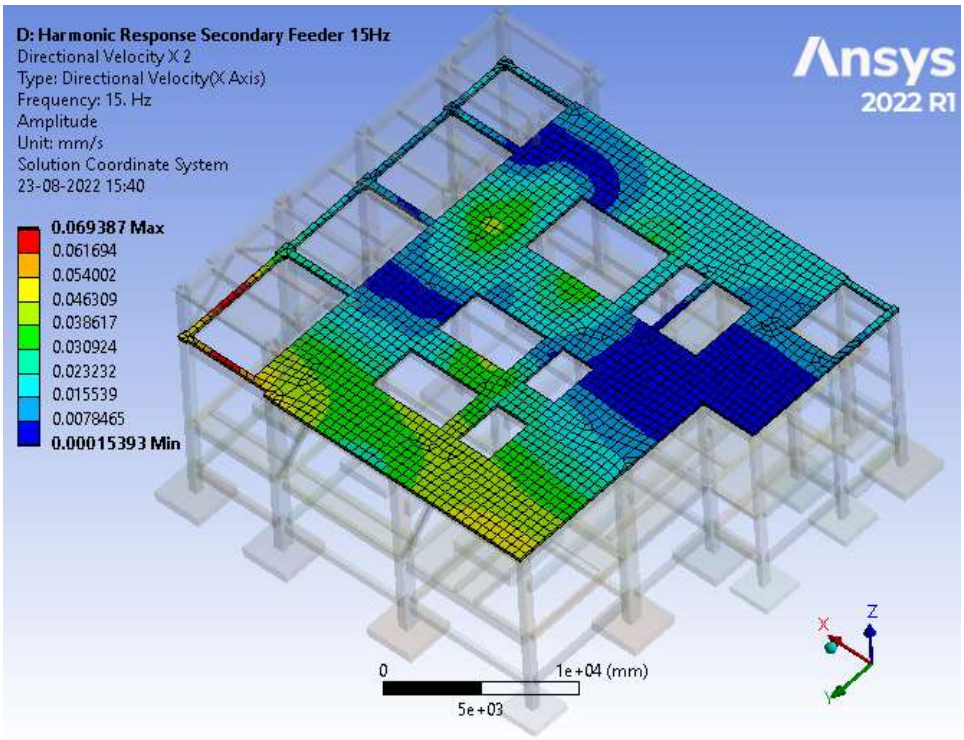


Figure 4t. Peak Velocity in X direction on Screen floor (Secondary feeder running at 15Hz)

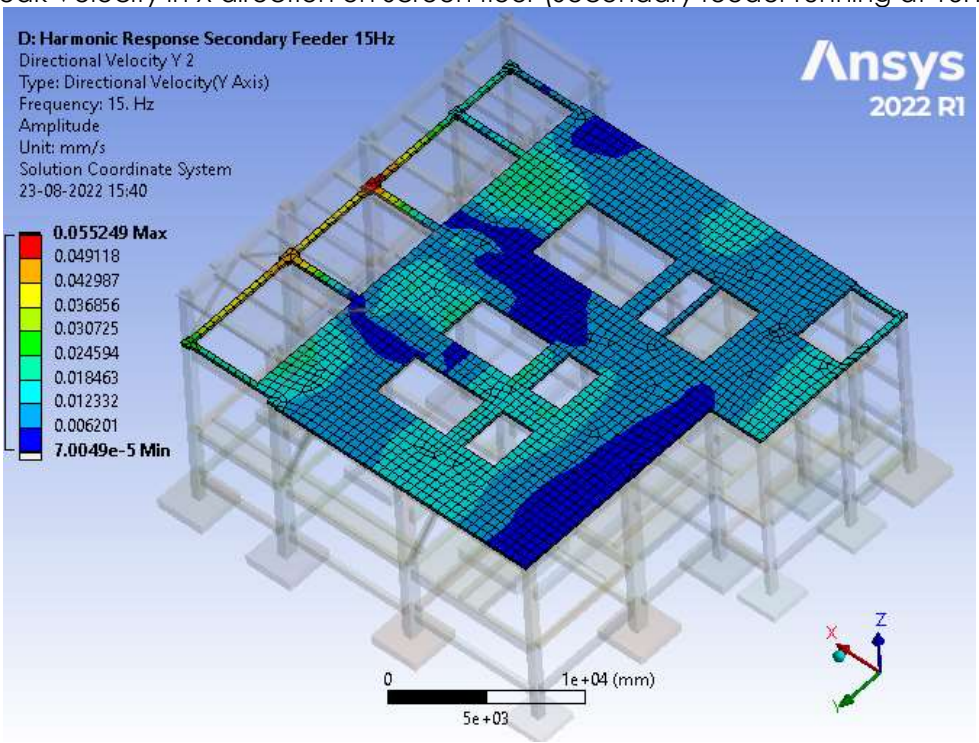


Figure 4u. Peak Velocity in Y direction on Screen floor (Secondary feeder running at 15Hz)

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

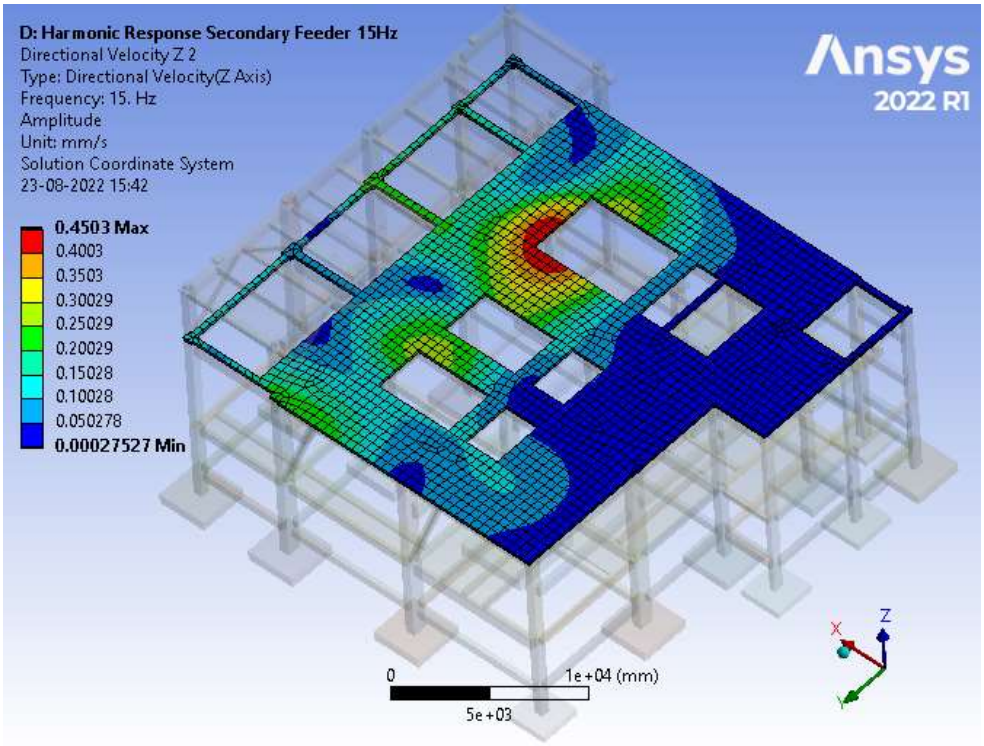


Figure 4v. Peak Velocity in Z direction on Screen floor (Secondary feeder running at 15Hz)

For all other cases Refer ***Appendix 4***

5. Conclusions

The structure has been analyzed with reduced dynamic load from tertiary Vibrofeeder as the same will be mounted on vibration isolation system.

The following are our observations

- To reduce the vibration occurring in case K (feeders running at natural frequency of building) the outer tertiary screen supporting beam has been strengthened through bracing arrangement. **(Refer Appendix 1**).
- The maximum RMS velocity of vibration at feeder floor level at all the measuring points after modification has been found as 2.951mm/sec for Case K.
- The maximum RMS velocity of vibration at screen floor level at all the measuring points has been found as 2.376mm/sec for Case K.
- The computed vibration velocity for start/Stop operation as well as other natural frequencies with high mode participation factor is found to be well within the permissible limit of 3mm/sec. The details are presented in **Appendix 5.**
- The load of vibration isolation foundation is found to be lower than the start/stop dynamic load used for static analysis and design. Hence no static check is required for the feeder floor beams.
- For those column and beam joints subject to excessive vibration during the commissioning of original tertiary vibration feeder, non-destructive tests will be undertaken to make sure the concrete structure is in good condition. Strengthening of the joints will be necessary if the test results show possible defective joints.

PROPRIETARY & CONFIDENTIAL
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

6. References

[1] ANSYS Mechanical Software

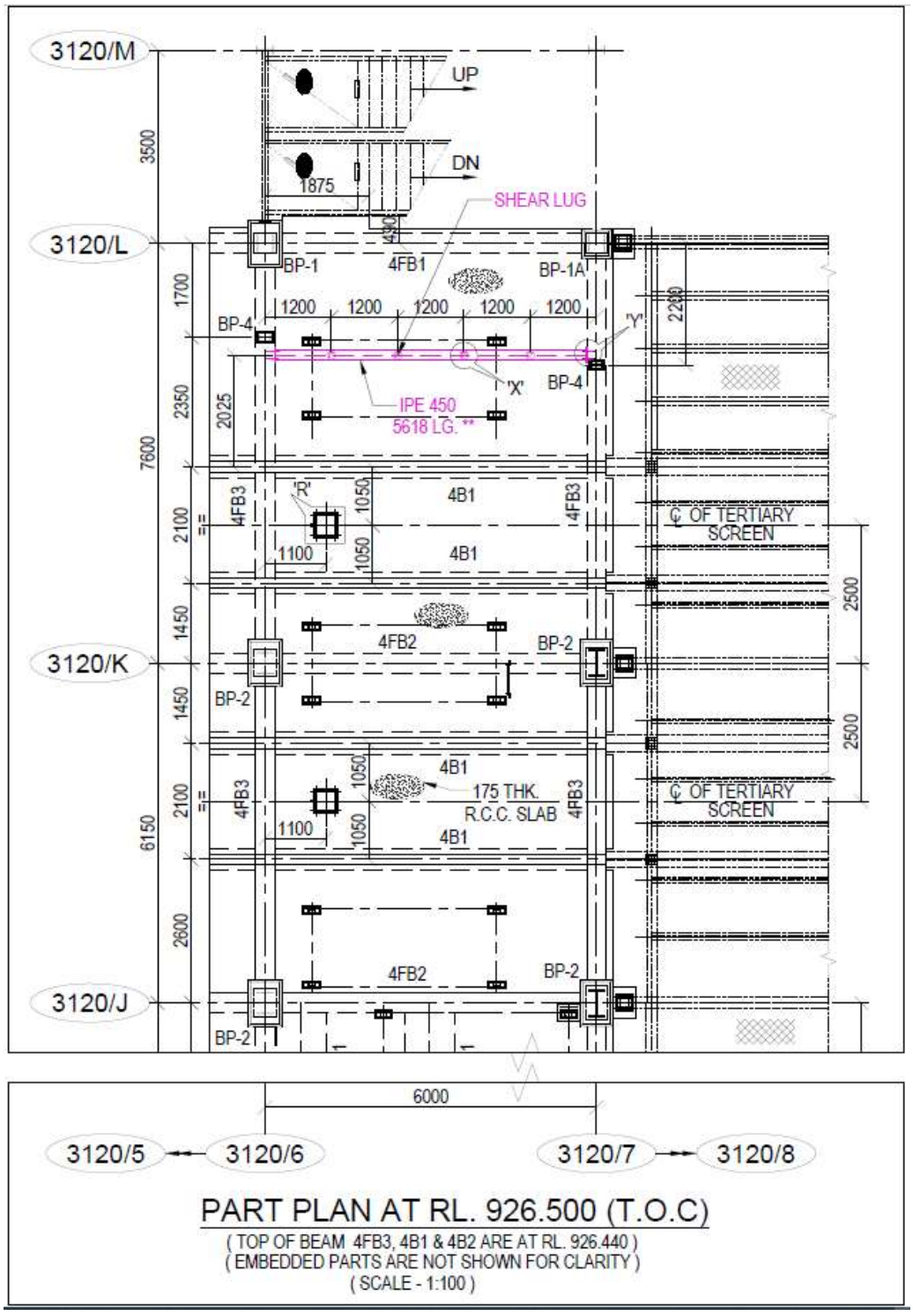
[2] MANSOURAH - MASSARAH GOLD PROJECT. GENERAL ARRANGEMENT DRAWING SECONDARY CRUSHING VIBRATING FEEDER 3120-FDR-001 (MD-216-3120-EG-ME-GAD-V500-1008-C01)

[3] MANSOURAH - MASSARAH GOLD PROJECT. GA TERTIARY CRUSHING VIBRATING FEEDER 3130-FDR-001-002 Model (MD-216-3130-EG-ME-GAD-V500-1008-C01)

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

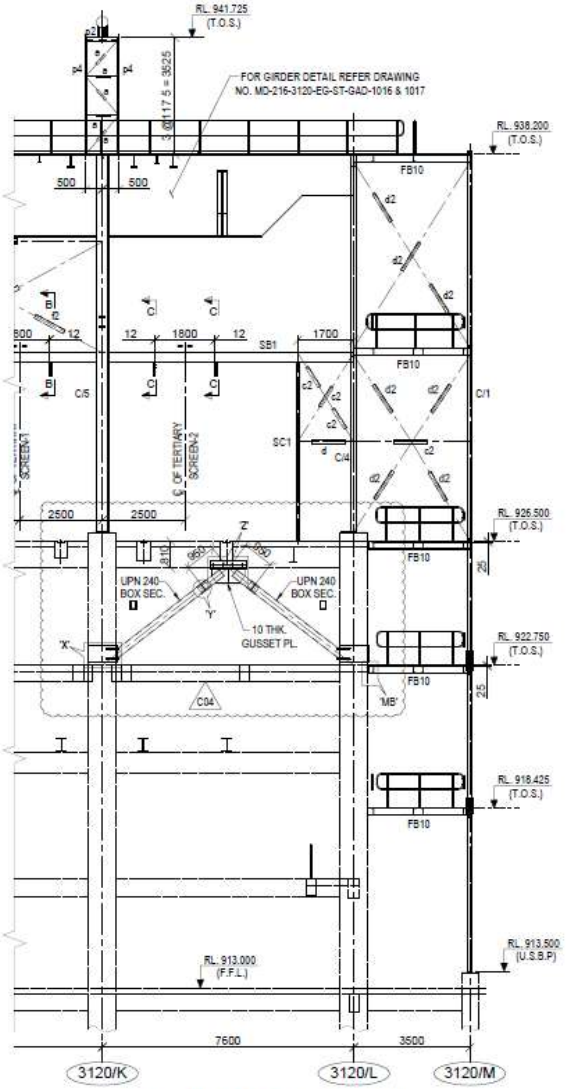
7. Appendix 1: Modified Feeder & Screen Floor



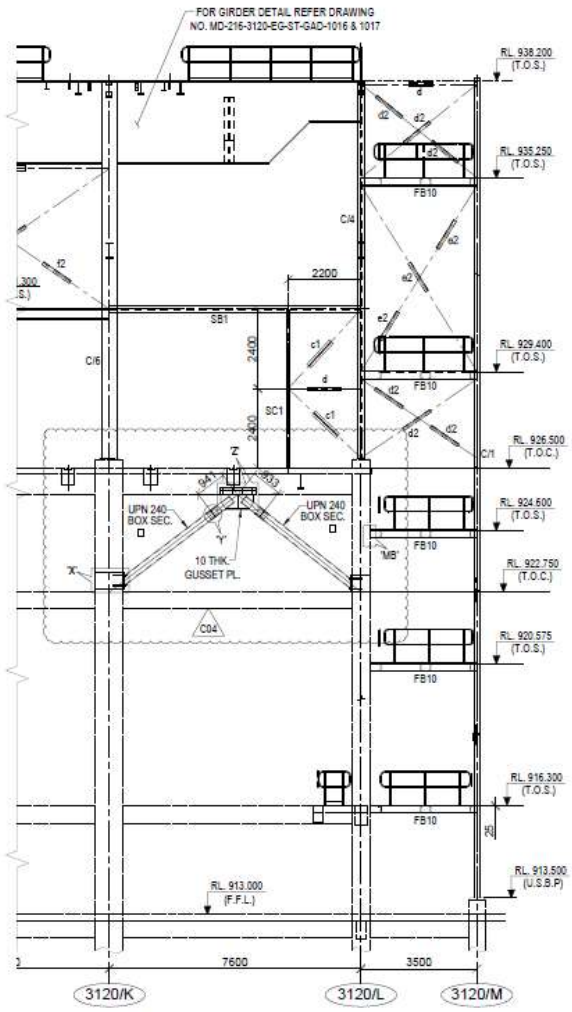
PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.



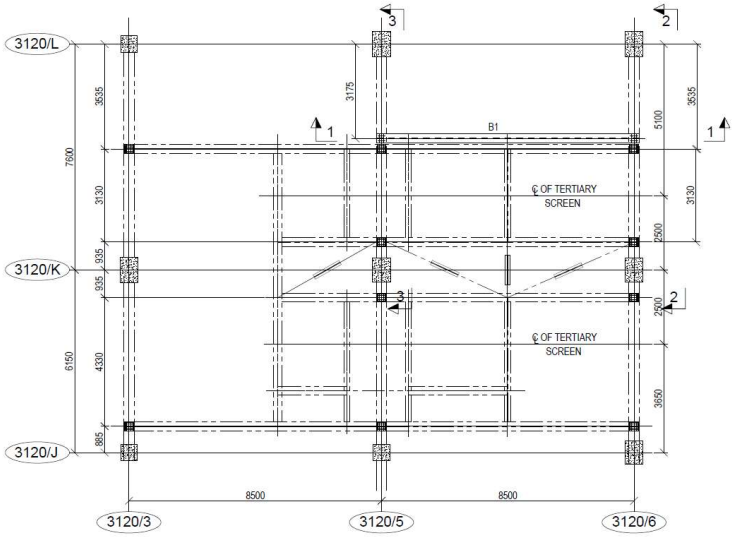
ELEVATION ALONG GRID 3120/6
(SCALE - 1:100)



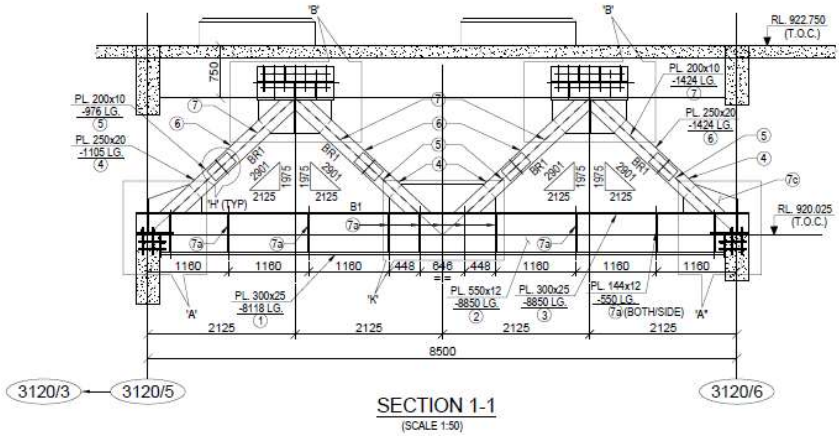
ELEVATION ALONG GRID 3120/7
(SCALE - 1:100)

PROPRIETARY & CONFIDENTIAL

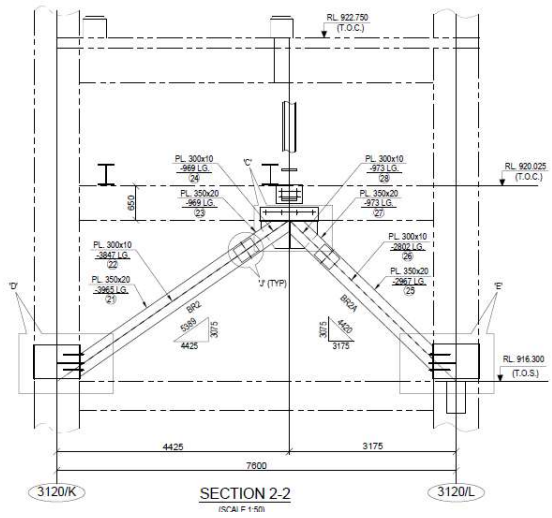
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.



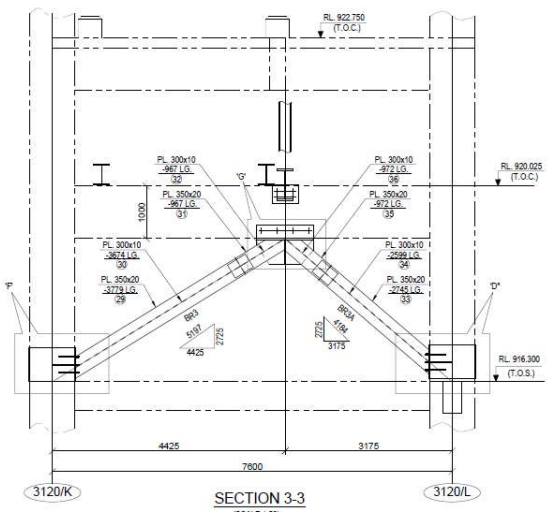
PART PLAN AT EL. 920.025(T.O.C)
(SCALE 1:100)



SECTION 1-1
(SCALE 1:50)



SECTION 2-2



SECTION 3-3

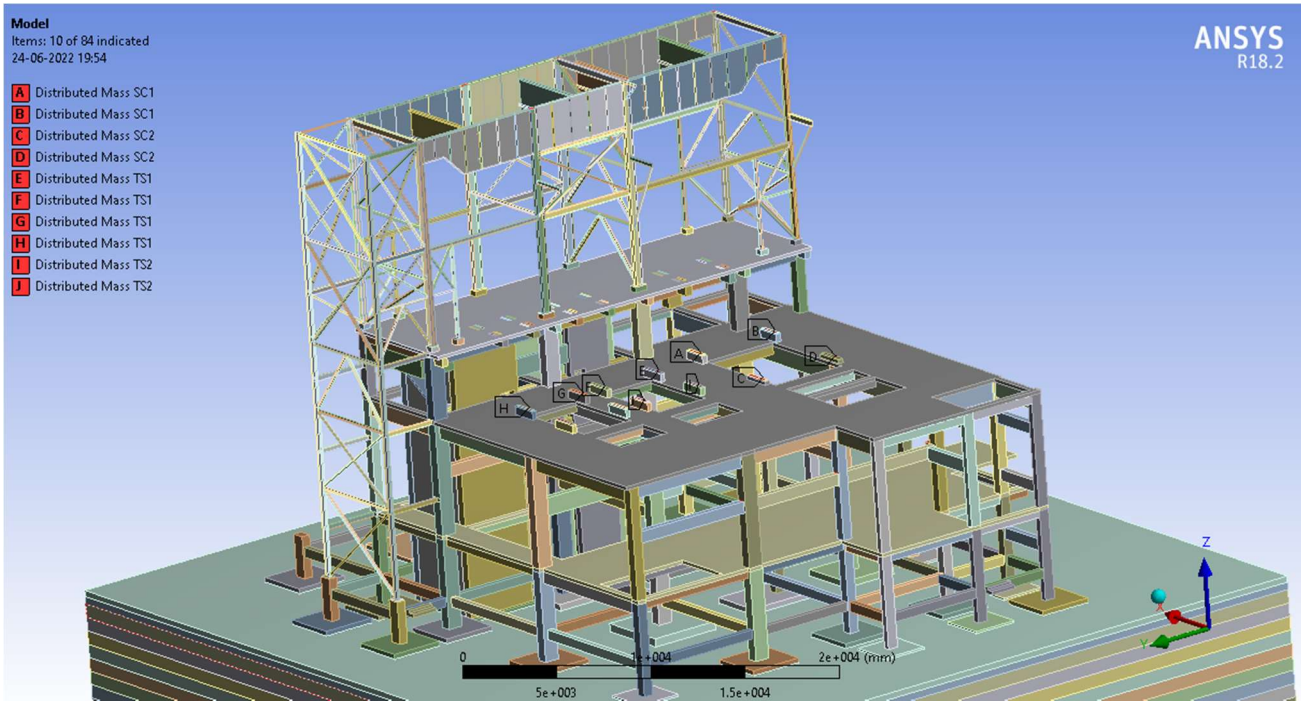
PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

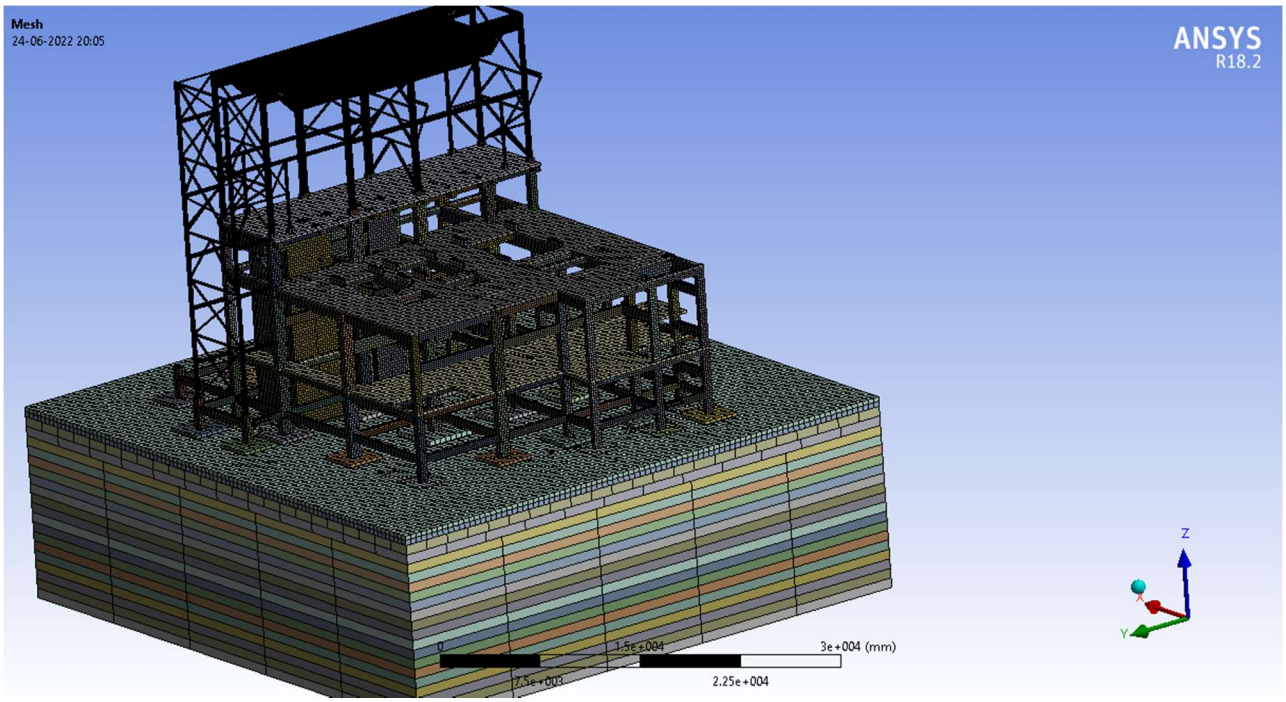
8. Appendix 2: Various Strengthening options

1. Complete Structure with Soil model – With additional walls below feeder floor

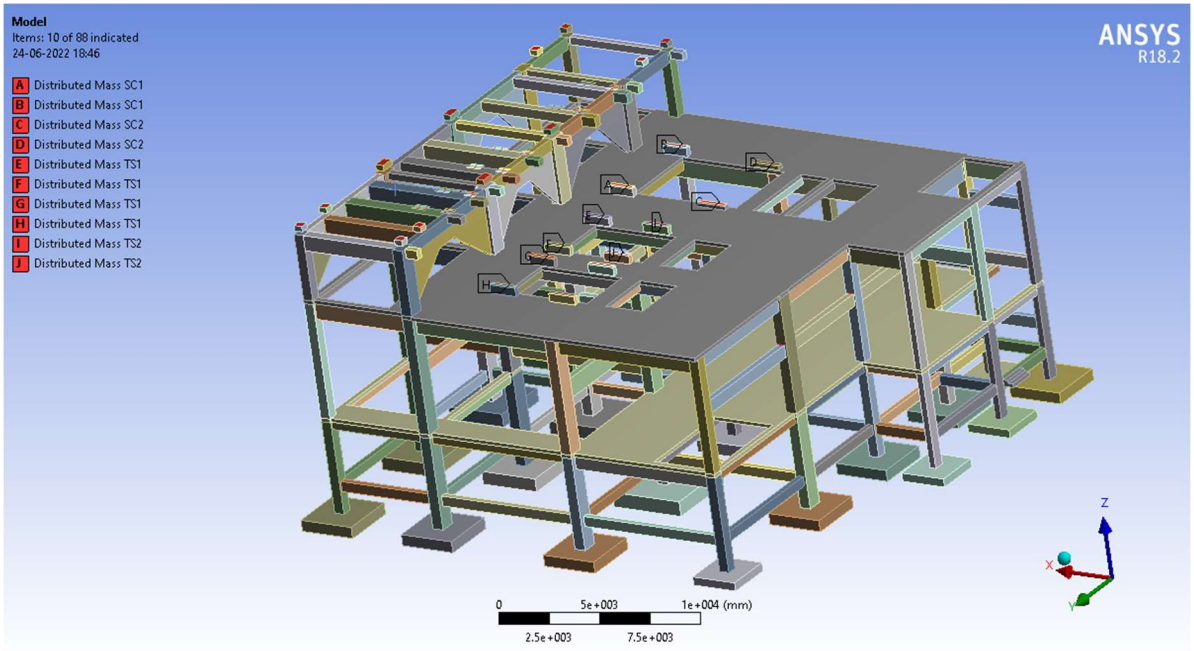


PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

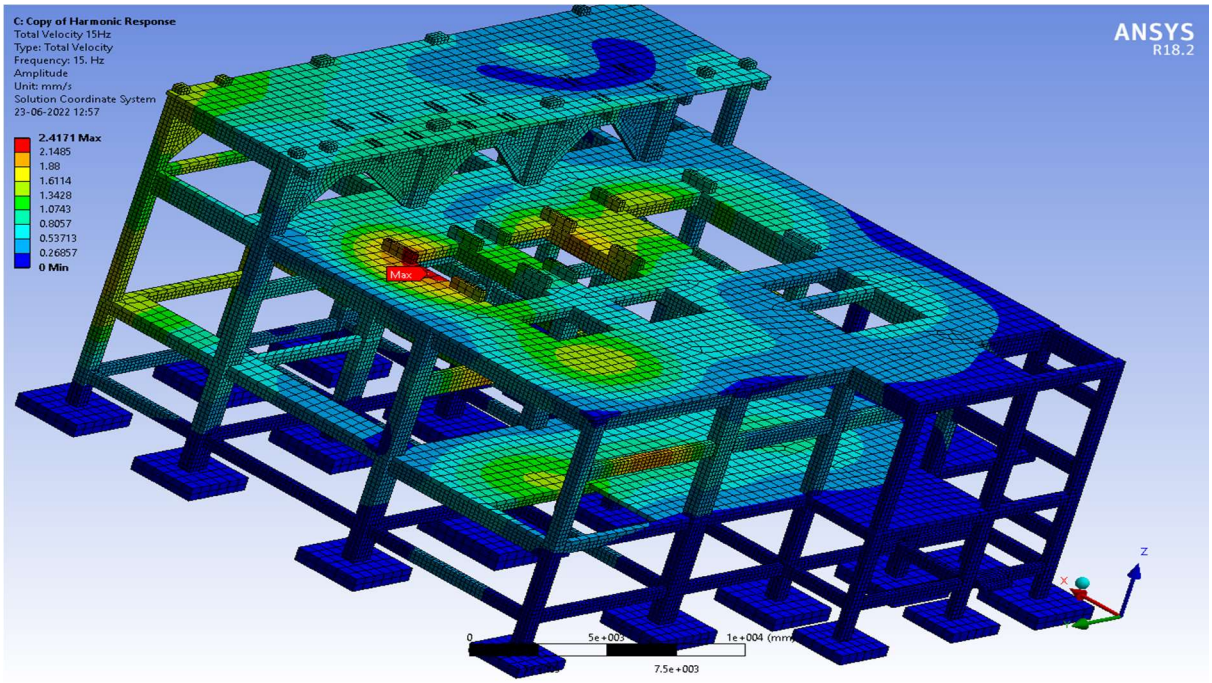


2. Additional walls below Feeder floor – 6 Walls and 6 number of beams

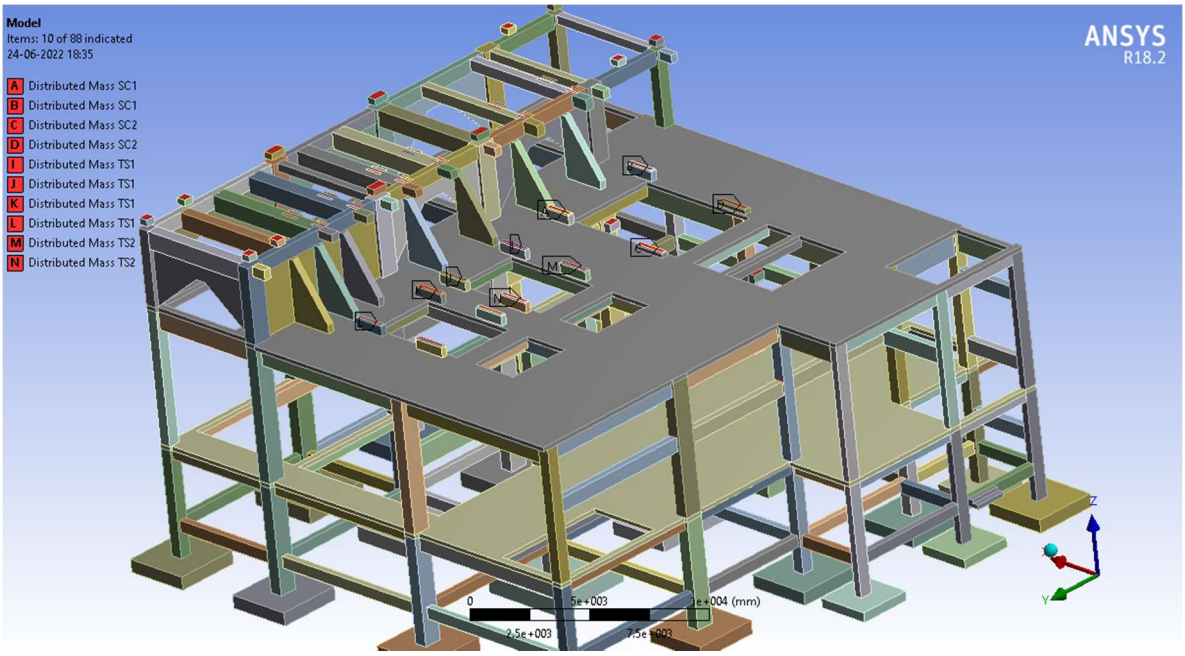


PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.



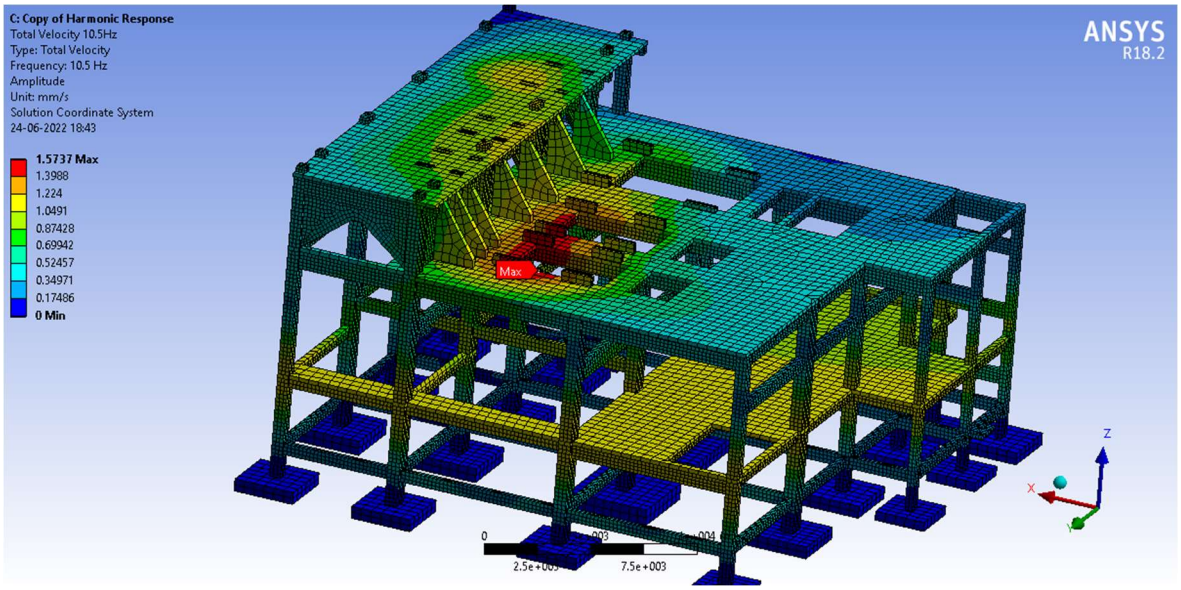
3. Additional walls below Feeder floor with Ribs



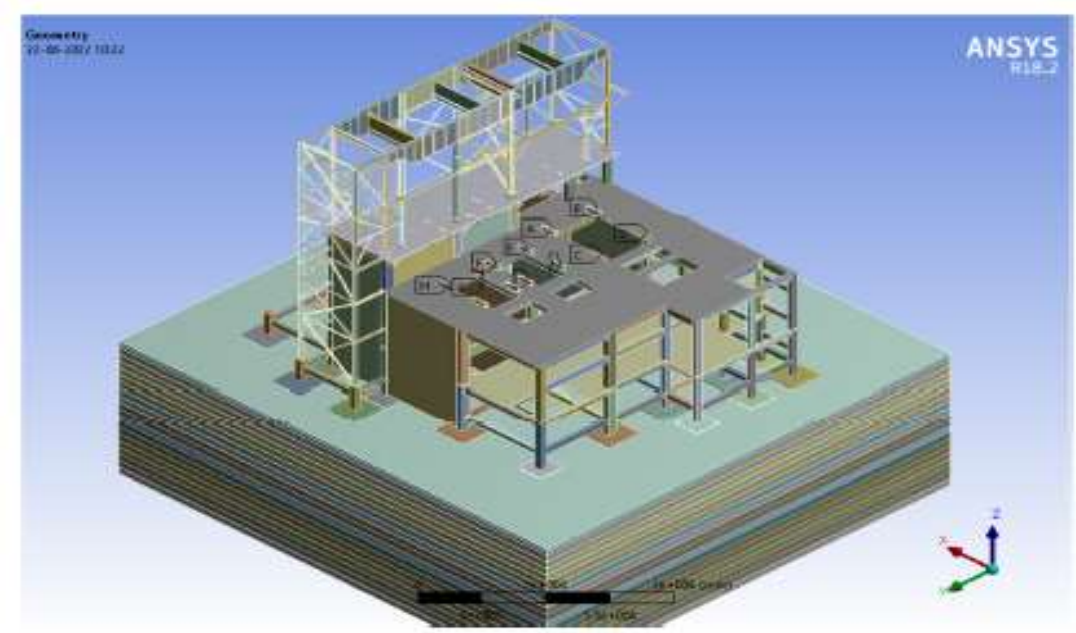
PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.



4. Additional walls below Feeder floor and Screen Supporting Floor

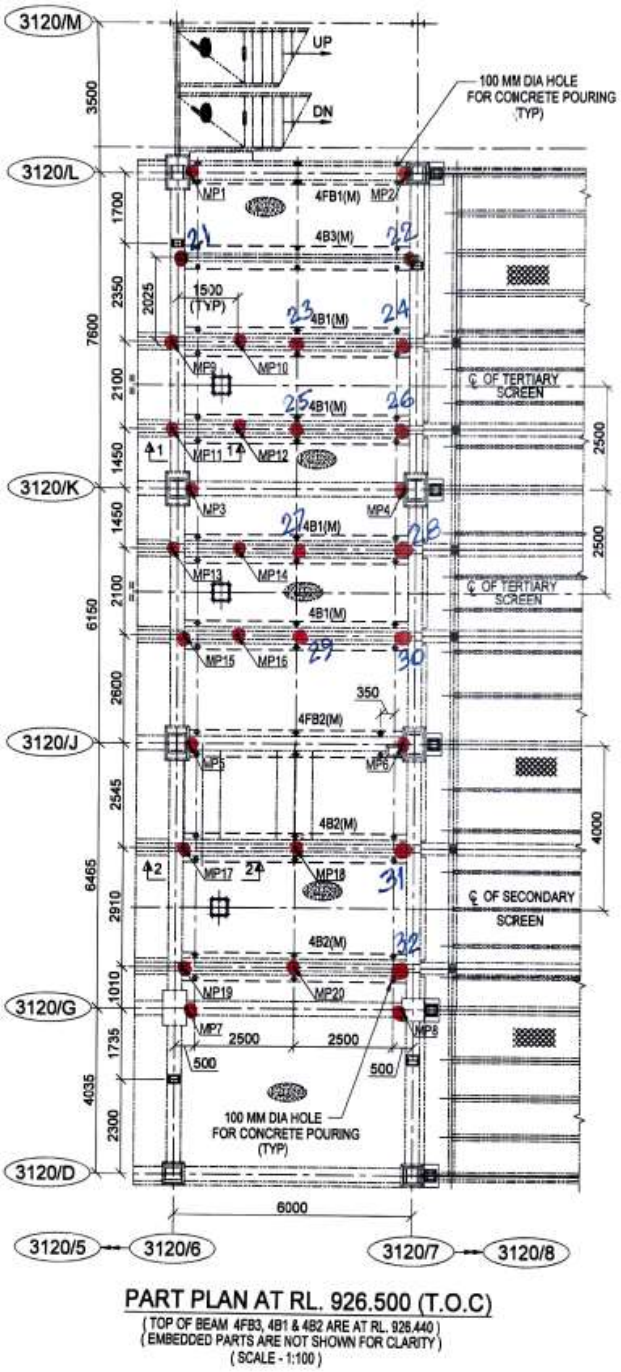


PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

9. Appendix 3: Measuring Points

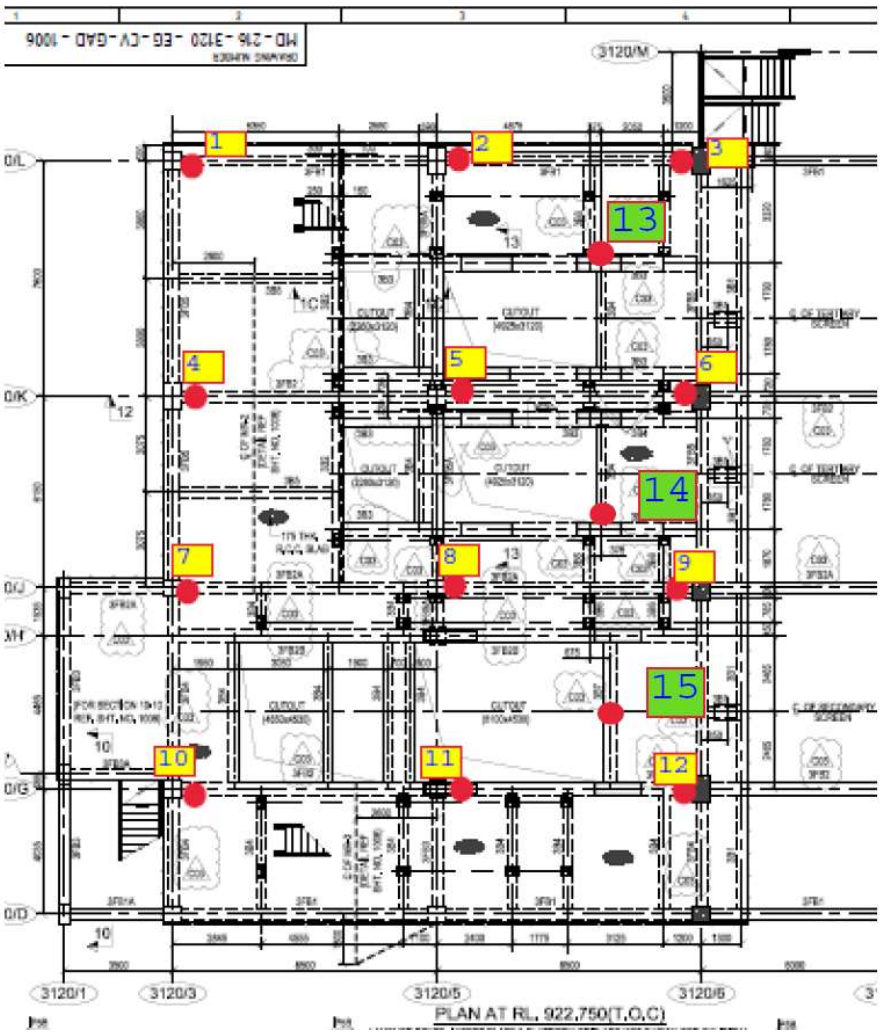
Vibrofeeder Floor



PART PLAN AT RL. 926.500 (T.O.C)
(TOP OF BEAM 4FB3, 4B1 & 4B2 ARE AT RL. 926.440)
(EMBEDDED PARTS ARE NOT SHOWN FOR CLARITY)
(SCALE - 1:100)

PROPRIETARY & CONFIDENTIAL
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Screen Floor



PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

10. Appendix 4: Summary of RMS Velocity at measuring points

Case A -Screens-14.25Hz & 13.75Hz, Ter Feed- 15 Hz & Sec Feed-15 Hz

Feeder Floor Points			
	X dir	Y dir	Z dir
MP1	0.0993	0.1129	0.7306
MP2	0.1053	0.1245	1.1173
MP3	0.0717	0.1568	1.1564
MP4	0.0985	0.1023	1.1201
MP5	0.1321	0.1942	0.9983
MP6	0.1173	0.0781	1.1050
MP7	0.0637	0.0794	0.6866
MP8	0.0556	0.1118	0.4671
MP9	0.0647	0.1128	0.9094
MP10	0.0579	0.1000	1.1215
MP11	0.0564	0.1221	0.7982
MP12	0.0406	0.1059	0.8133
MP13	0.0738	0.1456	1.4539
MP14	0.0635	0.1137	2.1217
MP15	0.0629	0.1037	1.6049
MP16	0.0543	0.0619	1.7723
MP17	0.1488	0.0681	1.5827
MP18	0.0842	0.0609	2.1055
MP19	0.0674	0.1054	0.8451
MP20	0.0467	0.0671	1.1676
MP21	0.0718	0.0970	1.0717
MP22	0.0791	0.0880	1.3074
MP23	0.0462	0.0929	1.1768
MP24	0.0659	0.1099	1.1595
MP25	0.0370	0.0939	0.9209
MP26	0.0611	0.1092	1.0231
MP27	0.0622	0.0959	2.2720
MP28	0.1286	0.0796	1.2605
MP29	0.0862	0.0431	1.7958
MP30	0.1641	0.0471	1.2617
MP31	0.1172	0.0902	1.0900
MP32	0.0706	0.1079	0.6328

Screen Floor Points			
	X dir	Y dir	Z dir
MP1	0.1255	0.1499	0.3342
MP2	0.1287	0.1547	0.3773
MP3	0.1275	0.0930	0.4562
MP4	0.0922	0.1379	0.3586
MP5	0.1304	0.1297	1.2956
MP6	0.1881	0.0810	0.7110
MP7	0.0302	0.1489	0.3243
MP8	0.1068	0.1852	0.8559
MP9	0.1491	0.1273	0.6568
MP10	0.2026	0.1447	0.3402
MP11	0.3109	0.1422	0.6375
MP12	0.2039	0.1214	0.6132
MP13	0.1571	0.1062	1.0566
MP14	0.1634	0.0714	1.4065
MP15	0.1460	0.0766	1.0495

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Case B -Screens-14.25Hz & 13.75Hz, Ter Feed- 9.16 Hz & Sec Feed-10.83 Hz

Feeder Floor Points			
	X dir	Y dir	Z dir
MP1	0.1750	0.2175	0.9619
MP2	0.1867	0.1375	1.0847
MP3	0.1728	0.1503	1.8814
MP4	0.1333	0.1134	1.1458
MP5	0.1756	0.2254	1.7494
MP6	0.1669	0.0602	0.9734
MP7	0.1127	0.1231	0.9799
MP8	0.0644	0.1341	0.5346
MP9	0.0951	0.1381	1.5325
MP10	0.0894	0.1113	1.8064
MP11	0.1288	0.1104	1.5424
MP12	0.0941	0.0883	1.7293
MP13	0.1863	0.1460	2.2592
MP14	0.1541	0.1035	2.9314
MP15	0.1797	0.1091	2.5047
MP16	0.1606	0.0676	2.5998
MP17	0.1423	0.1284	2.0940
MP18	0.1119	0.0672	2.4585
MP19	0.0879	0.1572	1.2477
MP20	0.0372	0.1064	1.4788
MP21	0.1175	0.1546	1.4446
MP22	0.1424	0.1184	1.3380
MP23	0.0858	0.0868	1.8382
MP24	0.0996	0.1347	1.2483
MP25	0.0680	0.0765	1.7480
MP26	0.0786	0.1309	1.0952
MP27	0.1252	0.0806	2.8841
MP28	0.1396	0.0808	1.3037
MP29	0.1433	0.0507	2.5241
MP30	0.1609	0.0556	1.3663
MP31	0.1565	0.0860	1.0911
MP32	0.0610	0.1386	0.7140

Screen Floor Points			
	X dir	Y dir	Z dir
MP1	0.0996	0.1579	0.2880
MP2	0.1307	0.1521	0.3958
MP3	0.1047	0.0975	0.6546
MP4	0.0910	0.1423	0.3190
MP5	0.1254	0.1308	1.2646
MP6	0.1808	0.0783	1.3042
MP7	0.0518	0.1491	0.3590
MP8	0.1026	0.1771	0.7633
MP9	0.1535	0.1175	1.2971
MP10	0.2385	0.1474	0.3583
MP11	0.2945	0.1380	0.5642
MP12	0.2581	0.1103	0.8730
MP13	0.1439	0.0974	1.2520
MP14	0.1728	0.0646	1.8684
MP15	0.1770	0.0557	1.3569

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Case C -Screens-14.25Hz & 13.75Hz, Ter Feed- 9.16 Hz & Sec Feed-12.2 Hz

Feeder Floor Points			
	X dir	Y dir	Z dir
MP1	0.2626	0.1601	0.7437
MP2	0.8887	0.1310	1.0337
MP3	0.3451	0.1247	1.1169
MP4	0.3974	0.1116	0.8848
MP5	0.1925	0.1448	1.3270
MP6	0.1137	0.0456	0.8650
MP7	0.0886	0.0823	0.8674
MP8	0.6920	0.0979	0.4533
MP9	0.7456	0.0965	1.1028
MP10	0.7339	0.0962	1.3671
MP11	0.8364	0.0824	0.9813
MP12	0.9201	0.0846	1.1061
MP13	1.1495	0.1246	1.3060
MP14	0.8876	0.0980	1.7329
MP15	1.1552	0.0748	1.5301
MP16	0.1261	0.0373	1.3110
MP17	0.1870	0.0630	1.8087
MP18	0.0961	0.0505	2.2423
MP19	0.0873	0.1166	1.0775
MP20	0.5842	0.0756	1.2977
MP21	0.2987	0.1121	1.1190
MP22	0.7309	0.0972	1.2611
MP23	0.3156	0.0993	1.4684
MP24	0.8152	0.1157	1.1205
MP25	0.3048	0.0840	1.1840
MP26	1.1013	0.1141	0.9124
MP27	0.3771	0.0876	1.7747
MP28	1.2096	0.0858	0.9615
MP29	0.3890	0.0231	1.2694
MP30	0.1468	0.0334	1.0056
MP31	0.1050	0.0715	1.0169
MP32	0.0544	0.1036	0.6018

Screen Floor Points			
	X dir	Y dir	Z dir
MP1	0.1367	0.1704	0.2934
MP2	0.1596	0.1628	0.4083
MP3	0.1367	0.1312	0.7656
MP4	0.0956	0.1561	0.3273
MP5	0.1132	0.1485	1.2712
MP6	0.1863	0.1239	1.3145
MP7	0.0489	0.1647	0.3909
MP8	0.1091	0.1954	0.7671
MP9	0.1542	0.1519	1.2009
MP10	0.2345	0.1632	0.3912
MP11	0.2921	0.1472	0.5633
MP12	0.2551	0.1578	0.8074
MP13	0.1697	0.1241	1.3425
MP14	0.1651	0.1194	1.6428
MP15	0.1802	0.0946	1.4762

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Case D -Screens-14.25Hz & 13.75Hzz, Ter Feed- 9.16 Hz & Sec Feed-15 Hz

Feeder Floor Points			
	X dir	Y dir	Z dir
MP1	0.0996	0.1626	1.0407
MP2	0.1177	0.1248	1.1365
MP3	0.1434	0.1544	1.6488
MP4	0.1083	0.0954	1.1315
MP5	0.1362	0.2113	1.2406
MP6	0.1148	0.0693	1.0113
MP7	0.0627	0.0957	0.6746
MP8	0.0565	0.0912	0.4450
MP9	0.0803	0.1210	1.4857
MP10	0.0726	0.1028	1.7164
MP11	0.1111	0.1077	1.3494
MP12	0.0810	0.0927	1.4766
MP13	0.1456	0.1450	2.0067
MP14	0.1176	0.1068	2.7098
MP15	0.1249	0.1222	2.1723
MP16	0.1040	0.0773	2.2924
MP17	0.1266	0.1004	1.4612
MP18	0.0790	0.0619	1.9395
MP19	0.0519	0.1181	0.8365
MP20	0.0360	0.0712	1.1480
MP21	0.0695	0.1193	1.5189
MP22	0.0911	0.0991	1.3882
MP23	0.0603	0.0872	1.7300
MP24	0.0753	0.1138	1.2680
MP25	0.0596	0.0833	1.5102
MP26	0.0713	0.1100	1.0885
MP27	0.0885	0.0848	2.7266
MP28	0.1135	0.0719	1.2589
MP29	0.0938	0.0550	2.2556
MP30	0.1318	0.0563	1.2409
MP31	0.1192	0.0713	1.0086
MP32	0.0639	0.0856	0.6072

Screen Floor Points			
	X dir	Y dir	Z dir
MP1	0.1236	0.1443	0.3015
MP2	0.1435	0.1542	0.4049
MP3	0.1246	0.0886	0.6955
MP4	0.0886	0.1308	0.3185
MP5	0.1265	0.1278	1.2302
MP6	0.1745	0.0802	1.0999
MP7	0.0281	0.1446	0.3177
MP8	0.0978	0.1742	0.7791
MP9	0.1428	0.1161	0.8850
MP10	0.2077	0.1410	0.3327
MP11	0.3054	0.1399	0.5766
MP12	0.2197	0.1200	0.5706
MP13	0.1626	0.0976	1.1974
MP14	0.1629	0.0719	1.4819
MP15	0.1569	0.0533	1.0781

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Case E -Screens-14.25Hz & 13.75Hz, Ter Feed- 12 Hz & Sec Feed-10.83 Hz

Feeder Floor Points			
	X dir	Y dir	Z dir
MP1	0.1756	0.2182	1.0464
MP2	0.1763	0.1642	1.1876
MP3	0.0958	0.1357	1.7131
MP4	0.1267	0.1237	1.0544
MP5	0.1648	0.1724	1.7620
MP6	0.1596	0.0507	0.9379
MP7	0.1184	0.1301	1.1428
MP8	0.0643	0.1454	0.5884
MP9	0.0793	0.1308	1.6247
MP10	0.0753	0.1148	1.8808
MP11	0.0780	0.1016	1.5628
MP12	0.0549	0.0862	1.6848
MP13	0.1023	0.1381	1.9050
MP14	0.0911	0.1053	2.3342
MP15	0.1081	0.0910	1.9331
MP16	0.1034	0.0481	1.6985
MP17	0.1616	0.1032	2.4577
MP18	0.1070	0.0638	2.8186
MP19	0.1048	0.1729	1.4698
MP20	0.0383	0.1147	1.6933
MP21	0.1187	0.1525	1.5454
MP22	0.1323	0.1239	1.4552
MP23	0.0717	0.1033	1.8911
MP24	0.0934	0.1362	1.3167
MP25	0.0460	0.0800	1.6769
MP26	0.0733	0.1303	1.1013
MP27	0.1021	0.0891	2.3007
MP28	0.1476	0.0924	1.1455
MP29	0.1321	0.0324	1.6838
MP30	0.1786	0.0422	1.1284
MP31	0.1403	0.0873	1.2021
MP32	0.0614	0.1515	0.7852

Screen Floor Points			
	X dir	Y dir	Z dir
MP1	0.1447	0.1616	0.2890
MP2	0.1508	0.1592	0.3939
MP3	0.1493	0.1432	0.7072
MP4	0.1180	0.1490	0.3356
MP5	0.1106	0.1479	1.3403
MP6	0.2142	0.1177	1.2538
MP7	0.0602	0.1612	0.3927
MP8	0.1036	0.1990	0.8293
MP9	0.1771	0.1456	1.2854
MP10	0.2466	0.1569	0.3949
MP11	0.2944	0.1517	0.6241
MP12	0.2754	0.1313	1.0218
MP13	0.1657	0.1166	1.3339
MP14	0.1780	0.1163	1.9737
MP15	0.1912	0.0849	1.6060

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Case F -Screens-14.25Hz & 13.75Hz, Ter Feed- 12 Hz & Sec Feed-12.2 Hz

Feeder Floor Points			
	X dir	Y dir	Z dir
MP1	0.1072	0.2160	1.2077
MP2	0.1244	0.1704	1.3111
MP3	0.0661	0.1348	1.7248
MP4	0.1014	0.1328	1.0882
MP5	0.1336	0.1566	1.6581
MP6	0.1121	0.0550	0.9191
MP7	0.0968	0.1210	1.0534
MP8	0.0571	0.1175	0.5115
MP9	0.0676	0.1204	1.8367
MP10	0.0608	0.1188	2.1415
MP11	0.0718	0.0870	1.6938
MP12	0.0481	0.0910	1.8469
MP13	0.0573	0.1404	1.8355
MP14	0.0464	0.1130	2.2435
MP15	0.0447	0.1058	1.8140
MP16	0.0394	0.0533	1.5418
MP17	0.1661	0.0840	2.2786
MP18	0.0769	0.0677	2.6577
MP19	0.0898	0.1656	1.3311
MP20	0.0369	0.1028	1.5292
MP21	0.0743	0.1427	1.7716
MP22	0.0991	0.1222	1.6216
MP23	0.0552	0.1222	2.1571
MP24	0.0827	0.1364	1.4531
MP25	0.0397	0.0942	1.8529
MP26	0.0759	0.1311	1.1895
MP27	0.0688	0.1033	2.2315
MP28	0.1235	0.1035	1.1261
MP29	0.0898	0.0310	1.5409
MP30	0.1545	0.0415	1.0829
MP31	0.1043	0.0840	1.1496
MP32	0.0592	0.1236	0.6784

Screen Floor Points			
	X dir	Y dir	Z dir
MP1	0.1819	0.1741	0.2944
MP2	0.1797	0.1699	0.4064
MP3	0.1813	0.1769	0.8182
MP4	0.1226	0.1628	0.3439
MP5	0.0985	0.1656	1.3469
MP6	0.2197	0.1633	1.2641
MP7	0.0573	0.1768	0.4246
MP8	0.1102	0.2173	0.8331
MP9	0.1778	0.1799	1.1893
MP10	0.2427	0.1727	0.4278
MP11	0.2920	0.1609	0.6232
MP12	0.2725	0.1788	0.9562
MP13	0.1915	0.1433	1.4244
MP14	0.1704	0.1711	1.7481
MP15	0.1944	0.1238	1.7253

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Case G -Screens-14.25Hz & 13.75Hz, Ter Feed- 12 Hz & Sec Feed-15 Hz

Feeder Floor Points			
	X dir	Y dir	Z dir
MP1	0.1002	0.1633	1.1251
MP2	0.1072	0.1516	1.2394
MP3	0.0663	0.1398	1.4805
MP4	0.1016	0.1056	1.0402
MP5	0.1254	0.1582	1.2532
MP6	0.1075	0.0598	0.9758
MP7	0.0683	0.1026	0.8375
MP8	0.0564	0.1025	0.4988
MP9	0.0645	0.1136	1.5778
MP10	0.0584	0.1064	1.7909
MP11	0.0604	0.0989	1.3698
MP12	0.0418	0.0906	1.4322
MP13	0.0616	0.1371	1.6524
MP14	0.0545	0.1086	2.1126
MP15	0.0534	0.1041	1.6008
MP16	0.0468	0.0578	1.3911
MP17	0.1459	0.0752	1.8248
MP18	0.0742	0.0585	2.2995
MP19	0.0688	0.1339	1.0586
MP20	0.0371	0.0795	1.3626
MP21	0.0707	0.1172	1.6197
MP22	0.0810	0.1045	1.5054
MP23	0.0463	0.1037	1.7828
MP24	0.0691	0.1153	1.3364
MP25	0.0376	0.0867	1.4391
MP26	0.0660	0.1094	1.0946
MP27	0.0654	0.0934	2.1432
MP28	0.1215	0.0834	1.1007
MP29	0.0825	0.0366	1.4153
MP30	0.1495	0.0429	1.0030
MP31	0.1030	0.0725	1.1196
MP32	0.0643	0.0986	0.6784

Screen Floor Points			
	X dir	Y dir	Z dir
MP1	0.1688	0.1480	0.3025
MP2	0.1636	0.1613	0.4030
MP3	0.1692	0.1344	0.7481
MP4	0.1156	0.1375	0.3351
MP5	0.1117	0.1450	1.3059
MP6	0.2080	0.1195	1.0496
MP7	0.0365	0.1568	0.3514
MP8	0.0988	0.1961	0.8451
MP9	0.1664	0.1441	0.8733
MP10	0.2158	0.1506	0.3693
MP11	0.3054	0.1536	0.6365
MP12	0.2370	0.1410	0.7194
MP13	0.1844	0.1169	1.2793
MP14	0.1681	0.1236	1.5872
MP15	0.1711	0.0825	1.3272

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Case H -Screens-14.25Hz & 13.75Hz, Ter Feed- 15 Hz & Sec Feed-10.83Hz

Feeder Floor Points			
	X dir	Y dir	Z dir
MP1	0.1748	0.1679	0.6518
MP2	0.1743	0.1372	1.0655
MP3	0.1011	0.1527	1.3889
MP4	0.1236	0.1203	1.1344
MP5	0.1715	0.2084	1.5070
MP6	0.1694	0.0690	1.0671
MP7	0.1137	0.1069	0.9918
MP8	0.0635	0.1547	0.5567
MP9	0.0795	0.1299	0.9562
MP10	0.0747	0.1085	1.2115
MP11	0.0740	0.1248	0.9912
MP12	0.0538	0.1016	1.0660
MP13	0.1144	0.1466	1.7065
MP14	0.1001	0.1104	2.3433
MP15	0.1176	0.0907	1.9372
MP16	0.1109	0.0523	2.0797
MP17	0.1646	0.0960	2.2155
MP18	0.1170	0.0663	2.6245
MP19	0.1035	0.1444	1.2563
MP20	0.0478	0.1023	1.4984
MP21	0.1197	0.1323	0.9973
MP22	0.1304	0.1073	1.2573
MP23	0.0716	0.0925	1.2851
MP24	0.0902	0.1309	1.1398
MP25	0.0453	0.0872	1.1586
MP26	0.0684	0.1301	1.0297
MP27	0.0989	0.0917	2.4295
MP28	0.1547	0.0886	1.3053
MP29	0.1357	0.0388	2.0643
MP30	0.1932	0.0464	1.3871
MP31	0.1545	0.1050	1.1725
MP32	0.0677	0.1608	0.7397

Screen Floor Points			
	X dir	Y dir	Z dir
MP1	0.1015	0.1636	0.3207
MP2	0.1158	0.1527	0.3682
MP3	0.1076	0.1019	0.4153
MP4	0.0945	0.1495	0.3591
MP5	0.1293	0.1326	1.3300
MP6	0.1944	0.0791	0.9153
MP7	0.0540	0.1533	0.3656
MP8	0.1115	0.1881	0.8401
MP9	0.1597	0.1287	1.0689
MP10	0.2334	0.1510	0.3657
MP11	0.2999	0.1404	0.6251
MP12	0.2423	0.1117	0.9156
MP13	0.1384	0.1060	1.1112
MP14	0.1733	0.0641	1.7931
MP15	0.1660	0.0790	1.3283

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Case I -Screens-14.25Hz & 13.75Hz, Ter Feed- 15 Hz & Sec Feed-12.2 Hz

Feeder Floor Points			
	X dir	Y dir	Z dir
MP1	0.1064	0.1656	0.8132
MP2	0.1224	0.1433	1.1890
MP3	0.0715	0.1519	1.4006
MP4	0.0983	0.1294	1.1681
MP5	0.1404	0.1926	1.4031
MP6	0.1218	0.0733	1.0483
MP7	0.0921	0.0978	0.9024
MP8	0.0563	0.1268	0.4798
MP9	0.0678	0.1195	1.1683
MP10	0.0602	0.1125	1.4722
MP11	0.0677	0.1102	1.1222
MP12	0.0470	0.1064	1.2280
MP13	0.0694	0.1489	1.6371
MP14	0.0554	0.1181	2.2526
MP15	0.0541	0.1055	1.8181
MP16	0.0469	0.0575	1.9230
MP17	0.1690	0.0768	2.0364
MP18	0.0869	0.0701	2.4637
MP19	0.0884	0.1371	1.1176
MP20	0.0465	0.0904	1.3342
MP21	0.0753	0.1225	1.2235
MP22	0.0971	0.1056	1.4236
MP23	0.0551	0.1114	1.5510
MP24	0.0795	0.1310	1.2762
MP25	0.0391	0.1014	1.3346
MP26	0.0710	0.1309	1.1180
MP27	0.0656	0.1059	2.3603
MP28	0.1306	0.0997	1.2859
MP29	0.0934	0.0375	1.9214
MP30	0.1691	0.0457	1.3416
MP31	0.1185	0.1017	1.1200
MP32	0.0656	0.1329	0.6328

Screen Floor Points			
	X dir	Y dir	Z dir
MP1	0.1386	0.1761	0.3261
MP2	0.1448	0.1633	0.3807
MP3	0.1396	0.1355	0.5263
MP4	0.0992	0.1633	0.3674
MP5	0.1172	0.1504	1.3365
MP6	0.1999	0.1247	0.9256
MP7	0.0511	0.1689	0.3975
MP8	0.1181	0.2065	0.8439
MP9	0.1604	0.1631	0.9728
MP10	0.2295	0.1668	0.3987
MP11	0.2975	0.1495	0.6242
MP12	0.2393	0.1593	0.8500
MP13	0.1642	0.1327	1.2016
MP14	0.1656	0.1189	1.5675
MP15	0.1692	0.1179	1.4476

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Case J -Screens-14.25Hz & 13.75Hz, Ter & Sec Feed- 10.914 Hz

Feeder Floor Points			
	X dir	Y dir	Z dir
MP1	0.3344	0.2697	1.0138
MP2	0.3337	0.2084	1.1250
MP3	0.1647	0.1700	2.0661
MP4	0.1815	0.1612	1.1167
MP5	0.2877	0.1868	1.8084
MP6	0.2774	0.0743	0.9808
MP7	0.1513	0.1643	0.8150
MP8	0.1260	0.1882	0.4135
MP9	0.1282	0.1857	1.6762
MP10	0.1201	0.1203	1.9655
MP11	0.1039	0.1554	1.7122
MP12	0.0769	0.1063	1.9108
MP13	0.1866	0.1663	2.3850
MP14	0.1683	0.1065	2.8620
MP15	0.2117	0.0932	2.6691
MP16	0.2016	0.0611	2.6947
MP17	0.1662	0.1415	1.9086
MP18	0.1635	0.0755	2.1911
MP19	0.0790	0.1737	1.0484
MP20	0.0485	0.0987	1.2294
MP21	0.2223	0.2128	1.5524
MP22	0.2355	0.1726	1.3942
MP23	0.1154	0.0813	1.9325
MP24	0.1359	0.1895	1.2900
MP25	0.0617	0.0728	1.8875
MP26	0.0842	0.1753	1.1280
MP27	0.1744	0.0732	2.8007
MP28	0.2203	0.1106	1.2405
MP29	0.2306	0.0473	2.4340
MP30	0.2743	0.0585	1.3007
MP31	0.1938	0.1203	0.9722
MP32	0.0518	0.1846	0.5817

Screen Floor Points			
	X dir	Y dir	Z dir
MP1	0.1287	0.1722	0.2853
MP2	0.1384	0.1569	0.3616
MP3	0.1353	0.1275	0.6966
MP4	0.1263	0.1553	0.3351
MP5	0.1257	0.1357	1.2612
MP6	0.2318	0.0786	1.4633
MP7	0.0783	0.1571	0.3728
MP8	0.1217	0.1682	0.7897
MP9	0.1949	0.0973	1.3863
MP10	0.2561	0.1551	0.3641
MP11	0.3129	0.1392	0.5366
MP12	0.2723	0.1011	0.7714
MP13	0.1538	0.0961	1.3434
MP14	0.1833	0.0578	2.2123
MP15	0.1961	0.0514	1.3963

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Case K -Screens-14.25Hz & 13.75Hz, Ter & Sec Feed- 11.172 Hz

Feeder Floor Points			
	X dir	Y dir	Z dir
MP1	0.3390	0.2129	1.0257
MP2	0.3234	0.2213	1.1264
MP3	0.1337	0.1632	2.0392
MP4	0.1888	0.1545	1.1083
MP5	0.2669	0.1708	1.8222
MP6	0.3086	0.0830	0.9890
MP7	0.1302	0.1299	0.8398
MP8	0.1001	0.2111	0.4639
MP9	0.1363	0.1626	1.6882
MP10	0.1246	0.1135	1.9711
MP11	0.0856	0.1472	1.7579
MP12	0.0586	0.0998	1.8860
MP13	0.1456	0.1514	2.4232
MP14	0.1386	0.0989	2.9510
MP15	0.1729	0.0966	2.6961
MP16	0.1729	0.0603	2.6540
MP17	0.1433	0.1287	1.8396
MP18	0.1667	0.0789	2.0911
MP19	0.0788	0.1268	0.9915
MP20	0.0381	0.0978	1.1725
MP21	0.2311	0.1695	1.5461
MP22	0.2266	0.1656	1.4154
MP23	0.1174	0.0858	1.9950
MP24	0.1282	0.1832	1.3061
MP25	0.0596	0.0678	1.8910
MP26	0.0920	0.1642	1.1076
MP27	0.1747	0.0758	2.9027
MP28	0.2359	0.1043	1.2986
MP29	0.2323	0.0467	2.4545
MP30	0.2854	0.0464	1.3633
MP31	0.2023	0.1317	0.9569
MP32	0.0452	0.2111	0.5964

Screen Floor Points			
	X dir	Y dir	Z dir
MP1	0.1265	0.1735	0.2848
MP2	0.1352	0.1564	0.3625
MP3	0.1324	0.1214	0.7008
MP4	0.1243	0.1565	0.3314
MP5	0.1211	0.1355	1.2561
MP6	0.2235	0.0828	1.4822
MP7	0.0783	0.1614	0.3858
MP8	0.1158	0.1664	0.8635
MP9	0.1987	0.0955	1.3866
MP10	0.2591	0.1583	0.3697
MP11	0.3114	0.1405	0.5483
MP12	0.2750	0.0981	0.7473
MP13	0.1524	0.0949	1.3612
MP14	0.1842	0.0594	2.3760
MP15	0.1993	0.0542	1.4161

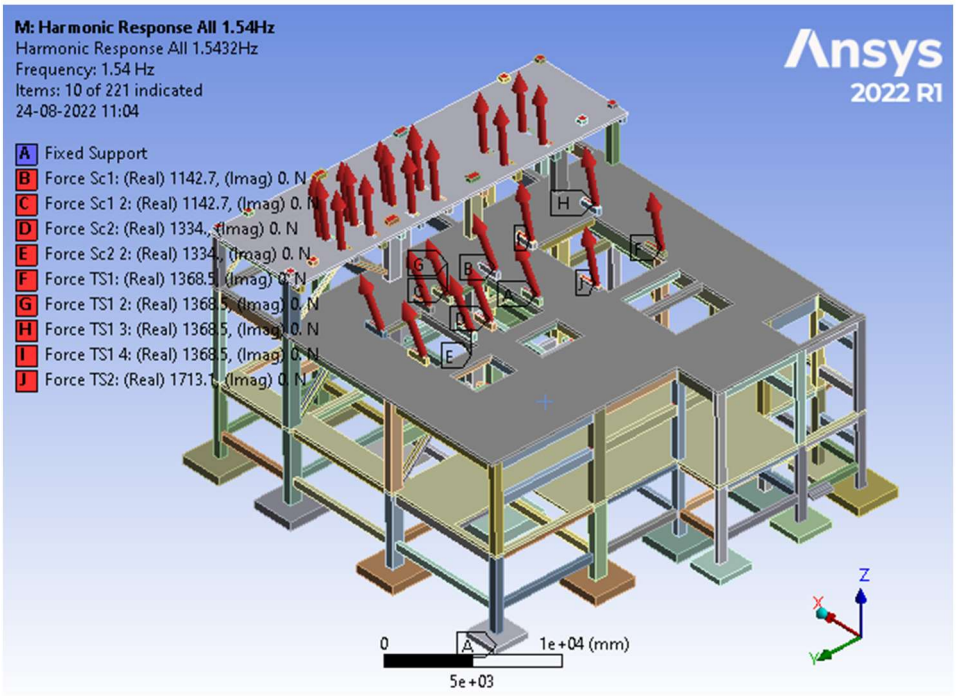
PROPRIETARY & CONFIDENTIAL

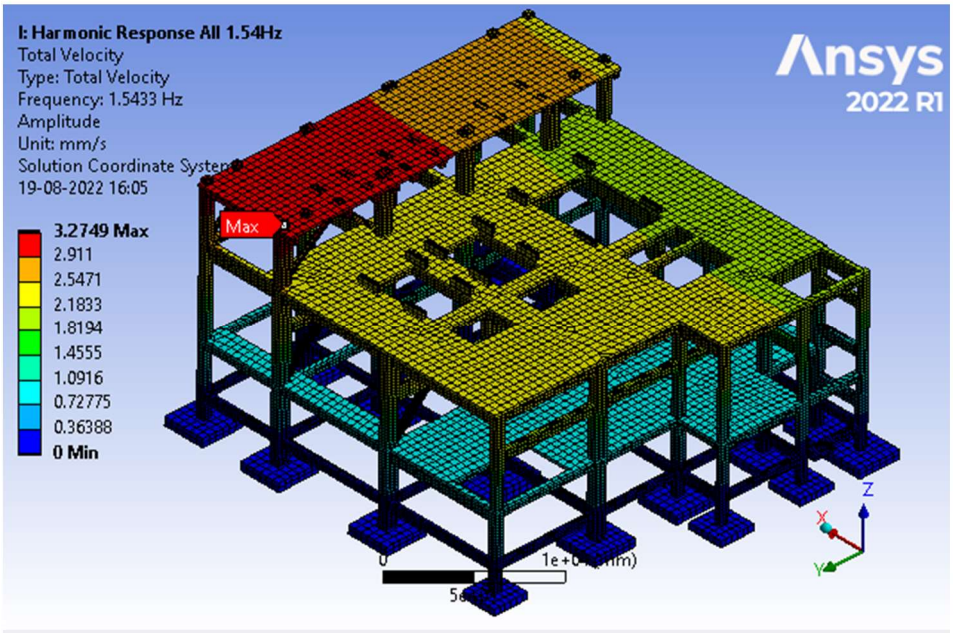
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

11. Appendix 5: Analysis with additional frequency

The following analysis is performed for frequencies at start/stop conditions and for natural frequencies with high mass participation in modal analysis.

ALL EQUIPMENTS RUNNING AT	1.543 HZ





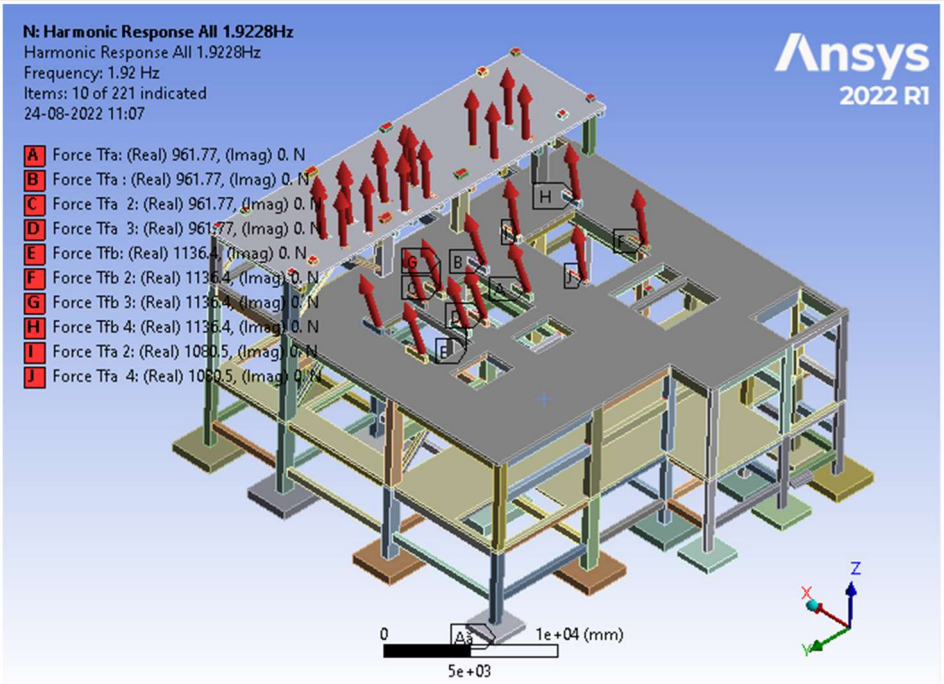
RMS Velocity	2.315 mm/sec

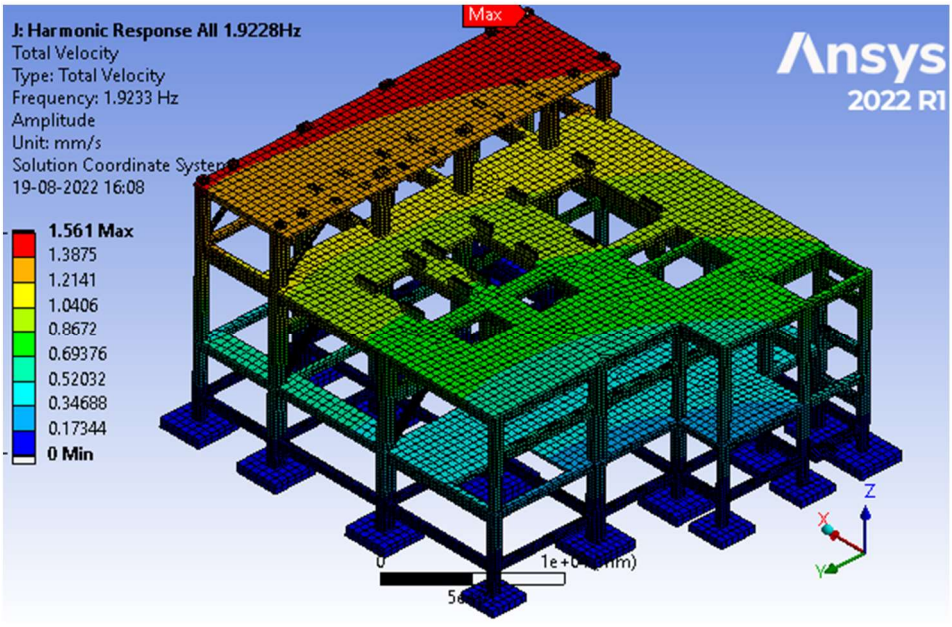
PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

ALL EQUIPMENTS RUNNING AT 1.922 HZ



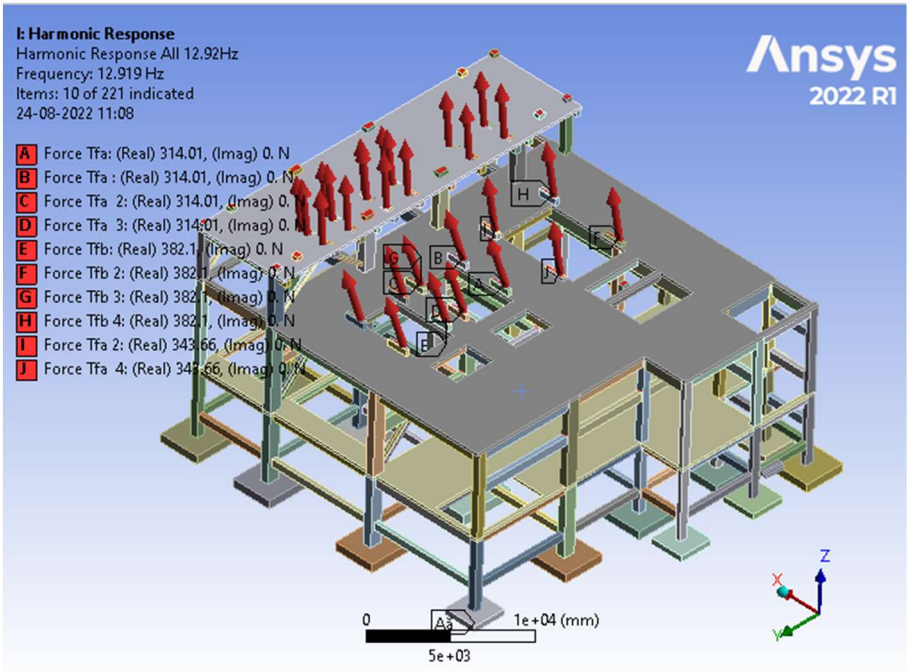


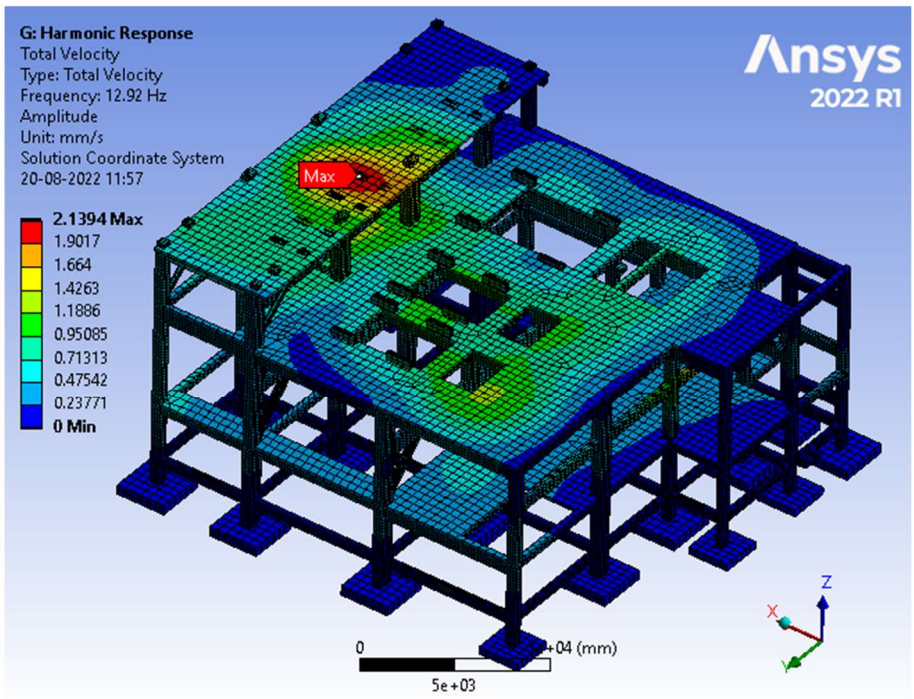
RMS Velocity 1.104 mm/sec

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

ALL EQUIPMENTS RUNNING AT	12.92 HZ





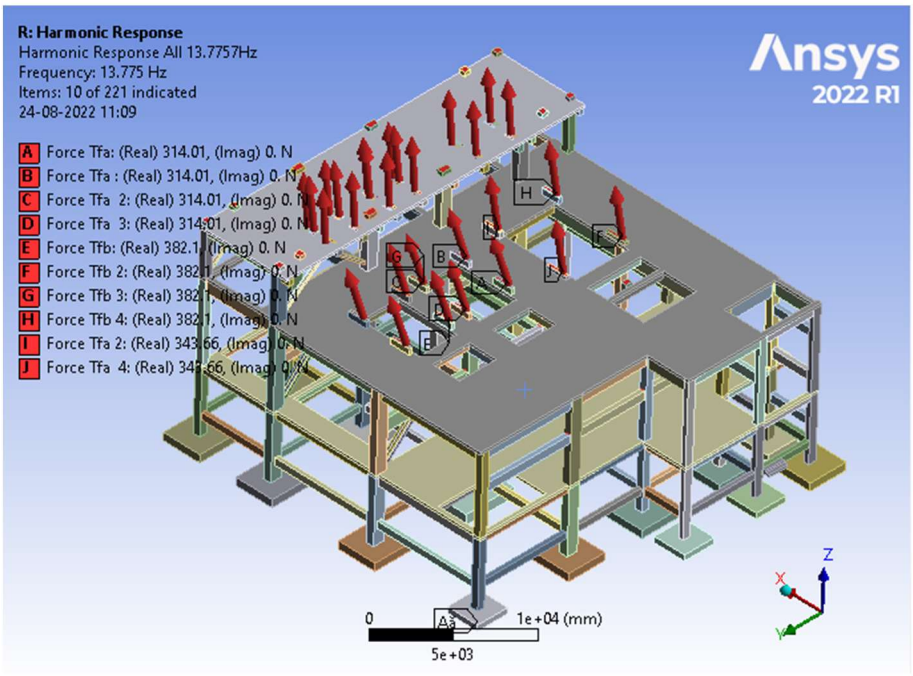
RMS Velocity	1.512 mm/sec

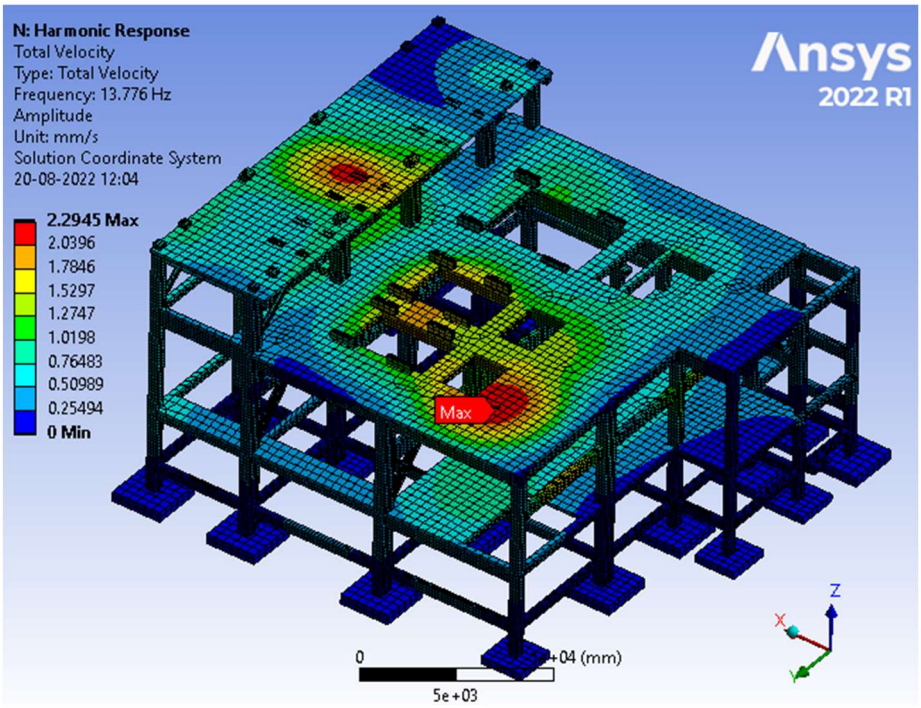
PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.

This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

ALL EQUIPMENTS RUNNING AT 13.776 HZ



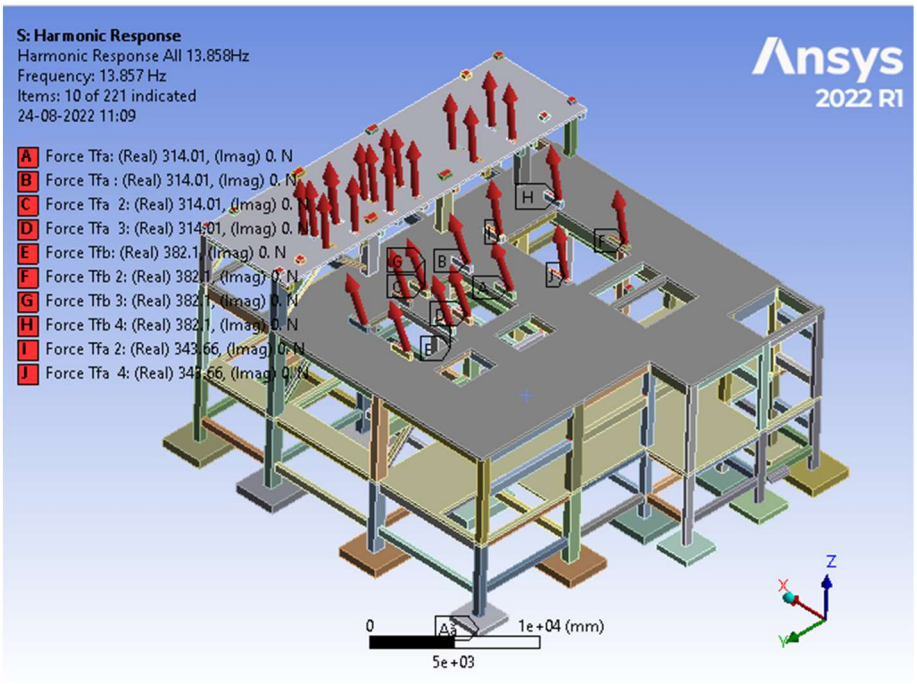


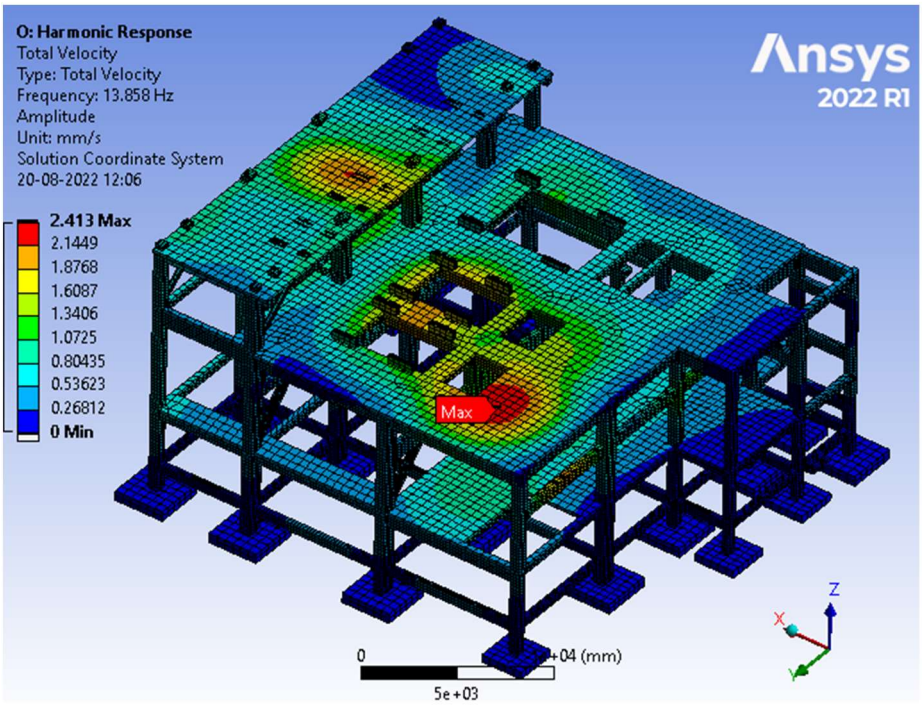
RMS Velocity 1.619 mm/sec

PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

ALL EQUIPMENTS RUNNING AT 13.858 HZ





RMS Velocity 1.706 mm/sec

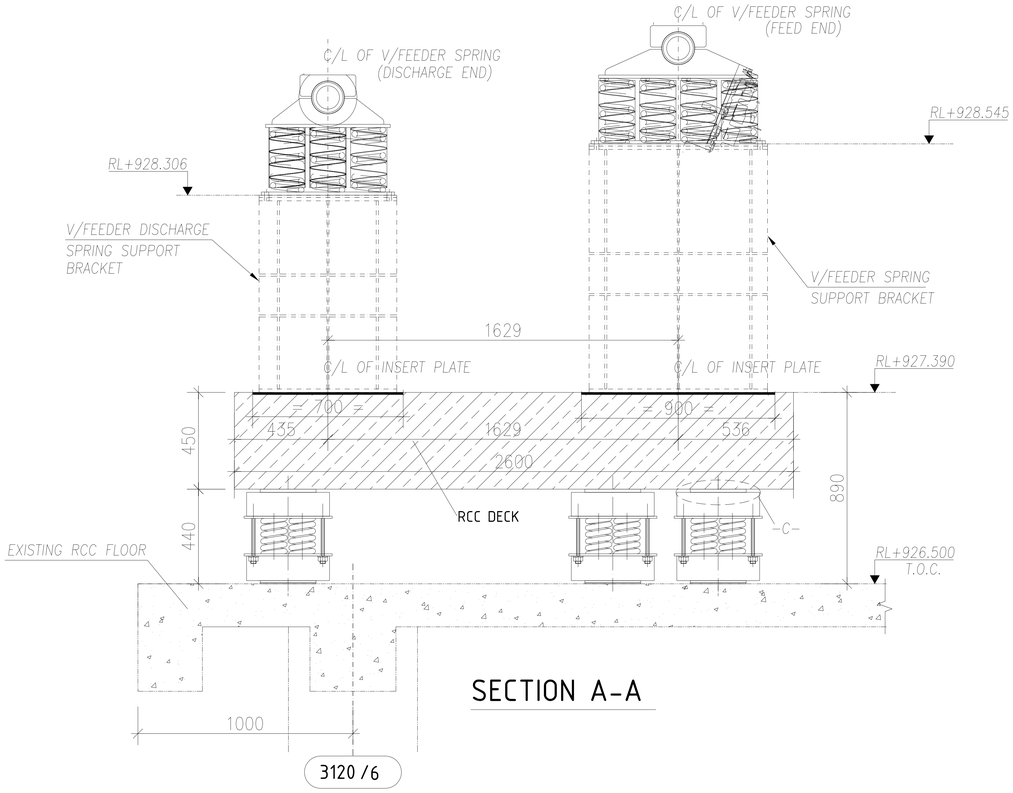
PROPRIETARY & CONFIDENTIAL

The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

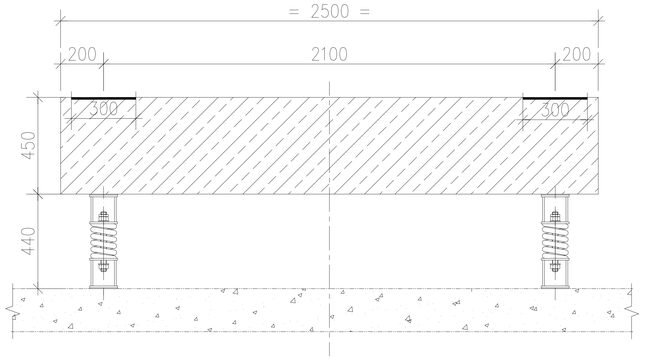
12. Appendix 6: Vibration Isolation System and Dynamic & Static Analysis Report

PROPRIETARY & CONFIDENTIAL

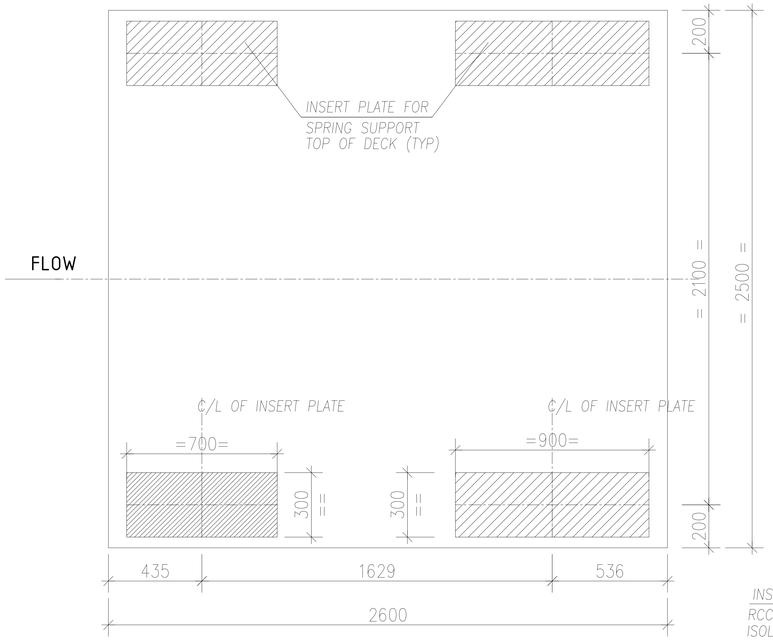
The contents of this document are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without the prior written permission of Outotec is prohibited.
This document is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.



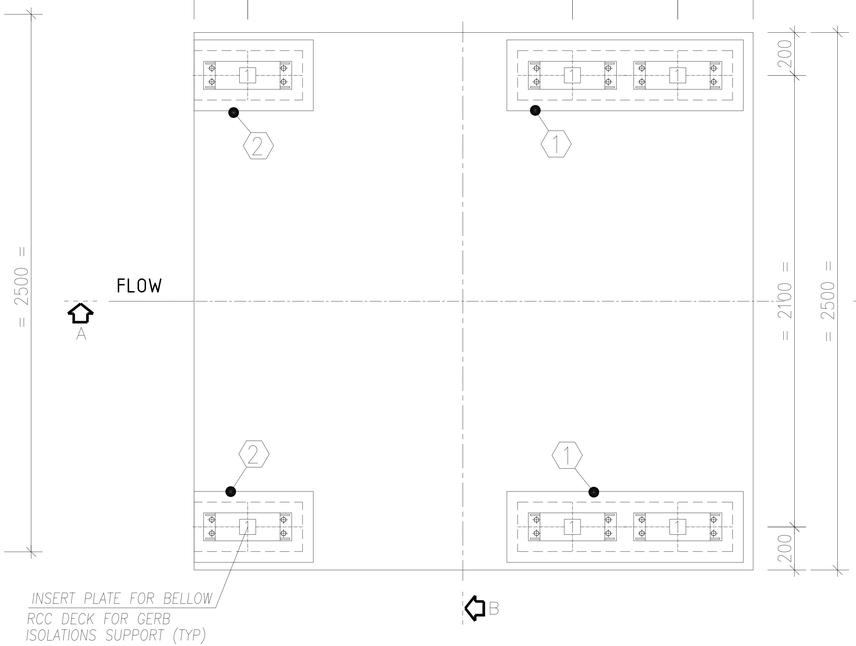
SECTION A-A



SECTION B-B



VIBRO FEEDER SPRING SUPPORTS LOCATIONS

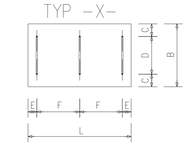


GERB ISOLATORS LOCATION

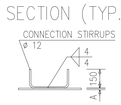
STEEL COVER PLATES (mm, kg)

MARK	TYP	PCS.	L x B	A	C	D	E	F	G	H	SNGL WEIGHT	TOTAL WEIGHT	CONNECTION STIRRUPS PSC.	BAR LENGTH	TOTAL LENGTH	TOTAL WEIGHT
①	-X-	2	1100x330	12	40	250	100	300	–	–	34.194	68.389	2x4=8	550	4400	3.894
②	-X-	2	505x330	12	40	250	27.5	250	–	–	17.252	34.505	2x3=6	550	3300	2.920
NOTE: UNDERSIDE OF STEEL COVER PLATES TO BE PAINTED											TOTAL :	102.894			7700	6.814

TYP -X-



SECTION (TYP.)



NOT TO SCALE

STATIC LOAD ON SUBSTRUCTURE

45 KN \ SPRING ELEMENT VERTICAL
±4.5 KN \ SPRING ELEMENT HORIZONTAL

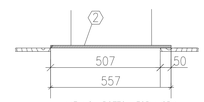
(THESE ARE STATIC LOADS TO BE CONSIDERED FOR DESIGN OF SUPPORTING STRUCTURE THAT IS 90% OF DYNAMIC LOADS ARE ISOLATION.)

NOTE :

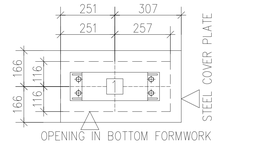
1. HORIZONTAL LOADS GIVEN ARE INCLUSIVE OF SEISMIC AS WELL AS OPRERATIONAL LOADS
2. THE DEFLECTION IN THE SUBSTRUCTURE AT THE POINT OF SPRING UNITS SHOULD NOT EXCEED 5.5 MM.

REFERENCE DRAWING : (3130-FDR-001/002) TERTIARY SCREEN VIBRATING FEEDER

DETAIL -C- (NOT FOR SCALE)
OPENING ABOVE SPRING ELEMENTS

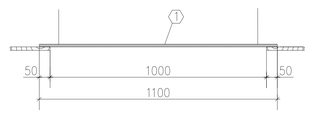


VIEW ON BOTTOM FORMWORK

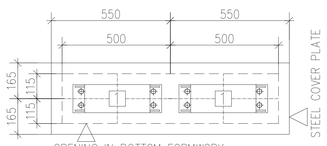


OPENING IN BOTTOM FORMWORK

DETAIL -C- (NOT FOR SCALE)
OPENING ABOVE SPRING ELEMENTS

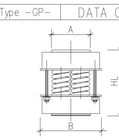


VIEW ON BOTTOM FORMWORK



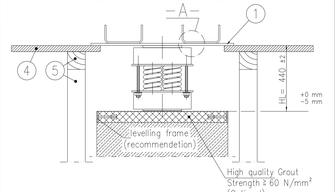
OPENING IN BOTTOM FORMWORK

Type -GP- DATA OF SPRING ELEMENTS



ITEM	PCS.	TYPE	A mm	B mm	C mm	D mm	H_L mm	K_V kN/mm	K_H kN/mm
1	6	GP-2-771	260	410	130	130	440	0.69	0.24

H_L = LOADED HEIGHT | W = WEIGHT PER ELEMENT | A x D = PRESSURE AREA



Setting details for deck support springs

① Steel cover plate with anchors
② Stack of shims as reqd (*)
③ Self-adhesive pad (*)
④ Soffit shuttering
⑤ Support

(*) Gerb supply

Detail -A- (typical)



NOTES

ALL DIMENSIONS ARE SUBJECT TO VERIFICATION AGAINST THE FINAL DRAWING OF THE MACHINE MANUFACTURER AND THE ACTUAL ON-SITE SITUATION.

GENERAL ARRANGEMENT OF MACHINE FOUNDATION IN THE BUILDING BY CUSTOMER.

THE CORRECT HEIGHT OF THE FORMWORK FOR THE FOUNDATION BLOCK MUST BE RESPONSIBLY CHECKED BY THE SUPERVISOR ON SITE. ELEVATIONS REFER TO OPERATING CONDITION. FOLLOW GERB CONSTRUCTION MANUAL!

TOLERANCES:
A) CONCRETE
- GENERAL : ± 5 mm
- COLUMN HEADS
  - FLATNESS : 2 mm
  - ELEVATION : +0/-5 mm
  - VERT. SPACE BETWEEN SURFACES OF CONTACT : 3 mm

B) ANCHOR BOLTS, SLEEVES, HOLES, PIPES, ETC.,
- IF NOT NOTED OTHERWISE :
- HORIZONTAL DISTANCE TO REFERENCE LINE : ±3 mm
- TILTING OF ANCHOR BOLTS, PIPES : 3 mm/m

EMBEDDED PARTS ARE NOT SUPPLIED BY MACHINE MANUFACTURER UNLESS OTHERWISE NOTED. EMBEDMENTS TO BE PLACED AND CHECKED ACCORDING TO THIS DRAWING. THEY SHALL BE SECURED SAFELY AGAINST SHIFTING.

NOT SHOWN CONDUITS OR OTHER OPENINGS (E.G. FOR POWER OR OIL SUPPLY) WHEN NEEDED TO BE DEFINED BY CUSTOMER.

ALL CONCRETE EDGES TO BE CHAMFERED.

CONCRETE OF INERTIA BLOCK TO BE PLACED WITHOUT INTERRUPTION.

CONCRETE SURFACES TO BE PROTECTED AGAINST DETRIMENTAL OIL WHERE NECESSARY.

ELECTRICAL GROUNDING OF THE BLOCK SHALL BE PROVIDED IN ACCORDANCE WITH THE MACHINE MANUFACTURER'S INSTRUCTIONS.

SPRING ELEMENTS ARE ATTACHED BY ADHESIVE PADS AT TOP AND BOTTOM.

MANUFACTURER OF THE VIBRATION ISOLATION SYSTEM:
GERB VIBRATION CONTROL SYSTEMS Pvt. Ltd. / INDIA

SCOPE OF DELIVERY:
- SPRING ELEMENTS
- ADHESIVE PADS
- STEEL SHIMS FOR HEIGHT ADJUSTMENT

ALL MEASURES ARE "mm".
ALL ELEVATIONS ARE "m".

WEIGHTS

FOUNDATION WEIGHT APPROX.	73	kN
MACHINE WEIGHT APPROX.	69	kN

FOR APPROVAL STATUS: 13.07.2022

Index/Rev.	Datum/Date	Aenderung/Modification	ges./dm.	gepr./chd.

GERB VIBRATION CONTROL SYSTEMS PVT. LTD.
2403, 4th Floor ETT-2, Noida Sector-132, Noida-201 301, (U.P.), India
Phone 0120-4751 600, 2975 501
Fax 0120-2973500
E-mail gerbindia@vsnl.com

Kunde/Client : M/s. L&T

Titel/Dwg. title : LAYOUT DRAWING

Projekt : TERTIARY SCREEN VIBRATING FEEDER

SPRING SUPPORTED FOUNDATION FOR VIBRO FEEDER (3130-FDR-001/002)

Bauteil/Structural Member : FOUNDATION BLOCK

This drawing is the property of GERB and subject to copyright protection. It must not be used, especially copied or passed on to third parties, wholly or partly or to be made available to third parties without permission.

Massstab/Scale : 1:10

	Datum/Date	Name
gezeichnet/drawn	13.07.2022	KRS
bearbeitet/coord.	13.07.2022	BKS
geprueft/checked	13.07.2022	JPS

Zeichnung Nr./Drawing No.

Objekt Nr.:	Plan Nr./Format	Index
Job No. :	Dwg.No. Size	Rev.
N1280122	0	0

DOCUMENT NUMBER	REFERENCE DOCUMENTS	REV	DATE	REVISION DESCRIPTION	DRN	CHK	ENG	APP	EMP
		A01	XX/XX/XX	XXXX	--	XX	--	XX	

VENDOR LOGO

LARSEN & TOUBRO

This drawing is the property of LARSEN & TOUBRO and must not be passed on to any person or body not authorized by us to receive it nor be copied or otherwise made use either in full or in part by such person or body without our prior permission in writing.

Outotec

The content of these documents are the sole property of Outotec Oyj. Any rights not expressly granted herein are reserved. Reproduction, transfer or distribution of part or all of the contents in any form without prior written permission of Outotec is prohibited.

MA'ADEN معادن
Saudi Arabian Mining Company شركة التعدين العربية السعودية

This document and the information contained herein is Ma'aden proprietary know-how. This document or any part thereof should not be reproduced, stored in a retrieval system or transmitted by any means, electronic or otherwise, or divulged without prior written permission of Ma'aden.

DRAWING : XXXX

GERB PROJECT DRAWING NUMBER	REVISION
N1280122	0

DATE	DRAWN	DATE	CHECKED	DATE	APPROVED
13/07/2022	KRS	13/07/2022	BKS	13/07/2022	JPS

LARSEN & TOUBRO DRAWING NUMBER	REVISION
O19045_100_H_V_XXXXX_001	0

OUTOTEC DRAWING NUMBER	DOCUMENT ID	REVISION
XXX _XXX _XX _XXXX _XXXXX	OU100668994	0

DRAWN	--	--
CHECKED	XXX	XX/XX/XX
ENGINEER	--	--
APPROVED	XXX	XX/XX/XX
EMPLOYER		
SCALE	1:20	A1

MANSOURAH - MASSARAH GOLD PROJECT
GENERAL ARRANGEMENT
TERTIARY CRUSHING
XXXXXXXXXXXXX

DRAWING NUMBER	REV
MD-216-XXXX-EG-ME-GAD-V500-XXXX	A01

DYNAMIC & STATIC

ANALYSIS

Foundation for a

Vibrating Feeder

Rev.	Datum / Date	Beschreibung / Description	Aufsteller / Author	Geprüft / Checked
2	05.08.2022	Revised as per customer comments	S. Lukes	T. Straten
1	03.08.2022	Revised as per customer comments	S. Lukes	T. Straten
0	15.07.2022	**First Issue / For Approval**	S. Lukes	T.Straten

GERB Engineering GmbH
Ruhrallee 311, D-45136 Essen
Phone: +49-(0)201-26604-20
Fax: +49-(0)201-26604-50
Email: eng@gerb.de

Kunde / Client

M/s Larsen & Toubro Limited

Bestellnummer / Order No.
OE-

Bestelldatum / Order date
-

Projekt / Project

Mansourah – Massarah Gold Project

Vibration Isolation of a Tertiary Screen Vibrating Feeder

Kunden Dok.-Nr. / Client Doc.-No

Dok.-Nr. / Doc.-No.
22-1668-11

Seite / Page
1/49

© GERB 2022. All rights reserved

CONTENTS

Page

1 S U M M A R Y 4
1.1 General 4
1.2 Building Materials 5
1.3 Reference Documents 5
1.4 Main System Data 6
1.5 Machine Data 6
1.6 Vibration Isolation System 6
1.7 Load on Substructure 7
1.8 Discussion of Dynamic Characteristic 7
1.9 Theoretical Transmission 8
1.10 Literature 12
1.11 Calculation Fundamentals and Programs 12
1.12 Calculation Models 13
1.13 Determination of Cross Section Properties 15

2 D Y N A M I C A N A L Y S I S 16
2.1 Foundation and Machine Masses 16
2.2 Results of Dynamic Analysis 17
2.2.1 General 17
2.2.2 Comments on the Results 17
2.2.3 Plot of the Mode Shapes 17
2.3 Determination of Vibrational Amplitudes 23
2.3.1 Excitation by Unbalanced Forces 23
2.3.2 Frequency Depending Load Response 24
2.4 Static Equivalent Load 25

3 S T A T I C A N A L Y S I S 26
3.1 General 26
3.2 Compilation of Load Cases 26
3.3 Description of the Single Load Cases 27
3.3.1 Load Case 1 - Dead Load Foundation 27
3.3.2 Load Case 2 – Machine Load 27
3.3.3 Load Case 3 – Live Load 28
3.3.4 Load Case 4 – Maximum Static Load of Material 29
3.3.5 Load Case 5 – Static Equivalent Load / Vertical (±) 30
3.3.6 Load Case 6 – Static Equivalent Load / Vertical Alternating (±) 31
3.3.7 Load Case 8 – Static Equivalent Load / Horizontal (±) 32
3.3.8 Load Case 9 – Static Equivalent Load / Horizontal Alternating (±) 33
3.4 Superposition of Load Cases 34
3.5 Design Forces 37
3.6 Design of Reinforcement 38
3.6.1 Building Materials 38
3.6.2 Calculation Method 38
3.6.3 Required Reinforcement 39

3.6.4 Chosen Reinforcement 41

4 S P R I NG F O R C E S 42

4.1 Spring Forces and Deflections 42

5 A P P E N D I X 43

5.1 Files of Dynamic Analysis 44
5.1.1 Input File of Modal Extraction (Extract) 44
5.1.2 Results of Modal Extraction Analysis 46
5.2 Files of Static Analysis 47
5.2.1 Input File of Static Calculation (Extract) 47
5.2.2 Output File of Reinforcement Calculation (Extract) 49

1 SUMMARY

1.1 GENERAL

The present analysis deals with the design of a vibration controlled reinforced concrete foundation supporting a screen vibrating feeder.

The SUMMARY (1st part) gives general information, compiles and comments on all relevant project data including the most important results from the dynamic calculation. References are made to the corresponding clauses of the calculation where details are shown.

The 2nd part represents the DYNAMIC ANALYSIS giving evidence about the dynamic behaviour of the structure. Eigenfrequencies are calculated and dynamic velocities considering operational load. Static equivalent loads are defined to be used within the structural analysis.

The 3rd part covers the STATIC ANALYSIS considering all relevant load cases. The reinforcement design follows the regulations of DIN EN 1992-1-1 and DIN 4024 allowing for the below mentioned concrete and steel qualities.

Finally (4th part), as the substructure below the spring elements is not subject to this paper, design loads are given to be used by the design engineer in charge of the substructure.

Note: The analysis of the theoretical model is performed on the assumption of a system dynamically decoupled from the adjacent structure. Accordingly the spring elements are expected to be placed on a relatively stiff substructure, i. e. the stiffness of the supporting beams shall be at least 10 times the stiffness of the spring system (comparison of static deflections) acc. to DIN 4024.

The seismic design of the structure is not scope of this paper.

1.2 BUILDING MATERIALS

Concrete:	(acc. to local standard):	C35
	compressive strength	$f_{ck,cube}$ = 30 N/mm²
	Modulus of elasticity	E = 32000 MN/m²
	Density of reinforced concrete	γ = 25 kN/m³
	Poisson's ratio	ν = 0.20
Steel:	(acc. to ASTM A 615)	Grade 60
	Yield strength:	420 MN/m²
	Design value	365 MN/m²
	Modulus of elasticity	200000 MN/m²

The design was performed according to DIN EN 1992-1-1 and DIN 4024, chapter 6, April 1988.

1.3 REFERENCE DOCUMENTS

Drawings and documents of the machine manufacturer:

L&T Ltd.:

dwg. no. 019045_100_H_V_16510_001	MANSOURAH - MASSARAH GOLD PROJECT GENERAL ARRANGEMENT TERTIARY CRUSHING VIBRATING FEEDER 3130-FDR-001/002
dwg. no. 019045_100_H_D_16436_001	MANSOURAH - MASSARAH GOLD PROJECT DETAIL DRAWING TERTIARY CRUSHING TERTIARY SCREEN VIBRATING FEEDER TROLLEY FRAME

Construction drawings of GERB:

dwg. no. N1280122	LAYOUT DRAWING -FOUNDATION BLOCK-
dwg. no. 22-1668-13 ff.	REINFORCEMENT -FOUNDATION BLOCK-

1.4 MAIN SYSTEM DATA

Machine weight:	G_m	=	82.8	kN
Concrete weight:	G_f	≈	73.1	kN
Rating:	G_f / G_m	=	0.9	
Total spring supported weight:	G_t	≈	155.9	kN
Vertical mode:	n_e	=	2.4	Hz
Relevant operation frequency:	f_m	=	550 - 900	rpm
			(9.1 – 15.0	Hz)

1.5 MACHINE DATA

Vibrating Screen

moving weight:	L	=	59.0	kN
rated speed:	f_m	=	550 - 900	min^{-1}

Screen Stool

static weight:	G	=	23.8	kN
total machine weight:	**G_m**	**=**	**82.8**	**kN**

1.6 VIBRATION ISOLATION SYSTEM

The foundation is supported on spring elements consisting of helical steel springs. Manufacturer: GERB Vibration Control Systems Pvt. Ltd. - Bangalore/India.

- 6 Spring elements type GP-2-771:

spring rates:	k_v =	0.69 kN/mm	k_h =	0.24 kN/mm
Total spring rate:	k_v =	4.14 kN/mm	k_h =	1.44 kN/mm
Static deflection (mean value):	u =	37.7 mm		
Natural vertical system frequency	f =	2.43 Hz		

Nominal bearing capacity of 1 element:

GP-2-771:	F_{max} =	45 kN
GP-2-771:	F_{max} =	4.5 kN

1.7 LOAD ON SUBSTRUCTURE

To consider short-time loadings resulting from extraordinary load conditions at maintenance, emergency etc. the nominal load capacity of the spring units should be used for the design of the substructure. 10 % of the static load should be taken into account in horizontal direction with alternating signs.

GP-2-771:

$F_{vert.}$ = 45 kN
$F_{horiz.,operation}$ = 0.10 · 45 kN = ±4.5 kN

1.8 DISCUSSION OF DYNAMIC CHARACTERISTIC

Modes 1 to 9 are representing the rigid body natural frequencies. Higher modes depend on the elasticity of the structure. They are virtually decoupled from the rigid body modes.

Evaluation of the results in respect of DIN 4024 – Part 1, 5.3.2:

1. f_1 = 1.09 Hz ≤ 0.80 · f_m = 0.80 ·9.17 Hz = 7.33 Hz

2a.) f_9 = 6.45 Hz ≤ 0.90 · f_m = 0.90 · 9.17 Hz = 8.25 Hz

f_{10} = 68.56 Hz ≥ 1.10 · f_m = 1.10 · 15.00 Hz = 16.50 Hz

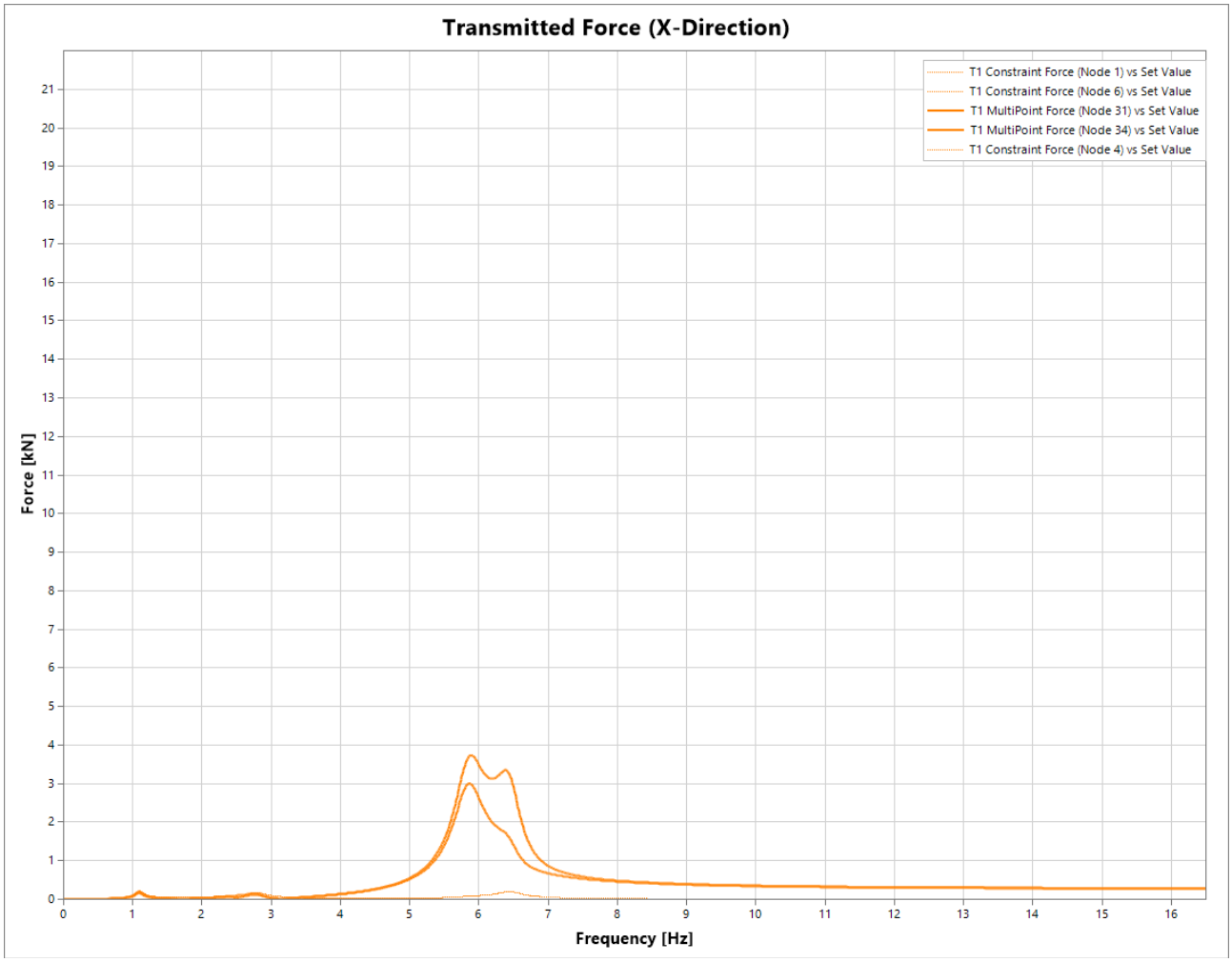
The six rigid body natural frequencies have to be lower than 80% of the service frequency (condition 1.). The next higher frequency has to be 10% below and above the service frequency. The requirements of DIN 4024 – Part 1, 5.3.2 therefore are fulfilled.

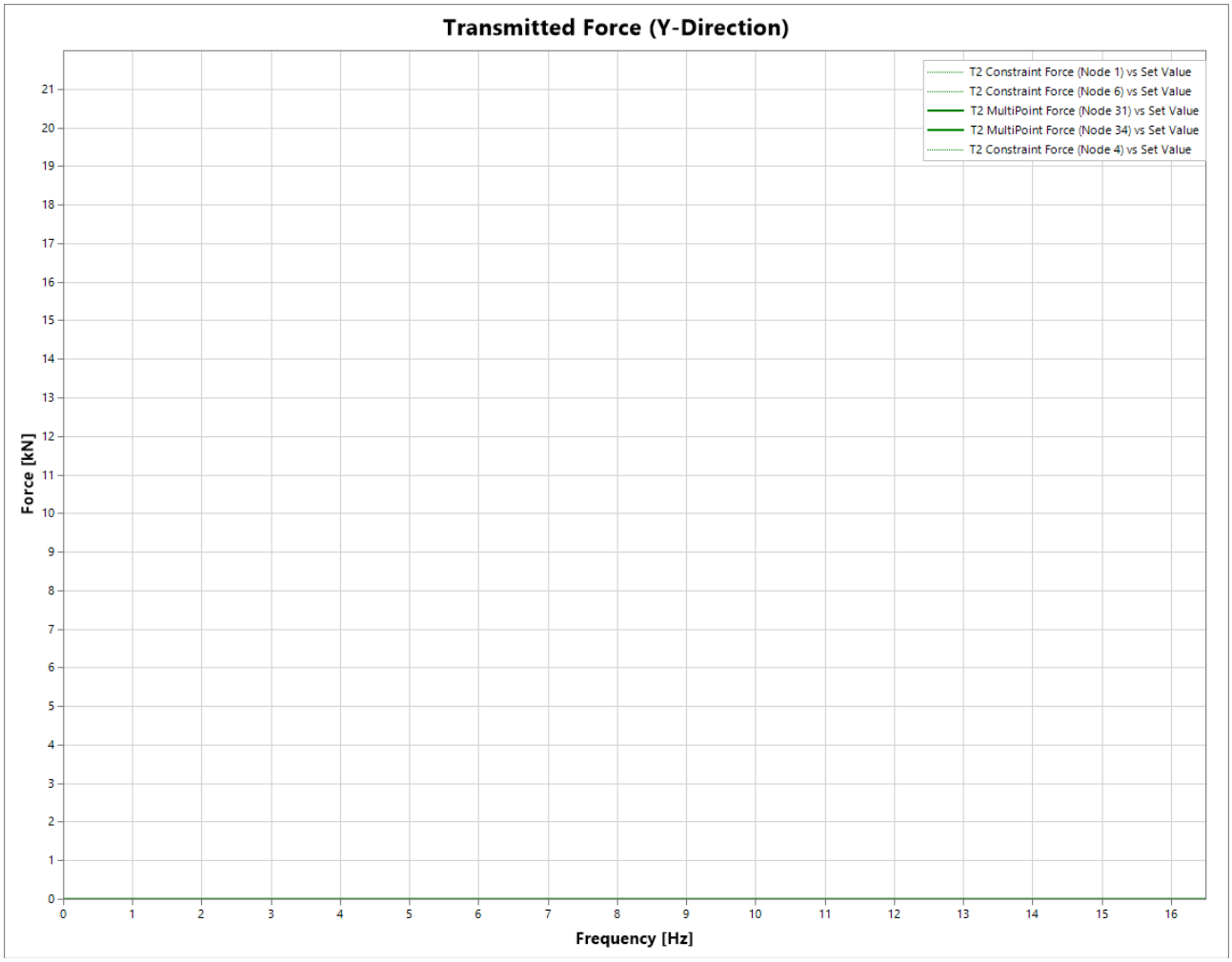
1.9 THEORETICAL TRANSMISSION

Theoretical amplitudes are calculated due to unbalance caused by the machine at normal operation. The unbalanced forces are given by customer (ref. to cl. 1.3). Vibrational amplitudes from the single-force excitation at the c.o.g of the screen (node numbers ref. to cl. 1.12). The calculated values are only theoretical based on a simplified an idealized model. In the below mentioned table a scaled value (to the input force) is shown, to give an overview about the load reduction.

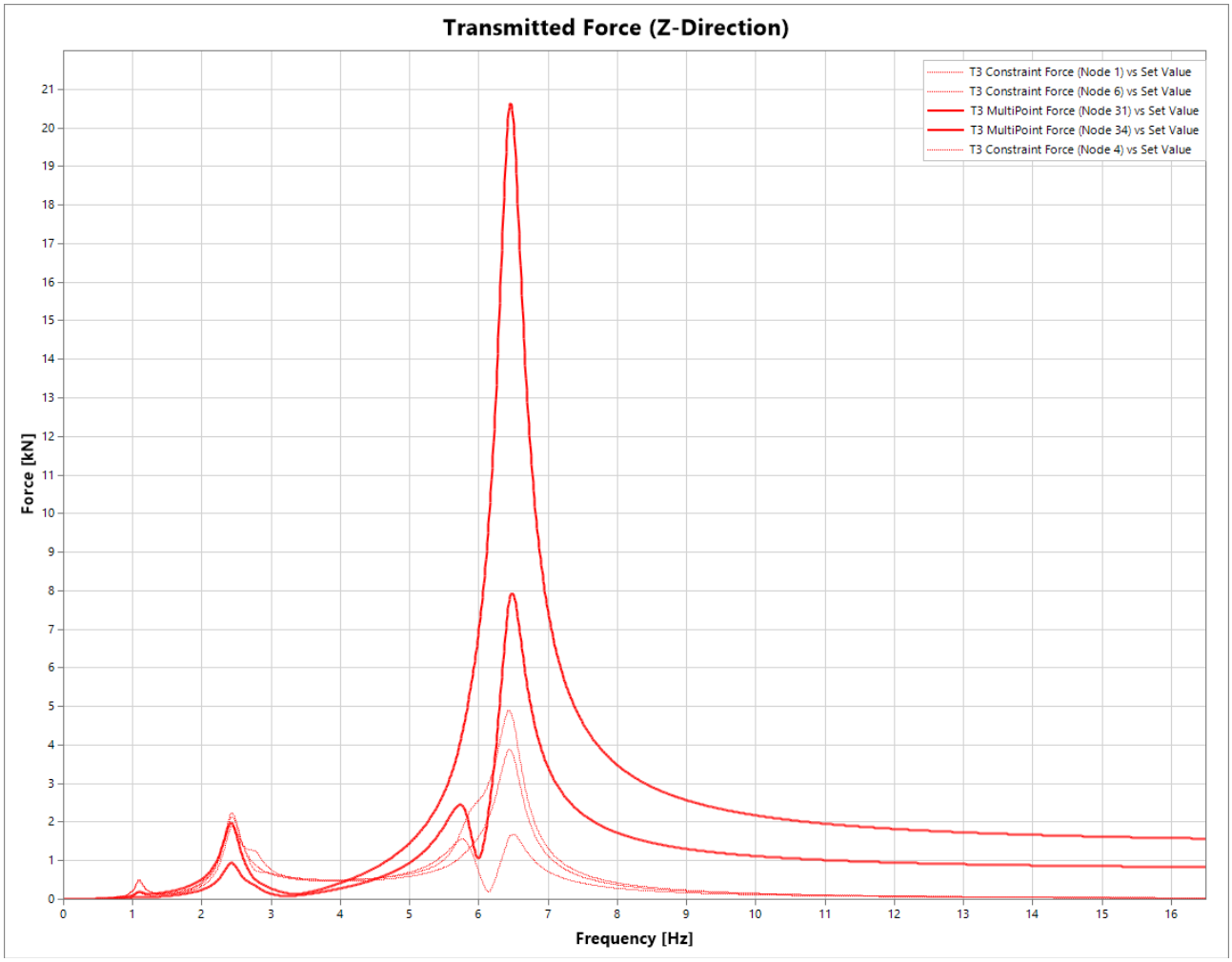
Direction	Frequency [Hz]	Location	Node	Load [kN]	Location	Node	Load [kN]
X	9	Feeder	31	4.375	Below Spring	1	0.12
X	9	Feeder	34	5.825	Below Spring	4	0.15
X	9	Feeder	34	5.825	Below Spring	6	0.15
Z	9	Feeder	31	8.75	Below Spring	1	1.13
Z	9	Feeder	34	11.65	Below Spring	4	0.95
Z	9	Feeder	34	11.65	Below Spring	6	1.07
X	13	Feeder	31	4.375	Below Spring	1	0.05
X	13	Feeder	34	5.825	Below Spring	4	0.06
X	13	Feeder	34	5.825	Below Spring	6	0.06
Z	13	Feeder	31	8.75	Below Spring	1	0.51
Z	13	Feeder	34	11.65	Below Spring	4	0.42
Z	13	Feeder	34	11.65	Below Spring	6	0.47
X	15	Feeder	31	4.375	Below Spring	1	0.04
X	15	Feeder	34	5.825	Below Spring	4	0.05
Z	15	Feeder	34	5.825	Below Spring	6	0.25
Z	15	Feeder	31	8.75	Below Spring	1	0.38
Z	15	Feeder	34	11.65	Below Spring	4	0.31
Z	15	Feeder	34	11.65	Below Spring	6	0.34

The calculated load transmission is related to the operation frequency. The speed range of the machine is shown in the figure below (conservatively printed for machine speed of 0-16.5 Hz).


Transmitted Force (X-Direction)
T1 Constraint Force (Node 1) vs Set Value
T1 Constraint Force (Node 6) vs Set Value
T1 MultiPoint Force (Node 31) vs Set Value
T1 MultiPoint Force (Node 34) vs Set Value
T1 Constraint Force (Node 4) vs Set Value
Force [kN]
Frequency [Hz]



The calculated transmitted horizontal loads (transvers) are negligible.


Transmitted Force (Z-Direction)
T3 Constraint Force (Node 1) vs Set Value
T3 Constraint Force (Node 6) vs Set Value
T3 MultiPoint Force (Node 31) vs Set Value
T3 MultiPoint Force (Node 34) vs Set Value
T3 Constraint Force (Node 4) vs Set Value
Force [kN]
Frequency [Hz]

1.10 LITERATURE

The present calculation refers to the following literature:

(1) GERB, Vibration Isolation Systems, 11th Edition 2002
(2) DIN EN 1992-1-1, Eurocode 2: Design of concrete structures, January 2011
(3) DIN 4024 - part 1, Machine foundations; elastic supporting constructions for machines with rotating masses, April 1988

1.11 CALCULATION FUNDAMENTALS AND PROGRAMS

The Dynamic and Structural Analysis of the foundation is performed by the program system 'NX Nastran' from Siemens Product Lifecycle Management Software Inc.

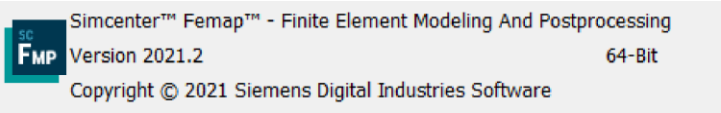
To idealize the structure plate elements with five d.o.f. per node and beam elements with six d.o.f are used.

The calculation model is created using 'FEMAP', a finite-element modelling and postprocessing system by Siemens PLM Software GmbH.

The determination of the eigenvalues and eigenvectors in the Dynamic Analysis is made using the 'LANCZOS Modal Extraction'. The LANCZOS procedure was first described by C. Lanczos (1950) with error analysis added by J. Wilkinson (1965). It was successfully demonstrated in large finite element programs by I. Ojalvo and M. Newman (1967 - 1970). The description of the method can be found in NASA document CR-2731 (1976).

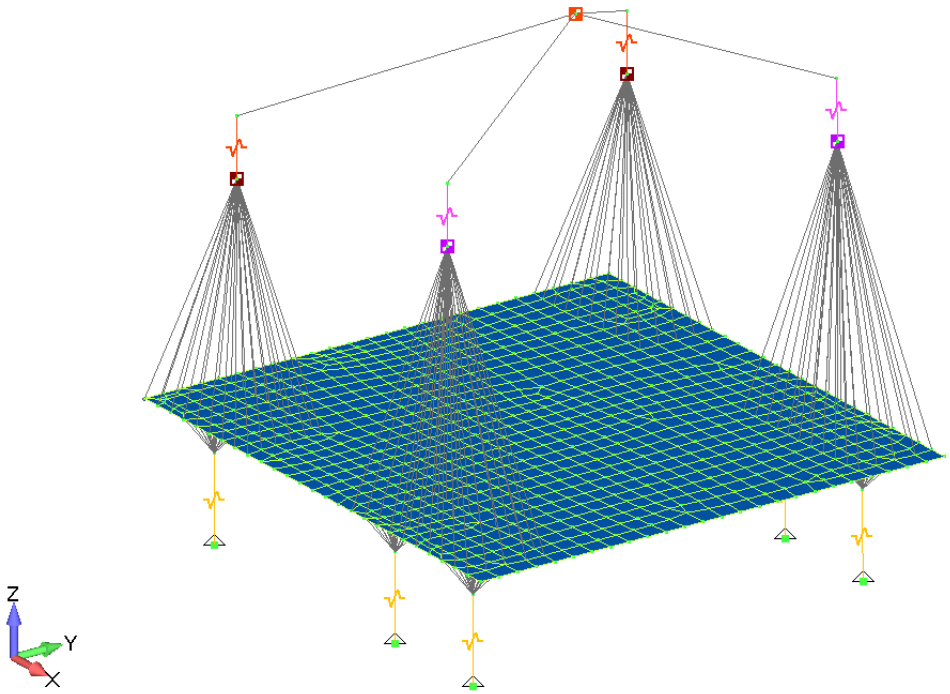
The nodes of the structural model are determined by supports, loading points and cross section cuts.

Software vendor:

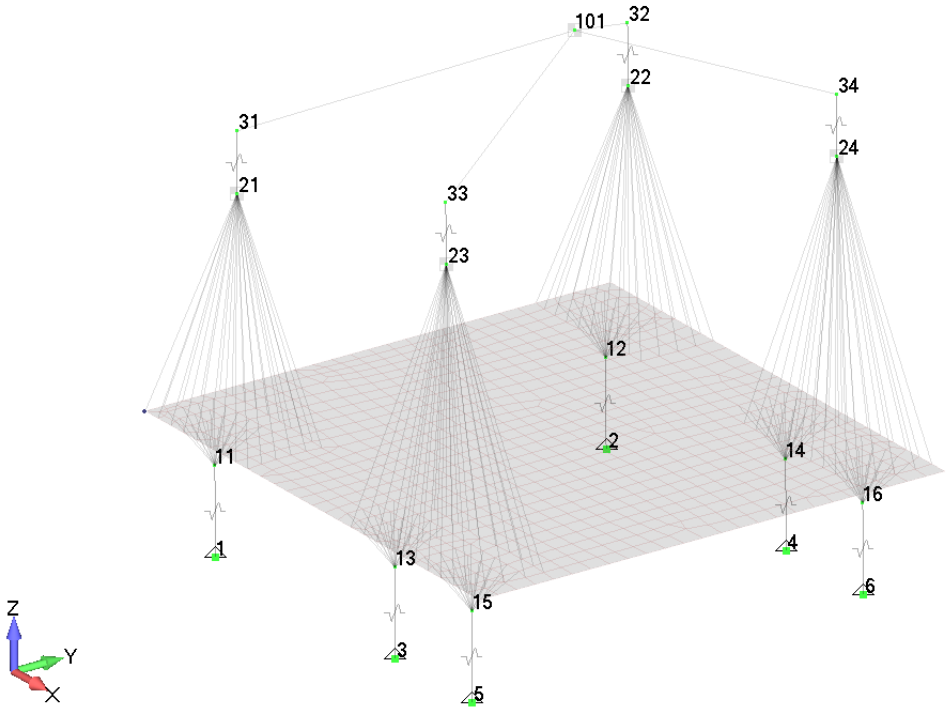


1.12 CALCULATION MODELS

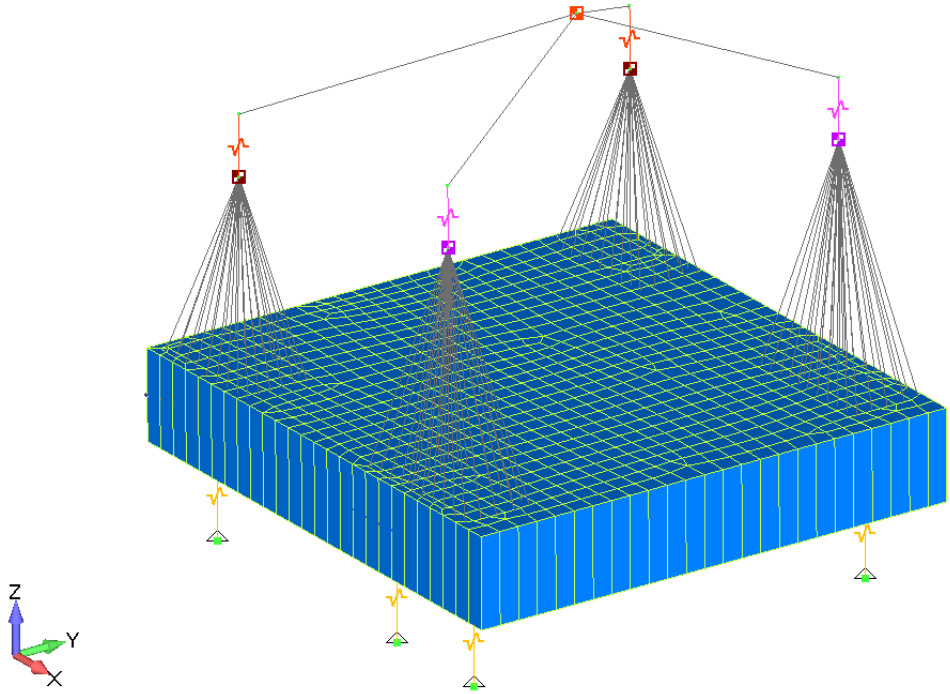
Calculation Model for Static and Dynamic Analysis



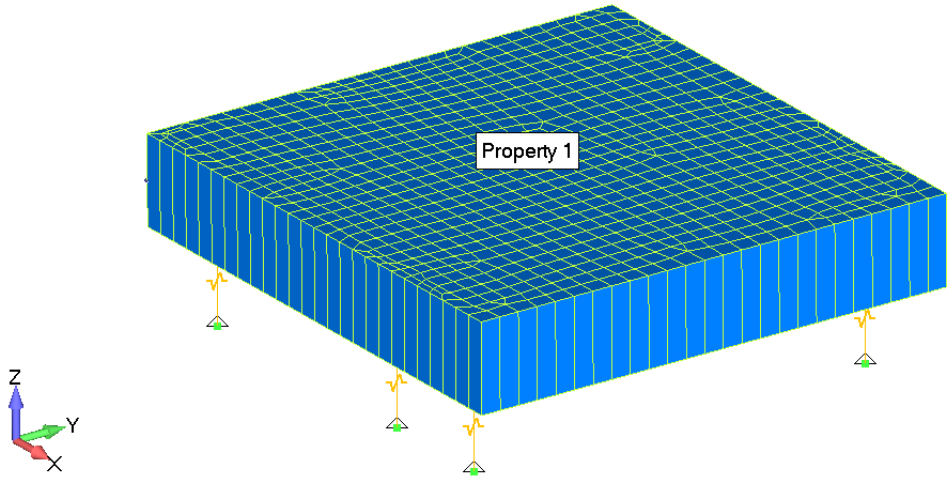
Node Numbers (selected Nodes)



Cross Section View



Element Properties



1.13 DETERMINATION OF CROSS SECTION PROPERTIES

The different parts of the structure are comprehended in several property groups.

The thickness of the plate elements is represented by property group 1

```
1 Property(s) Selected...
Property 1 - D=450
  Type PLATE           Color 110   Layer 1          Material 1          #Elem 847
 Bending Matl 0            TrShear Matl 0            CouplingMatl 0
    Thickness 0.45
    Top Fiber    - Bottom Surf OFF 0.
    Bottom Fiber - Top Surf    OFF 0.
 NS Mass/Area 0.               12I/T**3 0.              Tshear/T 0.
```

The mass of the concrete structure will be automatically calculated by the program taking plate element area, plate element thickness and specific weight into account.

To connect the machine masses, shaft and bearing points and the spring elements to the concrete structure so-called rigid links (RBE 3) are used to describe their correct height and position.

2 DYNAMIC ANALYSIS

2.1 FOUNDATION AND MACHINE MASSES

a) mass of foundation

The mass of the foundation is automatically calculated by 'NX NASTRAN' taking cross section area, beam element length or plate element area, plate thickness respectively and specific mass of material into account.

b) machine masses

The machine masses are applied as given by the machine manufacturer. The single machine masses are distributed on nodal points of the calculation model as follows:

Machine	Node	Mass [t]
Screen	101	5.90
Stool 1.1	102	0.49
Stool 1.2	103	0.49
Stool 2.1	104	0.70
Stool 2.2	105	0.70
Total	Σ =	8.28

2.2 RESULTS OF DYNAMIC ANALYSIS

2.2.1 General

The eigenvalues of the foundation are calculated up to 69 Hz. There are found 10 frequencies up to this limit.

Table of eigenvalues starting from 0.00 Hz:

n	**f_n** [Hz]
1	1.09
2	1.20
3	2.30
4	2.43
5	2.77
6	3.03
7	5.84
8	6.09
9	6.45
10	68.56

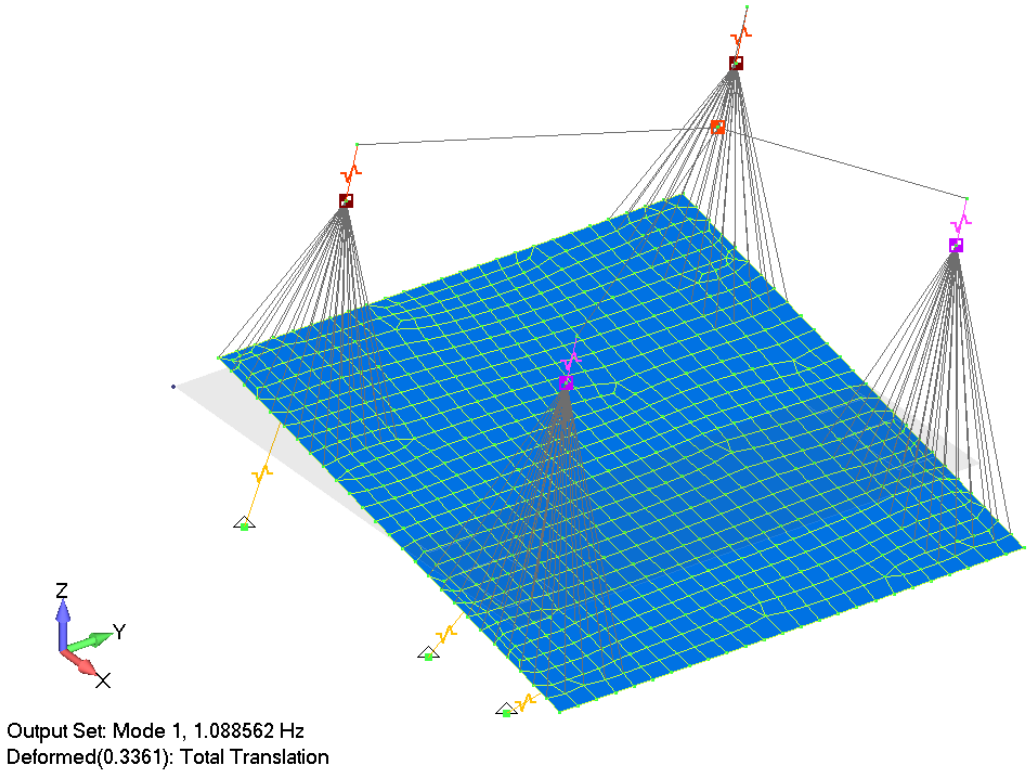
The FEA-program output covers the print of the natural frequencies, the generalized weights, the modal weights as well as the translational degree of freedom with its corresponding node representing the direction of the maximum translation for the single eigenvector.

2.2.2 Comments on the Results

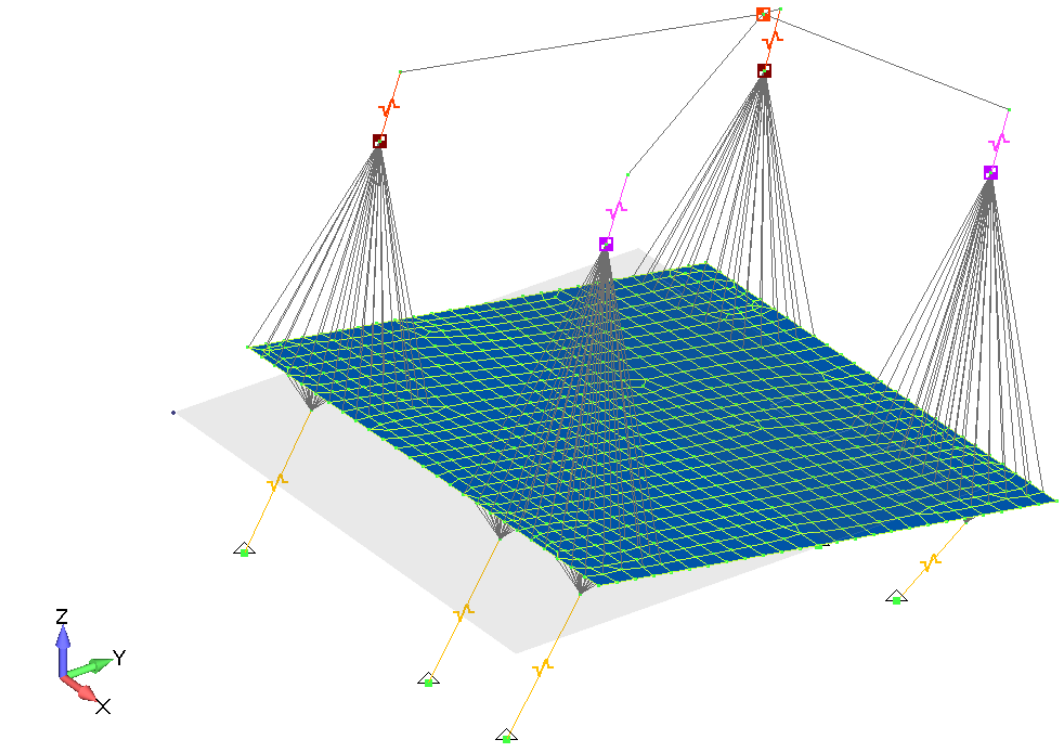
Refer to clause (1.7) of this paper.

2.2.3 Plot of the Mode Shapes

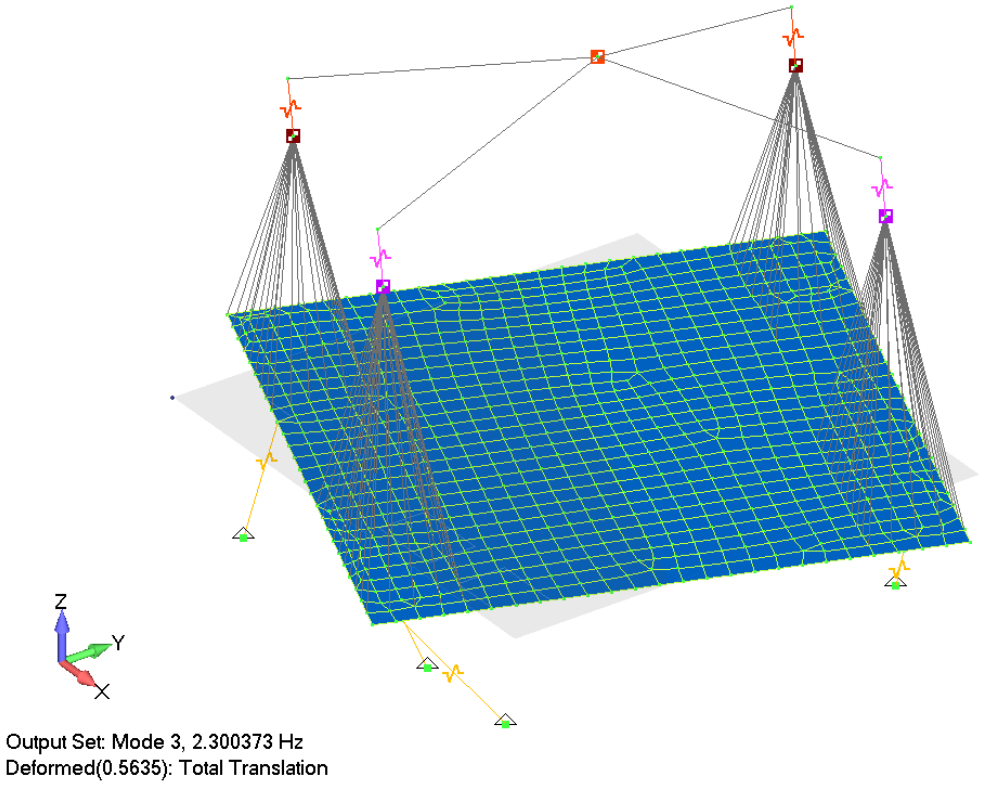
The mode shapes of the most relevant natural frequencies are shown by plots on the following pages.



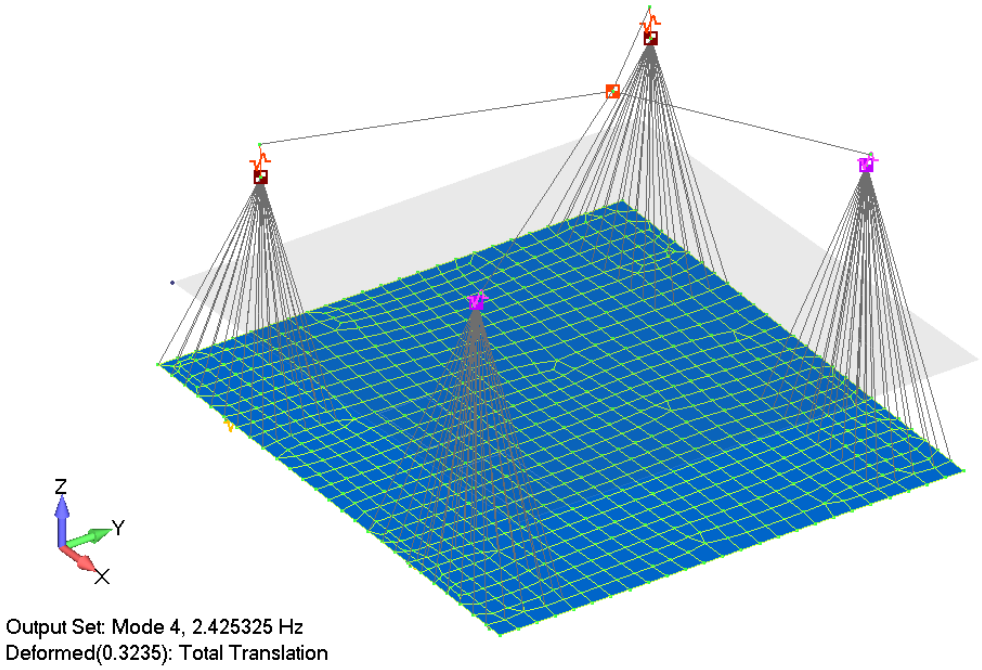
Output Set: Mode 1, 1.088562 Hz
Deformed(0.3361): Total Translation



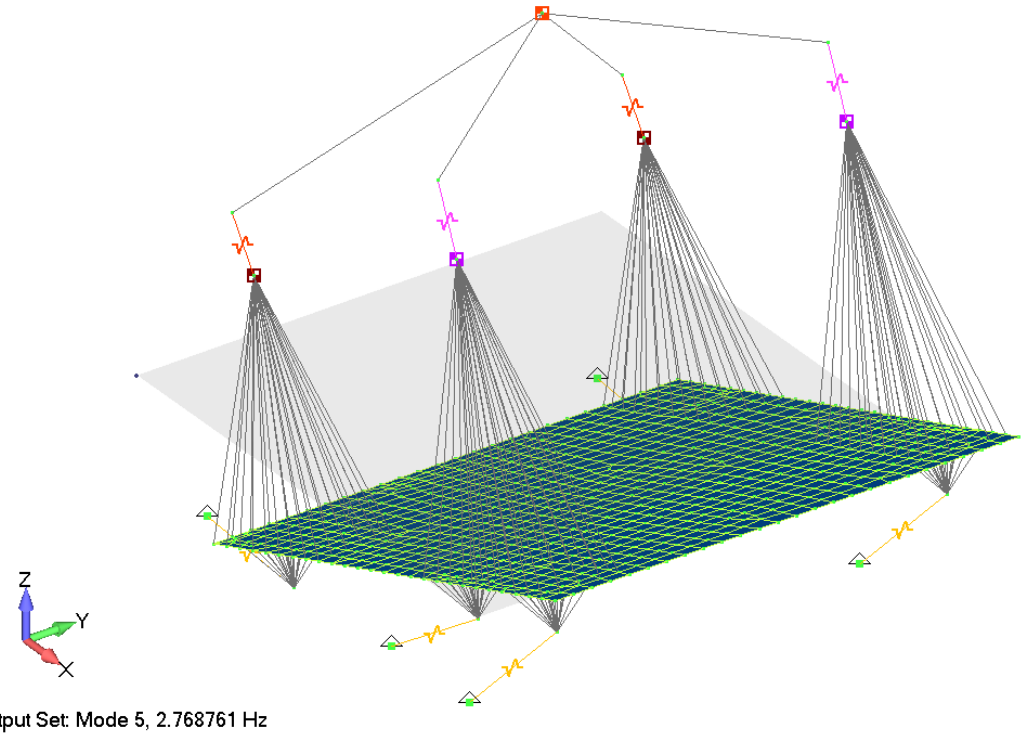
Output Set: Mode 2, 1.200371 Hz
Deformed(0.3277): Total Translation



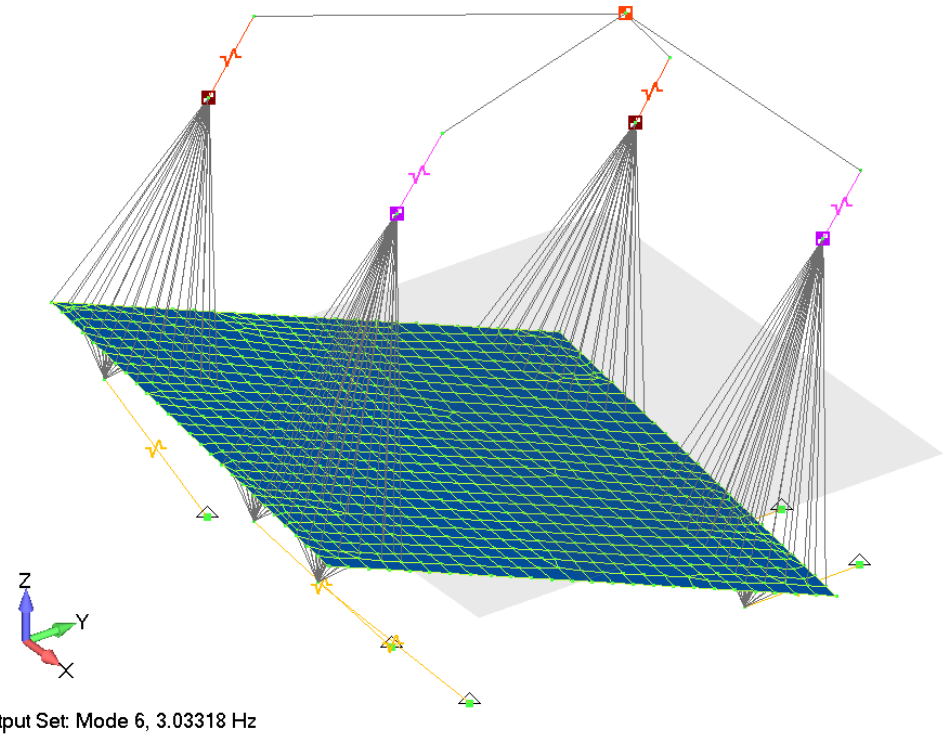
Output Set: Mode 3, 2.300373 Hz
Deformed(0.5635): Total Translation



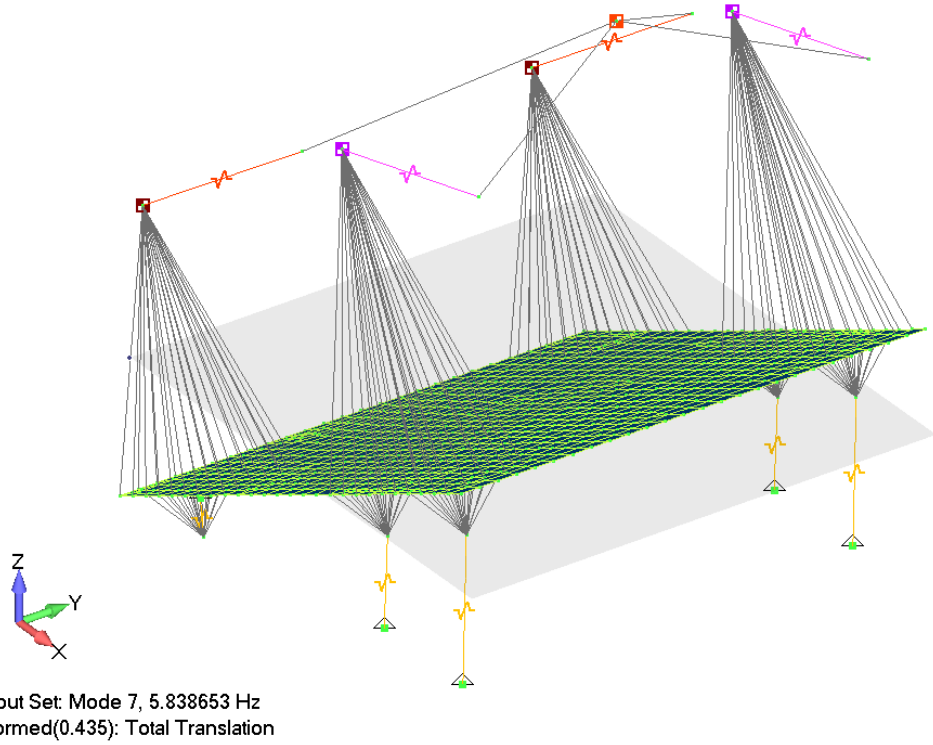
Output Set: Mode 4, 2.425325 Hz
Deformed(0.3235): Total Translation



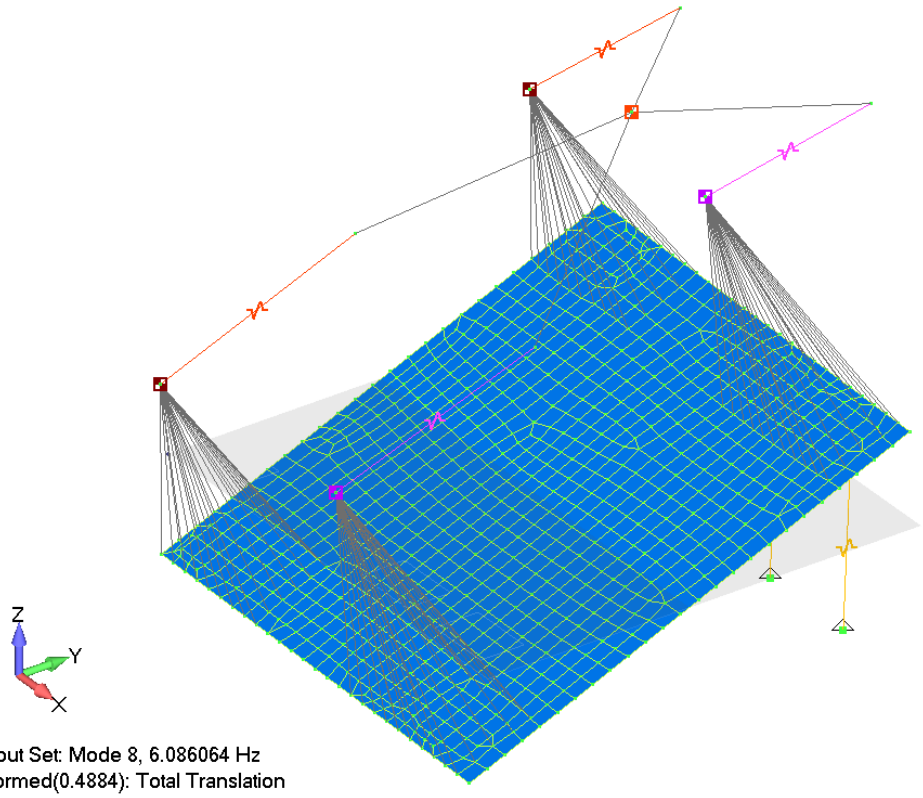
Output Set: Mode 5, 2.768761 Hz
Deformed(0.43): Total Translation



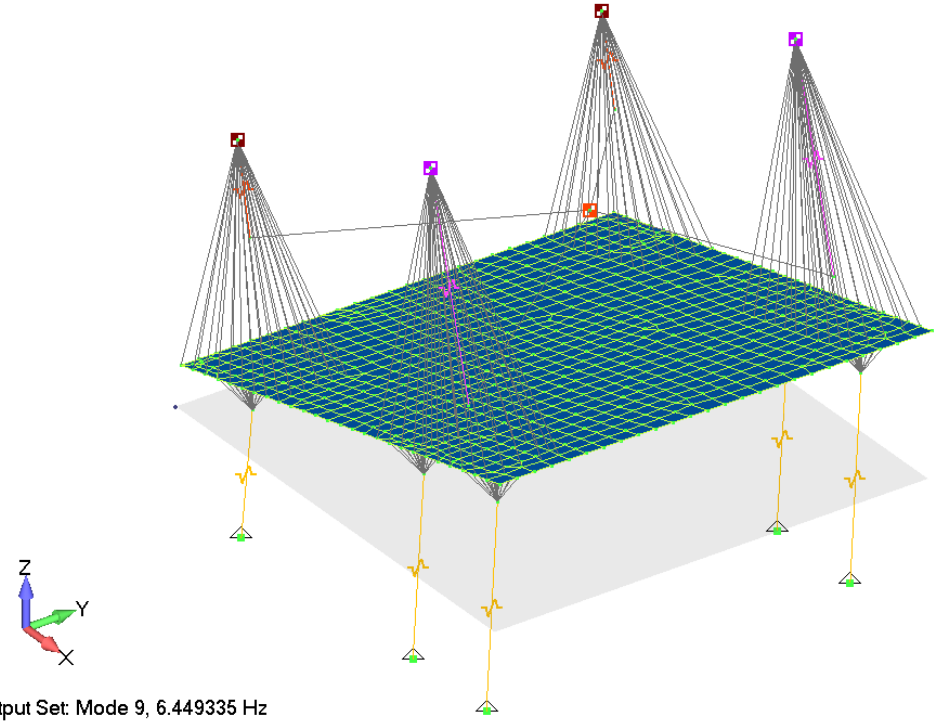
Output Set: Mode 6, 3.03318 Hz
Deformed(0.4487): Total Translation



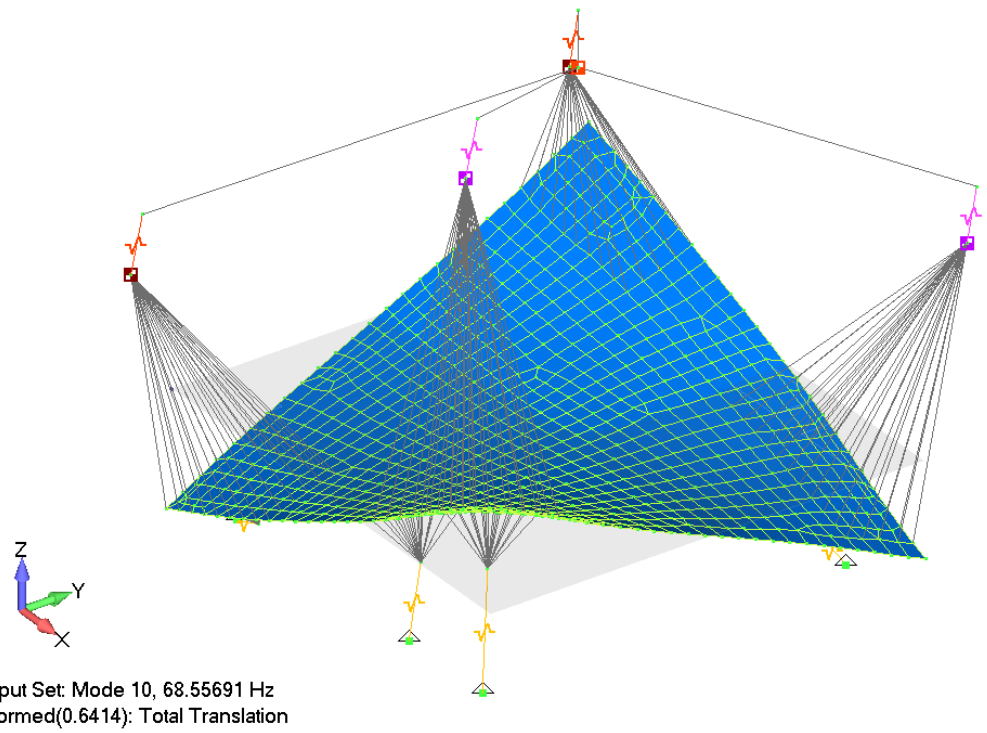
Output Set: Mode 7, 5.838653 Hz
Deformed(0.435): Total Translation



Output Set: Mode 8, 6.086064 Hz
Deformed(0.4884): Total Translation



Output Set: Mode 9, 6.449335 Hz
Deformed(0.3438): Total Translation



Output Set: Mode 10, 68.55691 Hz
Deformed(0.6414): Total Translation

2.3 DETERMINATION OF VIBRATIONAL AMPLITUDES

Theoretical amplitudes are determined due to unbalance caused by the machine at normal operating condition.

The amplitudes are determined using the steady-state technique.

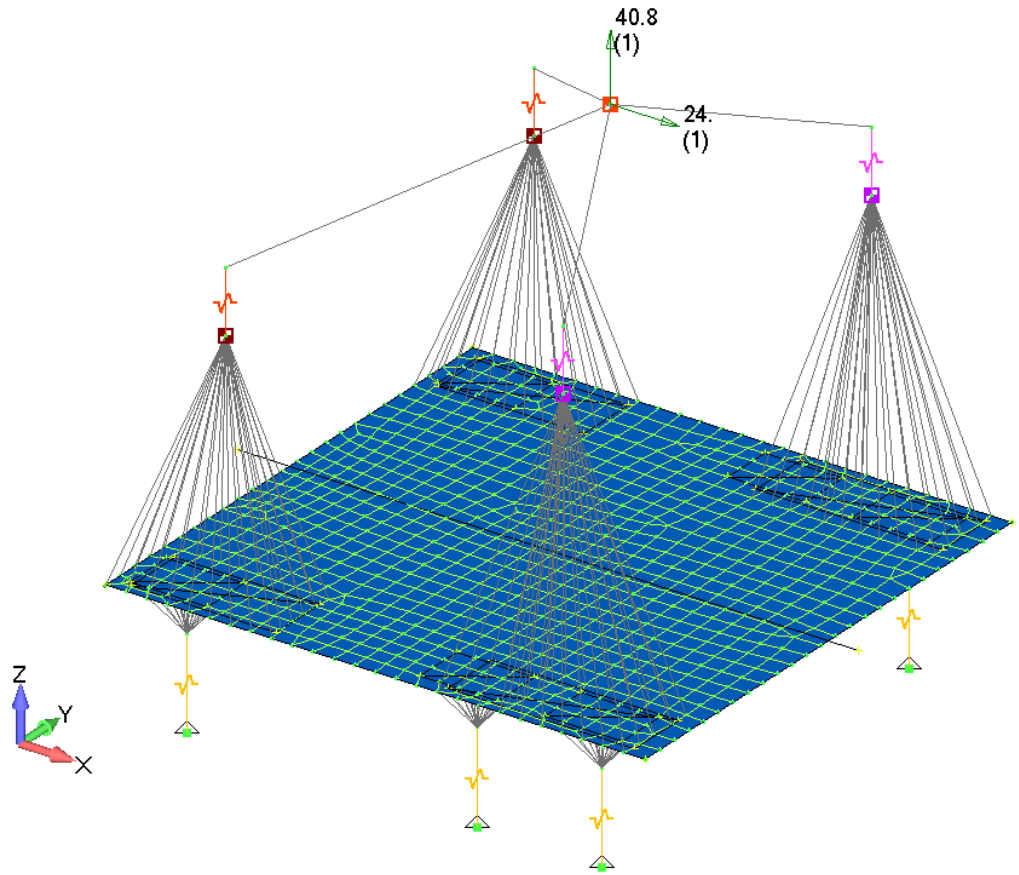
2.3.1 Excitation by Unbalanced Forces

The unbalance load is given by the machine manufacturer

List Load Definitions

```
Load Set 1 - SCREEN
  Load Definition 1 - X-Force 20.4 kN
Nodal Forces (on Node)
        ID  Color      Layer     Def CS       X          Y          Z          Phase
        101    10          9          0        20.4                                0.
                                             1-15 Hz

Load Set 1 - SCREEN
  Load Definition 2 - Z-Force 40.8 kN
Nodal Forces (on Node)
        ID  Color      Layer     Def CS       X          Y          Z          Phase
        101    10          9          0         0.         0.       40.8         90.
                                             1-15 Hz
```



If eigenvalues are located within a range of ± 5% of the operating frequency, acc. to DIN 4024 forced vibrations have to be calculated in resonance for the both adjacent natural frequencies. Due to the speed range of the fan the results due to unbalanced forces are superimposed for 0 Hz to 16.50 Hz (15.00 Hz + 10%).

2.3.2 Frequency Depending Load Response

The excitation at the different bearing points from 0.00 Hz to 16.50 Hz in steps of 0.01 Hz results in the frequency responses of amplitudes at the above mentioned nodal points.

The frequency dependent unbalance force is calculated as follows:

$$U_{per}(f) = e_{per} \cdot \omega^2 \cdot L / g = U_{per}(f_m) \cdot (f/f_m)^2$$

For the plot of the resulting force responses refer to clause (1.9) of this paper.

2.4 STATIC EQUIVALENT LOAD

To cover all dynamic loads and as a conservative assumption, a static equivalent load (Acc. to DIN 4024, part 1, edition April 1988) is calculated.

Determination of Static Equivalent Load acc. to DIN 4024, part 1, cl. 5.4.4 :

Static equivalent load: $F = K / |(1- eta^2)| \leq 15 * K$

where $K = \alpha * L * f_m / 50$

Note: The factor α depends on the balance quality grade of the machine group!

α = 1.2	balance quality grade G 2.5
α = 3.0	balance quality grade G 6.3
α = 7.5	balance quality grade G 16
α = 18.8	balance quality grade G 40

Designation of machine: **Screen**

operating frequency:	f_m =	**15.00**	[Hz]
adjacent natural frequency:	f_n =	**68.56**	[Hz]
rotating weight:	L =	**59.00**	[kN]
factor α:	α =	**18.8**	[-]

eta = f_m / f_n = **0.219**

K = **332.76** [kN]

min. F = **1.05** · K

chosen: F = **1.21** · K

= **404.00** [kN]

3 STATIC ANALYSIS

3.1 GENERAL

The structural analysis of the foundation is performed on the following pages.

For idealisation of the foundation in the structural analysis the calculation model used for the dynamic analysis is adapted concerning loading points, elemental and nodal numbering.

3.2 COMPILATION OF LOAD CASES

The following load cases will be considered:

Load Case	Description
1	Dead Load
2	Machine Load
3	Live Load 5 kN/m²
4	Max. Static Load of Material
5	Static Equiv. Load Vertical [+/-]
6	Static Equiv. Load Vertical Alternating [+/-]
7	Static Equiv. Load Horiz. [+/-]
8	Static Equiv. Load Horiz. Alternating [+/-]

Load cases 1 to 4 represent operating conditions and load cases 5 to 8 consider dynamic effects.

3.3 DESCRIPTION OF THE SINGLE LOAD CASES

3.3.1 Load Case 1 - Dead Load Foundation

The dead weight of the concrete structure will be automatically calculated by program taking cross-sectional area, beam element length, plate element area and thickness as well as specific weight of corresponding material group into account.

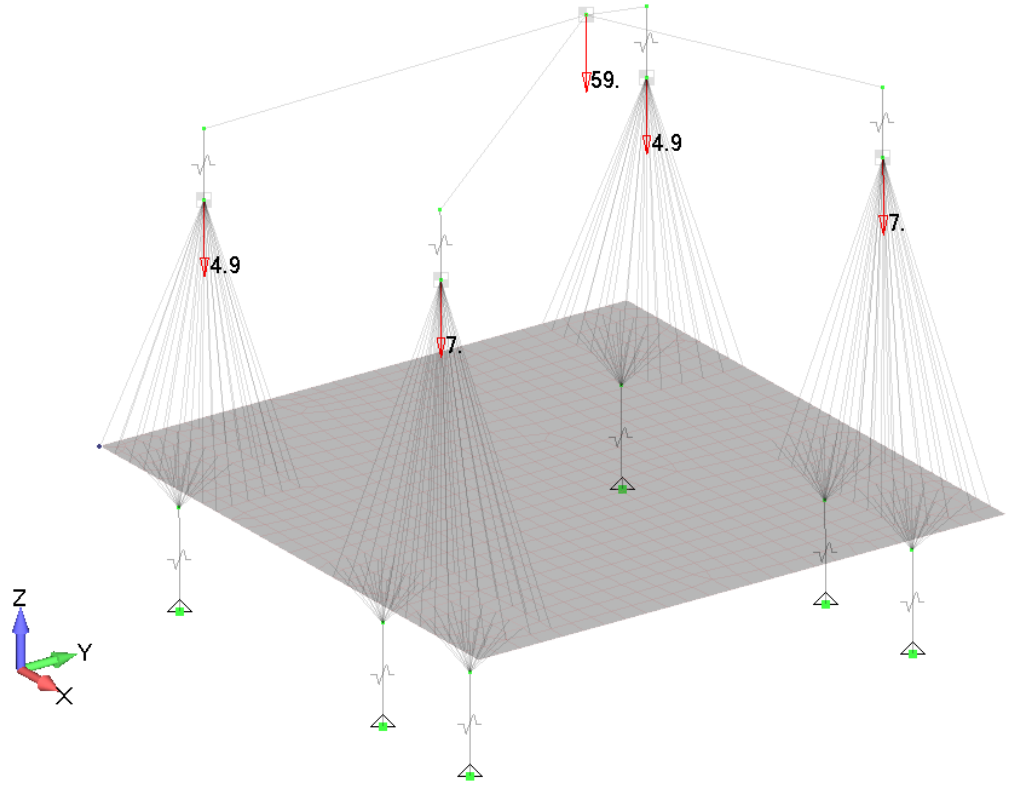
Total weight of concrete: G = -73.1 kN

3.3.2 Load Case 2 – Machine Load

The loads are adopted according to the customer drawings (ref. to cl. 1.3).

Screen		=	-59.0 kN
Stool	= 2 ·(-4.9) + 2 · (-7)	=	-23.8 kN
Total			82.8 kN

They are distributed as single loads acting on nodal points in global z-direction.

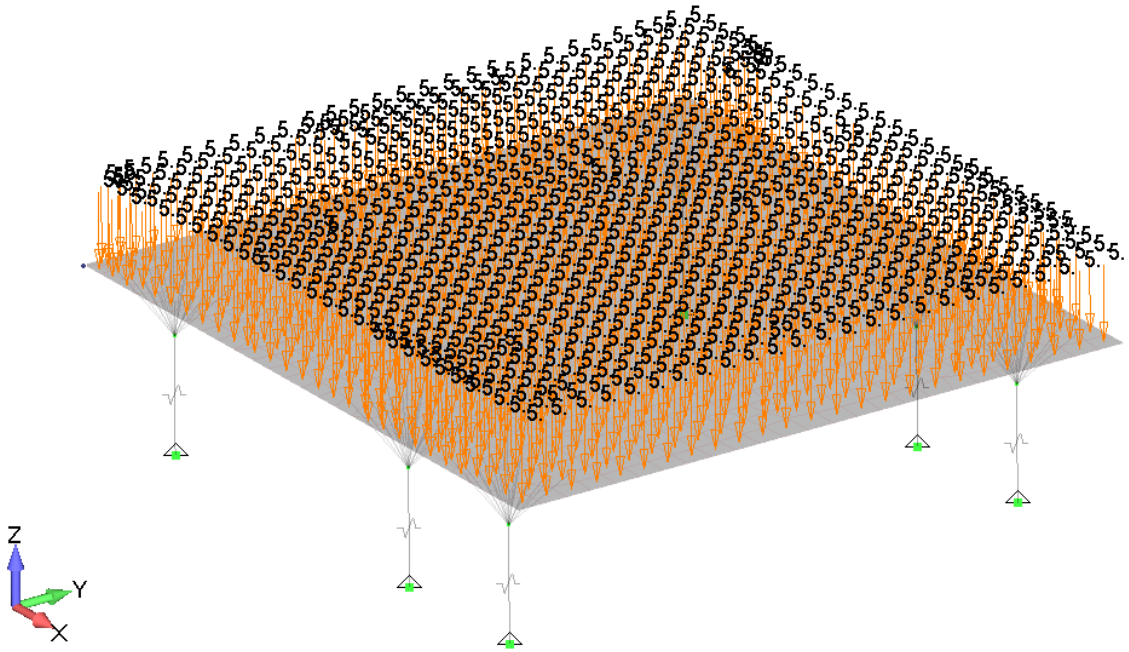


3.3.3 Load Case 3 – Live Load

A live load of p = 5.0 kN/m² is considered acc. to DIN 4024.

load resultant: ΣF_z = -32.5 kN

The load is applied to all plate elements as an uniformly distributed pressure load in negative z-direction.

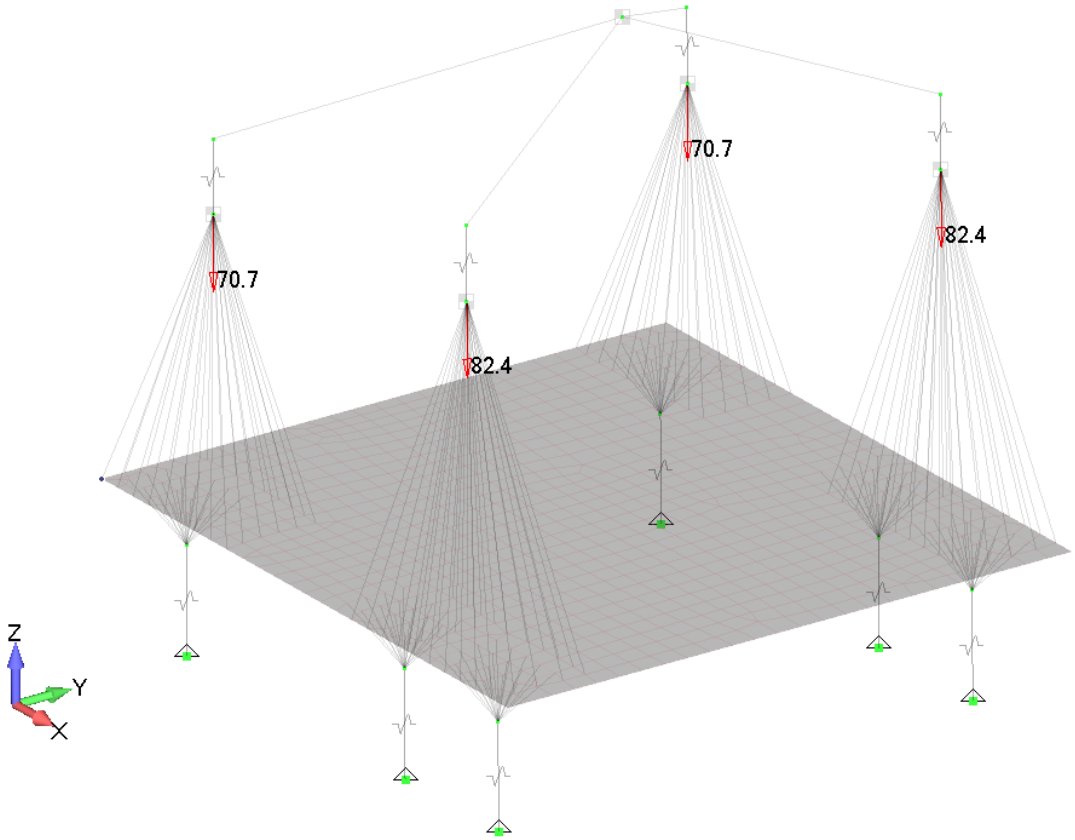


3.3.4 Load Case 4 – Maximum Static Load of Material

The loads are adopted according to the customer drawing: 019045_100_H_V_16510_001 (ref. to cl. 1.3)

Stool A: -82.4 kN
Stool B: -70.7 kN

They are distributed as single loads acting on nodal points in global directions.



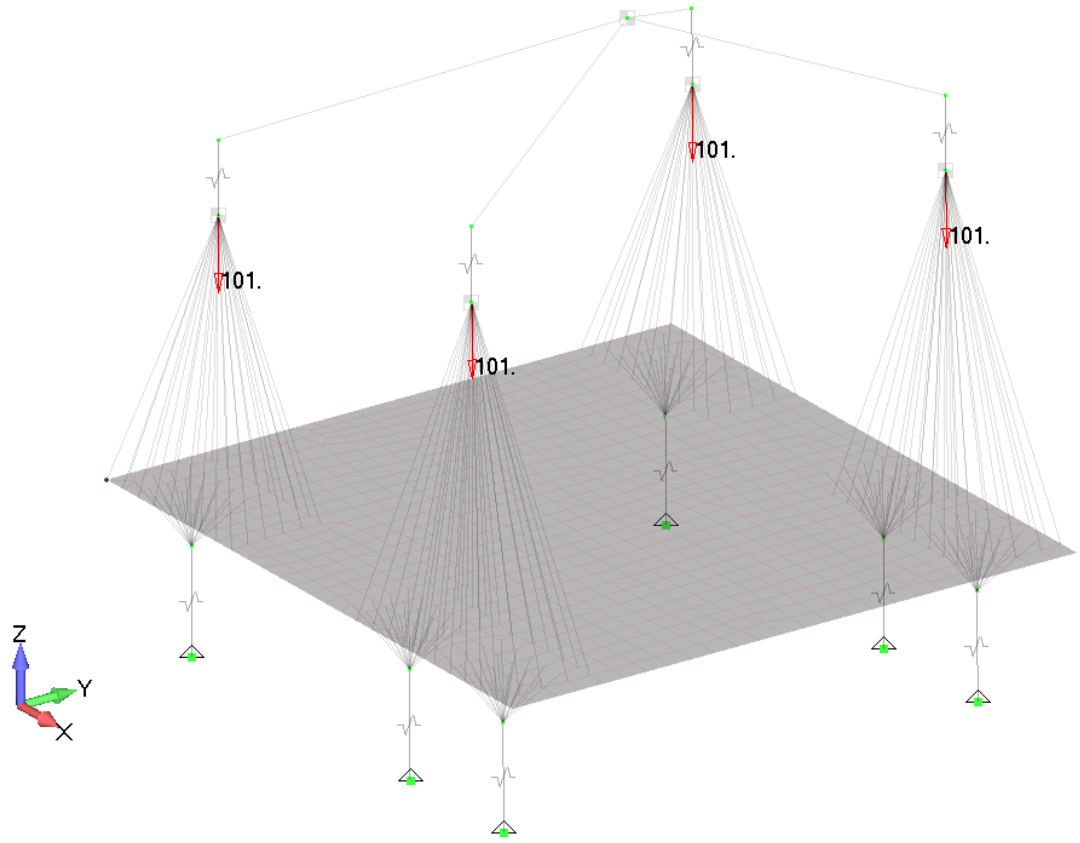
3.3.5 Load Case 5 – Static Equivalent Load / Vertical (±)

For determination of static equivalent loads (acc. to DIN 4024, part 1) refer to section [2.4] of this document.

The load is distributed on nodal points (stool) acting in global z-direction :

$$F_z = 4 \cdot (-101.0\ \text{kN}) = -404.0\ \text{kN}$$

For the determination of design forces the internal forces caused by this load case have to be considered with alternating signs.



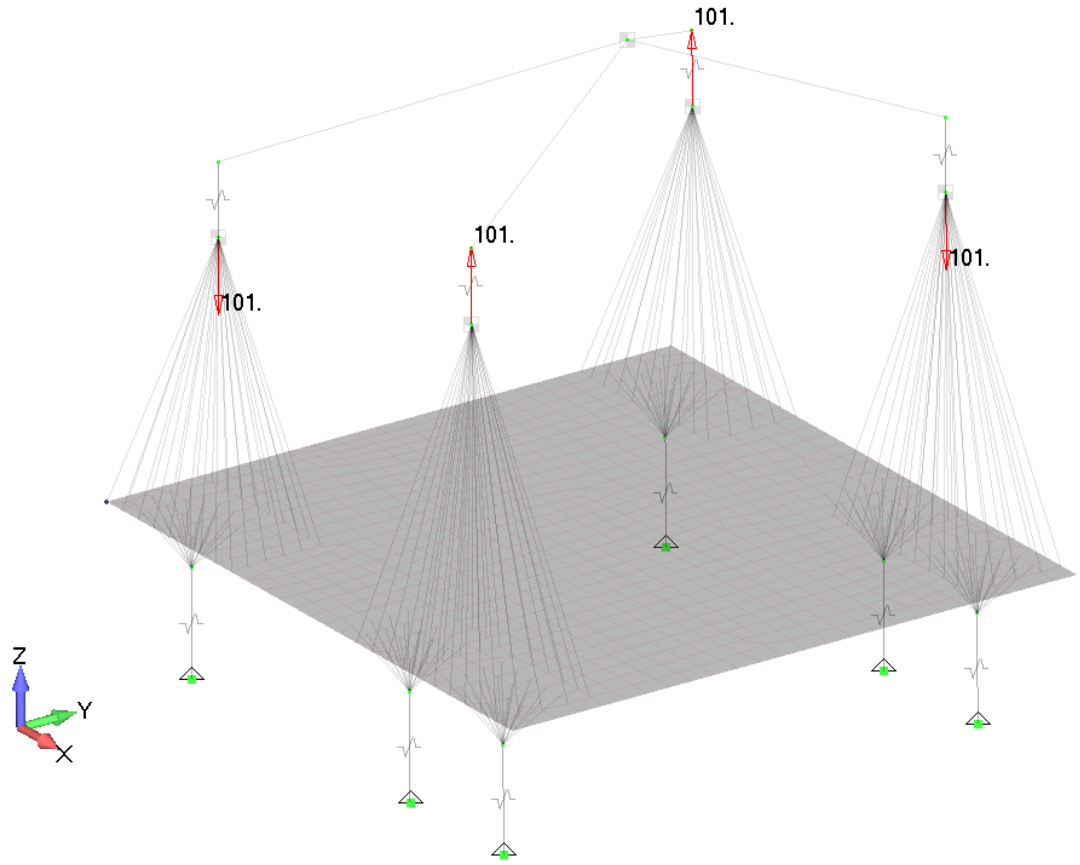
3.3.6 Load Case 6 – Static Equivalent Load / Vertical Alternating (±)

For determination of static equivalent loads (acc. to DIN 4024, part 1) refer to section [2.4] of this document.

The load is distributed on nodal points (stool) acting in global z-direction :

$$F_z = 4 \cdot (\pm 101.0 \text{ kN})$$

For the determination of design forces the internal forces caused by this load case have to be considered with alternating signs.



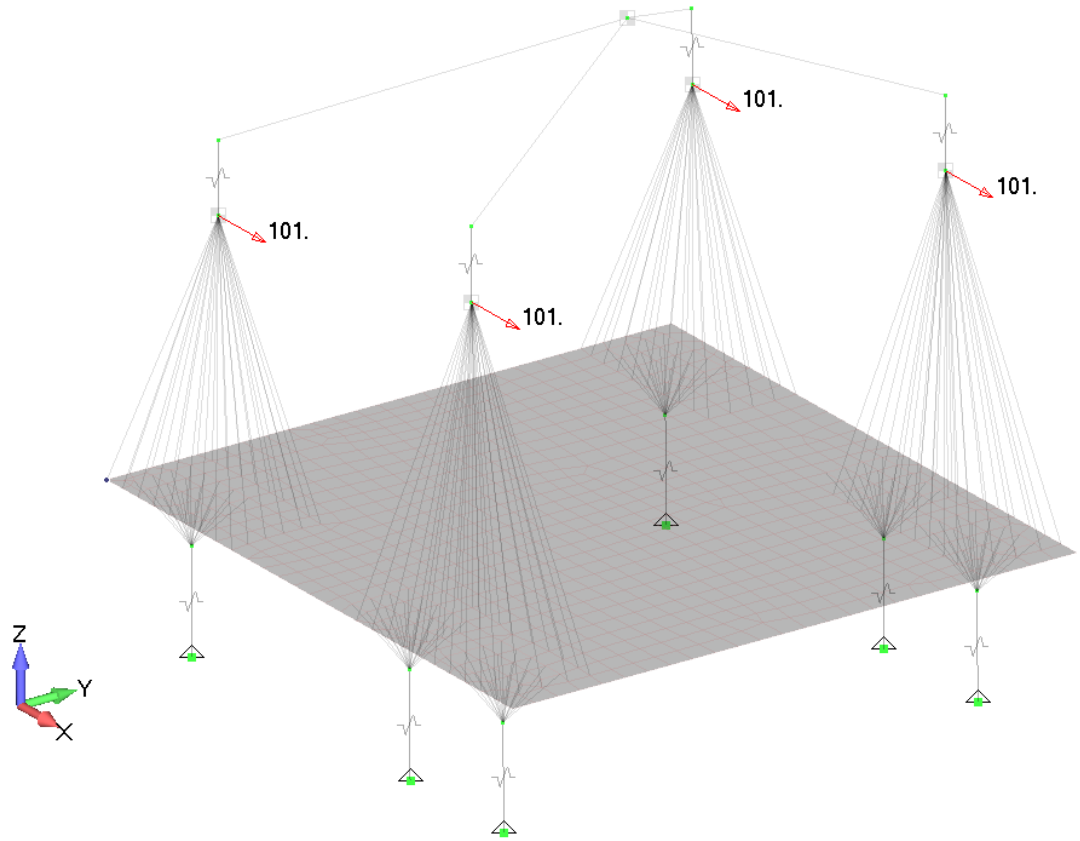
3.3.7 Load Case 8 – Static Equivalent Load / Horizontal (±)

For determination of static equivalent loads (acc. to DIN 4024, part 1) refer to section [2.4] of this document.

The load is distributed on nodal points (stool) acting in global x-direction :

$$F_z = 4 \cdot (-101.0\ \text{kN}) = -404.0\ \text{kN}$$

For the determination of design forces the internal forces caused by this load case have to be considered with alternating signs.



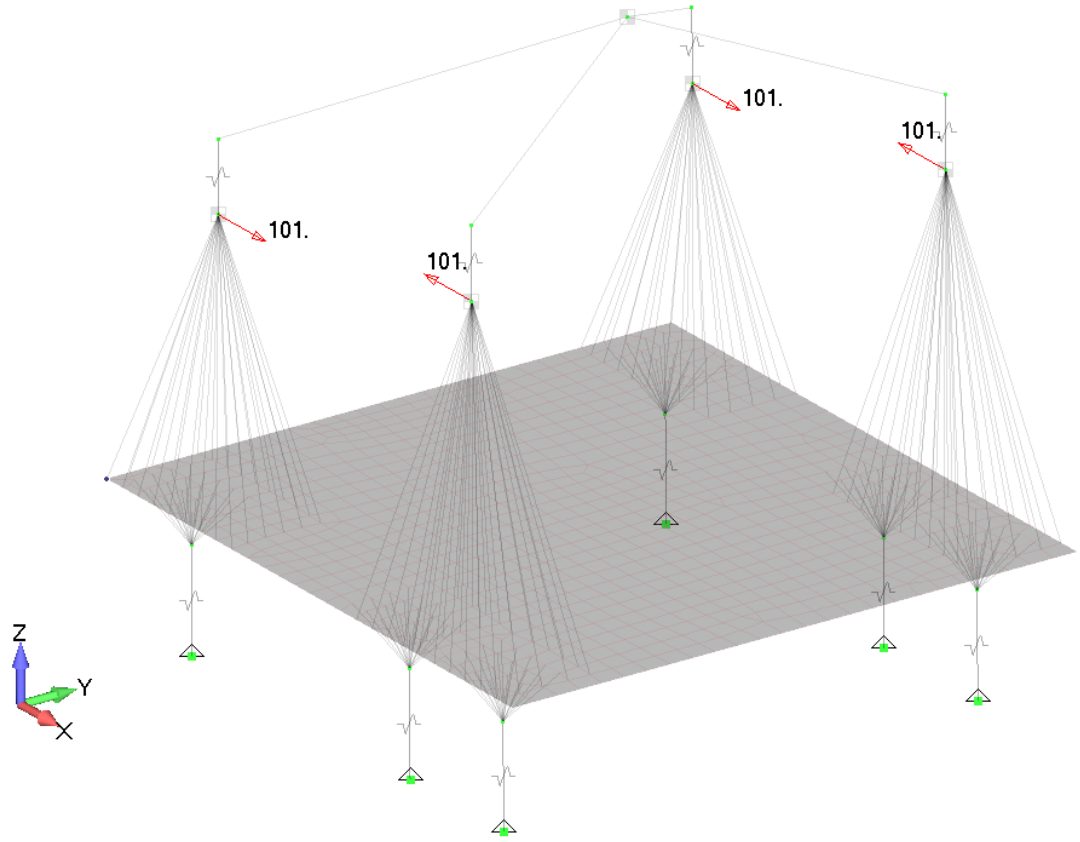
3.3.8 Load Case 9 – Static Equivalent Load / Horizontal Alternating (±)

For determination of static equivalent loads (acc. to DIN 4024, part 1) refer to section [2.4] of this document.

The load is distributed on nodal points (stool) acting in global x-direction :

$$F_z = 4 \cdot (\pm 101.0 \text{ kN})$$

For the determination of design forces the internal forces caused by this load case have to be considered with alternating signs.



3.4 SUPERPOSITION OF LOAD CASES

For the determination of design load cases the single load cases are taken into account acc. to the following superposition schema.
Permanent loads are considered with a safety factor of γ = 1.35.
For operational loads a safety factor of γ = 1.50 has been used. For dynamic effects and emergency load cases a reduced factor of γ = 1.00 can be considered acc. to DIN EN 1992-1-1.

Load Case	Description	Type	γ
1	Dead Load	G	1.35
2	Machine Load	G	1.35
3	Live Load 5 kN/m²	Q	1.50
4	Max. Static Load of Material	Q	1.00
5	Static Equiv. Load Vertical [+/-]	A	1.00
6	Static Equiv. Load Vertical Alternating [+/-]	A	1.00
7	Static Equiv. Load Horiz. [+/-]	A	1.00
8	Static Equiv. Load Horiz. Alternating [+/-]	A	1.00

load case types:
- G permanent load
- Q non-permanent load
- W non-permanent load with alternating signs (±)
- A alternative load with alternating signs (±)

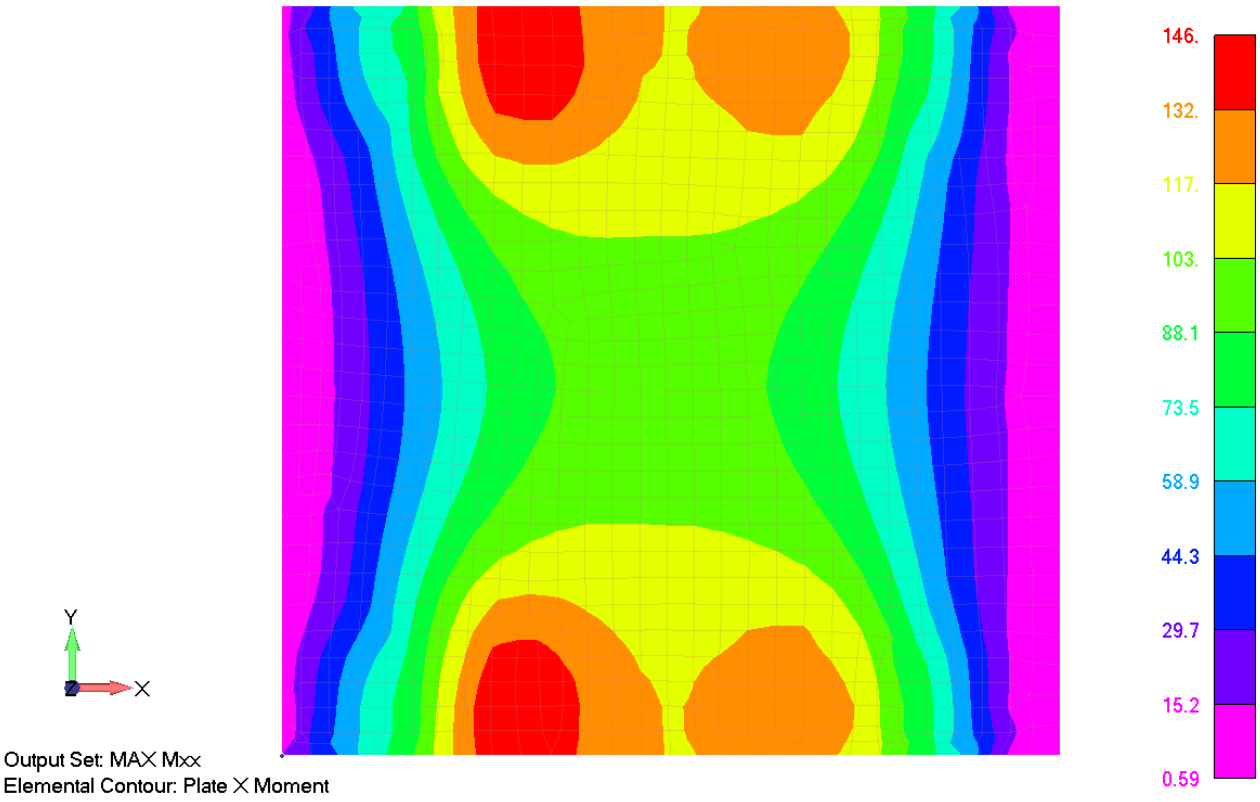
For each plate element of the foundation those load case combinations are determined which lead to the maximum and minimum internal forces and moments. Within the superposition load cases the maximum and minimum forces or moments are stored together with the corresponding values of the remaining forces and moments of each element.

This procedure leads to the following design load cases:

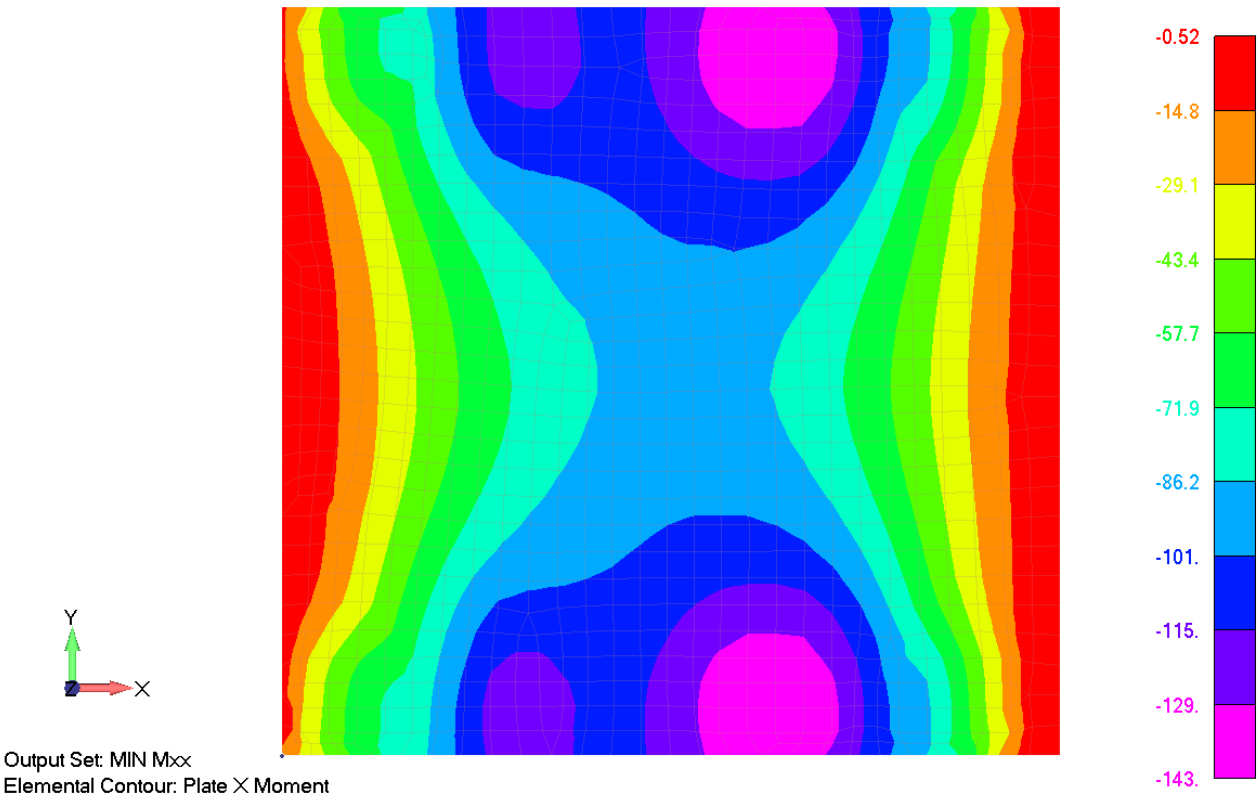
max n_{xx}	min n_{xx}
max n_{yy}	min n_{yy}
max n_{xy}	min n_{xy}
max q_{xz}	min q_{xz}
max m_{xx}	min m_{xx}
max m_{yy}	min m_{yy}
max m_{xy}	min m_{xy}
max q_{yz}	min q_{yz}

The distributions of the calculated maximum and minimum plate moments m_{xx} and m_{yy} are shown exemplarily on the following pages.

plate moments m_{xx}

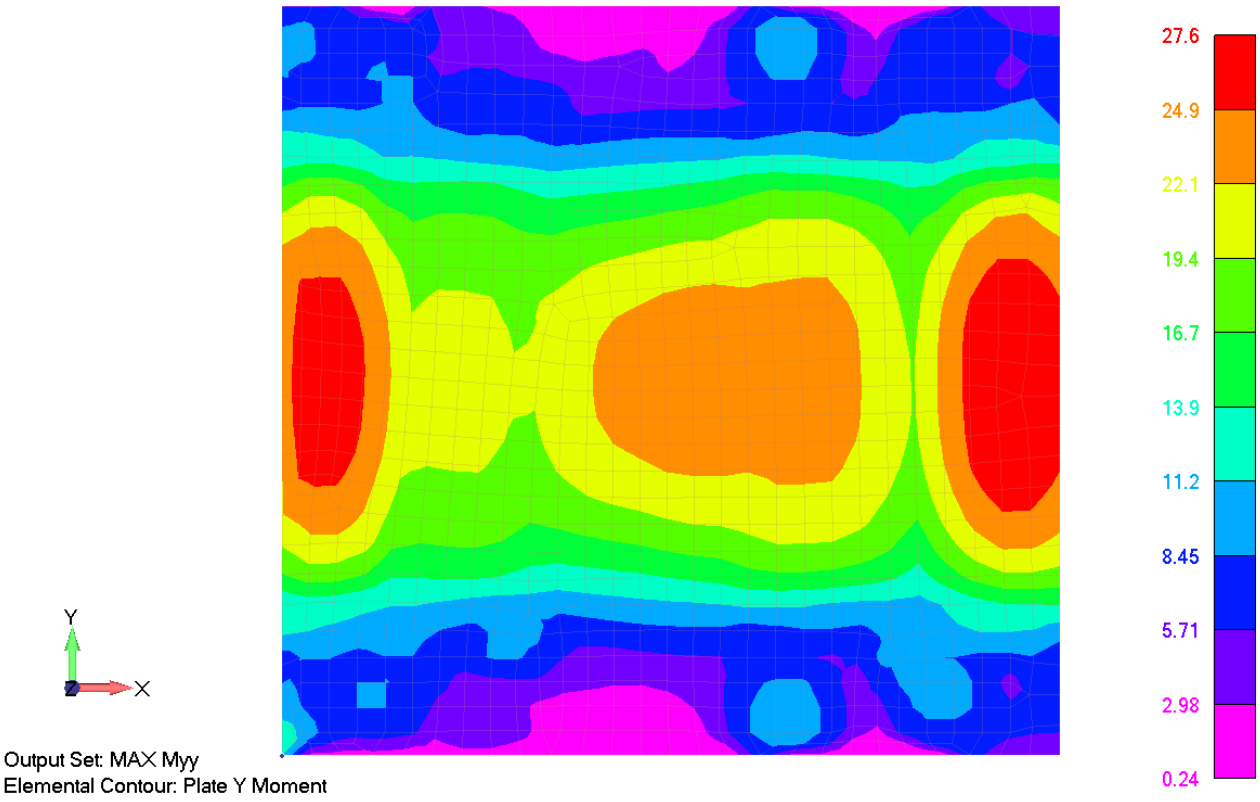


Output Set: MAX Mxx
Elemental Contour: Plate X Moment

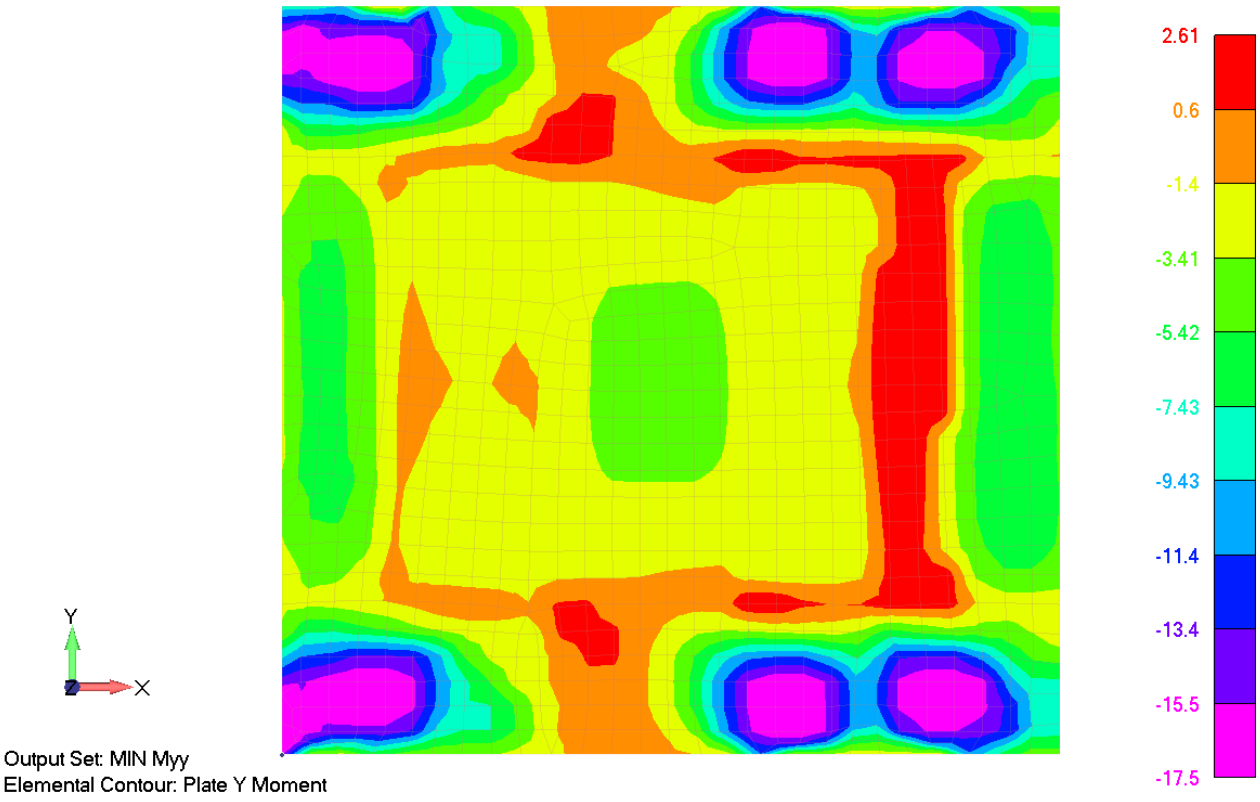


Output Set: MIN Mxx
Elemental Contour: Plate X Moment

plate moments m_{yy}



Output Set: MAX Myy
Elemental Contour: Plate Y Moment

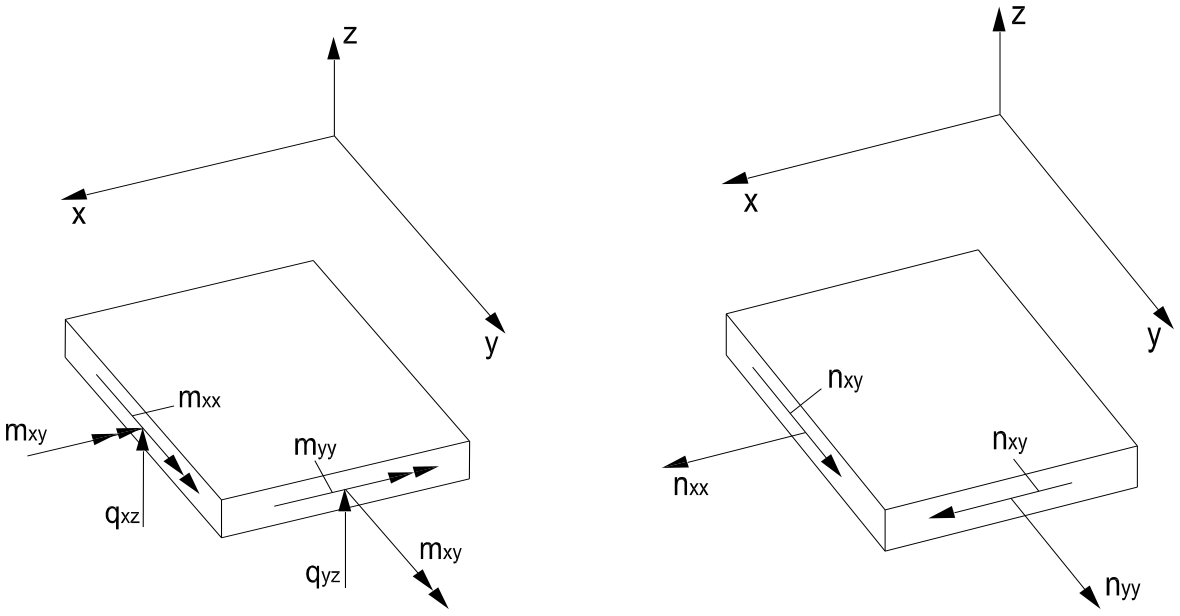


Output Set: MIN Myy
Elemental Contour: Plate Y Moment

3.5 DESIGN FORCES

- The reinforcement design is performed within the local coordinate system of the plate elements.
- The local coordinate systems corresponds to the global coordinates, i.e. the local z-axis is directed upwardly.
- The maximum required reinforcement for the load cases mentioned in clause (3.4) will be determined.
- The sign convention of the internal forces for plate elements is shown below (acc. to DIN 1080).

Sign convention of internal plate forces & moments:



3.6 DESIGN OF REINFORCEMENT

The foundation plate is designed for the maximum values of internal forces, which are determined as described in clause (3.4).

3.6.1 Building Materials

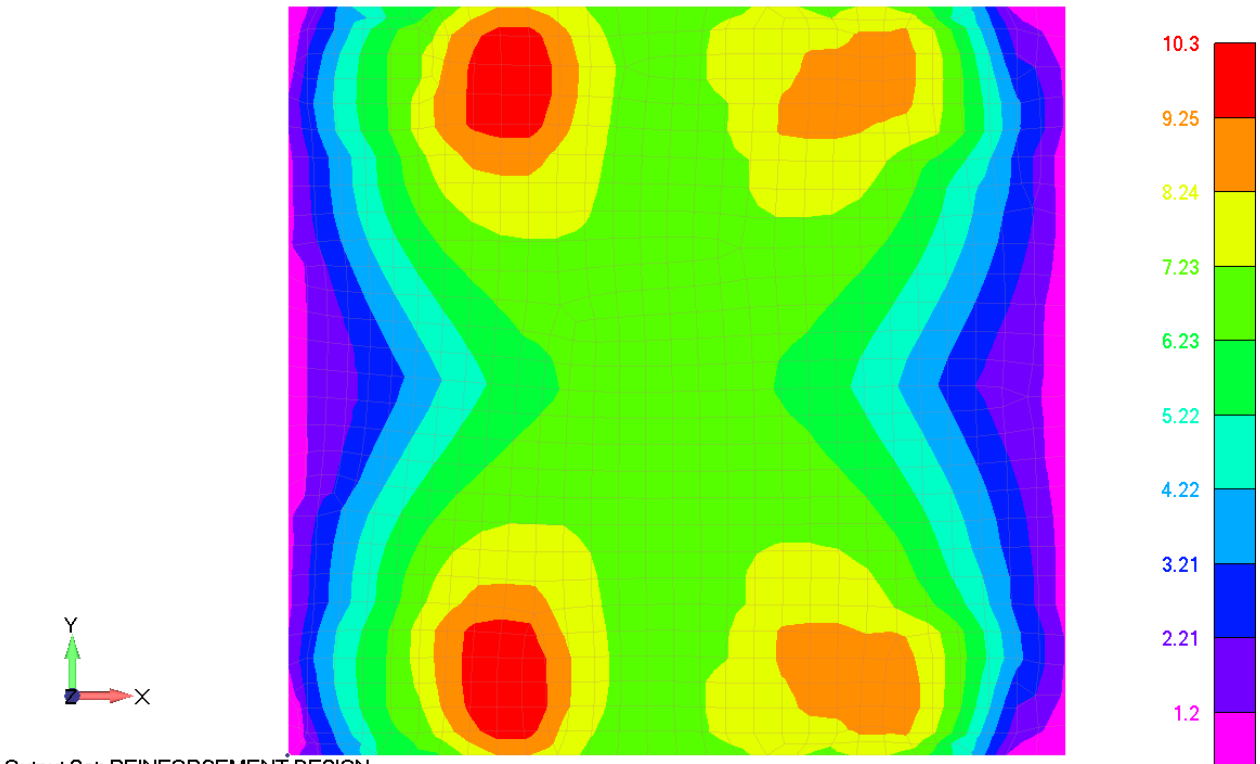
Refer to clause (1.2)

3.6.2 Calculation Method

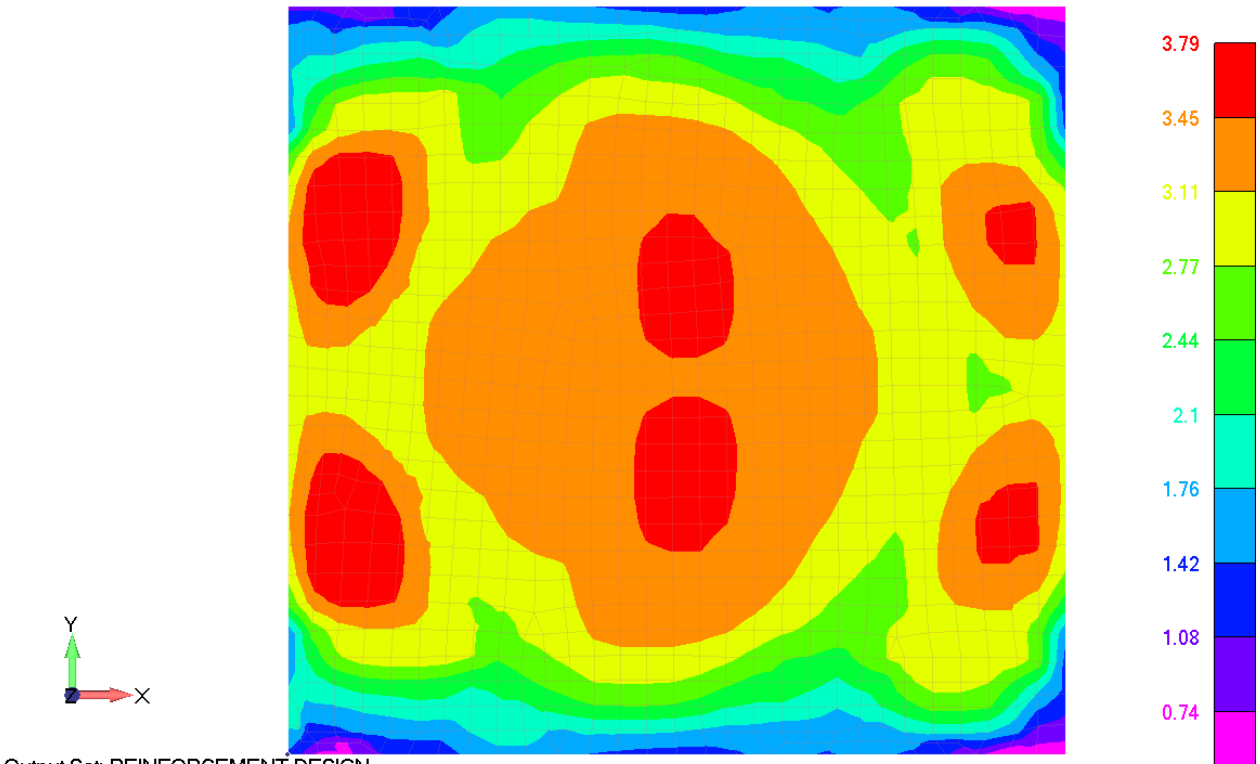
The reinforcement design is performed according to the German standard DIN EN 1992-1-1. The design forces of the different reinforcement layers are calculated acc. to the method of Baumann.

The results of the calculation showing the required reinforcement for each element of the foundation plate are graphically shown by plots on the following pages.

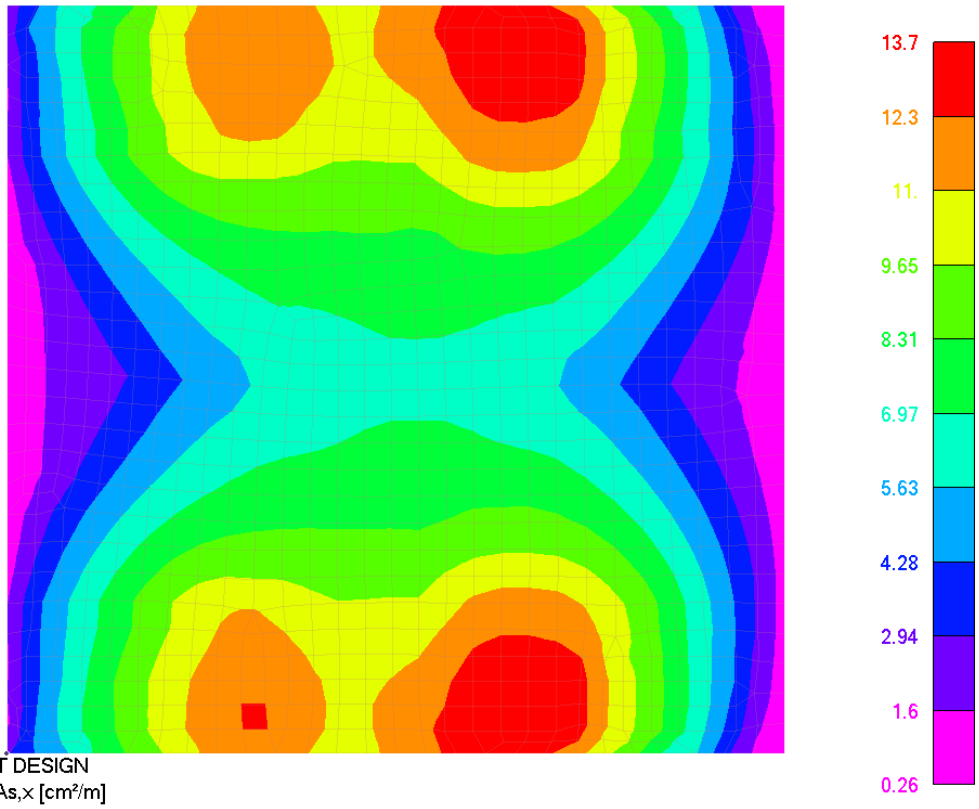
3.6.3 Required Reinforcement



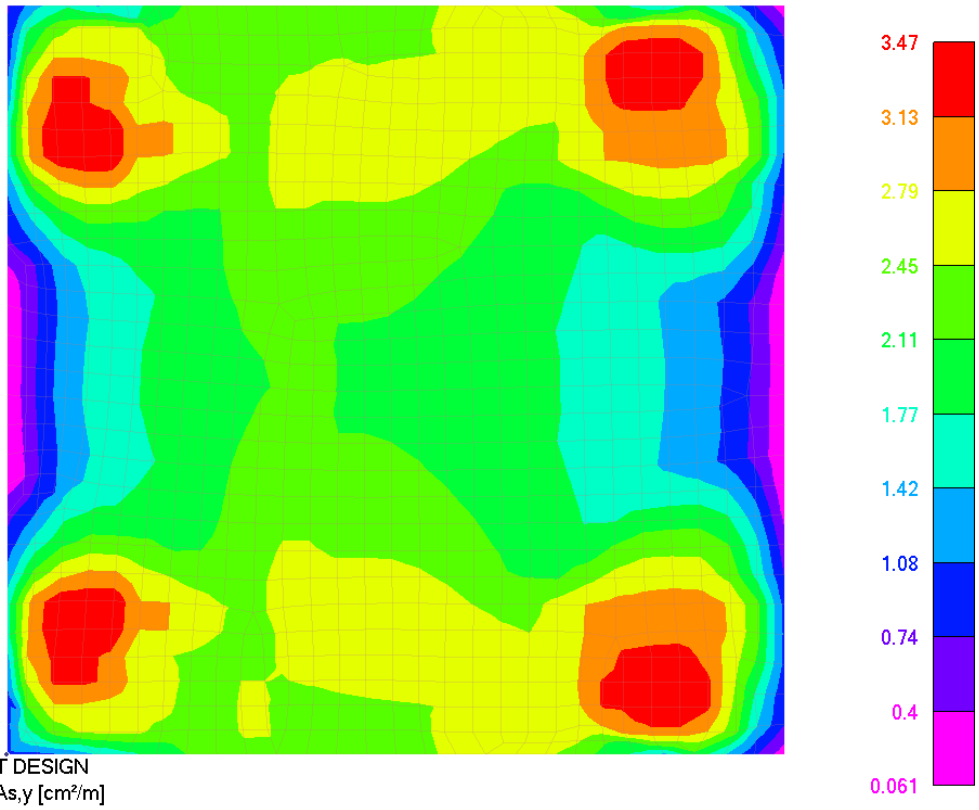
Output Set: REINFORCEMENT DESIGN
Elemental Contour: Bottom layer As,x [cm²/m]



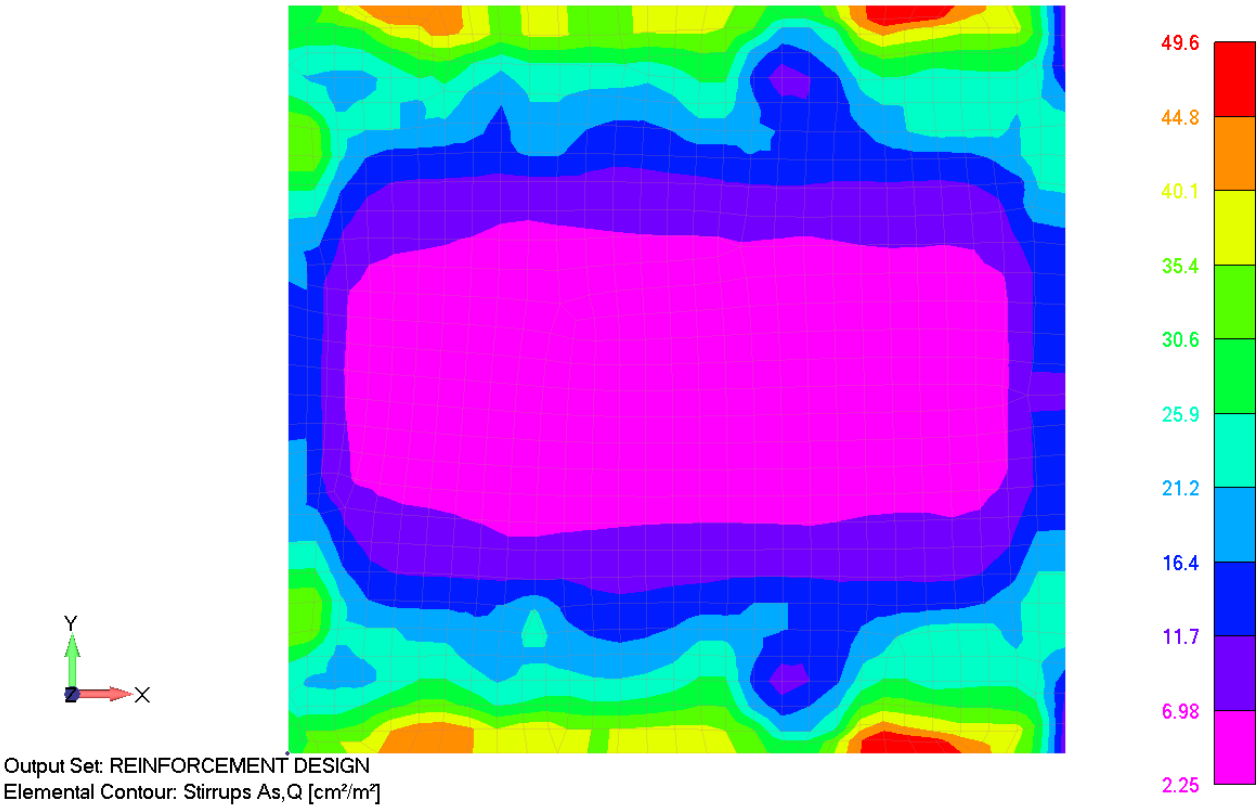
Output Set: REINFORCEMENT DESIGN
Elemental Contour: Bottom layer As,y [cm²/m]



Output Set: REINFORCEMENT DESIGN
Elemental Contour: Top layer $A_{s,x}$ [cm²/m]



Output Set: REINFORCEMENT DESIGN
Elemental Contour: Top layer $A_{s,y}$ [cm²/m]



Output Set: REINFORCEMENT DESIGN
Elemental Contour: Stirrups As,Q [cm²/m²]

3.6.4 Chosen Reinforcement

Min. longitudinal bottom reinforcement	d = 20 mm, e = 15 cm, a_s = 20.9 cm²/m
Min. longitudinal top reinforcement	d = 20 mm, e = 15 cm, a_s = 20.9 cm²/m
Min. transverse bottom reinforcement	d = 16 mm, e = 15 cm, a_s = 13.4 cm²/m
Min. transverse top reinforcement	d = 16 mm, e = 15 cm, a_s = 13.4 cm²/m
Min. crosswise reinforcement on vertical faces	d = 16 mm, e = 15 cm, a_s = 13.4 cm²/m
Min. double shear stirrup reinforcement	d = 16 mm, e = 15 cm, a_s = 26.8 cm²/m

(additional bars where required)

4 SPRING FORCES

4.1 SPRING FORCES AND DEFLECTIONS

Load case 1 (dead load foundation) and load case 2 (machine loads) are taken into account.

Support No.	Nodal Point	Spring Stiffness	LC 1 Dead Load		LC 2 Machine Load		LC 1+2		Load Capacity
		Kv	F_z	u	F_z	u	F_z	u	Fmax
		KN/mm	kN	mm	kN	mm	kN	mm	kN
1	0	0.69	13.8	20.0	11.1	16.1	24.9	36.1	45
2	0	0.69	13.8	20.0	11.1	16.1	24.9	36.1	45
3	0	0.69	11.8	17.1	14.4	20.9	26.2	38.0	45
4	0	0.69	11.8	17.1	14.4	20.9	26.2	38.0	45
5	0	0.69	10.9	15.9	15.9	23.0	26.8	38.9	45
6	0	0.69	10.9	15.9	15.9	23.0	26.8	38.9	45
	Total	**4.1**	**73.1**		**82.8**		**155.9**		**270**

Description of the used abbreviations in the upper table:

k_v = vertical stiffness per spring supported node
F = bearing force
u = static deflections

5 APPENDIX

5.1 FILES OF DYNAMIC ANALYSIS

5.1.1 Input File of Modal Extraction (Extract)

```
INIT MASTER(S)
NASTRAN SYSTEM(442)=-1,SYSTEM(319)=1
ID F2,FEMAP
SOL SEMODES
CEND
  TITLE = EF
  ECHO = NONE
  DISPLACEMENT(PLOT) = ALL
  SPCFORCE(PLOT) = ALL
  ESE(PLOT) = ALL
  METHOD = 1
  SPC = 1
BEGIN BULK
$ ***************************************************************************
$   Written by : Femap
$   Version    : 2021.2.0
$   Translator : Simcenter Nastran
$   From Model : D:\22-1668_Mansourah_SANDWICK_RUE\10\22-1668-10_DYN_EF-1-1.modfem
$   Date       : Wed Aug  3 15:05:32 2022
$   Output To  : D:\22-1668_Mansourah_SANDWICK_RUE\10\
$ ***************************************************************************
$
PARAM,PRGPST,NO
PARAM,POST,-1
PARAM,OGEOM,NO
PARAM,AUTOSPC,NO
PARAM,K6ROT,100.
PARAM,MAXRATIO,1.+7
PARAM,GRDPNT,0
EIGRL           1                       10       0                     MASS
CORD2C          1       0      0.       0.      0.      0.      0.      1.+FEMAPC1
+FEMAPC1      1.      0.      1.
CORD2S          2       0      0.       0.      0.      0.      0.      1.+FEMAPC2
+FEMAPC2      1.      0.      1.
$ Femap Constraint Set 1 : LC1
SPC1            1     123       1
SPC1            1     123       2
SPC1            1     123       3
SPC1            1     123       4
SPC1            1     123       5
SPC1            1     123       6
$ Femap Property 1 : D=450
PSHELL          1       1     .45       1               1              0.
$ Femap Material 1 : C35
MAT1            1   3.2+7              .2     2.5    1.-5      0.
GRID            1       0.2495404      .2   -.665       0
GRID            2       0.2495404     2.3   -.665       0
...
GRID         1891       02.343447.2416667      0.       0
CQUAD4          1       1    1016    1014    1015    1017
CQUAD4          2       1    1017    1012    1013    1016
...
CQUAD4        847       1    1890    1886    1124    1123
CELAS2        848    240.      11       1       1       1      0.
CELAS2        849    240.      11       2       1       2      0.
...
CELAS2        865    690.      16       3       6       3      0.
RBE3          866              11     123      1.     123    1150    1151+
+            1152    1153    1154    1155    1156    1157    1158    1159+
+            1160    1161    1162    1163    1874    1875    1876    1877+
+            1878    1879
RBE3          867              13     123      1.     123    1136    1137+
```

```
+           1138    1139    1140    1141    1142    1143    1144    1145+
+           1146    1147    1148    1149    1880    1881    1882    1883+
+           1884    1885
RBE3         868              15     123      1.     123    1122    1123+
+           1124    1125    1126    1127    1128    1129    1130    1131+
+           1132    1133    1134    1135    1886    1887    1888    1889+
+           1890    1891
RBE3         869              12     123      1.     123    1000    1001+
+           1002    1003    1004    1005    1006    1007    1008    1009+
+           1010    1011    1012    1013    1014    1015    1016    1017+
+           1018    1019
RBE3         870              14     123      1.     123    1178    1179+
+           1180    1181    1182    1183    1184    1185    1186    1187+
+           1188    1189    1190    1191    1862    1863    1864    1865+
+           1866    1867
RBE3         871              16     123      1.     123    1164    1165+
+           1166    1167    1168    1169    1170    1171    1172    1173+
+           1174    1175    1176    1177    1868    1869    1870    1871+
+           1872    1873
RBE3         872              22  123456      1.     123    1101    1100+
+           1099    1098    1097    1096    1095    1094    1093    1092+
+           1091    1090    1089    1384    1617    1616    1393    1394+
+           1395    1195    1255    1357    1354    1349    1233
RBE3         873              24  123456      1.     123    1081    1080+
+           1079    1078    1077    1076    1075    1074    1073    1072+
+           1631    1640    1234    1848    1849    1659    1725    1726+
+           1727    1718    1700    1730    1678    1748    1632    1242+
+           1856    1854    1236
RBE3         874              23  123456      1.     123    1811    1812+
+           1774    1793    1240    1858    1859    1239    1297    1039+
+           1038    1037    1036    1040    1041    1042    1043    1044+
+           1045    1237    1852    1853    1851    1641    1798    1804+
+           1808    1809
RBE3         875              21  123456      1.     123    1118    1119+
+           1120    1121    1020    1021    1022    1023    1024    1025+
+           1026    1027    1028    1251    1602    1399    1580    1583+
+           1584    1561    1572    1213    1273    1315    1314    1214
CELAS2       876    585.      31       1      21       1      0.
CELAS2       877    585.      32       1      22       1      0.
...
CELAS2       887   1548.      34       3      24       3      0.
RBE3         888             101  123456      1.     123      31      32+
+             34      33
CONM2        889     101       0     5.9      0.      0.      0.
CONM2        890      22       0     .49      0.      0.      0.
CONM2        891      21       0     .49      0.      0.      0.
CONM2        892      24       0      .7      0.      0.      0.
CONM2        893      23       0      .7      0.      0.      0.
ENDDATA 244a0429
```

5.1.2 Results of Modal Extraction Analysis

R E A L E I G E N V A L U E S

MODE NO.	EXTRACTION ORDER	EIGENVALUE	RADIANS	CYCLES	GENERALIZED MASS	GENERALIZED STIFFNESS
1	1	4.678059E+01	6.839634E+00	1.088562E+00	1.000000E+00	4.678059E+01
2	2	5.688410E+01	7.542155E+00	1.200371E+00	1.000000E+00	5.688410E+01
3	3	2.089085E+02	1.445367E+01	2.300373E+00	1.000000E+00	2.089085E+02
4	4	2.322201E+02	1.523877E+01	2.425325E+00	1.000000E+00	2.322201E+02
5	5	3.026431E+02	1.739664E+01	2.768761E+00	1.000000E+00	3.026431E+02
6	6	3.632086E+02	1.905803E+01	3.033180E+00	1.000000E+00	3.632086E+02
7	7	1.345814E+03	3.668534E+01	5.838653E+00	1.000000E+00	1.345814E+03
8	8	1.462288E+03	3.823987E+01	6.086064E+00	1.000000E+00	1.462288E+03
9	9	1.642062E+03	4.052237E+01	6.449335E+00	1.000000E+00	1.642062E+03
10	10	1.855505E+05	4.307558E+02	6.855691E+01	1.000000E+00	1.855505E+05

1 EF AUGUST 3, 2022 SIMCENTER NASTRAN 11/ 1/19 PAGE 12

5.2 FILES OF STATIC ANALYSIS

5.2.1 Input File of Static Calculation (Extract)

```
INIT MASTER(S)
NASTRAN SYSTEM(442)=-1,SYSTEM(319)=1
ID F2,FEMAP
SOL SESTATIC
CEND
  TITLE = STAT
  ECHO = NONE
  DISPLACEMENT(PLOT) = ALL
  SPCFORCE(PLOT) = ALL
  OLOAD(PLOT) = ALL
  FORCE(PLOT,CORNER) = ALL
  STRESS(PLOT,CORNER) = ALL
SUBCASE 1
  SUBTITLE = LC1 - DEAD_LOAD
  SPC = 1
  LOAD = 1
SUBCASE 2
  SUBTITLE = LC1 - MACHINE_LOAD
  SPC = 1
  LOAD = 2
SUBCASE 3
  SUBTITLE = LC1 - LIVE_LOAD
  SPC = 1
  LOAD = 3
SUBCASE 4
  SUBTITLE = LC1 - MAX_MATERIAL_LOAD
  SPC = 1
  LOAD = 4
SUBCASE 5
  SUBTITLE = LC1 - ST_EQ_VER_LOAD
  SPC = 1
  LOAD = 5
SUBCASE 6
  SUBTITLE = LC1 - ST_EQ_VER_LOAD_ALTER
  SPC = 1
  LOAD = 6
SUBCASE 7
  SUBTITLE = LC1 - ST_EQ_HOR_LOAD
  SPC = 1
  LOAD = 7
SUBCASE 8
  SUBTITLE = LC1 - ST_EQ_HOR_LOAD_ALTER
  SPC = 1
  LOAD = 8
BEGIN BULK
$ ************************************************************************
$   Written by : Femap
$   Version    : 2021.2.0
$   Translator : Simcenter Nastran
$   From Model : D:\22-1668_Mansourah_SANDWICK_RUE\10\22-1668-10_STAT-1-1.modfem
$   Date       : Wed Aug  3 16:48:51 2022
$   Output To  : D:\22-1668_Mansourah_SANDWICK_RUE\10\
$ ************************************************************************
$
PARAM,PRGPST,NO
PARAM,POST,-1
PARAM,OGEOM,NO
PARAM,AUTOSPC,NO
PARAM,K6ROT,100.
PARAM,MAXRATIO,1.+7
PARAM,GRDPNT,0
CORD2C          1       0      0.      0.      0.      0.      0.      1.+FEMAPC1
+FEMAPC1      1.      0.      1.
CORD2S          2       0      0.      0.      0.      0.      0.      1.+FEMAPC2
+FEMAPC2      1.      0.      1.
$ Femap Load Set 1 : DEAD_LOAD
FORCE           9     101       0      1.      0.      0.     59.
FORCE           9      22       0      1.      0.      0.     4.9
FORCE           9      21       0      1.      0.      0.     4.9
```

```
FORCE           9      24       0      1.      0.      0.      7.
FORCE           9      23       0      1.      0.      0.      7.
GRAV           10       0      1.      0.      0.    -10.
LOAD            1      1.      1.       9      1.      10
$ Femap Load Set 2 : MACHINE_LOAD
FORCE           2     101       0      1.      0.      0.    -59.
FORCE           2      22       0      1.      0.      0.    -4.9
FORCE           2      21       0      1.      0.      0.    -4.9
FORCE           2      24       0      1.      0.      0.     -7.
FORCE           2      23       0      1.      0.      0.     -7.
$ Femap Load Set 3 : LIVE_LOAD
PLOAD4          3       1     -5.
PLOAD4          3       2     -5.
...
PLOAD4          3     847     -5.
$ Femap Load Set 4 : MAX_MATERIAL_LOAD
FORCE           4      22       0      1.      0.      0.   -70.7
FORCE           4      21       0      1.      0.      0.   -70.7
FORCE           4      24       0      1.      0.      0.   -82.4
FORCE           4      23       0      1.      0.      0.   -82.4
$ Femap Load Set 5 : ST_EQ_VER_LOAD
FORCE           5      22       0      1.      0.      0.   -101.
FORCE           5      21       0      1.      0.      0.   -101.
FORCE           5      24       0      1.      0.      0.   -101.
FORCE           5      23       0      1.      0.      0.   -101.
$ Femap Load Set 6 : ST_EQ_VER_LOAD_ALTER
FORCE           6      22       0      1.      0.      0.    101.
FORCE           6      21       0      1.      0.      0.   -101.
FORCE           6      24       0      1.      0.      0.   -101.
FORCE           6      23       0      1.      0.      0.    101.
$ Femap Load Set 7 : ST_EQ_HOR_LOAD
FORCE           7      22       0      1.    101.      0.      0.
FORCE           7      21       0      1.    101.      0.      0.
FORCE           7      24       0      1.    101.      0.      0.
FORCE           7      23       0      1.    101.      0.      0.
$ Femap Load Set 8 : ST_EQ_HOR_LOAD_ALTER
FORCE           8      22       0      1.    101.      0.      0.
FORCE           8      21       0      1.    101.      0.      0.
FORCE           8      24       0      1.   -101.      0.      0.
FORCE           8      23       0      1.   -101.      0.      0.
$ Femap Constraint Set 1 : LC1
SPC1            1     123       1
SPC1            1     123       2
SPC1            1     123       3
SPC1            1     123       4
SPC1            1     123       5
SPC1            1     123       6
$ Femap Property 1 : D=450
PSHELL          1       1     .45       1               1              0.
$ Femap Material 1 : C35
MAT1            1   3.2+7              .2     2.5    1.-5      0.
GRID            1       0.2495404      .2   -.665       0
GRID            2       0.2495404     2.3   -.665       0
...
GRID         1891       02.343447.2416667      0.       0
CQUAD4          1       1    1016    1014    1015    1017      0.
CQUAD4          2       1    1017    1012    1013    1016    180.
...
CQUAD4        847       1    1890    1886    1124    1123    180.
CELAS2        848    240.      11       1       1       1      0.
CELAS2        849    240.      11       2       1       2      0.
CELAS2        850    690.      11       3       1       3      0.
...
CONM2         893      23       0      .7      0.      0.      0.
ENDDATA 5eb0b168
```

5.2.2 Output File of Reinforcement Calculation (Extract)

```
************************************
******** B A U M E S S . E X E *******
******** DESIGN OF SHELLS ************
****** Windows Release 6.50 **********

Modelfile: D:\22-1668_Mansourah_SANDWICK_RUE\10\22-1668-10_STAT-1-1.modfem
Date:      Mittwoch, 3. August 2022 16:50:49

*** SINGLE LOAD CASES ***
  LC  DESCRIPTION                                        TYPE   GAM
   1  LC1 - DEAD_LOAD, Simcenter NASTRAN Case 1             G  1.35
   2  LC1 - MACHINE_LOAD, Simcenter NASTRAN Case 2          G  1.35
   3  LC1 - LIVE_LOAD, Simcenter NASTRAN Case 3             Q  1.50
   4  LC1 - MAX_MATERIAL_LOAD, Simcenter NASTRAN Case 4     Q  1.50
   5  LC1 - ST_EQ_VER_LOAD, Simcenter NASTRAN Case 5       A1  1.00
   6  LC1 - ST_EQ_VER_LOAD_ALTER, Simcenter NASTRAN Case   A1  1.00
   7  LC1 - ST_EQ_HOR_LOAD, Simcenter NASTRAN Case 7       A1  1.00
   8  LC1 - ST_EQ_HOR_LOAD_ALTER, Simcenter NASTRAN Case   A1  1.00



*** DESIGN PARAMETER ***
  Steel grade:          Grade 60 ASTM A 615
  Yield strength:       FYD  = 378.38 MPa

  Concrete grade:       fc 35 (5000 PSI) ACI 318M-05
  Compression strength: FCD  =  18.67 MPa

  Bottom spacing:       d1,L =    8.0 cm
  Bottom spacing:       d1,T =    6.0 cm
  Bottom spacing:       d1,m =    7.0 cm
  Upper spacing:        d2,L =    8.0 cm
  Upper spacing:        d2,T =    6.0 cm
  Upper spacing:        d2,m =    7.0 cm

*** MODEL VALUES ***
  Load cases:           NLF  =      8
  Plate elements:       NE   =    847

*** PLATE PROPERTIES ***
  Plate thickness plate property 1:    45.0 cm

*** OUTPUT OF 16 DESIGN LOADCOMBINATIONS ***

*** MAX nxx AND CORRESPONDING INTERNAL FORCES ***

BEM LC  EL NR           nxx       nyy       nxy       qxz       qyz       mxx       myy       mxy
     1      1         61.15    -10.19    -10.49    201.61    -30.69    -31.38     -2.54     27.47
     1      2         47.94    -26.07     -0.06    193.84    -79.26    -18.04     -3.64     18.36
...
    16    847         58.47     -3.60    -23.60   -335.45     83.87     22.89      4.69    -24.78

*** OUTPUT OF REINFORCEMENT RESULTS ***

      ELNR       H        asx+      asx-      asy+      asy-       asw
         1    0.45        5.23      3.99      2.70      1.75     18.46
         2    0.45        3.28      2.21      2.02      1.74     19.87
...
       847    0.45        5.42      3.62      2.78      1.77     30.03
ENDE
```

Mansourah Massarah Gold Project
Comment Resolution Log Sheet / Compliance Report

PROJECT: MANSOURAH MASSARAH GOLD PROJECT **PROJECT NO.:** 216

STAGE: 4 - PROJECT EXECUTION. **REVIEW LOG DOC. NO. & REV.:**

ACONEX REF. NO. & DATE:

STATUS CODE: A - APPROVED, C - APPROVED WITH COMMENTS, R - REVISE & RESUBMIT & O - OTHERS (INFORMATION, REFERENCE, REVIEWED)

DOCUMENT NUMBER	Area / Title	Comment No.	PMC/PMT COMMENT DATE: 2 September 2022	Comment by	EPC Response DATE: 01 February 2024	Status	Remarks
			General Comments (cc)				
MD-216-3120-EG-CV-CAL-1008-A04	3120	1	Please show floor plans with indication of column/beam joints subject to excessive vibration during the commissioning stage	Mr. Hong Su	Noted.	Closed	
MD-216-3120-EG-CV-CAL-1008-A04	3120	2	Please show part plan of screen floor showing equipment layout similar to that in page 9	Mr. Hong Su	Noted.	Closed	
MD-216-3120-EG-CV-CAL-1008-A04	3120	3	Harmonic analysis is performed to determine the steady-state response of the structure subject to sinusoidal loads from the startup/close down and operation of equipment. Harmonic analysis is done from: 1. two critical low frequencies matching the natural frequencies of lateral vibration modes to assess the structural response for startup/close down (see page 55 & page 57); and 2. 50%, 70% & 100% working speeds together with other critical frequencies matching other natural frequencies falling between the working speeds (page 59).	Mr. Hong Su	Noted.	Closed	
MD-216-3120-EG-CV-CAL-1008-A04	3120	4	Please show analysis setting for case B as case A shown in page 64	Mr. Hong Su	Noted.	Closed	
MD-216-3120-EG-CV-CAL-1008-A04	3120	5	Please draw a conclusion from appendix 5	Mr. Hong Su	Noted.	Closed	